SOFTWARE

PACKAGE

FOR

SP12 TURBOFAN

PARAMETERS SIMULATION

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NOMENCLATURE

A	turbine corrected mass flow rate, coefficients of Jacobean matrix;
a	sonic velocity;
C_p	specific heat of the air or gas at constant pressure;
d	moisture content;
F	area;
G	secondary mass flow rate;
H	flight altitude; residuals of mathematical model
i	specific enthalpy;
J	impulse, mass moment of inertia;
k	adiabatic index;
L	specific work;
L_0	stoichiometric ratio of air to fuel masses;
m	bypass ratio; mass;
M	Mach number;
n	rotational speed;
N	power;
p	pressure;
Q	heat flow; independent variables of mathematical model;
q	relative combustion products content of fossil fuel;
R	thrust; gas constant;
S	specific entropy;
T	temperature;
t	time;
U	input parameters;
u	radial clearance;
V	volume;
v	velocity;
X	state parameters;
Y	modeled parameters;
Z	parameter to characterize operating point position on compressor performance map;
π	pressure ratio;
η	efficiency, combustion efficiency;
α	excess air ratio, angular position; heat transfer factor;
Δ	absolute deviation;
ΔT	temperature deviation from standard temperature at this altitude;
ε	relative error;
μ	flow coefficient;
φ	velocity factor;
λ	gas dynamic function of velocity (ratio between current and critical velocities);
σ	pressure recovery;
ω	angular velocity of rotor.

INDEXES

ad	adiabatic value;
base	base conditions;
a	air
f	fan and station at fan discharge;
HP	high pressure rotor
wat	water vapor;
in	inlet and station at inlet discharge, parameters at unit inlet;
int	intake and station at intake discharge;
dis	parameters at unit discharge;
g	gas;
d	parameter of disk;
giv	given value;
c	compressor and station at compressor discharge;
cr	critical values of parameters at the station;
cc	combustion chamber and station at combustion chamber discharge;
bl	parameter of blade;
mech	mechanical losses;
h	Parameters of undisturbed flow before engine inlet, ambient atmosphere;
LP	low pressure rotor;
bleed	air bleed;
fl	flight;
trans	transition duct;
supl	supply of working substance;
cor	corrected parameters;
mode	mode parameter;
calc	calculated value of parameter;
I	impeller;
n	jet nozzle, station at jet nozzle discharge;
NV	nozzle vane;
IP	intermediate pressure rotor;
mix	mixer and station at mixer discharge;
stat	static value of state parameter of working substance; parameter valuve at steady mode; parameters of engine stator;
T	turbine;
f	fuel, products of fuel combustion;
aft	afterburner and station at afterburner discharge;
perf	parameter value, calculated by unit performance map;
Σ	total value;
0	initial estimation;
I	primary flow of TFE;
II	secondary flow of TFE;
$-$	relative value;
$\rightarrow$	vector of values;
*	total parameter of a flow;

ABBREVIATIOINS

QLDM	linear (quick proceeding) dynamic model;
IGV	inlet guide vane;
GTE	gas turbine engine;
HPC	high pressure compressor;
LPC	low pressure compressor;
IPC	intermediate pressure compressor;
ABV	air bleeding valve;
EFF	efficiency;
CC	combustion chamber;
SE	structural element;
AIRCR	aircraft;
LDM	linear dynamic model;
LMB	left mouse button;
OL	operating line;
max	maximum value;
NMM	non-linear mathematical model;
PC	personal computer;
RMB	right mouse button;
HPR	high pressure rotor;
LPR	low pressure rotor;
IPR	intermediate pressure rotor;
HPT	high pressure turbine;
LPT	low pressure turbine;
TFE	turbofan engine;
TJE-1	one-spool turbojet engine;
TJE-2	two-spool turbojet engine;
TFE-2	two-spool turbofan engine;
TFE-3	three-spool turbofan engine;
TFE-2 mix	two-spool turbofan engine with first and second flows mixing;
TFE-3 mix	three-spool turbofan engine with first and second flows mixing;
GTE-1	turboshaft engine with single-spool gas generator;
GTE-2	turboshaft engine with two-spool gas generator;

1 MATHEMATICAL MODELLING OF GTE

Equation Chapter 1 Section 1

1.1 Non-linear steady-state component level model for GTE parameters calculating on steady operating modes

1.1.1 General characteristic of modeling method

Let`s consider GTE non-linear steady-state component level model, which ensures working substance parameters calculation for each module and main engine parameters calculation (thrust, specific fuel consumption, etc.) $\vec{Y}$, using engine parameters of airflow at engine inlet, operating mode parameters U_{om} and unit parameters $\vec{\upsilon}$

$$\vec{Y} = F\left(\vec{U}_{fl}, \vec{U}_{om}, \vec{\nu}\right). \tag{1.1}$$

Information about units is given in a form of component maps, like “black box”, which neglects module structure, but expresses only empirical-analytical correlations between input and output ($\vec{\nu}$) parameters (structure of files with units performances is represented in Appendix A).

Parameters of modules (efficiencies, loss coefficients, etc.) are determined by operating point position on a corresponding unit performance map; operating point position in its turn depends on joint operation of all gas-path units.

Nowadays there are some different concepts for structural arrangement of this model type, as a calculating procedure of gas-path modules parameters matching, are known and applied.

Expounded next model has a lot in common with these concepts and was formed to a large degree under their influence. Model development was focused to ensure short proceeding time, because gas-path diagnostics, identification, transient processes simulating require parameters of all units to be matched. Matching procedure needs big number of iterations (10^4 and more).

Base principles of the model are:

- using of generic gas-path scheme of GTE for data input-output;
- hierarchical modular concept of program structure (e.g. subroutine, which performs all necessary calculations for a single shaft gas-generator is used for all engine types). Subroutines, which are applied to model parameters behavior, are described in Appendix B;
- Unique main program module is formed for each engine type to calculate parameters of gas-path (this allows effective proceeding).

Fig 1.1 shows generic gas-path scheme. It corresponds to afterburning turbofan engine, which comprises of fan with boosters, two-spool gas generator with first and second flows mixing, and afterburners, arranged downstream mixer and in bypass duct. There is a configuration of air bleeds from compressor and air taps to turbine in the upper part of the figure.

Decomposing this scheme and when necessary complementing it with energy consumers of power, produced by free turbine, such as propeller, gas supercharger or electric generator, allows getting model for practically all newly designed gas turbine engines and engines, which are in maintenance.

Gas-path stations, shown in the figure, are fixed, i.e. they are preserved in decomposed model (thus, "unnecessary" stations are excluded from consideration). Also there is a possibility to add stations (to their maximum number, equal to 50) for taking into account unique features of each particular engine.

Parameters of working substance, such as total pressures and temperatures, flow rates, humidity and relative content of fuel combustion products, correspond to each of specified stations.

To calculate them and also to get other parameters of engine working substance, it is necessary to meet the requirement of units' joint operation at steady modes. They are law of matter continuity in gas-path, energy and momentum conservation requirements.

Considered above requirement of units' joint operation cannot be expressed ana-

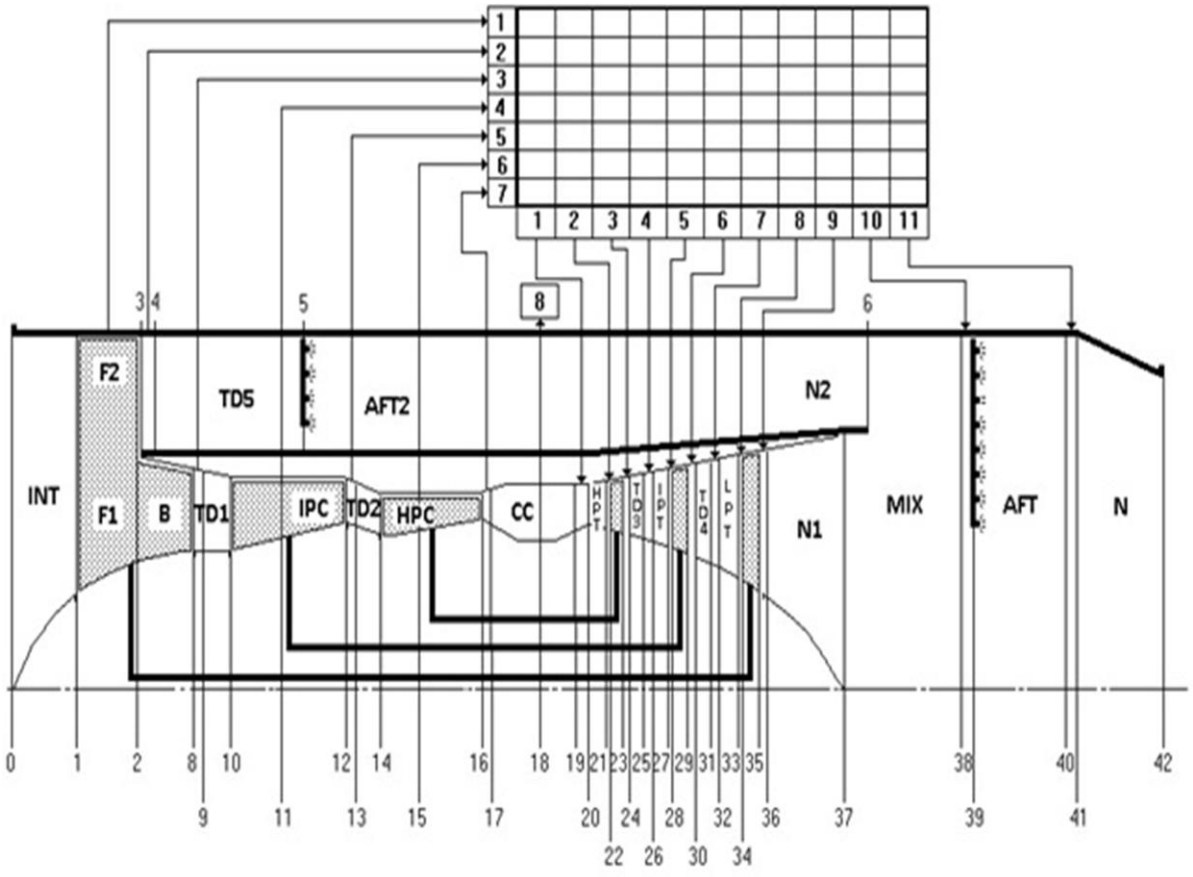


Figure 1.1 – Generic scheme of GTE

lytically, because performances of units have tabular format and physical properties of working substance vary along the gas path relating substance temperature and content. Therefore considered model has an algorithmic form.

The base of specified above algorithm is the procedure of gas-path parameters calculation under given conditions at engine inlet. Calculations are sequentially performed for each module according to mass flow heading.

To provide calculating parameters of working substance at each unit discharge, it is necessary to get parameters $\vec{\nu}$ of this module from its performance map. Arguments of unit performance, which are needed to calculate unit parameters, can be unknown, because they comprise the vector of engine parameters, which need to be calculated. To cope with this problem, needed for sequential calculation variables are given random initial values $\vec{Q}^0$ before the calculation and the variables are united in vector of independent variables $\vec{Q}$. Requirement of units' joint operation is not met after the first iteration, because initial values of variables are given randomly; so the values of variables need correction and in the next iteration; this iteration also may not result in requirement of units joints' operation meeting, so new correction of initial values of variables must be done etc. Thereby, calculating of gas-path parameters is an iterative procedure. It can be expressed as an algorithm of new iteration making, basing on the results of previous one:

$$\vec{Q}^i = f\left(\vec{Q}^{i-1}\right),$$

where i is a number of iteration.

Purposive correction of independent variables is possible only in case of using information about their influence on requirement of units' joint operation. Residuals $\vec{H}$ show the level of requirement of units` joint operation meeting (e.g. airflow continuity equation through compressor spools is expressed in a form of residual between airflow rates through spools with taking into account bleeds). These residuals $\vec{H}$ depend on independent variables $\vec{Q}$, ambient conditions $\vec{U}_{fl}$, operating mode parameters $\vec{U}_{om}$, and parameters of units $\vec{\nu}$. Equating residuals to zeroes, the following equation appears:

$$\vec{H}\left(\vec{Q}, \vec{U}_{fl}, \vec{U}_{om}, \vec{\nu}\right) = 0. \qquad (1.2)$$

As it will be shown next, when residuals calculating by given independent variables, it is necessary to calculate practically all unknown engine parameters, united in vector $\vec{Y}$.

Thereby, model (1.1) has non-linear algebraic equation set form (1.2). Obviously, residuals number must equate to independent variables number to meet the requirement of compatibility.

Newton-Raphson method is applied to solve the equation set (1.2). Iterative procedure is organized in the way to calculate additive correction $\Delta\vec{Q}^i$ to vector of independent variables: $\vec{Q}^i = \vec{Q}^{i-1} + \Delta\vec{Q}^i$, additive correction $\Delta\vec{Q}^i$ is the solution of linear algebraic equation.

$$\vec{H}^i = A^i \Delta\vec{Q}^i,$$

where $A^i = \left(\frac{\partial\vec{H}}{\partial\vec{Q}}\right)^i$ - Jacobian matrix.

As it was previously considered, this model cannot be expressed analytically.

That is why, calculation of Jacobian matrix elements is done numerically by consecutive varying of each independent variable Q_j and further calculating of corresponding residuals $\vec{H}_j$. One side variation is used to minimize calculations:

$$\frac{\partial H_j^i}{\partial Q_j^i} = \frac{H_j(Q_1^i,...,Q_l^i,...,Q_n^i) - H_j\left(\vec{Q}^{i-1}\right)}{\Delta Q_j^i},$$

where *n* – number of residuals.

Calculation finishes when

$$N = \sqrt{\frac{H_1 + ... + H_n}{n}} \prec \varepsilon_R,$$

where N is a standard residual; ε_R is a given calculations tolerance.

1.1.2 Calculating parameters of working process in main units of GTE

1.1.2.1 Calculating physical properties of working substance according to its given compound and temperature (CPFT subroutine)

Unknown parameters are gas constant *R*, enthalpy *i*, entropy *S* heat capacity C_p.

Initial data is humidity *d*, relative combustion products content of fossil fuel *q* and working substance temperature *T*.

Physical properties of working substance components were approximated by 8th degree polynomial.

Calculations are run according to next algorithm:

$$R = \frac{R_a + qR_f + dR_{wat}}{1+q+d};\ C_p = \sum_{j=1}^{8} a_j \left(\frac{T}{1000}\right)^{j-1};\ i = \frac{qC}{1+q} + T\sum_{j=1}^{a} \frac{a_j}{j}\left(\frac{T}{1000}\right)^{j-1};$$

$$S = a_1 1nT + \frac{C_1 - qC_2}{1+q} + \sum_{j=2}^{8} \frac{a_j}{j-1}\left(\frac{T}{1000}\right)^{j-1},$$

where $a_j = \frac{a_{aj} + qa_{fj} + da_{watj}}{1+q+d}$ - coefficients of a polynomials (see. table 1.1) $C = -74282{,}206$; $C_1 = 991{,}24216$; $C_2 = 2221{,}2649$; R is a gas constant ($R_a = 287{,}0406$, $R_f = 311{,}7504$, $R_{wat} = 461{,}589$).

For further considerations this calculation is conventionally denoted as $i(T)$, $S(T)$, or $C_p(T)$, depending on parameter, used in following calculations.

Table 1.1 – Coefficients of polynomial used to approximate physical properties of working substance

	a_a	a_f	a_{wat}
1	999,74634	1536,9851	1967.8267
2	-37,717087	1609,1306	-1342.3066
3	80,897812	798,14526	5007.9883
4	555,63574	-1514,6443	7237.0625
5	-796,18994	1102,8777	6491.7109
6	434,56421	-471,50879	-3471.6755
7	-108,22488	107,71439	992.50928
8	10,272973	-9,940177	-116.13098

1.1.2.2 Calculating temperature and physical properties of working substance according to given enthalpy (subroutine TFI)

Unknown parameters are working substance temperature T, entropy S and heat capacity C_p.

Initial data is humidity d, relative combustion products content of fossil fuel q and enthalpy i_{giv}.

Temperature is calculated by solving the non-linear algebraic equation $i(T) = i_{giv},$, where $i(T)$ is the considered above procedure to calculate physical properties of working substance depending on its temperature. This equation is solved iteratively; herewith temperature for the next iteration is calculated as follows:

$$T_{k+1} = T_k - \frac{i(T_k) - i_{giv}}{C_p},$$

and the following temperature is given as initial approximation:

$$T_0 = \begin{cases} \dfrac{(1+q)\cdot i_{giv} + e + q\cdot f}{g + q\cdot h}, & i_{giv} \le i_{max}; \\ \dfrac{(1+q)\cdot i_{giv} + e_1 + q\cdot f}{g_1 + q\cdot h}, & i_{giv} > i_{max}, \end{cases}$$

where $e=17793{,}9$, $f=1224639$, $h=3571{,}34$, , $e_1 = 180116{,}1$, $g=1050{,}886$ $g_1 = 1214{,}172$ are constants, $i_{max} = 1034139$.

Calculation finishes when the following requirement is met:

$$\left| \frac{i(T_k) - i_{giv}}{i_{giv}} \right| \le \varepsilon_T,$$

where ε_T is given tolerance of physical properties calculating.

1.1.2.3 Calculating temperature and physical properties of working substance according to given entropy (subroutine TFS)

Unknown parameters are working substance temperature T, enthalpy i and heat capacity C_p.

Initial data is humidity d, relative combustion products content of fossil fuel q, and entropy value S_{giv}.

Temperature is calculated by solving the non-linear algebraic equation $S(T) = S_{giv}$, where $S(T)$ is the considered above procedure to calculate physical properties of working substance depending on its temperature. This equation is solved iteratively, herewith temperature value for next iteration is calculated as follows:

$$T_{k+1} = T_\kappa \left(1 - \frac{S(T_k) - S_{giv}}{C_p(T_k)} \right),$$

and the following temperature is given as initial approximation:

$$T_0 = e^{(S_u \cdot W)},$$

where $S_u = \begin{cases} \dfrac{(1+q)\cdot S_{giv} - E + q\cdot F}{G + q\cdot H}, & S_{giv} \ge S_{min}; \\ \dfrac{(1+q)\cdot S_{giv} + E_1 + q\cdot F}{G_1 + q\cdot H}, & S_{giv} < S_{min}, \end{cases}$

$E = 761{,}705$, $F = 12670{,}97$, $G = 0{,}24889$, $H = 0{,}79188$, $E_1 = 432{,}161$, $G_1 = 0{,}28996$, $W = 0{,}238846 \cdot 10^{-3}$ are constants.

Calculation finishes when the following requirement is met:

$$\left| \frac{S(T) - S_{giv}}{S_{giv}} \right| \le \varepsilon_T.$$

1.1.2.4 Calculating parameters of an airflow at engine inlet (subroutine ATMOSX)

Unknown parameters are P_0^* and T_0^* – airflow total pressure and temperature at engine inlet.

Initial data is flight conditions. They can be given in any of considered next parameters sets:

$$1)\, p_0,\ T_0,\ v_{fl};\quad 2)\, p_0,\ T_0,\ M;\quad 3)\, H,\ M,\ \Delta T;\quad 4)\, H,\ v_{fl},\ \Delta T$$

In first two cases, static enthalpy and entropy of airflow, its heat capacity and gas constant are foremost to be calculated: $i_0 = i(T_0),\ S_0 = S(T_0), C_p = C_p(T_0),\ R = R(T_0).$

The next parameters to be calculated are airflow velocity and its total enthalpy:

$$v_{fl} = M\sqrt{\frac{C_p}{C_p - R}RT_0}; \quad i_0^* = i_0 + \frac{v_{fl}^2}{2}.$$

after that, according to known enthalpy i_0^*, airflow total temperature and entropy are calculated:

$$T_0^* = T\left(i_0^*\right); \quad S_0^* = S\left(i_0^*\right). \tag{1.3}$$

And finally, by the known entropy S_0^*, total pressure of airflow is calculated:

$$p_0^* = p_0^{\frac{S_0^* - S_0}{R}}. \tag{1.4}$$

In the last two cases static temperatures and pressures are calculated by the equations that approximate parameters of international standard atmosphere (subroutine ATMOS):

$$H_g = \frac{H}{\left(1 + \frac{H}{6{,}35 \cdot 10^6}\right) \cdot 10^3};$$

When $H_g < 11$

$$p_0 = 1{,}013233 \cdot 10^5 \left(1 - 0{,}0225576 \cdot H_g\right)^{5{,}25};$$
$$T_0 = 288{,}15 - 6{,}5H + \Delta T;$$

when $11 \le H_g \le 20$

$$p_0 = 2{,}2632 \cdot 10^4 \cdot e^{-0{,}157689 \cdot \left(H_g - 11\right)};$$
$$T_0 = 216{,}65 + \Delta T; \tag{1.5}$$

when $H_g \ge 20$

$$p_0 = \frac{5470}{\left[1 + 4{,}615 \cdot 10^{-3} \cdot \left(H_g - 20\right)\right]^{34{,}16}};$$
$$T_0 = 216{,}65 + \left(H_g - 20\right) + \Delta T.$$

Total pressure and temperature are calculated according to considered above algorithm, by known values of static pressure and temperature, calculated in equations (1.5).

1.1.2.5 Calculating critical values of parameters at given temperature (subroutine CRIT)

Unknown parameters are T_{cr}, π_{cr}, v_{cr}.

Initial data is airflow total temperature T_{in}^*, velocity factor φ, humidity d and relative combustion products content of fossil fuel q.

Physical properties are initially calculated by known total temperature of working substance and its compound:

$$i_{in}^{*} = i\left(T_{in}^{*}\right);\quad C_{p\,in} = C_p\left(T_{in}^{*}\right);\quad R_{in} = R\left(T_{in}^{*}\right).$$

The next step is getting T_{cr} by solving the following equation

$$i_{cr} = i_0^{*} - \frac{a\left(T_{cr}\right)^2}{2}, \tag{1.6}$$

where $a\left(T_{cr}\right) = \sqrt{\frac{C_{p\,cr}}{C_{p\,cr} - R_{in}} R_{in} T_{cr}}$ is sonic velocity; $i_{cr} = i\left(T_{cr}\right)$ is an enthalpy, that corresponds to and given content of working substance.

Nonlinear algebraic equation (1.6) is iteratively solved with initial approximation value calculated by the following equation.

$$T_{cr\,0} = \frac{2 \cdot T_{in}^{*}}{1 + \frac{C_{p\,in}}{C_{p\,in} - R_{in}}}.$$

Precising values $T_{cr\,n+1}$ at $\left(n+1\right)$ iteration is done according to the following equation

$$T_{cr\,n+1} = T_{cr\,n} + \frac{i_{in}^{*} - i\left(T_{cr\,n}\right) - \frac{\left[a\left(T_{cr\,n}\right)\right]^2}{2}}{C_p\left(T_{cr\,n}\right)\frac{1 + C_p\left(T_{cr\,n}\right)}{2 \cdot \left(C_p\left(T_{cr\,n}\right) - R_{in}\right)}}.$$

Calculation finishes when the following requirement is met

$$i_{in}^{*} - i\left(T_{cr\,n}\right) - \frac{\left[a\left(T_{cr\,n}\right)\right]^2}{2} < \frac{C_p\left(T_{cr\,n}\right)}{100}.$$

After these, $S_{cr} = S\left(i_{cr}\right)$ and $T_{cr} = T\left(i_{cr}\right)$ are calculated. In these equations $i_{cr} = i_{in}^{*} - \frac{\left[\frac{a\left(T_{cr}\right)}{\varphi}\right]^2}{2}$.

And finally, it is necessary to calculate $\pi_{cr} = \frac{S_{in}^{*} - S_{cr}}{R_{in}}$.

1.1.2.6 Calculating velocities ratio (λ) of working substance (subroutine LAM)

Unknown parameter is λ .

Initial data is parameters of working substance: stagnation temperature T^*, and pressure p^* , mass flow G , humidity d , relative combustion products content of fossil fuel q and cross-section area of gas-path duct F .

Knowing T^*, q and d, and using considered above algorithm it is necessary to calculate critical velocity v_{cr}, enthalpy $i\left(T^*\right)$ and entropy $S\left(T^*\right)$ of working substance. Then airflow velocity v is obtained as a result of solving the following equation

$$i^* = i_{stat}\left(v\right) + \frac{v^2}{2}. \tag{1.7}$$

Nonlinear algebraic equation (1.7) is iteratively solved; herewith initial approximation is calculated as follows

$$v_0 = \frac{G \cdot R\left(T^*\right) \cdot T^*}{p \cdot F}.$$

Preciesed values of v_n at the nth iteration are calculated by the next algorithm:

$$i_{stat\ n} = i^* - \frac{v_n^2}{2};$$

$$S_{stat_n} = S\left(i_{stat_n}\right);$$

$$p_{stat\ n} = \frac{p^*}{e^{\frac{S(T^*) - S_{stat\ n}}{R}}};$$

$$v_{n+1} = \frac{G \cdot R\left(T^*\right) \cdot T^*}{p_{stat\ n} \cdot F}.$$

Calculation finishes when the following requirement is met

$$\left| \frac{v_{n+1}}{v_n} - 1 \right| < \varepsilon_T.$$

Finally velocities ratio is calculated by the next equation

$$\lambda = \frac{v_n}{v_{cr}}.$$

1.1.2.7 Calculating parameters at compressor spool discharge (subroutine KOMPR)

Unknown parameters are total temperature at compressor discharge $T^*_{dis\ C}$, and specific work of compressor L_C

Initial data is parameters of working substance at compressor inlet $T^*_{in\,C}$, q, d and parameters that characterize compressor operation π^*_C and η^*_C.

Physical properties of working substance are initially calculated according to its known parameters at compressor inlet: $S^*_{in\,C} = S\left(T^*_{in\,C}\right); R = R\left(T^*_{in\,C}\right)$.

The following calculations are performed to get total entropy and corresponding total adiabatic enthalpy at compressor discharge: $S^*_{dis\,C} = S^*_{in\,C} + Rln\pi^*_C$; $i^*_{dis\,C\,ad} = i\left(S^*_{dis\,C}\right)$.

And finally unknown parameters are:

$$L_C = \left(i^*_{dis\,C\,ad} - i^*_{in\,C}\right)\frac{1}{\eta^*_C};$$

$$i^*_{dis\,C} = i^*_{in\,C} + L_C; \quad T^*_{dis\,C} = T\left(i^*_{dis\,C}\right).$$

1.1.2.8 Calculating parameters at turbine spool discharge (subroutine TURBIN)

Unknown parameters are temperature at turbine discharge $T^*_{dis\,T}$ and specific work of a turbine L_T

Initial data is parameters of working substance $T^*_{in\,T}$, q, d at turbine inlet and parameters that characterize turbine operation π^*_T and η^*_T.

Physical properties of working substance are initially calculated according to its known parameters at turbine inlet: $S^*_{in\,T} = S\left(T^*_{in\,T}\right); \; R = R\left(T^*_{in\,T}\right)$.

The following calculations are performed to get total entropy and corresponding total adiabatic enthalpy at compressor discharge: $S^*_{dis\,T} = S^*_{in\,T} + Rln\pi^*_T; \; i^*_{dis\,T\,ad} = i\left(S^*_{dis\,T}\right)$.

And finally unknown parameters are:

$$L_T = \left(i^*_{in\,T} - i^*_{dis\,T\,ad}\right)\eta^*_T; \quad i^*_{dis\,T} = i^*_{in\,T} - L_T; \; T^*_{dis\,T} = T\left(i^*_{dis\,T}\right).$$

1.1.2.9 Calculating parameters of combustion chamber (subroutine KAMSG)

Unknown parameters are relative fuel consumption q_f and excess air/fuel ratio α_{CC}.

Initial data is parameters of working substance at combustion chamber inlet $q_{in\,CC}$, d, $T^*_{in\,CC}$, temperature at combustion chamber discharge $T^*_{dis\,CC}$ and combustion efficiency η_{CC} .

Stagnation enthalpy is initially calculated according to known parameters of working substance at combustion chamber inlet and discharge: $i^*_{in\,CC} = i(T^*_{in\,CC})$, $i^*_{dis\,CC} = i\left(T^*_{dis\,CC}\right)$.

Then, fuel enthalpy is calculated at combustion chamber discharge according to known polynomial correlation $i^*_{f\,dis} = i_C + T^*_{dis\,CC}\sum_{j=1}^{8}\frac{a_{f\,j}}{j}\left(\frac{T^*_{dis\,CC}}{1000}\right)^{j-1}$ (coefficients $a_{f\,j}$ are shown in Table 1.1, $i_c = -74282{,}2$).

And, finally, unknown parameters are:

$$q_{dis} = \frac{\left(1+q_{in}+d\right)\left(i^*_{dis}-i^*_{in}\right)}{H_u\eta_{CC}-i^*_{f\,dis}+i_o}; \quad \alpha_{CC} = \frac{1}{L_o\left(q_{dis\,CC}-q_{in\,CC}\right)},$$

where H_u – lower caloric value of the fuel, i_o – fuel enthalpy at combustion chamber inlet, L_o – stoichiometric ratio (ratio between mass flow and fuel consumption in case of complete fuel burning $\left(\alpha_{CC}=1\right)$). These input parameters are given in files with performances of units (Appendix A).

1.1.2.10 Calculating parameters of working substance in conical nozzle (subroutine SOPLO1)

Unknown values are nozzle discharge area F_N, pressure ratio π_N, and parameters of working substance p^*_N, T^*_N, v_N , λ_N at nozzle discharge.

Initial data is parameters of working substance at nozzle inlet $p^*_{in\,N}$, $T^*_{in\,N}$, $G_{in\,N}$ velocity factor φ_N and static pressure at nozzle discharge p_{dis}.

Critical parameters of working substance $T_{N\,cr}$, $\pi_{N\,cr}$, $V_{N\,cr}$ are initially calculated. Considered above algorithm is used for this. Then, $\pi_N = \frac{p^*_{in\,N}}{p_{dis\,N}}$ is calculated and the following condition is checked.

$$\pi_{N\ min} \le \pi_N \le \pi_{N\,cr}, \tag{1.8}$$

where $\pi_{C\ min} = 1{,}005$ - is limiting value, which is aimed to prevent obtaining too big nozzle discharge cross-section areas or reverse gas flow.

If condition (1.8) is met, then gas flow at nozzle discharge is subcritical and next calculations are performed:

$S^*_{in} = S\left(T^*_{in}\right)$; $i^*_{in} = i\left(T^*_{in}\right)$; $R_{in} = R\left(T^*_{in}\right)$ - physical properties of working substance at nozzle inlet;

$S_{dis} = S^*_{in} - R\cdot ln\,\pi_N$; $i^*_{dis} = i\left(S_{dis}\right)$ - physical properties of working substance at nozzle discharge neglecting velocity losses;

$v_N = \varphi\sqrt{2\left|i^*_{in\ N} - i^*_{dis\ N}\right|}$; $\lambda_N = \frac{v_N}{v_{N\ cr}}$; $i_{dis} = i^*_{in} - \frac{v_N^2}{2}$ - enthalpy of working substance at nozzle discharge with taking into account velocity losses;

$T_{N\ stat} = T(i_{dis})$; $p_{N\ stat} = p_{dis}$ – static temperature and pressure at nozzle discharge;

$$F_N = \frac{G_{in}\ R\ T_{stat}}{v_N\ p_0\ \mu_N}.$$

If condition (1.8) is not met and $\pi_N > \pi_{N\ cr}$, then gas flow at nozzle discharge is supercritical and next calculations are performed:

$$T_{stat} = T_{N\ cr};\quad \pi_N = \pi_{N\ cr};\ p_{dis} = \frac{p^*_{in}}{\pi_{N\ cr}};\quad v_N = v_{N\ cr};\ \lambda_N = 1;$$

$$F_N = \frac{G_N\ \cdot R\ \cdot T_{N\ cr}}{v_N\ \cdot p_{dis}}.$$

Situation, when $\pi_N > \pi_{N\ min}$ may appear at intermediate iterations when calculating the condition of units joint operation. To ensure normal iterative process, next accessory procedure is applied:

– $\pi_N = \pi_{N\ min}$;;

– all parameters at nozzle discharge are calculated according to algorithm for subcritical flow (except equation for static enthalpy calculation $i_{dis} = i^*_{in} + \frac{v_N^2}{2}$);

– values of area and velocities ratio are corrected by specially calculated adjustments:

$$F_N = F_N\left[1 + \left(-C_1\left(\frac{p^*_{in}}{p_{dis}}\right)^2 - C_2\frac{p^*_{in}}{p_{dis}} + C_3\right)\right];$$

$$\lambda_N = C_4\left(\frac{p^*_{in}}{p_{dis}}\right)^3 + C_5\left(\frac{p^*_{in}}{p_{dis}}\right)^2 - C_6\frac{p^*_{in}}{p_{dis}} + C_7,$$

$$V_N = \lambda_N \cdot V_{N\ cr},$$

where C_1= -7,3088866, C_2=-0,05,C_3=7,4303882, C_4=-9,04056704, C_5=27,257309, C_6=-18,1798103, C_7 =0,01.

1.1.2.11 Calculating of working substance parameters in bypass duct, taking into account bleed air from compressor (subroutine RP5)

For correct calculation of TFE thrust at modes with open ABVs, it is necessary to calculate parameters of secondary airflow, taking into account parameters of air, bleed from core duct through ABVs.

Unknown parameters are pressure p^*_2, temperature T^*_2 and airflow rate G_2 in bypass downstream air bleeding valves.

Initial data is parameters p_1^*, T_1^*, G_1 of bypass duct upstream ABVs and bleed air parameters $p_{dis\,ABV}^*$, $T_{dis\,ABV}^*$, G_{ABV}.

Mass flow through bypass duct downstream ABVs is $G_2 = G_1 + G_{ABV}$.

Temperature downstream ABVs is calculated from energy conservation law:

$$i_1^* = i(T_i^*);\ i_{dis\,ABV}^* = i(T_{dis\,ABV}^*);\ i_2^* = i_1^* + i_{dis\,ABV}^*;\ T_2^* = T(i_2^*).$$

1.1.2.12 Calculating parameters of working substance taking into account cooling air taps

Unknown values are compound parameters q_3, d_3, temperature T_3^* and airflow G_3 downstream cooling air tap station.

Initial data is parameters of working substance q_1, d_1, T_1^*, G_1 upstream cooling air tap station; cooling air parameters q_2, d_2, T_2^*, G_2.

Pressure change of working substance is negligibly small, that is why it is neglected for this calculation.

Mass flow and content of working substance are calculated regarding to mass conservation law:

$$d_3 = \frac{d_1 G_1 + d_2 G_2}{G_3};\ q_3 = \frac{q_1 G_1 + q_2 G_2}{G_3};\ G_3 = G_1 + G_2.$$

Temperature is calculated regarding energy conservation law:

$$i_1^* = i\left(T_1^*\right);\quad i_2^* = i\left(T_2^*\right);\quad i_3^* = i_1^* + i_2^*;\quad T_3^* = T\left(i_3^*\right).$$

1.1.2.13 Calculating parameters of working substance in bleed stations (subroutine TAU)

Unknown parameter is temperature at discharge T_{bleed}^*.

Initial data is compressor parameters π_C^* and η_C^*, parameters of working substance in the air bleeding station $T_{in\,C}^*$, q, d and indicators, which specify the number of air bleeding station K_π and K_η.

Physical properties of working substance $S_{in\,C}^* = S\left(T_{in\,C}^*\right);\ R = R\left(T_{in\,C}^*\right)$. are initially calculated by the parameters of at compressor inlet.

Then parameters in bleed stations are calculated by the following equations:

$$\pi_{bleed}^* = (\pi_C^* - 1)\cdot K_\pi + 1;$$

$$\eta_{bleed}^* = \eta_C^* \cdot K_\eta.$$

The following calculations are performed to get total entropy and corresponding total adiabatic enthalpy at bleed station: $S_{bleed}^* = S_{in\,C}^* + R ln \pi_{bleed}^*;\ i_{bleed\,ad}^* = i\left(S_{bleed}^*\right)$.

And finally, unknown parameters equal to:

$$i_{bleed}^* = i_{in\,C}^* + \left(i_{bleed\,ad}^* - i_{in\,C}^*\right)\frac{1}{\eta_{bleed}^*};$$

$$T^*_{bleed} = T\left(i^*_{bleed}\right).$$

1.1.2.14 Calculating static parameters (subroutine STATPAR)

Unknown parameters are static temperature T_{st} and pressure p_{st} flight velocity V_{fl} or Mach number M.

Initial data is humidity d, relative combustion products content of fossil fuel q, total pressure p^* and temperature T^*, crossectional area F of the station where it is necessary to calculated static parameters and mass flow G.

First to be calculated is the velocity factor (see subroutine *LAM*) at known parameters (T^*, p^*, G, d, q, F) of working substance.

Then according to given content (d and q) and temperature (T^*) of working substance, the following parameters are calculated: heat capacity c_p, gas constant R and entropy $S\left(T^*\right)$ (subroutine *CPFT*).

Next to be calculated is adiabatic index $k\frac{c_p}{c_p - R}$.

Then gas-dynamic functions $\pi(\lambda) = \left(1 - \frac{K-1}{K+1}\lambda^2\right)^{\frac{k}{k-1}}$ and temperatures $\tau(\lambda) = 1 - \frac{k-1}{k+1}\lambda^2$ must be calculated.

Next static pressure $p_{st} = p^* \cdot \pi(\lambda)$ and temperature $T_{st} = T^* \cdot \tau(\lambda)$ must be calculated.

After these it is necessary to calculate entropy for the static temperature: $S_{st} = S(T^*) - ln\left(\frac{1}{\pi(\lambda)}\right) \cdot R$.

According to known d, q and S_{st} it is needed to calculate static enthalpy i_{st}. And finally, we are to calculate flight velocity and Mach number

$$V = \sqrt{2 \cdot \left(i^* - i_{st}\right)};\ M = \frac{V}{\sqrt{k \cdot R \cdot T_{st}}}.$$

1.2 Specific features of each particular engine type modeling, basing universal mathematical model

1.2.1 List of independent variables

As it was noted in the subsec. 1.1.1, the independent variables list $\vec{Q}$ of the model is chosen in the way to ensure all gas path parameters calculation from inlet to discharge. All necessary independent variables, which are needed for modeling the two spool turbofan without primary and secondary flows mixing, are specified in table 1.2.

Table 1.2 – List to independent variables of two spool TFE

Name	Designation
Corrected rotational speed of low pressure rotor	$n_{LPC\,cor}$
Factor that determines operating point position at constant speed line in performance map of bypass part of fan	$Z_{LPC\,1}$
Factor that determines operating point position at constant speed line in performance map of core part of fan and boosters	$Z_{LPC\,2}$
Corrected rotational speed of high pressure rotor	$n_{HPC\,cor}$
Factor that determines operating point position at constant speed line in performance map of HPC	Z_{HPC}
Turbine inlet temperature	T_G^*
Pressure ratio of HPT	π_{HPT}^*
Pressure ratio of LPT	π_{LPT}^*
Mass flow rate at CC discharge	G_{CC}
Fuel flow in CC	$G_{f\,CC}$
Air temperature at secondary nozzle discharge	$T_{N\,2}^*$
Gas flow rate at secondary nozzle discharge	$G_{N\,2}$
Gas temperature at primary nozzle discharge	$T_{N\,1}^*$
Gas flow rate at primary nozzle discharge	$G_{N\,1}$
Temperature of HPT stator	$T_{st\,HPT}$
Temperature of HPT blade	$T_{bl\,HPT}$
Temperature of HPT disk	$T_{d\,HPT}$
Temperature of LPT stator	$T_{st\,LPT}$
Temperature of LPT blade	$T_{bl\,LPT}$
Temperature of LPT disk	$T_{d\,LPT}$

If necessary, list of independent variables can be complemented. But user should always remember that this will result in model dimension increasing, which in its turn will increase the amount of calculations.

1.2.2 Residuals list

If n is the number of independent variables and L is the number of freedom degrees (it equates number of control actions), then considered above number of residuals for particular engine type equates $(n-L)$. Rest L residuals are formed according to parameters of operating mode $\vec{U}_{om}$, which are given and calculated according to current independent variables values $\vec{Q}$.

SP12 turbofan (TFE without primary and secondary airflows mixing has twenty independent variables and one degree of freedom) has the following equations set:

– residuals of compressor and turbine power, formed for each spool:

$$H_{N\ LP}(\vec{Q}) = \frac{N_{F2}(\vec{Q}) + N_{F1+BOOSTER}(\vec{Q})}{N_F(\vec{Q}) \cdot \eta_{mech\ F} - \Delta N_{takeoff\ F}} - 1 = 0; \quad (1.9)$$

$$H_{N\ HP}(\vec{Q}) = \frac{N_{HPC}(\vec{Q})}{N_{HPT}(\vec{Q}) \cdot \eta_{mech\ HP} - \Delta N_{takeoff\ HP}} - 1 = 0; \quad (1.10)$$

– residual of corrected mass flows through nozzle vanes of turbine spool, calculated according to working substance parameters at turbine spool inlet and turbine performance map:

$$H_{A\ LPT}(\vec{Q}) = \frac{A_{FT\ calc}(\vec{Q})}{A_{FT\ perf}(\vec{Q})} - 1 = 0; \quad (1.11)$$

$$H_{A\ HPT}(\vec{Q}) = \frac{A_{HPT\ calc}(\vec{Q})}{A_{HPT\ perf}(\vec{Q})} - 1 = 0; \quad (1.12)$$

- residuals of nozzle area of primary flow at mixer inlet, nozzle area of secondary flow at mixer inlet and area of common nozzle, and corresponding structural areas:

$$H_{F\ N1}(\vec{Q}) = \frac{F_{N\ 1}(\vec{Q})}{F_{N\ 1\ struct}} - 1 = 0; \quad (1.13)$$

$$H_{F\ N2}(\vec{Q}) = \frac{F_{N\ 2}(\vec{Q})}{F_{N\ 2\ struct}} - 1 = 0; \quad (1.14)$$

- residual of mass flows, calculated according to spool maps of different compressors:

$$H_{G\ LP-IP}(\vec{Q}) = \frac{G_{dis\ LPC}(\vec{Q}) - \Delta G_{bleed}(\vec{Q})}{G_{in\ IPC}(\vec{Q})} - 1 = 0; \quad (1.15)$$

- residuals, formed according to flow continuity equation and energy balance in specified volume (we have formed three volumes, which are CC, transition duct of secondary flow and transition duct between fan turbine and primary nozzle discharge):

$$H_{G\ CC}(\vec{Q}) = \frac{G_{in\ CC}(\vec{Q}) + G_{f\ CC}(\vec{Q}) + \Delta G_{supl\ CC}}{G_{dis\ CC}(\vec{Q})} - 1 = 0; \quad (1.16)$$

$$H_{I\ CC}(\vec{Q}) = \frac{G_{in\ CC}(\vec{Q}) i\left(T_{in\ CC}\right) + G_{f\ CC}(\vec{Q})(H_u \cdot \eta_{CC} + i_0) + \Delta Q_{supl\ CC}}{G_{dis\ CC}(\vec{Q}) \cdot i\left(T_{dis\ CC}\right)} - 1 = 0; \quad (1.17)$$

$$H_{G\ N2}(\vec{Q}) = \frac{G_{in\ N\ 2}(\vec{Q}) + \Delta G_{supl\ N\ 2}}{G_{dis\ N\ 2}(\vec{Q})} - 1 = 0; \quad (1.18)$$

$$H_{I\ N2}(\vec{Q}) = \frac{G_{in\ N\ 2}(\vec{Q}) \cdot i\left(T_{in\ N\ 2}\right) + \Delta Q_{supl\ N\ 2}}{G_{dis\ N\ 2}(\vec{Q}) \cdot i\left(T_{dis\ N\ 2}\right)} - 1 = 0; \quad (1.19)$$

$$H_{G\,N}(\vec{Q}) = \frac{G_{in\,N}(\vec{Q}) + \Delta G_{supl\,N}}{G_{dis\,N}(\vec{Q})} - 1 = 0; \quad (1.20)$$

$$H_{I\,N}(\vec{Q}) = \frac{G_{in\,N}(\vec{Q}) \cdot i\left(T_{in\,N}\right) + \Delta Q_{supl\,N}}{G_{dis\,N}(\vec{Q}) \cdot i\left(T_{dis\,N}\right)} - 1 = 0; \quad (1.21)$$

$$H_{G\,N1}(\vec{Q}) = \frac{G_{in\,N\,1}(\vec{Q}) + \Delta G_{supl\,N\,1}}{G_{dis\,N\,1}(\vec{Q})} - 1 = 0; \quad (1.22)$$

$$H_{I\,N\,1}(\vec{Q}) = \frac{G_{in\,N\,1}(\vec{Q}) \cdot i\left(T_{in\,N\,1}\right) + \Delta Q_{supl\,N\,1}}{G_{dis\,N\,1}(\vec{Q}) \cdot i\left(T_{dis\,N\,1}\right)} - 1 = 0; \quad 1.23$$

where $\Delta G_{supl\,CC}$, $\Delta G_{supl\,N\,2}$, $\Delta G_{supl\,N\,1}$, $\Delta Q_{supl\,CC}$, $\Delta Q_{supl\,N\,2}$, $\Delta Q_{supl\,N\,1}$ are additional supplying of working substance or heat energy to combustion chamber, transition duct of bypass and transition duct between mixer and discharge of common nozzle correspondingly (these are instrumental parameters, which function will be explained in subsection 1.4);

- residuals of heat flows in structural elements of turbine

$$H_{Q\,stat\,i}(\vec{Q}) = \frac{\alpha_{stat\,i} \cdot F_{stat\,i} \cdot \left(T^*_{in\,GV\,i} - T_{stat\,i}\right) + \Delta Q_{stat\,i}}{\alpha_{stat\,cool\,i} \cdot F_{stat\,cool\,i} \cdot \left(T_{stat\,i} - T^*_{cool\,stat\,i}\right)} - 1 = 0; \quad (1.24)$$

$$H_{Q\,bl\,i}(\vec{Q}) = \frac{\alpha_{bl\,i} \cdot F_{bl\,i} \cdot \left(T^*_{in\,IMP\,i} - T_{bl\,i}\right) + \Delta Q_{bl\,i}}{\alpha_{bl\,cool\,i} \cdot F_{bl\,cool\,i} \cdot \left(T_{bl\,i} - T^*_{cool\,bl\,i}\right) + k_{lock} \cdot T_{lock} \cdot \left(T_{bl\,i} - T_{d\,i}\right)} - 1 = 0; \quad (1.25)$$

$$H_{Q\,d\,i}(\vec{Q}) = \frac{k_{lock} \cdot F_{lock} \cdot \left(T_{bl\,i} - T^*_{d\,i}\right) + \Delta Q_{d\,i}}{\alpha_{d\,cool\,i} \cdot F_{d\,cool\,i} \cdot \left(T_{d\,i} - T^*_{cool\,d\,i}\right)} - 1 = 0; \quad (1.26)$$

where i is the number of rotor (1 is HPR, 2 – LPR); $\Delta Q_{bl\,i}$, $\Delta Q_{stat\,i}$, $\Delta Q_{d\,i}$ are additional heat supplies to i[th] structural element, used as instrumental variable (its function will be explained next)

- residual, formed according to given control parameter and control parameter, calculated by current independent variables:

$$H_{Uom}(\vec{Q}) = \frac{U_{om}(\vec{Q})}{U_{om\,giv}} - 1 = 0,, \quad (1.27)$$

where $\vec{Q}$ is the vector of independent variables, which are presented in the Table 1.2.

1.2.3 List of calculated parameters

As it is seen from considered above residuals analysis, to calculate them according to given $\vec{Q}$, it is necessary to calculate all main parameters of working process. That is why, almost all needed unknown parameters of working process $\vec{Y}$ are calculated automatically as a result of equation set solving (1.2). Exclusions are some parameters, which are derived from parameters of working process (e.g. total pressure ratio in com-

pressor, thrust, different parameters, used to control engine transient operation and diagnostics, etc.); let`s name them optional parameters. For convenient data exchange, engine parameters are united in some groups (for exchange between program components), and all groups form one global structure (for simple information outputting and for arbitrary setting parameters of operating mode). Global structure is shown in Appendix D.

1.2.4 Program implementation of the static model

Software structure for GTE parameters modeling at steady modes is shown in fig 1.2. It qualitatively expresses software content and interaction of its components.

Programs for throttle, altitude-velocity or climatic performance calculating can be main program. These programs comprise the software.

To calculate mode parameters of TFE at steady modes, the caller refers to the program, which solves set of non-linear algebraic equations (RSUMN). This program expresses unknown variables $\vec{Q}$ in normalized form. All variables are numbered and numbered in succession from 1 to 20. Universal programs of thermodynamic calculations have implement independent variables in dimensional form. That is why normalized unknown variables must be transformed to dimensional form and also it is necessary to specify the correlation between independent variables in specified program and program that calculates residuals (PDST2). Main program also calls subroutine that runs the thermodynamic calculation of GTE gas-path.

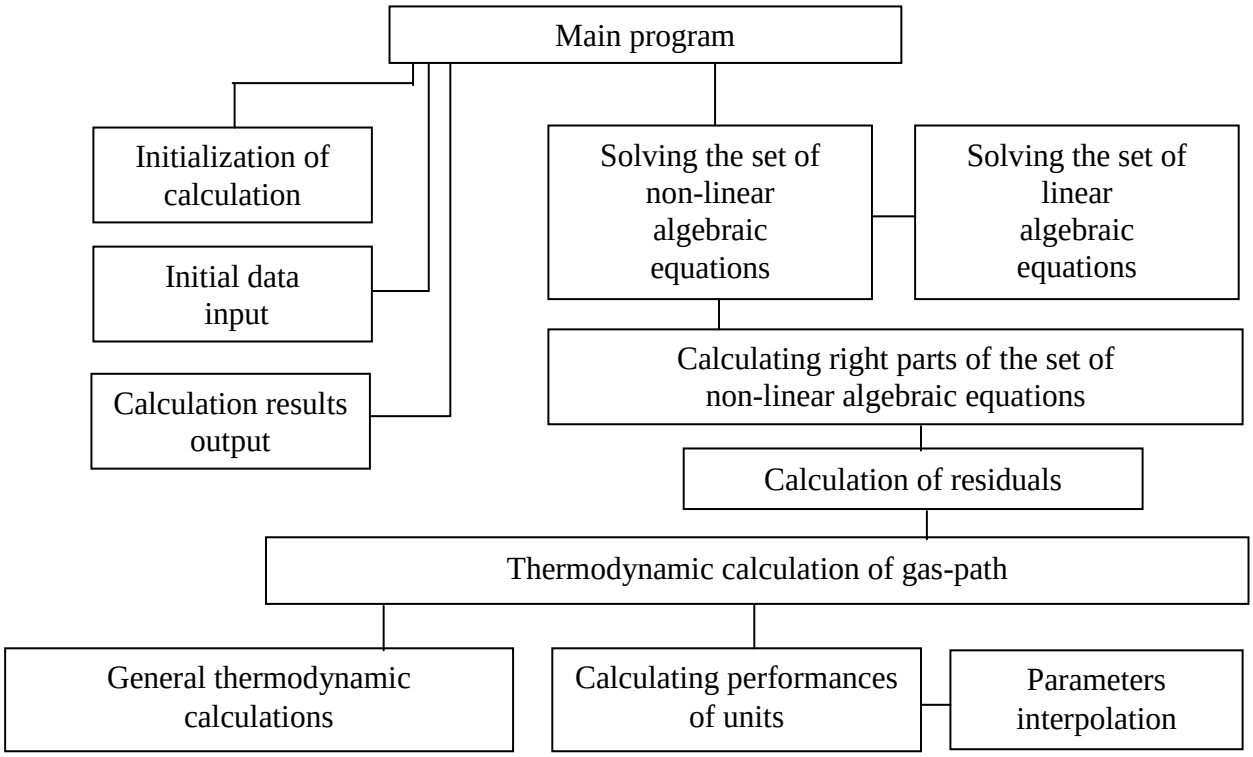


Figure 1.2 – Structure of software that implements unit-level mathematical model of GTE operation at steady modes

Program of thermodynamic calculations gets the parameters of two-shaft TFE which correspond to given independent variables. These programs consist of next universal modules:

- thermodynamic calculation of single-spool gas generator (GG1);
- thermodynamic calculation of two-spool gas generator (GG2).

Appendix C comprises of information about purposes, calling, input-output parameters of each subroutine.

Calling order of subroutines, which are used to calculate three-spool TFEmix, is expressed in fig. 1.3.

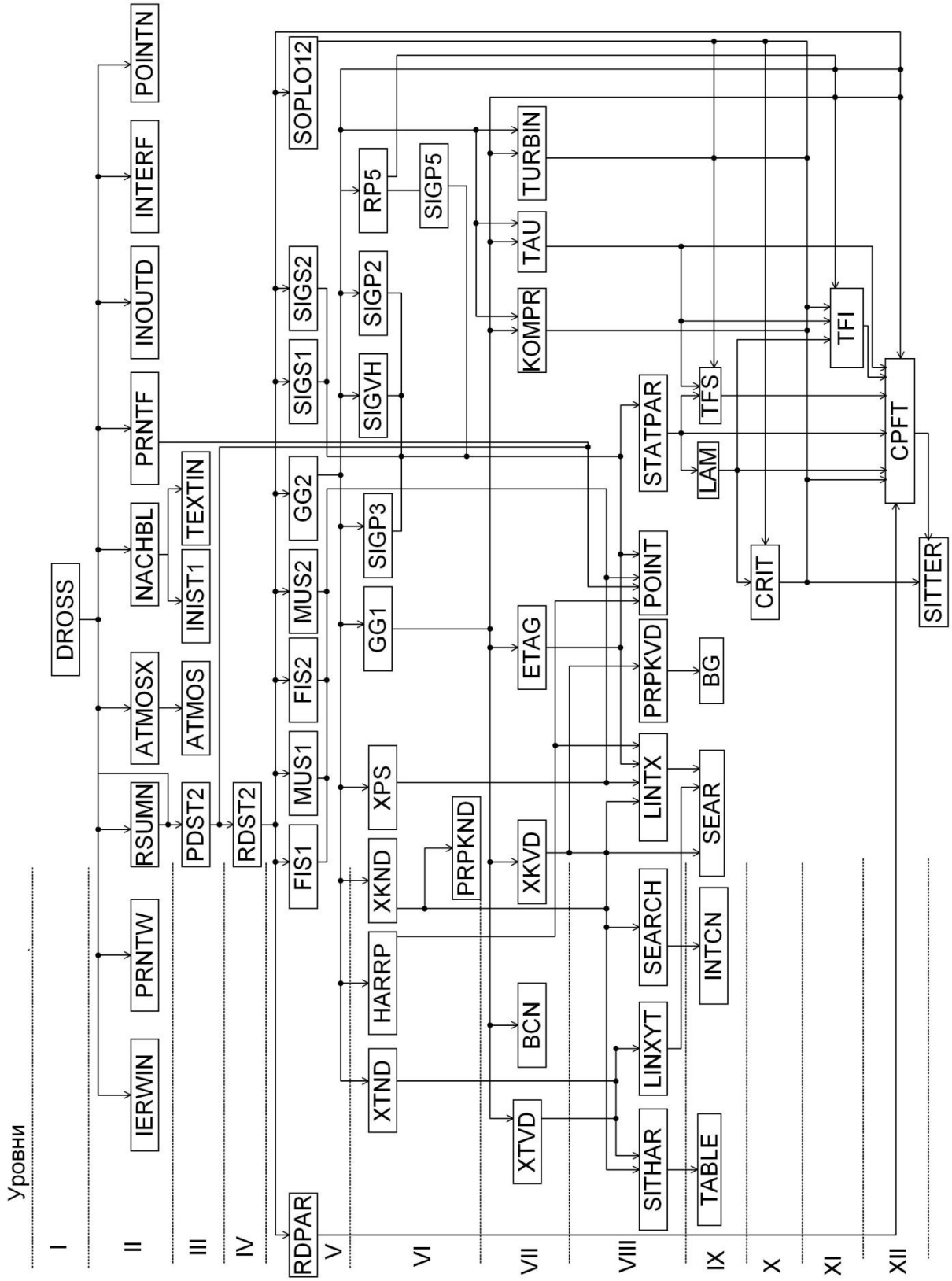


Figure 1.3 –Calling order of subroutines, which take part in static performance of 2 spool TF

1.3 Non-linear unit-level dynamic model of GTE

Changing of control and ambient actions violate the condition of units` joint operation that in its turn creates transient processes in the engine. These processes are limited in time because of some reasons, which we will name dynamic factors.

Non-linear unit-level dynamic model in a form of an algorithm that calculates parameters of GTE at transient modes can be expressed as follows:

$$\vec{Y} = F\left(\vec{Y}_o, \vec{U}_{om}, \vec{U}_n, \vec{v}, t\right), \tag{1.28}$$

where $\vec{Y}_o$ are initial values of parameters at initial steady mode.

To form this algorithm, it is reasonable to use the considered above algorithm of GTE parameters calculation at steady modes and corresponding software (fig. 1.2), taking into account the changes in equations of working process and units performances done because of dynamic factors action. This approach allows calculating parameters at steady and transient modes by single model.

1.3.1 Rotors inertia effect on equations of working process

The most significant dynamic factor of GTEs is inertia of rotors. It appears because equation of power balance for steady modes is violated, because power, produced by turbine, is spent to drive not only compressor, but also to accelerate the rotor. Equation of rotor motion is shown next:

$$J\frac{d\omega}{dt} = \Delta M_{exc}, \;\; or \;\; \frac{dn}{dt} = \frac{N_T\eta_{mech} - N_C - \Delta N_{takeoff}}{n}\left(\frac{30}{\pi}\right)^2\frac{1}{J}, \tag{1.29}$$

where J is inertia moment of rotor rotating masses; N_T is power produced by turbine, N_C and $\Delta N_{takeoff}$ are correspondingly power consumed by compressor and devices, n is rotational speed of rotor; ΔM_{exc} is an excessive torque moment, transmitted through shaft.

1.3.2 Compressibility of working substance effect on equations of working process

Besides compressor power matching turbine power, also it is necessary to match air and gas flows through each engine unit. Mass flow balance at steady modes can be formulated as follows: mass flow at discharge of each engine unit $G_{dis\,i}$ equals to sum of mass flow at inlet of this engine unit $G_{in\,i}$ and mass flow supplied to this unit $G_{supl\,i}$.

$$G_{dis\,i} = G_{in\,i} + G_{supl\,i}.$$

For example, mass flow of air at combustion chamber discharge equals to sum of mass flow of air at combustion chamber inlet and mass flow of fuel, sprayed to combustion chamber ($G_a + G_f = G_g$).

These requirements can be violated at transient modes because of working substance compressibility. Compressibility shows up in the fact that it is necessary to change thermodynamic parameters of working substance (pressure and temperature) at all stations of gas-path to change engine operating mode; this is caused by finite volume of each gas-path unit, and to make this change, mass and energy of these volumes must be changed. Thereby, at transient mode mass flow at discharge of any engine unit is not equal to sum of mass flow at inlet of engine unit and mass flow supplied to this unit, because mass of working substance, stored inside unit volume changes. With the increase of an operating mode, mass flow at unit discharge becomes lower than the considered above sum, and inversely, with the decrease of an operating mode, the mass flow becomes higher. Unsteady balance equation of ith unit mass has the following form:

$$G_{dis\,i} = G_{in\,i} + G_{supl\,i} - \frac{dm_i}{dt}, \tag{1.30}$$

where m_i is mass of working substance, stored in ith specified volume;

$G_{in\,i}$, $G_{dis\,i}$ are mass flows at ith unit inlet and discharge;

$G_{supl\,i}$ is an additional supply of working substance to ith volume.

That fact, that working substance accumulating results in energy accumulating in the considered volume, must be taken into account by using the following energy balance equation for the volume:

$$i^*_{in\,i} \cdot G_{in\,i} + i^*_{supl\,i} \cdot G_{supl\,i} + \Delta Q_i = i^*_{dis\,i} \cdot G_{dis\,i} + C_p \cdot m_i \cdot \frac{dT^*_{dis\,i}}{dt}, \tag{1.31}$$

where $i^*_{supl\,i}$ is an enthalpy of supplied working substance to ith volume;

ΔQ_i is an additional heat energy, supplied to ith volume.

Let`s substitute the mass of working substance by its pressure and temperature ($m_i = \frac{p^*_{dis\,i} \cdot V_i}{R \cdot T^*_{dis\,i}}$, where V_i is the ith volume) and, having performed the necessary transformations of these equations, present them in the following form:

$$\frac{dm_i}{dt} = G_{in\,i} + G_{supl\,i} - G_{dis\,i}; \tag{1.32}$$

$$\frac{dT^*_{dis\,i}}{dt} = \frac{RT^*_{dis\,i}}{C_p V_i p^*_{dis\,i}} \left[i(T^*_{in\,i}) G_{in\,i} - i\left(T^*_{dis\,i}\right) G_{dis\,i} + i\left(T^*_{supl\,i}\right) G_{supl\,i} + \Delta Q_i \right]. \tag{1.33}$$

Analysis of equations (1.32) and (1.33) shows that intensity of transient processes, caused by compressibility of working substance, is inversely proportional to specific volumes of gas-path. Obviously, it is necessary to consider the biggest parts of gas-path

Volumes of all engine units are generally considered. These results in drastic increase of order of differential equations set that express engine dynamics. To simplify the model and to prevent the significant inaccuracy, dynamics of small gas-path volumes can be neglected or some small volumes can modeled as a single one. For example, compressor volume is modeled jointly with combustion chamber volume. So, combustion chamber volume, used in calculations equals to sum of combustion chamber actual volume and volume of compressor`s gas-path.

Before forming dynamic model of an engine, it is possible to estimate the necessity to take into account the dynamics of volumes. For this, time constant of the volume and characteristic time of modeled processes must be compared.

To make this comparison, user should know the characteristic time of processes that will be studied with the newly formed model. So, for example, time constant of engine transient behavior during fuel flow change, acceleration or deceleration processes is determined by the dynamics of rotor, and its value τ is few seconds (depending on rotor inertia). Time constant to model the frequency response of an engine or an automatic control system is determined by studying frequency range. Minimum time equals to minimum oscillatory period $1/f_{max}$, where f_{max} is the maximum frequency.

Equation to calculate time constant of volume τ_1 is deduced from equation (1.33) and is given below:

$$\tau_i = \frac{RT^*_{dis\,i}}{C_p V_i p^*_{dis\,i}}.$$

If $\tau > (3\ldots5)\tau_i$, than this volume can be neglected.

For example, combustion chamber, which has volume $V = 0{,}1\ m^3$, gas temperature and pressure at combustion chamber discharge $T = 1500\ K$, $p^*_{dis\,i} = 2 \cdot 10^6\ Pa$, has time constant $\tau_1 = 0{,}025\ s$. To study the transient behavior of an engine with characteristic time τ equal to some seconds, this volume dynamics can be neglected.

1.3.3 Heat exchange with structural elements effect on the equation of working process

When modeling steady modes of engine operation, equations of energy balance are formed in the way that energy at unit discharge must equate to the sum of energies at unit inlet and energy, supplied to working substance in this unit. So, it is supposed that unit has no heat exchange with the ambience, particularly with structural elements (SE) comprising rotor and stator. The suggestion about absence of heat exchange between these elements can be explained in the way that engine case and internal cavities of rotor are heat-insulated, there are no heat exchange with atmospheric air, and temperature of parts is equal to temperature of working substance in the place, where these parts are placed.

Temperature inertia of parts results in equality violating between temperature of working substance in gas-path and temperature of parts, which contact with working substance. Temperature inertia appears in energy exchange between working substance in gas-path and part, and this fact also must be accounted. In case of engine operating mode increasing, it is necessary to supply heat energy to parts, and otherwise, in case of engine operating mode decreasing, it is necessary to retract heat energy from parts to working substance in gas-path. This process takes some time, which is determined by heat capacity of parts. Finally, we can conclude that there is a delay in parameters changing at transient modes, caused by energy exchange between working substance and SEs.

The methodology of taking into account unsteady heat exchange in structural elements will be explained for turbine unit. From one side turbine is the most termostressed unit, from other side the significant increasing of temperature at turbine inlet needs intense cooling of its structural elements by air taken from compressor; these facts prove that heat exchange processes significantly affect turbine performances.

While steady modes modeling, working substance and structural elements are thermally balanced, which means that their temperatures are constant, and heat flow from gas to structural elements equates to heat flow from structural elements to cooling air, which is brought back to gas-path. This balance is violated at transient modes, and corresponding change of heat state can be expressed by equations of heat exchange at surfaces of the part and by equations of heat conductivity inside the part. Generally, these are partial differential equations, and their solution together with gas-path equations consumes significant calculating capacities. For this reason it is appropriate only for precise analysis of SE thermal state.

For solving tasks of control and gas-path parametric diagnostics, it is permissible to apply the simplest models of SEs heat state with following assumptions:

1) parameters of working substance in axial direction are assumed to equate to average values at unit inlet; parameters variation in radial and circumferential directions are neglected;

2) temperature change in volume is neglected;

3) heat convection coefficient is assumed to be constant for all surfaces of the part;

4) heat exchange between gas and SE is performed by convection; also it is assumed that because of high intensity of gas and cooling flows, mass forces make no significant influence on convection;

5) heat exchange by radiation means is neglected;

6) temperature of cooling air equates to its temperature at station, from which it was bleed.

Then each group of structural elements can be considered as single-volume thermal element, which dynamics equation of its heat exchange with working substance is next:

$$C_i m_{SE\,i} \frac{dT_{SE\,i}}{dt} = Q_{in\,i} - Q_{dis\,i}, \qquad (1.34)$$

where C_i is a specific heat capacity of ith SE; $m_{SE\,i}$ is a mass of ith structural element; $Q_{in\,i}$ is a heat flow to ith SE; $Q_{dis\,i}$ is a heat flow from ith SE to cooling air.

The scheme for an analysis of nonstationary heat exchange in turbine, formed with taking into account the considered above assumptions is expressed in fig. 1.4.

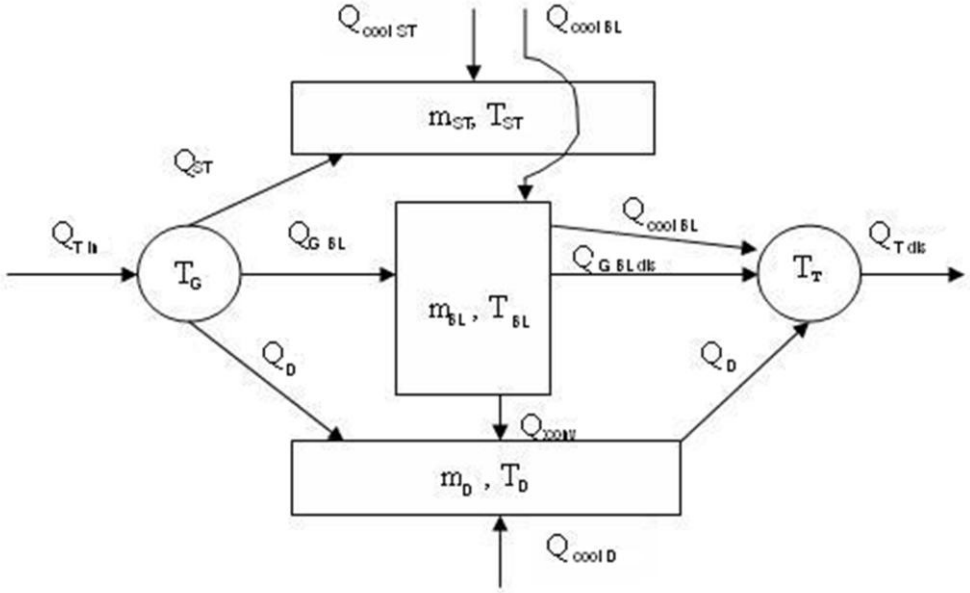


Figure 1.4 – Analysis scheme of heat exchange processes in turbine

It is appropriate that turbine consists of three parts (SEs), which are accumulators of heat energy: stator with nozzle vanes (mass m_{ST}), rotary blades (mass m_{BL}) and disk (mass m_D).

Heat flows, shown in diagram, express heat exchange between gas and SE, between cooling air and SE, and also between two adjacent SEs (e.g. blade and disk) by heat contact means.

Heat flows from gas to SE and from SE to cooling air are calculated according to the formula for the forced convection:

$$Q_i = \alpha_i \cdot F_i \cdot (T_{g\,i} - T_{SE\,i}), \tag{1.35}$$

where α_i is heat convection coefficient on ith surface of SE, F_i is an area of heat exchange surface of ith SE, $T_{g\,i}$ is temperature of gas, streamlining the ith SE, $T_{SE\,i}$ is temperature of ith SE. Generally, coefficient of heat convection depends on many physical parameters, like flow velocity, heat conductivity, viscosity, form and sizes of ducts. Its calculation is performed by the criterion dependence given next:

$$Nu = f(Fo,\ Re,\ Gr,\ Pr),$$

where Nu is the Nusselt number;

Fo is the Fourier number (expresses the influence of unsteadiness);

Re is the Reynolds number (expresses influence of the forced convection);

Gr is the Grashof number (expresses influence of free convection);

Pr is the Prandtl number (expresses physical properties of heat-transfer agent)

Grashof number can be neglected, because free convection contributes insignificant percent in total heat exchange between gas flow and cooling air flow and turbine SE. Prandtl number can be assumed constant for gas mixture from air and combustion products, because it varies insignificantly in the range of operating temperatures.

Neglecting Fourier number variation at transient processes and averaging heat convection coefficient, criterion dependence can be simplified to follows:

$$Nu = \alpha Re^{k}.$$

It is known, that index k, calculated for different heat exchange cases, is within the range 0,5…0,8. Variation of $7 \cdot 10^4 < Re < 3 \cdot 10^5$ results in 3…8% error in Nusselt number calculating, and variation of Reynolds number $10^4 < Re < 10^8$ results in 10…15% error. Error of unsteady heat exchange calculation is commensurable with error of Nusselt number calculation, so index k can be assumed constant for these calculations.

Heat flow between turbine blades and disk happens by contact heat exchange. This contact is considered as heat resistance with heat convection coefficient k_{lock} as follows:

$$Q = K_{lock} F_{lock} (T_{bl} - T_d). \tag{1.36}$$

For considered above heat energy accumulators (fig. 1.4), heat inertia of turbine spool, from equation (1.34), will be described by the following equations set:

$$\begin{aligned}
C_{st} m_{st} \frac{dT_{st}}{dt} &= \alpha_{st\,i} \cdot F_{st\,i} \cdot \left(T^{*}_{in\,NV\,i} - T_{st\,i}\right) - \\
&- \alpha_{st\,cool} \cdot F_{st\,cool} \cdot \left(T_{st} - T^{*}_{cool\,st}\right) + \Delta Q_{st}; \\
C_{bl} m_{bl} \frac{dT_{bl}}{dt} &= \alpha_{bl} \cdot F_{bl} \cdot \left(T^{*}_{in\,IMP} - T_{bl}\right) - \\
&- \alpha_{bl\,cool} \cdot F_{bl\,cool} \cdot \left(T_{bl} - T^{*}_{cool\,bl}\right) - k_{lock} \cdot T_{lock} \cdot \left(T_{bl} - T_d\right) + \Delta Q_{bl}; \\
C_d m_d \frac{dT_d}{dt} &= k_{lock} \cdot F_{lock} \cdot \left(T_{bl} - T_d\right) - \alpha_{d\,cool} \cdot F_{d\,cool} \cdot \left(T_d - T^{*}_{cool\,d}\right) + \Delta Q_d.
\end{aligned} \tag{1.37}$$

Operation of turbine airfoil array can be expressed in the following form:

$$L_T = L_0 - Q_{qk},$$

where Q_{qk} is the amount of heat, retracted from gas because of heat exchange,

L_0 theoretically calculated specific work, which gas may do in airfoil array in case of absent heat exchange.

Theoretical bases of heat exchange between SE and gas or cooling air modeling, and theoretical bases of heat state analysis of SEs of gas turbines and other GTE units are expounded in the papers [1 - 7]. Reliability of developed simplified models significantly depends on qualification and experience of engineer who synthesizes these models. Involving specialists in thermal calculations to model synthesizing process allows escaping rough errors.

But before starting synthesis it is reasonable to examine the necessity of taking into account heat exchange between gas and structural elements. To do this engineer must compare time constant of SE with characteristic time of investigated processes τ.

Equation for time constant of SE is deduced from equations (1.37):

$$\tau_i = \frac{C_i m_i}{\alpha_i F_i},$$

where C_i is specific heat capacity of SE material;

m_i is mass of SE;

α_i is heat convection coefficient;

F_i is an area of surface, where heat exchange happens.

If $\tau \ll \tau_i$ then heat exchange can be neglected.

For example, disk with parameters $C_i = 440\frac{J}{kg \cdot K}$, $m_i = 50kg$, $\alpha_i = 500\frac{W}{m^2}$, $F_i = 0{,}12m^2$, time constant equals to $\tau_1 \approx 370\ s$. Time of transient processes in gas-path is equal to some seconds. Temperature change during this time is very small, which allows neglecting dynamic heating of the disk while modeling transient processes in gas-path.

1.3.4 Heating degree of an engine effect on performances of GTE units

Except considered above dynamic factors, changes in performances of units effect transient processes in engine. These changes are caused by difference in geometrical parameters of an engine at steady and transient modes at the same value of parameter, which determines the operating mode of an engine [8, 9]. So, for example, when engine rotor accelerates, temperature of blades rises slower than rotational speed. That is why less blades elongations and hence higher radial clearances and lower turbine efficiency at transient modes correspond to π_T and n_T values at steady modes. To cope with mentioned above problem it is necessary to apply dynamic correction of efficiency, which can be expressed as follows:

$$\eta_T = \eta_{T0}\left(1 + k\frac{\delta\Delta}{l}\right), \tag{1.38}$$

where η_{T0} is an efficiency, calculated by performance of unit (this value corresponds to steady mode and is determined according to radial clearance at this steady mode);

$k = -1{,}8...-2{,}2$ is an influence factor that depends on blade length (higher k correspond to long blades, and "-" shows that the higher radial clearance is, the lower efficiency is and inversely, the lower radial clearance is, the more efficient this mode is);

l is blade length;

$\delta\Delta$ is a clearance deviation from clearance of warmed up engine (it is necessary to know the clearance value, in case engine operates at steady mode with current parameters).

To calculate $\delta\Delta$ let`s use the following formula:

$$\delta\Delta = (u_R - u_{CASE})_{\tau\to\infty} - (u_R - u_{CASE})_{\tau\to t}, \tag{1.39}$$

where $(u_R - u_{CASE})_{\tau\to\infty}$ is the clearance in turbine stage, after absolute warming-up (i.e. parts do not change their sizes because of warming because they have reached thermal equilibrium);

$(u_R - u_{CASE})_{\tau\to t}$ is a current clearance, corresponding to dynamic heating of structural elements;

u_R - radial displacement of rotor (disk + blade);

u_{cm} - stator displacement.

Non-stationary thermal finite element model of stator and impeller that takes into account cooling of all elements must be implemented to synthesize an appropriate model of clearance change between blade and case [10]. Analysis with this model gives the transient performances of check points displacing under temperature loading. Approximate shape of transient performances is expressed in fig 1.5.

Absolute clearance depends on some factors: check point displacement under centrifugal forces action, under gas dynamic forces action, under pressure and under thermal expansion. Displacements, caused by first three factors, are almost immediate, and can be neglected for dynamic clearance $\delta\Delta$ calculation. Only "warming up" displacements, must be taken into account.

Displacement of the check point (CP) caused by temperature action is calculated from transient performance, which in its turn is simulated for each separate part (stator, disk and blade):

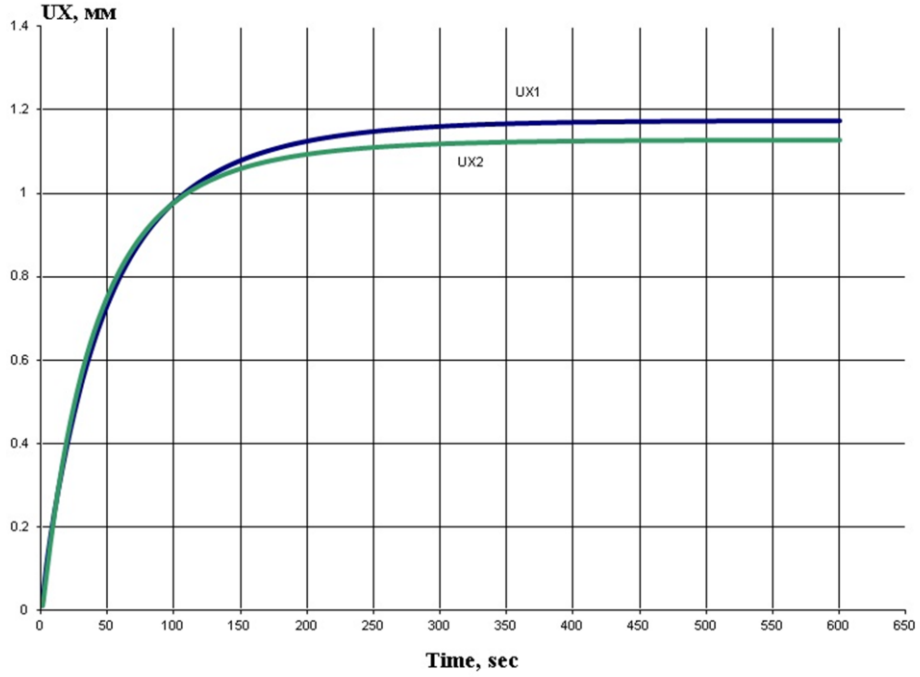


Figure 1.5 - Transient performance of check points displacement

1.3.5 Dynamic model at given ambient and control actions changing in time

1.3.5.1 Modelling the dynamics of clearance between HPT rotor blades and facing them casing

There are three factors affecting the HPT clearance dynamics. They are centrifugal forces due to rotation, heating due to contact with gases in the gas path and cooling by cooling air from active clearance control system.

The model of clearance dynamics calculates the deviation of current (dynamic) clearance from clearances at steady modes:

$$\delta\Delta = \Delta_{dyn} - \Delta_{st}, \quad (1.40)$$

where Δ_{dyn} is the current clearance (at current time), Δ_{st} – clearance at steady mode.

The deviation of clearance during transients $\delta\Delta$ is considered to be caused by heat inertia of engine parts. Heat inertia itself is time and gas path parameters dependent. Clearance dynamics due to centrifugal forces and pressures is neglected because it only depends on engine parameters and is time independent at constant gas path parameters.

Clearance dynamics is also considered to be the product of thermal expansion of three main engine parts: HPT disk, HPT blade, HPT casing. The considered parts of HPT are mechanically isolated.

Basing these assumptions, eq. 1.40 can be transformed to

$$\delta\Delta = (\Delta_{dyn\ \mathrm{cas}} - \Delta_{st\ \mathrm{cas}}) - (\Delta_{dyn\ \mathrm{disk}} - \Delta_{st\ \mathrm{disk}}) - (\Delta_{dyn\ \mathrm{bl}} - \Delta_{st\ \mathrm{bl}}), \quad (1.41)$$

where $\Delta_{dyn\ disk}$ – deviation of dynamic radial clearace due to thermal strains of turbine disk;

$\Delta_{dyn\ bl}$ – deviation of dynamic radial clearace due to thermal strains of turbine blade;

$\Delta_{dyn\ cas}$ – deviation of dynamic radial clearace due to thermal strains of turbine casing;

$\Delta_{st\ disk}$, $\Delta_{st\ bl}$, $\Delta_{st\ cas}$ – deviation of radial clearance at steady modes due to thermal strains of turbine disk, blade and casing respectively.

1.3.5.2 Identification of radial clearance performances

To model gas path with acive clearance control system, the gas path model must be complemented with the model of clearance dynamics. Using high level models (e.g. finite element model of heat and stressed state) for this purpose is unreasonable due high requirements to hardware. Hence, to meet the requirement concerning processing time clearance dynamics is modeled by simple model. The model was identified by the high level model. Thus, two models were considered: finite element high level model and simplified model, which was used to carry out the identification, i.e. approximate description of clearance dynamics. First one model (high level model) is an assembly of three parts that are responsible for clearance dynamics. Model to be tuned during the identification are simplified quick processing models.

Let`s consider the sources of thermal loading, because it is necessary to model the clearance dynamics due to thermal expansion of parts.

1.3.5.3 Factors and parameters of heat loading

Heat (thermal) loading factors are factors that result in thermal strains in parts due to temperature field change. The most important are:

– convective heat exchange between parts and working substance;

– convective heat exchange between parts and cooling air;
– convective heat exchange between parts.

Heat state of parts is determined by thermal loading parameters, which describe heat exchange between parts, cooling air and working substance. These parameters are

– distribution of "heating" and "cooling" temperatures on the surface;
– heat transfer coefficients on the surface.

Fluid temperature

Convection is described by the following equation

$$\alpha\,(t - T_{n=0}) = -\left(\lambda \frac{\partial t}{\partial n}\right)_{n=0}, \qquad (1.42)$$

where $T_{n=0} = T(x_s, y_s, z_s, \tau)$ – fluid temperature in the point with coordinates x_s, y_s, z_s in time τ;

$t=t(x_s,y_s,z_s,\tau)$ – temperature of surface;

$\alpha=\alpha(x_s,y_s,z_s,\tau)$ – heat transfer coeffiicient;

$\lambda=\lambda(x,y,z,\tau)$ – heat exchange coefficient;

n – surface normal.

Generally, fluid temperature is a local total temperature at relative motion T_i^*. The distribution of this temperature on surfaces, which are in contact with hot gasses from the gas path and cooling air from active clearance control system, is performed during special gasdynamic analysis. The results of gasdynamic analysis are validated or corrected according to temperatures on areas available for gaging during engine testing.

The temperature T_i^* distributioin at arbitrary modes is unknown. But it can be determined according to controlled parameters of the engine using the similarity theory. Steady modes are used as reference ones.

Let vector of controlled parameter consist of $R \equiv \left[r^T \; M_{fl} \; T_0^* \right]^T$:

– r – vector of internal thermodynamic parameters (including parameters that determine engine operating mode),
– M_{fl}, T_0^* – Mach number and total temperature at engine inlet.

Let`s present calculation order and gradation by accuracy of corrected fluid temperature by the following equations sequence:

$$\overline{T}_i \equiv \frac{T_i^*}{n^2} = \overline{T}_i(\mathbf{R}) \cong \overline{T}_i(\mathbf{r}) \approx \text{idem}, \qquad (1.43)$$

where n – rotational speed of rotor.

To get additional models, it is necessary to accept similar models for points with fluid temperature sensors $s1$ and $s2$:

$$\overline{\overline{T}}_{\text{i}} \equiv \frac{T_i^*}{T_{s1}^*} = \overline{\overline{T}}_i(\mathbf{R}) \cong \overline{\overline{T}}_i(\mathbf{r}) \approx \text{idem}; \qquad (1.44)$$

$$\overline{\Theta}_i \equiv \frac{T_i^* - T_{s1}^*}{n^2} = \overline{\Theta}_i(\mathbf{R}) \cong \overline{\Theta}_i(\mathbf{r}) \approx \text{idem}; \qquad (1.45)$$

$$\overline{\overline{\Theta}}_i \equiv \frac{T_i^* - T_{s1}^*}{T_{s2}^* - T_{s1}^*} = \overline{\overline{\Theta}}_i(\mathbf{R}) \cong \overline{\overline{\Theta}}_i(\mathbf{r}) \approx \text{idem} . \qquad (1.46)$$

Models (1.43) – (1.46) are relations that link local fluid temperature with measured parameters of the gas path:

$$T_i^* = \overline{T}_i \, n^2 = \overline{T}_i \, T_{s1}^* = T_{s1}^* + \overline{\Theta}_i \, n^2 = T_{s1}^* + \overline{\Theta}_i (T_{s2}^* - T_{s1}^*) . \qquad (1.47)$$

If to consider corrected temperatures in eq. (1.43) and (1.46) constant, the following equation that links local fluind temperature with its value at reference (base) mode can be obtained from eq. 1.47:

$$T_i^* = T_{i\,ref}^* \cdot k_t = T_{s1}^* + \left(T_{i\,ref}^* - T_{s1\,ref}^* \right) \cdot k_t \qquad (1.48)$$

where k_t – *coefficient to transfer to any mode according to fluid temperature*; it is calculated according to measured parameters of the engine:

$$k_t = \frac{n^2}{n_{ref}^2} = \frac{T_{s1}^*}{T_{s1\,ref}^*} = \frac{T_{s2}^* - T_{s1}^*}{T_{s2\,ref}^* - T_{s1\,ref}^*} \qquad (1.49)$$

where subscript "*ref*" corresponds to parameters at reference (base) mode.

Thus, depending on chosen model, parameters of thermal loading of fluid temperature models can be corrected temperatures (1.43) and (1.46) or one (some) of measured parameters T_{s1}^*, T_{s2}^*, n^2. If model (1.48) is chosen, then parameters of thermal loading are fluid temperature at base mode and coefficient to transfer to any mode according to fluid temperature k_t. Schemes of thermal loading of parts and components of engines are presented in fig. 1.6 and 1.7

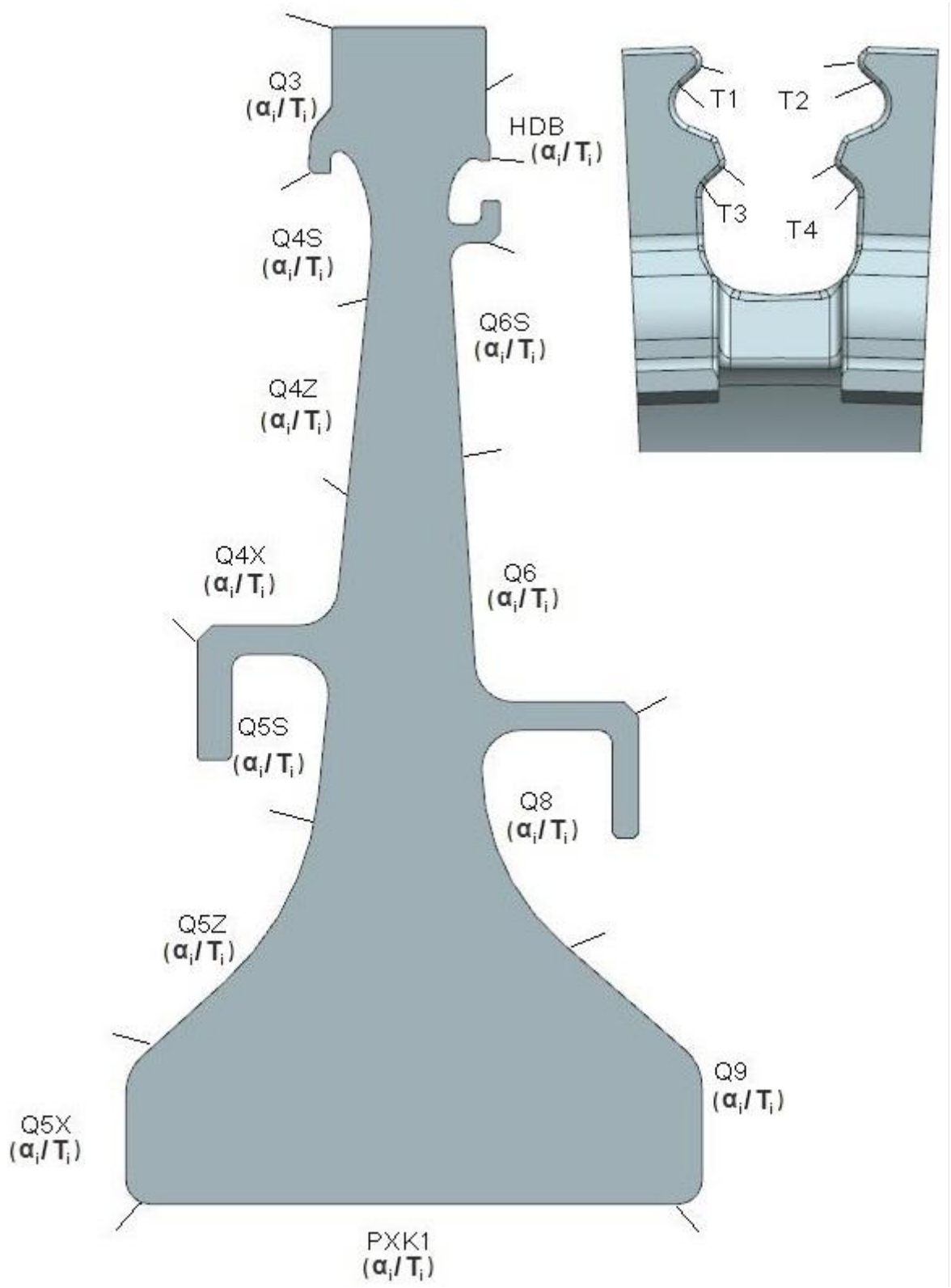


Figure 1.6 – Temperature loading of HPT disk
(heat transfer coefficient / fluid temperature)

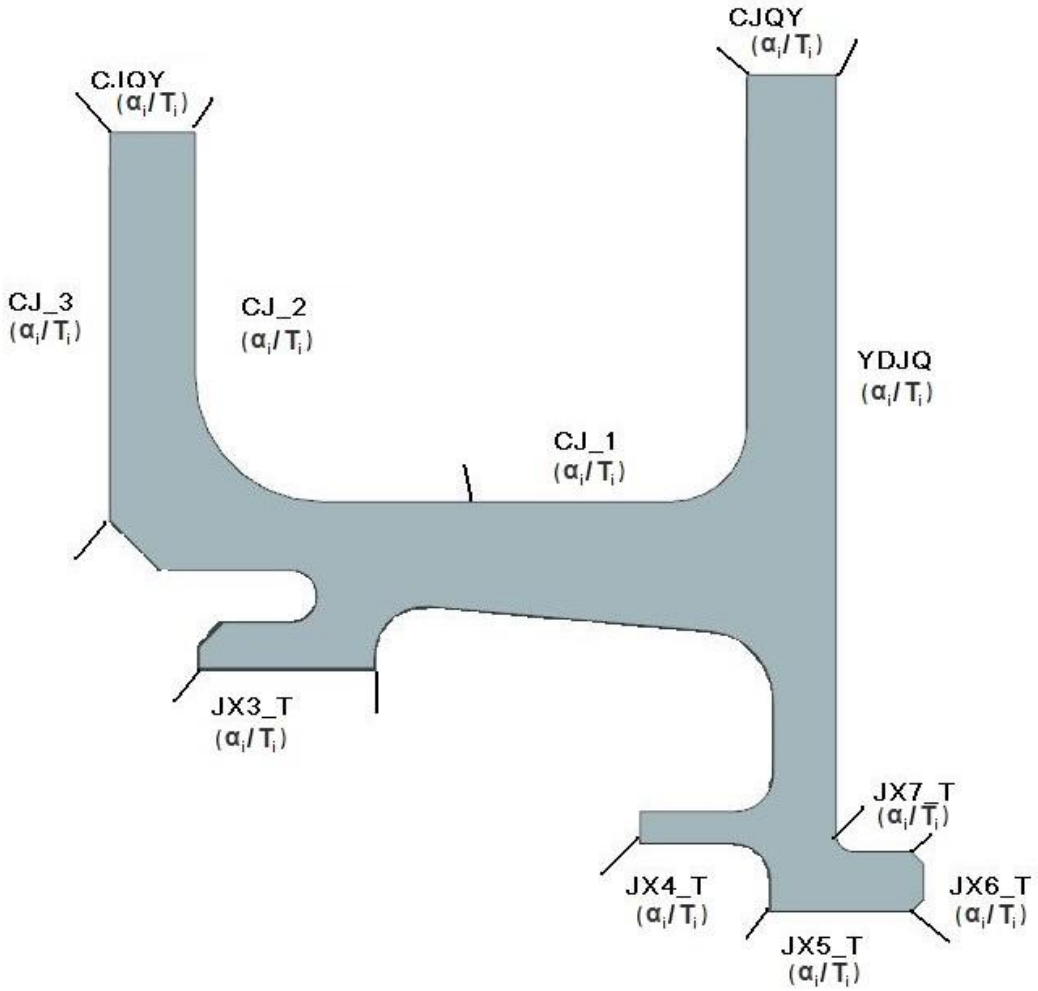


Figure 1.7 – Temperature loading of HPT casing
(heat transfer coefficient / fluid temperature)

Heat transfer model

Heat transfer at area of part surface is generally considered to be modeled by criterial equations like

$$\mathbf{Nu} = C \cdot \mathbf{Re}^{k} \cdot \mathbf{Pr}^{m}, \qquad (1.50)$$

where $\mathbf{Nu} \equiv \alpha r / \lambda$, $\mathbf{Re} \equiv \rho w r / \mu$, $\mathbf{Pr} \equiv c_{\mathrm{p}} \mu / \lambda$;

α, λ, μ, c_{p} – heat transfer, heat conductivity, dynamic viscosity, heat capacity at constant pressure;

ρ – flow density;

r – characteristic size of area;

w – relative velocity;

C, k, m – constants.

Let`s link parameters in eq. (1.50) with thermodynamic paraeters of the flow in the gas path using known relations. If to consider state equation and conditions of gas dynamics similarity we will get:

$$\rho \sim p/T, \qquad w \sim n, \quad T \sim T^{*}_{HPC}, \quad P \sim P^{*}_{HPC}.$$

Relations of thermal performances from temperature can be expressed by Satterland equations:

$$\lambda \sim T^{0.76},\ \mu \sim T^{0.64},\ c \approx const.$$

Let`s deduce the equations that expresse local heat transfer coefficient at arbitrary mode relates to similar coefficient at base (reference) mode for different parts (coefficient to transfer to any mode according to heat transfer coefficient). The deduction must be started from relation (1.50).

Active clearance control system:

$$k_{\alpha\ \text{упр}} = \frac{\alpha}{\alpha_б} = \left(\frac{w}{w_б}\right)^{k} \left(\frac{P^*_{contr}}{P_{contr\,ref}}\right)^{k} \left(\frac{T^*_{contr}}{T_{contr\,ref}}\right)^{0.76-1.64k-0.12m}, \quad (1.51)$$

where subscribe “contr” means that parameter belongs to control cavity of ACCS.

Gas path:

$$k_{\alpha\ \text{пр}} = \frac{\alpha}{\alpha_б} = \left(\frac{n}{n}\right)^{k} \left(\frac{P^*_{HPC}}{P_{HPC\,ref}}\right)^{k} \left(\frac{T^*_{G}}{T_{G\,ref}}\right)^{0.76-1.64k-0.12m}. \quad (1.52)$$

Disk of the turbine:

$$k_{\alpha\ disk} = \frac{\alpha}{\alpha_{ref}} = \left(\frac{n}{n_{ref}}\right)^{k} \left(\frac{P^*_{HPC}}{P_{HPC\,ref}}\right)^{k} \left(\frac{T^*_{HPC}}{T_{HPC\,ref}}\right)^{0.76-1.64k-0.12m} \quad (1.53)$$

All made considerations let us recommend the proportional heat transfer model for main operational modes:

$$\alpha = \alpha_{ref} \cdot k_{\alpha} \quad (1.54)$$

So, the analysis of heat exchange equations revealed that local heat transfer coefficients at base mode α_{ref} and coefficient to transfer to any mode according to heat transfer coefficient can be used as parameters of heat loading. The distribution of α_{ref} on part surface is the invariable performance of the engine. This performance is analytically obtained and experimentally accurated. Coefficient k_{α} depends on engine operating mode and ambient conditions. Coefficient k_{α} can be calculated by equations 1.52 – **Ошибка! Источник ссылки не найден.**.

The problem of identification of model that simulates strains of parts at steady and transient modes has a lot of aspects. One of the most important aspects is modeling temperature strains and the way they depend on parameters of temperature loading. This problem is considered in details in the following chapter.

1.3.5.4 Static model of temperature straining

The analysis revealed the most impotant factors to calculate the temperature straining of parts that form the radial clearance. The following parameters were chosen to calculate the strains of the disk at steady modes:

- T^*_{HPC} – temperature of cooling air;
- $k_{\alpha\ disk}$ – coefficient to transfer to any mode according to heat transfer coefficient;
- k_t – coefficient to transfer to any mode according to fluid temperature (T^*_{HPC}

and T_G^* were used to be characteristic temperatures)

The most important factor to affect temperature strains of the blade is T_G^* (temperature of gas in the gas path).

- T_{contr}^* – temperature of air in active clearance control system;
- $k_{\alpha\ cor}$ – coefficient to transfer to any mode according to heat transfer coefficient in the gas path;
- $k_{\alpha\ contr}$ – coefficient to transfer to any mode according to heat transfer coefficient in the ACCS;
- $k_{t\ cas}$ – coefficient to transfer to any mode according to fluid temperature (T_{HPC}^* and T_G^* were used to be characteristic temperatures)

The following relations were obrained:

$$\Delta_{st\ disk} = f\ T_{HPC}^*, k_{\alpha\ disk}, k_{t\ disk}\ ; \quad (1.55)$$

$$\Delta_{st\ bl} = f\ T_G^*\ ; \quad (1.56)$$

$$\Delta_{st\ cas} = f\ T_{contr}^*, T_G^*, k_{\alpha\ cor}, k_{\alpha\ contr}\ . \quad (1.57)$$

The following relations were obtained by linear interpolation of tabular data.

1.3.5.5 Dynamic model of temperature straining

To describe dynamic processes of temperature straining, the first and second order linear dynamic models were applied.

The first order equation with ***x*** (input) and ***y*** (output) parameters is the following:

$$T\frac{dy}{d\tau} + y = kx, \quad (1.58)$$

where k – gain coefficient;

T – time constant.

The following transfer function is equivalent to eq. (1.58)

$$W(s) = \frac{k}{Ts+1}, \quad (1.59)$$

and transfer performance (the response on unit step excitation) is

$$y(\tau) = k(1 - e^{-\frac{\tau}{T}}). \quad (1.60)$$

The second equation order is the following

$$T_1T_2\frac{d^2y}{d\tau^2} + (T_1+T_2)\frac{dy}{d\tau} + y = k_1(\frac{dx}{d\tau} + k_2x), \quad (1.61)$$

where k_1 k_2, – gain coefficients;

T_1, T_2 – time constants.

The following transfer function is equivalent to eq. (1.61)

$$W(s) = \frac{k_1s + k_1k_2}{T_1T_2s + (T_1 + T_2s) + 1}, \quad (1.62)$$

and transfer performance (the response on unit step excitation) is

$$y(\tau) = k_1 k_2 (1 - \frac{1}{k_2} e^{-\frac{\tau}{T_1}} - \frac{1}{k_1} e^{-\frac{\tau}{T_2}}). \quad (1.63)$$

Time constants of transients are assumed to be mostly depend on heat transfer intensity. The intensity is characterized by coefficient to transfer to any mode according to heat transfer coefficient for this part.

- time constants of turbine disk depend on $k_{\alpha\ disk}$;
- time constants of blade depend on $k_{\alpha\ cor}$;
- time constants of casing depend on $k_{\alpha\ cor}$ and $k_{\alpha\ contr}$.

To provide the robust simulation for second order lag element, it is recommended to simulate the transients in the elements separatly and only then together. This is due to specifities of solution of descrete equations set. The descrete equations are time dependant and comprise transfer-function coefficients.

The analysis of dynamic strains of disk and blade is carried out by eq. (1.62), using $\Delta_{st\ disk}$ and $\Delta_{st\ bl}$ as step input.

The analysis of dynamic strains of casing consists of two steps:

1. Calculating imaginary parameters of gas path $T^*_{contr\ imag}, T^*_{G\ imag}$, $k_{\alpha\ cor\ imag}$, $k_{\alpha\ contr\ imag}$ by the eq. (1.62). Input variables are $T^*_{contr}, T^*_G, k_{\alpha\ cor}, k_{\alpha\ contr}$;
2. Calculating dynamic strains of the casing basing calculated imaginary parameters (1.57).

$$\Delta_{dyn\ cas} = f\ (T^*_{cor\ imag}, T^*_{G\ imag}, k_{\alpha\ cor\ imag}, k_{\alpha\ contr\ imag})\ .$$

1.3.6 Implementation of dynamic model at given ambient and control actions changing in time

Thus, to form non-linear unit-level mathematical model of GTE it is necessary:

1) to take into account rotor dynamics with equations like (1.29);
2) to select specific volumes in gas-path and to take into account their dynamics by equations (1.32), (1.33);
3) to allocate character structural elements and to take into account their heat exchange with working substance by equations like (1.37);
4) to do necessary corrections in performances of units (1.38).

Comparing these equations with residuals for steady state model (1.9) – (1.27), it becomes clear that some residuals are not equal to zero at transient modes; they must be substituted by the corresponding differential equations:

– residuals by power (1.9) - (1.10) are not equal to zero, so corresponding algebraic equations are substituted by differential ones like (1.29);

– residuals by mass flow (1.16), (1.18), (1.20) are substituted by equations like (1.32) for corresponding specific volumes;

– residuals by energy balance in specific volume (1.17), (1.19), (1.21) are substituted by equations like (1.33);

– residuals by heat flows (1.24)-(1.26)are substituted by sets of equations like (1.37).

At the same time all other residuals are still require to be equal to zero at transient modes. So, mathematical model of GTE transient modes can be expressed as a common solving of nonlinear, differential and algebraic equations set.

So, the following set of equations describes transient operation of two spool turbofan engine without first and second flows mixing:

$$\frac{dn_F}{dt} = \left(\frac{30}{\pi}\right)^2 \frac{1}{J_F} \frac{N_{FT}\ \eta_{mech\ F} - N_{F\ 2} - N_{F\ 1\ +\ BOOSTERS} - \Delta N_{bleed\ F}}{n_F}; \tag{1.64}$$

$$\frac{dn_{HP}}{dt} = \left(\frac{30}{\pi}\right)^2 \frac{1}{J_{HP}} \frac{N_{HPT}\ \eta_{mech\ HP} - N_{HPC} - \Delta N_{bleed\ HP}}{n_{HP}}; \tag{1.65}$$

$$\frac{dm_{CC}}{dt} = G_{in\ CC} + G_{f\ CC} - G_{dis\ CC}; \tag{1.66}$$

$$\frac{dm_{N\ 2}}{dt} = G_{in\ N\ 2} - G_{dis\ N\ 2}; \tag{1.67}$$

$$\frac{dm_{N\ 1}}{dt} = G_{in\ N\ 1} - G_{dis\ N\ 1}; \tag{1.68}$$

$$\frac{dT^*_{dis\,CC}}{dt} = \frac{RT^*_{dis\,CC}}{C_p V_{CC} p^*_{dis\,CC}} \begin{bmatrix} G_{in\ CC} \cdot i\left(T^*_{in\ CC}\right) + G_{f\ CC} \cdot (H_u \cdot \eta_{CC} + i_0) - \\ -G_{dis\ CC} \cdot i\left(T^*_{dis\ CC}\right) \end{bmatrix}; \tag{1.69}$$

$$\frac{dT^*_{dis\,N2}}{dt} = \frac{RT^*_{dis\,N\ 2}}{C_p V_{C\ 2} p^*_{dis\,N\ 2}} \left[G_{in\ N\ 2}(\vec{Q}) \cdot i\left(T^*_{in\ N\ 2}\right) - G_{dis\ N\ 2}(\vec{Q}) \cdot i\left(T^*_{dis\ N\ 2}\right) \right]; \tag{1.70}$$

$$\frac{dT^*_{dis\,N\ 1}}{dt} = \frac{RT^*_{dis\ N\ 1}}{C_p V_{N\ 1} P^*_{dis\ N\ 1}} \left[G_{in\ N\ 1}(\vec{Q}) \cdot i\left(T^*_{in\ N\ 1}\right) - G_{dis\ N\ 1}(\vec{Q}) \cdot i\left(T^*_{dis\ N\ 1}\right) \right]; \tag{1.71}$$

$$\frac{dT_{st\ HPT}}{dt} = \frac{1}{C_{st\ HPT} m_{st\ HPT}} \begin{bmatrix} \alpha_{st\ HPT} F_{st\ HPT} \left(T^*_{in\ NV\ HPT} - T_{st\ HPT}\right) - \\ -\alpha_{st\ HPT\ cool} F_{st\ HPT\ cool} \left(T_{st\ HPT} - T^*_{cool\ st\ HPT}\right) \end{bmatrix}; \tag{1.72}$$

$$\frac{dT_{bl\ HPT}}{dt} = \frac{1}{C_{bl\ HPT} m_{bl\ HPT}} \begin{bmatrix} \alpha_{bl\ HPT} \cdot F_{bl\ HPT} \cdot \left(T^*_{in\ IMP\ HPT} - T_{bl\ HPT}\right) - \\ -k_{lock\ HPT} \cdot F_{lock\ HPT} \cdot \left(T_{bl\ HPT} - T_{d\ HPT}\right) - \\ -\alpha_{bl\ HPT\ cool} \cdot F_{bl\ HPT\ cool} \cdot \left(T_{bl\ HPT} - T^*_{cool\ bl\ HPT}\right) \end{bmatrix}; \tag{1.73}$$

$$\frac{dT_{d\ HPT}}{dt} = \frac{1}{C_{d\ HPT} m_{d\ HPT}} \begin{bmatrix} k_{lock\ HPT} \cdot F_{lock\ HPT} \cdot \left(T_{bl\ HPT} - T_{d\ HPT}\right) - \\ -\alpha_{d\ HPT\ cool} \cdot F_{d\ HPT\ cool} \cdot \left(T_{d\ HPT} - T^*_{cool\ d\ HPT}\right) \end{bmatrix}; \tag{1.74}$$

$$\frac{dT_{st\ FT}}{dt} = \frac{1}{C_{st\ FT} m_{st\ FT}} \begin{bmatrix} \alpha_{st\ FT} F_{st\ FT} \left(T^*_{in\ NV\ FT} - T_{st\ FT}\right) - \\ -\alpha_{st\ FT\ cool} F_{st\ FT\ cool} \left(T_{st\ FT} - T^*_{cool\ st\ FT}\right) \end{bmatrix}; \tag{1.75}$$

$$\frac{dT_{d\ FT}}{dt} = \frac{1}{C_{d\ FT} m_{d\ FT}} \begin{bmatrix} k_{lock\ FT} \cdot F_{lock\ FT} \cdot \left(T_{bl\ FT} - T_{d\ FT}\right) - \\ -\alpha_{d\ FT\ cool} \cdot F_{d\ FT\ cool} \cdot \left(T_{d\ FT} - T^*_{cool\ d\ FT}\right) \end{bmatrix}; \tag{1.76}$$

$$\frac{dT_{bl\,FT}}{dt}=\frac{1}{C_{bl\,FT}m_{bl\,FT}}\begin{bmatrix}\alpha_{bl\,FT}\cdot F_{bl\,FT}\cdot\left(T^{*}_{in\,IMP\,FT}-T_{bl\,FT}\right)-\\-k_{lock\,FT}\cdot F_{lock\,FT}\cdot\left(T_{bl\,FT}-T_{d\,FT}\right)-\\-\alpha_{bl\,FT\,cool}\cdot F_{bl\,FT\,cool}\cdot\left(T_{bl\,FT}-T^{*}_{cool\,bl\,FT}\right)\end{bmatrix}. \quad (1.77)$$

When synthesizing an algorithm that is aimed to solve the set of dynamic equations which describe TFE operation, it must be remembered that nowadays there are still no standard procedures for common differential and algebraic equations set solving. For this reason, this algorithm was formed to solve the set of differential equations:

$$\frac{d\vec{X}}{dt}=F\left(\vec{X},\,\vec{U}_{om}(t),\,\vec{U}_{fl}(t),\,\vec{v},\,t\right), \quad (1.78)$$

and, set of nonlinear algebraic equations is solved to calculate right parts of differential equations at each iteration

$$\vec{H}\left(\vec{Q},\,\vec{U}_{om}(t),\,\vec{U}_{fl}(t),\,\vec{v}\right)=0. \quad (1.79)$$

Matching $\vec{X}_{dif}$ values of $\vec{X}$ parameters, which are solutions of considered above differential equations, with $\vec{X}_{alg}$ values of the same parameters, calculated by algebraic equations solving, is ensured by new equation, like $H_{dif.alg}=\frac{X_{alg}}{X_{dif}}-1=0$, adding instead of algebraic equation, which was substituted by differential. So, in given example, equation $H_{dif\,alg\,LP}=\frac{n_{LP\,alg}}{n_{LP\,dif}}-1$ substitutes residual $H_{n\,LP}$ (see eq. 1.9), and such equations are applied for each rotor; equation $H_{dif\ alg\,m\,CC}=\frac{m_{CC\,alg}}{m_{CC\,dif}}-1$substitutes equation (1.16), etc.

Thus, in presented example nonlinear engine dynamic model is a procedure of numerical solution of differential equations set (fourteen differential equations), and to calculate their right parts, following equations set (twenty algebraic equations) must be resolved regarding the independent variables $\vec{Q}$:

$$\frac{n_F(\vec{Q})}{n_{F\,dif}}-1=0;\ \frac{n_{HP}(\vec{Q})}{n_{HP\,dif}}-1=0; \quad (1.80)$$

$$\frac{A_{FT\,calc}(\vec{Q})}{A_{FT\ perf}(\vec{Q})}-1=0;\ \frac{A_{HPT\,calc}(\vec{Q})}{A_{HPT\,perf}(\vec{Q})}-1=0; \quad (1.81)$$

$$\frac{F_{N\,1}(\vec{Q})}{F_{N\ 1\,struct}}-1=0;\ \frac{F_{N\,2}(\vec{Q})}{F_{N\ 2\,struct}}-1=0; \quad (1.82)$$

$$\frac{G_{dis\,BOOSTER}(\vec{Q})-\Delta G_{bleed}(\vec{Q})}{G_{in\,HPC}(\vec{Q})}-1=0; \quad (1.83)$$

$$\frac{m_{CC}(\vec{Q})}{m_{IPC}} - 1 = 0; \qquad \frac{T^*_{dis\,CC}(\vec{Q})}{T^*_{dis\,CC\ dif}} - 1 = 0; \tag{1.84}$$

$$\frac{m_{N\,2}(\vec{Q})}{m_{N\ 2\,dif}} - 1 = 0; \qquad \frac{T^*_{dis\,N\,2}(\vec{Q})}{T^*_{dis\ N\,2\,dif}} - 1 = 0; \tag{1.85}$$

$$\frac{m_{N\ 1}(\vec{Q})}{m_{N\ 1dif}} - 1 = 0; \qquad \frac{T^*_{dis\,N\,1}(\vec{Q})}{T^*_{dis\,N1\,dif}} - 1 = 0; \tag{1.86}$$

$$\frac{T_{st\,FT}(\vec{Q})}{T_{st\,FT\,d}} - 1 = 0; \quad \frac{T_{bl\,FT}(\vec{Q})}{T_{bl\,FT\,d}} - 1 = 0; \quad \frac{T_{d\,FT}(\vec{Q})}{T_{d\,FT\,d}} - 1 = 0; \tag{1.87}$$

$$\frac{T_{st\,HPT}(\vec{Q})}{T_{st\,HPT\,d}} - 1 = 0; \quad \frac{T_{bl\,HPT}(\vec{Q})}{T_{bl\,HPT\,d}} - 1 = 0; \quad \frac{T_{d\,HPT}(\vec{Q})}{T_{d\,HPT\,d}} - 1 = 0; \tag{1.88}$$

$$H_{Uom}(\vec{Q}) = \frac{U_{om}(\vec{Q})}{U_{om\ giv}} - 1 = 0. \tag{1.89}$$

Initial approximations of state parameters $\vec{X}_o$ are calculated at initial mode from static model. They are needed to solve considered above differential equations.

Structure of program software, which provides engine dynamic mathematical model implementation, is shown in fig. 1.8. Comparing with fig. 1.2 proves the significant compatibility of program modules used to model steady and transient modes. These models differ in two (three) program modules, one being module that calculates current values of input actions current including inlet influences on an engine.

Model university allows using any parameters of an engine as parameters of operating mode $\vec{U}_{om}$ and calculating the rest parameters at given $\vec{U}_{om}$ changing in time. But when parameter of operating mode is the component of state vector $\vec{X}$, there will appears two same equations (one equation is the residual of operating mode parameter and another one is also the residual, but this residual is calculated by joint solving of differential and algebraic equations set) in equations set (1.79), and Jacobian matrix of equations set will become singular.

This can be avoided in the next way: substituting residuals $H_{dif\ alg}$ of state parameters by the derivatives of these state parameters in algebraic equations set (1.79); these derivatives are solution of algebraic equations set. They are made an optional parameters and they are calculated in thermodynamic straight-through calculation as the right parts of differential equations (1.78).

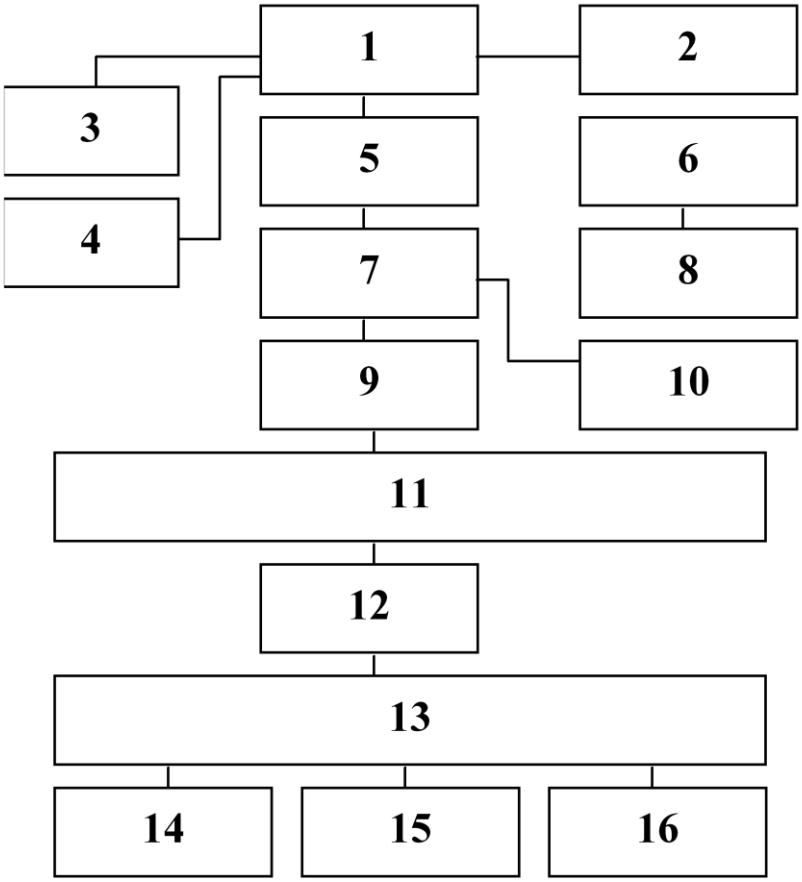


Figure 1.8 – Structure of software, which implements unit-level mathematical model of GTE transient modes:

1 – main program;
2 – calculating parameters at initial steady mode;
3 – initializing calculation;
4 – inputting initial data;
5 – solving the set of differential equation;
6 –managing the solution and results displaying;
7 – calculating the right parts of differential equations set;
8 – displaying the results of calculation;
9 – solving nonlinear algebraic equations set;
10 – calculating parameters of operating mode;
11 – calculating right parts of algebraic equations set;
12 – calculating residuals;
13 – thermodynamic calculation of gas-path;
14 – general thermodynamic calculations;
15 – calculating performances of units;
16 – data interpolation.

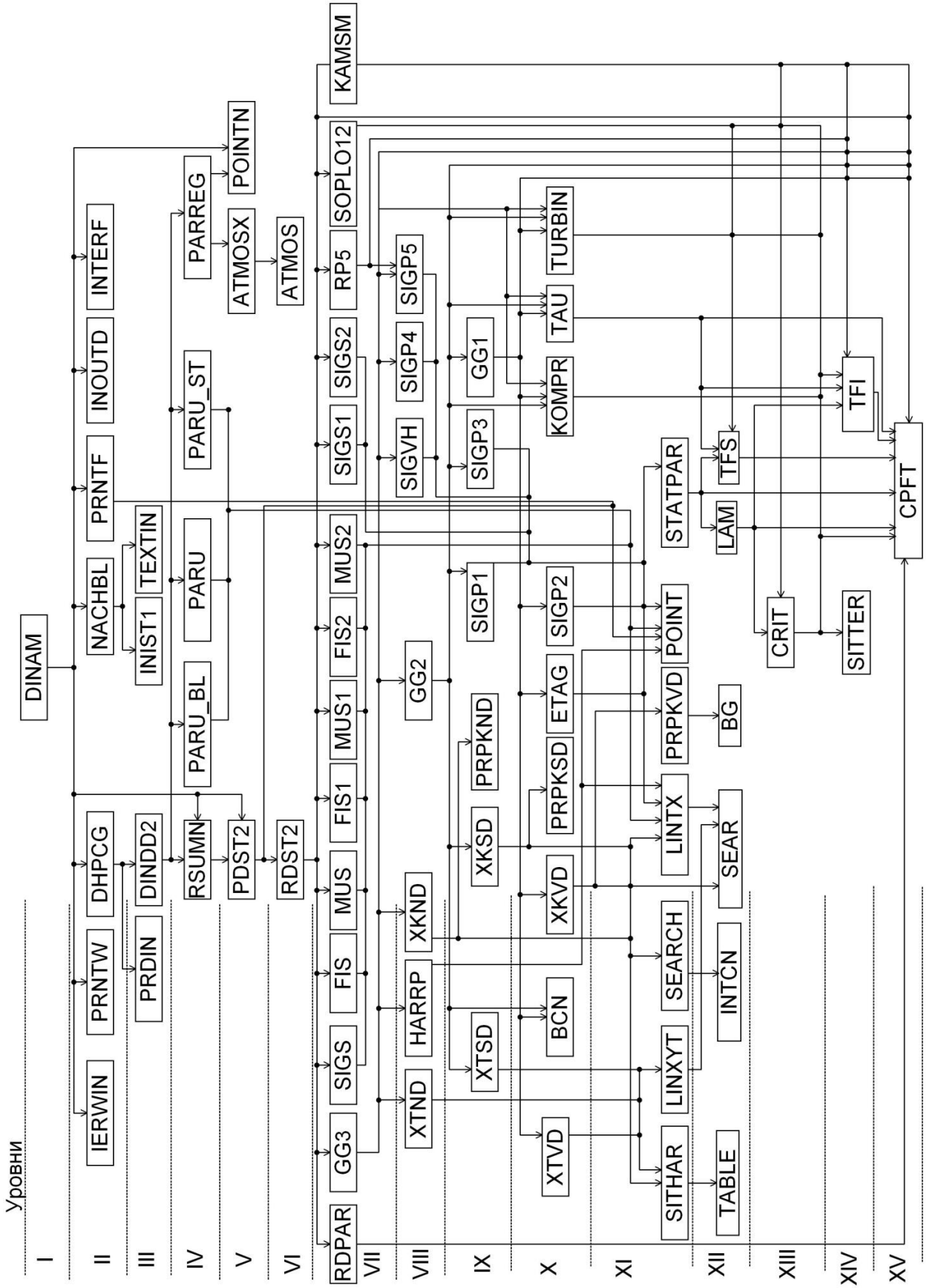


Figure 1.9 – Calling order of subroutines, needed for T FE-2 calculating

Calling order of program modules to calculate dynamics of 2 spool TFE is shown in fig. 1.9. Analysis of figs. 1.3 and 1.9 shows that subroutines of V – XV levels in dynamics calculating, are the same as IV – XIII levels in static performances calculating. Second levels are also similar. The only difference is presence of DHPCG subroutine (solves set of ordinary first-order differential equations by the Hamming method), which calls the subroutine RSUMN (solves set of non-linear equations by Newton method) and program modules that solve equations of heat state for structural elements of each turbine spool, like eq. **Ошибка! Источник ссылки не найден.** (PARU_BL, PARU, PARU_ST).

1.4 Quick calculating mathematical models

1.4.1 Structure of quick calculating mathematical model of an engine

Dynamic models are necessary to provide ACS optimal synthesis, multivariate calculations, real-time modeling and diagnostics by measured parameters of GTE at transient modes. Dynamic models are limited by available computing resources. Quick calculating dynamic mathematical models (QLDM) can cope with this problem.

There are three application area of QLDM for synthesizing automatic control systems:

− QLDM expresses dynamic features of an engine used for creating and checking algorithms of engine control;

− QLDM comprises control algorithms;

− QLDM is an engine simulator for semi-natural modeling to configure electronic control unit and execution units of ACS. Generally QLDM

Generally for all these application areas QLDM is implied by the means of universal or special programs (for example, VisSim, MathLab etc.), or directly by electronic control unit, what in another words means out of this software. That is why this software has means for QLDM synthesizing, but its implementation is performed by different means.

When running non-linear dynamic model (1.78), (1.79), the major part of processor operation time takes calculating of right parts of differential equations (1.78) and solving set of algebraic equations (1.79). So, to short operation time model must be simplified.

The simplest and the most universal way to simplify models in a form of equations set is to linearize them. Unfortunately linearized model may have significant inaccuracies when input parameters significantly differ from parameters used to linearize the model. To prevent considered above inaccuracy, only dynamic part of QLDM is linearized and static part of the model remains non-linear. Besides all modeled parameters are represented as a sum of static component y_{st} and dynamic deviation Δy:

$$y = y_{st} + \Delta y . \tag{1.90}$$

Static model has the following form

$$y_{st\,j} = f_j(\vec{U}_{om}, \vec{U}_{fl}), j=1,...,m, \quad (1.91)$$

where m is the amount of modeled parameters.

Dynamic model comprises from the set of m linear differential equations. Solving the dynamic model allows calculating changing of deviations $\Delta y(t)$ in time under given variation of mode parameter and ambient actions in time. The specific features of this model and methodology to form the model are considered in the next subsection. General structure of ALDM is shown in fig. 1.10.

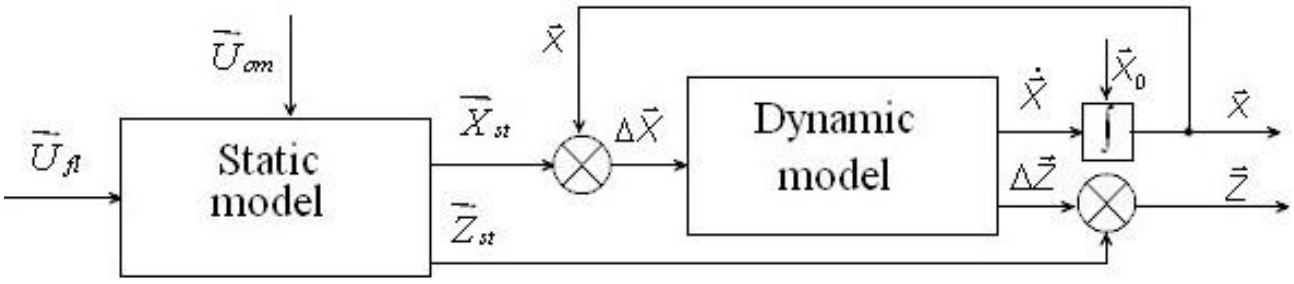


Figure 1.10 – General structure of QLDM

Vector of modeled parameters is expressed in the following form:

$$\vec{Y} = \begin{bmatrix} \vec{X} \\ \vec{Z} \end{bmatrix}, \quad (1.92)$$

where $\vec{X}$ - $(n\times 1)$-state vector (parameters of dynamic factors), which consists of parameters, which are unknown parameters in differential equations of engine operation;

$\vec{Z}$ - vector of additional parameters, which consists of other modeled engine parameters of engine.

Static model (1.90) gets parameters of operating mode $\vec{U}_{om}$ and flight conditions $\vec{U}_{fl}$ as input parameters. According to input parameters model calculates static values of modeled parameters y_{st}. Static values of state parameters $\vec{X}_{st}$ are used to calculate deviations $\Delta\vec{X}$, used as input data of dynamic model. Static values of additional parameters $\vec{Z}_{cm}$ are additives in calculating dynamic values of these parameters $\vec{Z}$.

Dynamic model gets deviations of state parameters $\Delta\vec{X}$, deviations of mode parameters $\Delta\vec{U}$ and flight conditions as input data. Dynamic model calculates derivatives of state parameters $\dot{\vec{X}}$ and dynamic deviations of additional parameters $\Delta\vec{Z}$. Obtained derivatives are integrated at given time slice to calculate dynamic values of these parameters.

To integrate derivatives correctly it is necessary to give initial approximations of state parameters. These initial values are calculated by static model at initial steady mode of engine operation. That is why initial values of mode parameters and ambient conditions are used as input data for static model.

User must rationally choose state parameters of operating mode and ambient conditions, which will be used as arguments of static and dynamic models, to form convenient structure of QLDM and to economy computational resources.

1.4.2 Synthesizing static model of an engine

Considered in previous subsection structure of QLDM does not constrain architecture of static model (1.91). Static model may consist of any algorithms, which provide calculating modeled static parameters y_{st} of an engine according to given initial parameters $\vec{U}_{om}$, $\vec{U}_{fl}$.

There are three variants to implement this algorithm practically:

– use static non-linear mathematical model;

– give static model in tabular form;

– approximate static model.

Static non-linear model (1.1), is based on solving equations of gas-path units` joint operation, which was considered above. Modern computers provide real-time simulating processes with QLDM with integrated unit-level static model. For mentioned above reason this configuration of QLDM can be used for semi natural simulation. This way is beneficial, because there is no necessity in simplifying static model and ensure its tolerance and good fit. The disadvantages of this way are the necessity to provide stability and convergence of solution (it requires correct performances of units) and relatively high proceeding time (this may become the significant problem, when PC is occupied by other auxiliary tasks that accompany simulation, like registrating and displaying results).

As it was shown in previous subsection QLDM is a package of throttle, climatic and altitude-velocity performances. That is why ***static model can be expressed as a set of points (table)*** or set of separate lines on performance maps of units. Static values of parameters y_{st} can be calculated by linear, squared or any other interpolation according to given argument-parameter.

The last variant is to approximate these performances. It is possible to use the same tabular format for inputting the data as in previous variant. An example of such an approximation is polynomial one. The following model

$$y_{st} = f(n_{HP}, p_0^*, T_0^*, M_{fl}) \tag{1.93}$$

is expressed in polynomial form:

$$\begin{aligned} y_{st} = c_0 + c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 + c_5x_1^2 + c_6x_2^2 + c_7x_3^2 + c_8x_4^2 + \\ + c_9x_1x_2 + c_{10}x_1x_3 + c_{11}x_1x_4 + c_{12}x_2x_3 + c_{13}x_2x_4 + c_{14}x_3x_4, \end{aligned} \tag{1.94}$$

where $x_1 = \dfrac{n_{HP}}{n_{HP\,0}}$; $x_2 = \dfrac{p_0^*}{p_{0\,0}^*}$; $x_3 = \dfrac{T_0^*}{T_{0\,0}^*}$; $x_4 = \dfrac{M_{fl}}{M_{fl\,0}}$ − are normalized values of parameters, which are used as arguments of the model;

$n_{HP\,0}, p_{0\,0}^*, T_{0\,0}^*, M_{fl\,0}$ − are reference values of parameters , which are used as arguments.

Degree and structure of polynomial are determined by the subject of modeling and requirements to model.

Using corrected parameters in static model allows taking into account flight conditions. Than the model simplifies to the following form:

$$y_{st\,cor} = f(U_{om\,cor}). \tag{1.95}$$

Correcting of parameters referring international standard atmosphere is performed by the following equations:

$$\begin{gathered} p^*_{i\,cor} = p^*_i \frac{p_{ref}}{p^*_0};\ T^*_{i\,cor} = T^*_i \frac{T_{ref}}{T^*_0};\ G^*_{i\,cor} = G^*_i \frac{p_{ref}}{p^*_0}\sqrt{\frac{T^*_0}{T_{ref}}};\ n_{cor} = n\sqrt{\frac{T_{ref}}{T^*_0}};\\ G_{f\,cor} = G_f \frac{p_{ref}}{p^*_0}\sqrt{\frac{T_{ref}}{T^*_0}};\ R^*_{cor} = R\frac{p_{ref}}{p^*_0};\ N_{cor} = N\frac{p_{ref}}{p^*_0}\sqrt{\frac{T_{ref}}{T^*_0}};\ C_{sp\,cor} = C_{sp}\sqrt{\frac{T_{ref}}{T^*_0}}. \end{gathered} \tag{1.96}$$

General view of correcting of parameter looks like

$$y_{j\,cor} = y_j K^0_j(p^*_0, T^*_0), \tag{1.97}$$

where $K^0_j(p^*_0, T^*_0)$ − is the correction coefficient of j[th] parameter that was calculated according to similarity theory.

It is obvious that correcting equations are based on similarity of working processes in the engine at different flight conditions. The similarity is followed with definite tolerance. Therefore the tolerance of static model, which is based on using correcting equations is also limited. So, for example, similarity equations do not incorporate Mach number. It is believed that similar operating modes have similar Mach numbers. But this is not true for different flight condition. That is why this static model does not take into account changing of flight velocity.

Other reasons for limited tolerance of similarity equations are discussed in books on theory of air-breathing engines.

The advantage of model (1.95), (1.96) is its consisting of two simple parts: first part is the correlation between corrected value of modeled parameter and corrected value of mode parameter (model of throttle performance), and second part is the part that performs correction. To keep the simple structure and improve the tolerance of the model, it is possible to supplement correction equations with adjustments, which depend on all flight parameters (p^*_0, T^*_0, M_{fl}) or selected influencing parameters, e.g. Mach number.

Then instead of using (1.97), the following equation is used to make the correction:

$$y_{j\,np} = y_j \cdot K^0_j(p^*_0, T^*_0) \cdot K^{correct}_j(p^*_0, T^*_0, M_{fl}), \tag{1.98}$$

where $K^{correct}_j(p^*_0, T^*_0, M_{fl})$ is an adjustment of j[th] parameter to add to correction equation.

Initial data to form adjustment coefficients for correction can be obtained from engine testing or from non-linear mathematical model. The form of model to calculate adjustment coefficients is determined by the developer of static model. The developer also chooses the software to calculate the coefficients of this model. For example, it is possible to use polynomial correlation like

$$K_j^{correct} = c_0 + c_1(p_0^* - p_{ref}) + c_2(T_0^* - T_{ref}) + c_3 M_{fl}$$

or correlation like (1.94) with arguments x_2, x_3, x_4.

1.4.3 Method to calculate coefficients of linear dynamic model

QLDM comprises of linear dynamic model (LDM). The model is linear, which means that it consists of set of linear differential equations. It is better not to set coefficients of model constant, but make them variable, means that these coefficients depend on operating mode and flight conditions. This allows increasing tolerance of modeling in wide range of operating mode and flight conditions. To get the dependence between coefficients of LDM and flight conditions, it is necessary to get these coefficients at different conditions, and then approximate them, using mode parameters and flight conditions as arguments of correlation. Let`s consider this problem at given mode parameters and flight conditions.

LDM shows dynamic features of an engine close to steady mode. There are two possible concepts to form LDM. They both are based of linearizing set of equations (1.78), (1.79). First concept consists in forming of one linear differential equation for each modeled parameter; the order of equation equals to number of modeled dynamic factors n. For random parameter y this equation has the following form:

$$a_n \frac{d^n y}{dt^n} + ... + a_1 \frac{dy}{dt} + a_0 \Delta y = \sum_{i=1}^{I} \left(b_{i\,m} \frac{d^m U_{om\,i}}{dt^m} + ... + b_{i\,1} \frac{dU_{om\,i}}{dt} + b_{i\,0} \Delta U_{om\,i} \right) + \\ + \sum_{j=1}^{J} \left(c_{j\,m} \frac{d^m U_{fl\,j}}{dt^m} + ... + c_{j\,1} \frac{dU_{fl\,j}}{dt} + c_{j\,0} \Delta U_{fl\,j} \right), \quad (1.99)$$

where $U_{om\,i}$ is the i[th] parameter, which determines the operating mode of an engine;

$U_{fl\,j}$ is the j[th] parameter of flight conditions;

a, b, c – coefficients of model;

$\Delta y = y - y_{st}$ is dynamic deviation of parameters from its static value y_{st}, which corresponds to steady operating mode of engine;

$\Delta U = U - U_0$ is dynamic deviation of parameter U from U_0, which corresponds to initial steady mode.

Model (1.99) is the model of high order and is very complex for practical application. That is why models, based on the second concept of equations set linearizing, are more abundant. According to this concept, set of non-linear differential and algebraic equations (1.78), (1.79) is substituted by the system of n first-order linear differential and linear algebraic equations. Let`s examine state parameters of an engine $\vec{Y}$ and input actions $\vec{U}$ (see equations (1.92), (**Ошибка! Источник ссылки не найден.**)). Then model (1.78), (1.79) can be expressed in ordinary form, which corresponds to state space model of linear system:

$$\Delta \dot{\vec{X}} = A \Delta \vec{X} + B \Delta \vec{U}; \quad (1.100)$$

$$\Delta \vec{Z} = C \Delta \vec{X} + D \Delta \vec{U}, \quad (1.101)$$

where A, B, C, D are matrixes of constant coefficients, which are parameters of linear dynamic model (LDM). In theory of control differential equation (1.100) is called state equation, and algebraic equation – watch equation (1.101).

Parameters of LDM can be calculated from non-linear static model. Let`s consider describing engine dynamics set of differential equations (1.78). If simulated transient process does not depend on internal parameters of an engine (for example, performances of units, fuel properties etc.), then transient processes is determined only by changing mode parameters and external actions. This allows exclude time from list of parameters, which explicitly comprise F-function in right part of equation (1.78), and also neglect changing of units performances $\vec{\upsilon}$. Then for model (1.78)

$$\frac{d\vec{X}}{dt} = F\left(\vec{X},\, \vec{U}_{om}(t),\, \vec{U}_{fl}(t),\, \vec{\nu},\, t\right)$$

may be simplified to the following

$$\frac{d\vec{X}}{dt} = F\left(\vec{X},\, \vec{U}(t)\right).$$

Initially system is at steady state, that is why $\vec{X} = \vec{X}_0$, $\frac{d\vec{X}}{dt}$=0.

Let`s perturb Δf_1 the right part of the first equation of the set at $\vec{U} = const$:

$$f_1\left(x_1, x_2, ..., x_n,\, \vec{U}(t)\right) = f_1\left(x_{10}, x_{20}, ..., x_{n0}, \vec{U}(t)\right) + \Delta f_1;$$

$$f_2\left(x_1, x_2, ..., x_n,\, \vec{U}(t)\right) = f_2\left(x_{10}, x_{20}, ..., x_{n0},\, \vec{U}(t)\right);$$

.....

$$f_n\left(x_1, x_2, ..., x_n,\, \vec{U}(t)\right) = f_n\left(x_{10}, x_{20}, ..., x_{n0},\, \vec{U}(t)\right);$$

Induced by inertia, state parameters does not change ($\vec{X} = \vec{X}_0$) at initial moment, therefore

$$\dot{x}_{11} = \Delta F_1;\ \dot{x}_{21} = \dot{x}_{31} = ... = \dot{x}_{n1} = 0,$$

where $\dot{x}_{ij}$ − is time derivative of ith state parameter of an engine, appeared because of disturbing the right part of jth equation.

When choosing disturbance in the way to apply it to static model in a form of changing one of initial parameters, then it is possible to calculate the changes

$$\Delta\vec{X}_1 = \left[\Delta x_{11}\ \Delta x_{21}\ ...\ \Delta x_{n1}\right]^{\mathrm{T}} = \left[(x_{11}\text{-}x_{10})\ (x_{21}\text{-}x_{20})\ ...\ (x_{n1} - x_{n0})\right]^{\mathrm{T}}$$

of state parameters at new mode, which will appear after transient process, which was initiated by the mentioned above disturbance. Let`s calculate corresponding $\dot{\vec{X}}$ and $\Delta\vec{X}_j$, by disturbing Δf_j one by one differential equations.

These values were calculated for initial set of non-linear differential equations (1.78) in case of small disturbances in right parts. From other hand, these values must correspond to linear equations (1.100) at $\Delta\vec{U}$=0. Substituting calculated values of $\dot{\vec{X}}$ and $\Delta\vec{X}_j$ the following equations set will be obtained:

$$\begin{cases} \Delta\dot{x}_{11} = a_{11}\Delta x_{11} + \ldots + a_{1n}\Delta x_{n1} \\ 0 = a_{21}\Delta x_{11} + \ldots + a_{2n}\Delta x_{n1} \\ \quad\vdots \qquad\qquad\quad \vdots \\ 0 = a_{n1}x_{11} + \ldots + a_{nn}x_{n1} \\ 0 = a_{11}x_{12} + \ldots + a_{1n}x_{n2} \\ \Delta\dot{x}_{22} = a_{21}x_{12} + \ldots + a_{2n}\Delta x_{n2} \\ \quad\vdots \qquad\qquad\quad \vdots \\ \Delta\dot{x}_{nn} = a_{n1}\Delta x_{1n} + \ldots + a_{nn}\Delta x_{nn} \end{cases}; \qquad (1.102)$$

or in vector-matrix form $P\vec{A} = \vec{Q}$, where $\vec{Q} = \begin{vmatrix} \Delta\dot{x}_{11} \\ 0 \\ \vdots \\ 0 \\ 0 \\ \Delta\dot{x}_{22} \\ \vdots \\ \Delta\dot{x}_{nn} \end{vmatrix}$; $\vec{A} = \begin{vmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \\ \vdots \\ a_{nn} \end{vmatrix}$;

$$P = \begin{vmatrix} \Delta x_{11} & \cdots & \Delta x_{n1} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \Delta x_{11} & \cdots & \Delta x_{n1} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \Delta x_{11} & \cdots & \Delta x_{n1} \\ \Delta x_{12} & \cdots & \Delta x_{n2} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \Delta x_{1n} & \cdots & \Delta x_{nn} \end{vmatrix}.$$

Order of matrix $P - (n^2 X n^2)$. Solving these equations for n^2 unknown coefficients a_{ij}, we will get the matrix A of the system (1.100). Elements of the matrix are coefficients of LDM. When forming the equations set (1.100) it is possible to calculate not only $\Delta\vec{X}$, but also the corresponding values of $\Delta\vec{Z}$. Substituting them in equation (1.101), we obtain the set from $m \times n$ equations to calculate coefficients of matrix C:

$$\begin{cases} \Delta z_{11} = c_{11}\Delta x_{11} + \ldots + c_{1n}\Delta x_{n1} \\ \Delta z_{21} = c_{21}\Delta x_{11} + \ldots + c_{2n}\Delta x_{n1} \\ \quad\vdots \qquad\qquad\qquad \vdots \\ \Delta z_{m1} = c_{m1}\Delta x_{1n} + \ldots + c_{mn}\Delta x_{nn} \\ \quad\vdots \qquad\qquad\qquad \vdots \\ \Delta z_{mn} = c_{mn}\Delta x_{1n} + \ldots + c_{mn}\Delta x_{nn} \end{cases}; \qquad (1.103)$$

or $E\vec{C} = \vec{F}$, where

$$F = \begin{vmatrix} \Delta z_{11} \\ \vdots \\ \Delta z_{m1} \\ \Delta z_{11} \\ \vdots \\ \Delta z_{mn} \end{vmatrix};\ \vec{C} = \begin{vmatrix} c_{11} \\ \vdots \\ c_{1n} \\ c_{21} \\ \vdots \\ c_{mn} \end{vmatrix};\ E = \begin{vmatrix} \Delta x_{11} & \cdots & \Delta x_{n1} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \Delta x_{11} & \cdots & \Delta x_{n1} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \Delta x_{11} & \cdots & \Delta x_{n1} \\ \Delta x_{12} & \cdots & \Delta x_{n2} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \Delta x_{1n} & \cdots & \Delta x_{nn} \end{vmatrix}.$$

Any parameter form the right parts of differential equations can be used as a disturbance, but it is better if this parameter has clear physical sense. As it was mentioned above, differential equations of GTE operation appear because of three dynamic factors: inertia of rotors, compressibility of working substance in the specified volumes of gas-path and heat exchange between working substance and structural elements.

Disturbances in motion equations of rotors can be made by power take-offs (see eq. (1.29)). Then $\Delta\dot{n}_i = \dfrac{\Delta N_{take-off\ i}}{n_i}\left(\dfrac{30}{\pi}\right)^2 \dfrac{1}{J_i}$

Disturbances in heat exchange equations can be made by additional heat supplies ΔQ_{st}, ΔQ_{bl}, ΔQ_d (see eq. (1.37)), then $\dot{T}_{SE\ j} = \dfrac{\Delta Q_{SE\ j}}{C_{SE\ j}}$.

There are two differential equations (1.32) and (1.33) applied, when modeling any volume of gas-path. Neglecting working substance supplying, we get:

$$\Delta\dot{m} = G_{supl};\ \Delta\dot{T}^*_{dis} = \frac{RT^*_{dis}\Delta Q}{C_p V p^*_{dis}}.$$

Mentioned above parameters in equations of mass flow supply and heat supply were specially added to set of differential equations and to static model for providing the possibility of LDM coefficients calculating.

Coefficients of matrix B and D can be calculated individually or separately for each column. This possibility is based on fact that model (1.100), (1.101) is correct close to reference point for both, transient and steady modes. That is why, calculating deviations $\Delta\vec{X}$, $\Delta\vec{Z}$ (all other acting factors are constant), which correspond to deviation ΔU_l by static model , it is necessary to put them in each kth equation of the set

(1.100) and each i[th] equation of the set (1.101), believing $\Delta\vec{X} = 0$, then:

$$b_{il} = \frac{1}{\Delta U_l}\left(a_{1l}\Delta x_{1l} + \ldots a_{in}\Delta x_{nl}\right); \quad k = 1,...,n;$$

$$d_{jl} = \frac{1}{\Delta U_l}\left(\Delta z_j - c_{j1}\Delta x_{1l} - \ldots - c_{jn}\Delta x_{nl}\right); \quad j = 1,m.$$

Calling scheme of program modules to calculate coefficients of linear dynamic model to simulate three-spool TFEmix operation is expressed in fig. 1.11

Comparing figs. 1.3 and 1.11 we can conclude, that calculating coefficients of QLDM requires the same program modules as calculating steady modes. The difference is the presence of subroutine that calculates coefficients of linear dynamic model (CFNLDM) between main program (LDM) and subroutine that calculates static performances of an engine (RSUMN)

Analogically to dynamics, it is not necessary to present structures to calculate coefficients of QLDM for all other engine types. Levels I-III will be the same for all engine types. The difference will be present in other levels (see Appendix B).

1.4.4 Method to form multimode dynamic model

Coefficients of matrixes A, B,C, D depend on engine operating mode and flight conditions. In case when the process of insignificant mode or flight conditions change is modeled then coefficients of QLDM (see fig. 1.10) are assumed to be constant (LDM with constant coefficients). But if operating conditions of an engine change significantly, then model with constant coefficients gives results with significant errors (time of transient processes may vary in some times).

The simplest form of model with simple structure, which takes into account mentioned actions, is linear dynamic model with variable coefficients. In practice this is realized by applying linear dynamic model (1.100), (1.101), but its coefficients depend on operating mode and ambient conditions:

$$a_{ij} = f(\vec{U}_{om}, \vec{U}_{fl}); \; b_{il} = f(\vec{U}_{om}, \vec{U}_{fl}); \; c_{kj} = f(\vec{U}_{om}, \vec{U}_{fl}); \; d_{kl} = f(\vec{U}_{om}, \vec{U}_{fl}) \; . \quad (1.104)$$

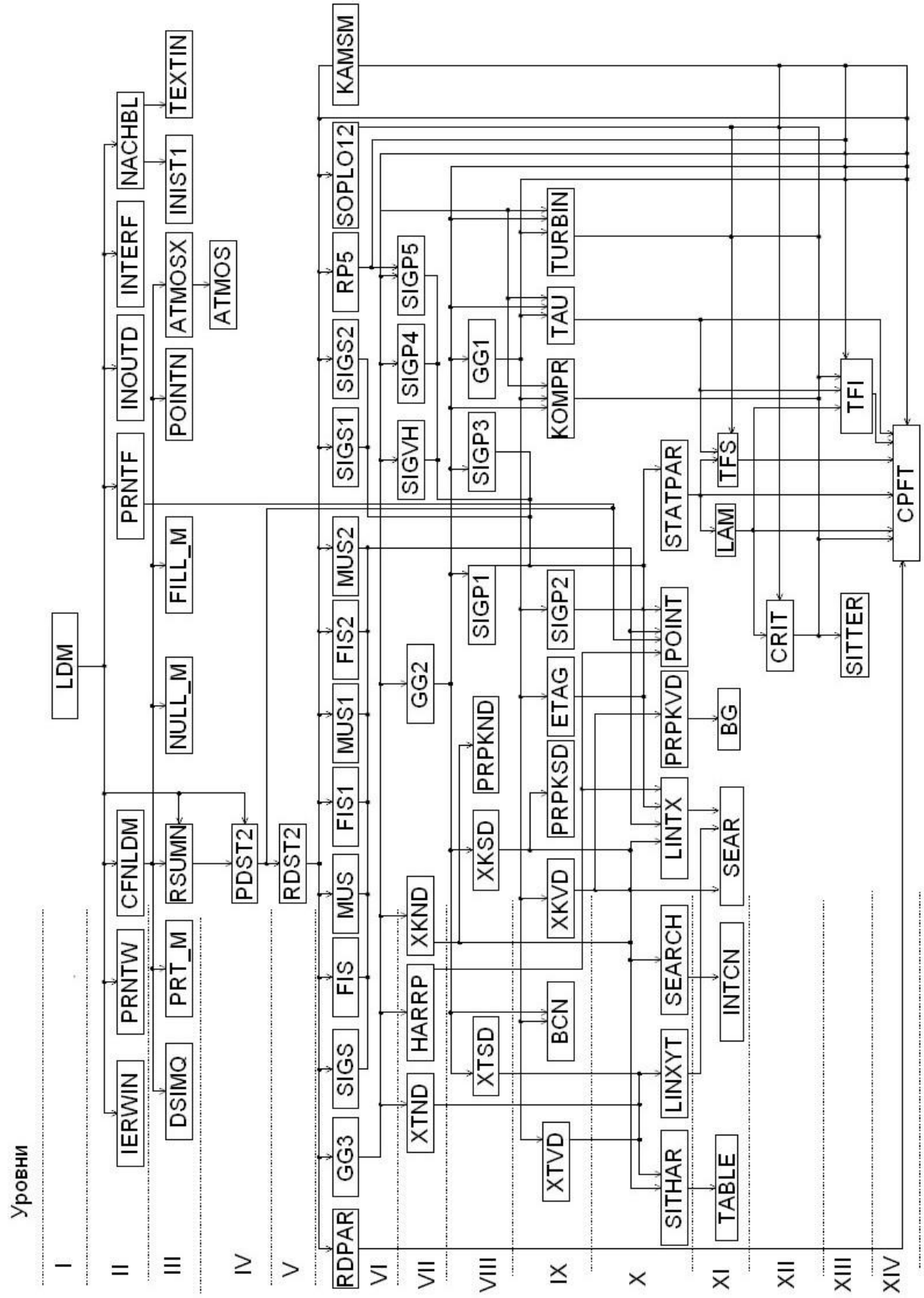


Figure 1.11 – Calling scheme of program modules to calculate coefficients of linear QLDM for TFE

All possible variants to form the correlation (1.104) are the same to static model, which was considered earlier:

- approximate variation of coefficients by all acting parameters;
- use corrective equations of LDM, and approximate variation of corrected coefficients by parameters of operating mode;
- corrective equations may be complemented by coefficients for more tolerant taking into account flight conditions.

2 FUNCTIONS AND APPOINTMENT OF PROGRAM COMPLEX

Equation Chapter (Next) Section 1

Program complex (next - "program") is used for throttle, altitude, climatic, velocity and dynamic performances calculating, LDM coefficients calculating, and also for making the identification.

Program implements algorithms of engine performances calculating according to component level non-linear mathematical model (NMM). Applied mathematical models take into account specific features of different engine designs. Main program functions are:

- calculating engine parameters at given operating point;
- calculating throttle performance (engine operating line);
- calculating altitude-velocity and climatic performances;
- calculating transients at given change of ambient conditions and control actions;
- visualizing and editing performances of engine components: geometrical parameters, performances of blade machines and their correction factors, power take-offs, mechanical efficiencies, absolute and relative air bleeds and taps of cooling air, correlations of losses along gas path, control laws of movable IGVs of compressor spools, momentum of rotor inertia;
- visualizing and editing specific volumes, heat capacities, masses, heat transfer coefficients, areas of all heat exchange surfaces, cooling temperatures of all SEs, which constitute modeled engine, transient strains of checkpoints at steady modes that depend on different factors, like time responses and gain factors of second order lag element that simulates transient heat process in SEs;
- inputting parameters to characterize aircraft flight and control factors;
- choosing dynamic factors to simulate engine dynamics and to calculate coefficients of LDM;
- managing the displaying mode to visualize calculation results and to provide its printing;
- displaying calculation results at performances maps of compressor and turbine;
- forming tables with simulation results and exporting them to files.

3 VERSION AND INSTALLATION, SOFTWARE CONTENT

Equation Chapter (Next) Section 1

3.1 Software version and installation

Program for mathematical modeling of GTE thermodynamic parameters is delivered on CD. In the beginning, it is necessary to insert the disk in CD-ROM and manually start installation process by running "*ModelsSP12.exe*". This will run installation wizard (fig. **3.1**). Installation happens in automatic mode. . User has an opportunity of choosing destination folder to install the program (default is *C:\Models*). After installation, it is necessary to finish installation wizard work.

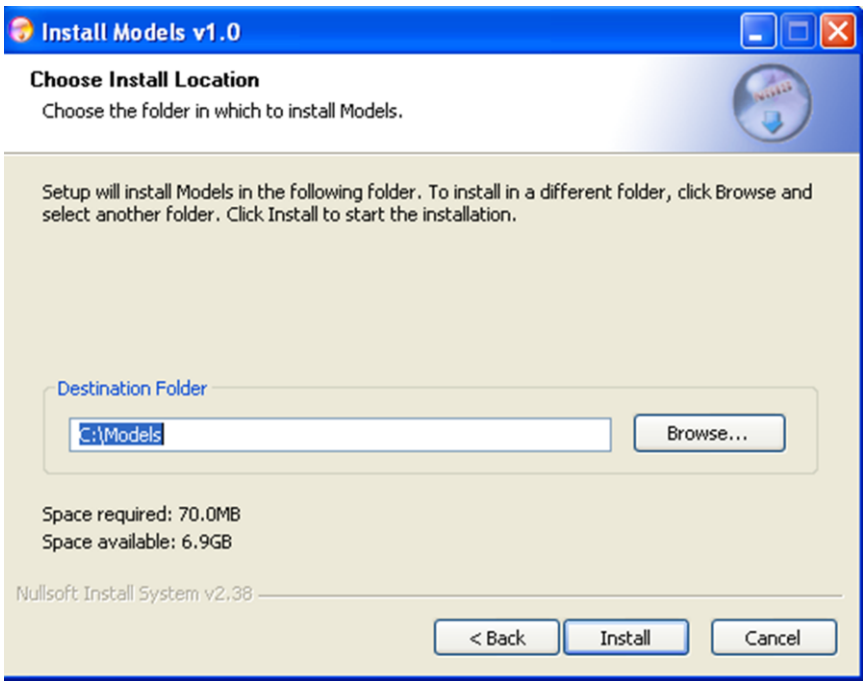


Figure 3.1 – Installing software in automatic mode

3.2 Software content

After installation, program is in folder named "MODELS". This folder contains the executive file to start the program (MODELS.EXE), file with current program settings (MODELS.INI), folders with different engine types and schemes, folders with appropriate language settings (EN – files of English interface, RU – files of Russian interface), and also folder INIT with files that contain general list of calculated (Appendix E) and state parameters (Appendix G).

The folder with simulated engine (SP12) has the following subfolders:

BIN contains all necessary modules for different tasks calculating;

DATA contains information about performances of engine units, initial approximations of independent variables, their measuring range;

DATAR contains files with input parameters of previously simulated tasks;

RESULT contains output files with results of an analysis.

Subfolders with interface language settings (their names correspond to language).

Brief descriptions of files are given in Table 3.1.

Table 3.1 - Names and short descriptions of files, present in the program of mathematical modeling of thermodynamic parameters of SP12 turbofan

File name	Application
Main folder (default Models)	
MODELS.EXE	Execute file to run the program
MODELS.INI	Current interface settings. File contains language settings for the next loading of the program, default engine type, and list of present interface languages.
UNINSTALL.EXE	Executive file to uninstall the program
Models\Ru (other language has the same list of files)	
INCLUDES.INI	Names of include files in Russian
JOBS.INI	Names of studies and configurations, projects descriptions in Russian
MENU.INI	Names of different menu items in Russian
NAMEHAR.TXT	Names and descriptions of state parameters in Russian
NAMEPAR.TXT	Names and descriptions of engine parameters in Russian
MANUAL_RU.PDF	User`s guide
NEW.PDF	What is new in current version of the program
WINDOWS.TXT	Text information which is applied for dialogue windows (Russian). They appear during the calculation.
Models\INIT	
NAMEHAR.TXT	Names and descriptions of state parameters in current language
NAMEPAR.TXT	Names of engine parameters and their descriptions in current language
MODELS\SP-12\BIN	
DROSS.EXE	Program to simulate static modes, throttle, altitude-velocity and climatic performances
DINAM.EXE	Program to simulate transient processes
KLDM.EXE	Program to calculate coefficients of QLDM
IDEN.EXE	Program to run the identification of the model according to test results
IDENTIF_RU.EXE	Program to start the interface for preparing data to identify model by test results
OWLETGR.EXE	Program to display the results of identification

Continuation of Table 3.1 – Names and short descriptions of files, present in the program of mathematical modeling of thermodynamic parameters of SP12 turbofan

MFTFE-2SHAFT\DATA	
CONTROLLERS.DAT	File with information about current settings of ACS controllers
MEH.DAT	Mechanical properties of rotors and mechanization
NEZPER0.DAT*	File with initial approximations of independent variables
PREDNP.DAT*	File with range of independent variables variation
RESULTSTN.DAT	Transient performance of LPT stator
RESULTSTV.DAT	Transient performance of HPT stator
RESULTTND.DAT	Transient performance of LPT disk
RESULTTNDBL.DAT	Transient performance of LPT blade
RESULTTVD.DAT	Transient performance of HPT disk
RESULTTVDBL.DAT	Transient performance of HPT blade
TEPL.DAT	Parameters of heat exchange
VBV.DAT	The performance of air bleeding valve mounted downstream the boosters. These valves bleed air to the bypass.
VOLUME.DAT	Characteristic volumes
XKND1.DAT*	Performances of primary flow part of fan and boosters
XKND2.DAT*	Performances of secondary flow part of fan
XKS.DAT*	Performance of main combustion chamber
XKVDC+3.dat	The performance of HPC if IGV positions is different from design one for +3 degrees.
XKVDC-0.dat	The performance of HPC if IGVs are at their design positions
XKVDC-3.dat	The performance of HPC if IGV positions is different from design one for –3 degrees.
XKSM.DAT*	Geometrical parameters of the engine.
XOTBOR1.DAT*	File with information about absolute and relative air bleeds and taps, and power takeoffs in case of air bleeding from the fifth stage of HPC for active clearance control system needs
XOTBOR2.DAT*	File with information about absolute and relative air bleeds and taps, and power takeoffs in case of air bleeding from the ninth stage of HPC for active clearance control system needs
XOTBOR3.DAT*	File with information about absolute and relative air bleeds and taps, and power takeoffs in case of air bleeding from both, fifth and ninth stage of HPC for active clearance control system needs
XPER*.DAT	Performances of transient ducts
XSOPL1.DAT*	Performance of primary nozzle
XSOPL2.DAT*	Performance of secondary nozzle

* File names may differ, but content is important. It determines file belonging to one or another engine unit (see Appendix A).

XTND.DAT*	Performance of LPT

Ending of Table 3.1 – Names and short descriptions of files, present in the program of mathematical modeling of thermodynamic parameters of SP12 turbofan

XTND1.DAT*	The performance of first stage of LPT
XTVD.DAT*	Performance of HPT
XVHOD.dat	Intake performance
MODELS\MFTFE-2SHAFT\RU	
PROJECT.BMP	Scheme of the modeled engine with specified stations and their numbers according to general list of parameters
*.bmp	Pictures for interface
GRAPHICS.INI	File with displaying settings in Russian
JOBS.INI	File with the study settings of the particular engine in Russian
MISC.INI	File with different settings of the particular engine in Russian
NAMEPAR1.TXT	File with names of unique parameters for particular engine and their numbers in general list

* File names may differ, but content is important. It determines file belonging to one or another engine unit (see Appendix A).

4 HARDWARE AND SOWTWARE REQUIREMENTS

Equation Chapter (Next) Section 1

4.1 Minimal and recommended hardware requirements

Recommended requirements to PC:

- any CPU Intel (Celeron, Pentium), AMD (Duron, Athlon) with clock 300MHz or higher;
- 64 Mb RAM or more;
- 17”of bigger SVGA display;
- more than 50 Mb of free space in HDD.

4.2 Software requirements

Program works with Windows 2000/XP/7/8 operation systems.

5 STARTING THE “MODEL”

Equation Chapter (Next) Section 1

To start working with the program it is necessary to run it by calling program MODELS.EXE from installation directory or by clicking twice on the link “MODELS” on a desktop.

After successful starting, main program window appears, shown in fig. 5.1

Program works at interactive mode using menu or toolbar.

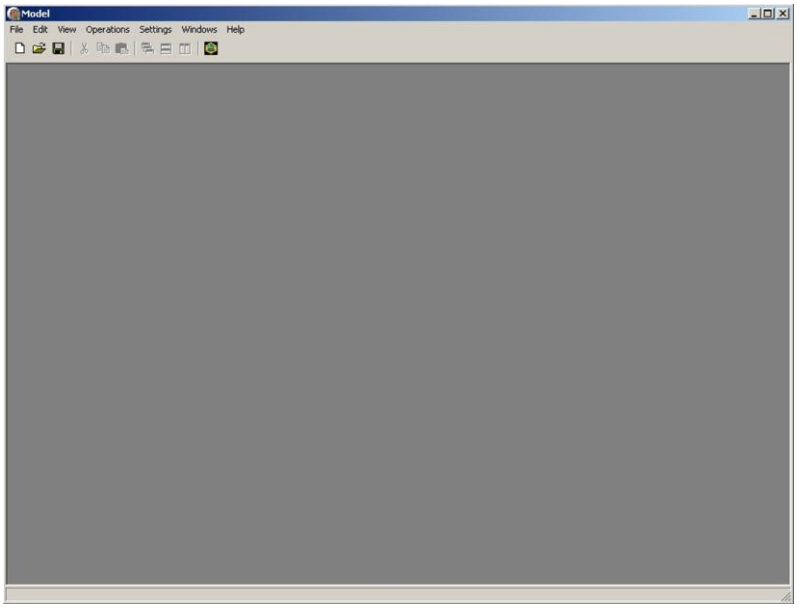


Figure 5.1 – Main window of the program

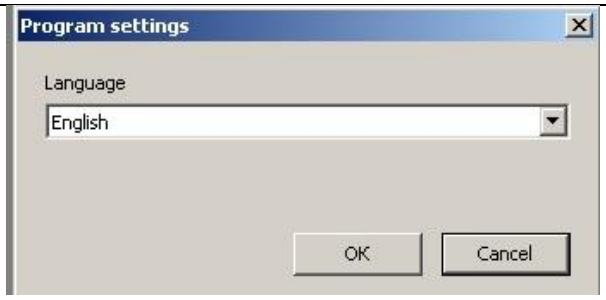


Figure 5.2 – Menu for choosing language

For user convenience, program has language choosing options: ***Settings*** **=>*Options***. Next appears window, where user can choose the language (fig. 5.2).

6 WORKING WITH PROGRAM

Equation Chapter (Next) Section 1

6.1 Loading initial data

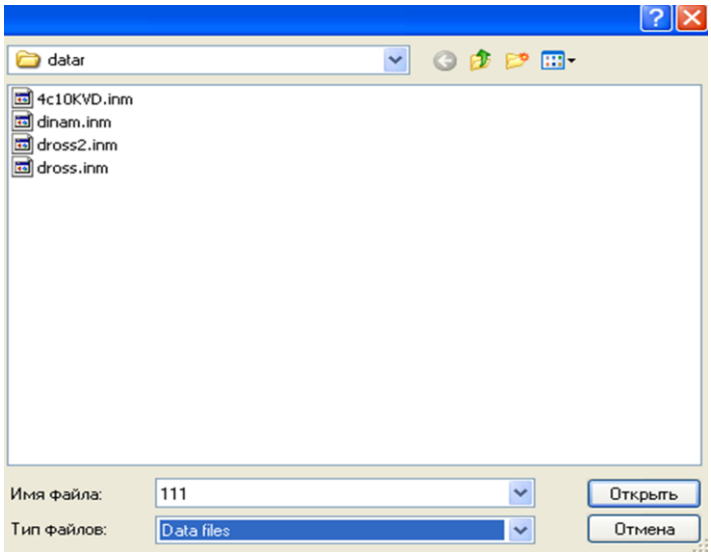


Figure 6.1 – Dialogue window for opening previously created project (input data)

Input data file of current project is in the DATAR folder.

Working with program starts from loading file with initial data. To do this, it is necessary to click next menu item ***File*** ⇒ ***Open*** or click the icon  on the toolbar (or use buttons combination Ctrl+O).

Then appears the interactive window where user chooses file with input data (fig. 6.1). The default location to be opened is the DATAR folder with files (*.inm), which have input data for the simulation. User can navigate folders of PC and choose any available file with input data (“*.inm”).

Input data file is a text file with all necessary input parameters to run the calculation of gas-path parameters.

After opening and loading file with input data appears the scheme of engine gas-path and menu for inputting initial data and outputting calculation results (fig. 6.2). This menu has following structure (it is duplicated in a form of anchors):

- Study:
  - Basic data.
- Initial data:
  - Initial settings;
  - Include files;
  - Control laws.
- Input parameters:
  - gas path;
  - units performances maps;
  - ICM;
  - control actions.
- Calculated parameters:
  - gas-path;
  - units performances maps.
- Calculation results:
  - Parameter diagrams;
  - Table;
  - Operating lines;

- Files-tables.

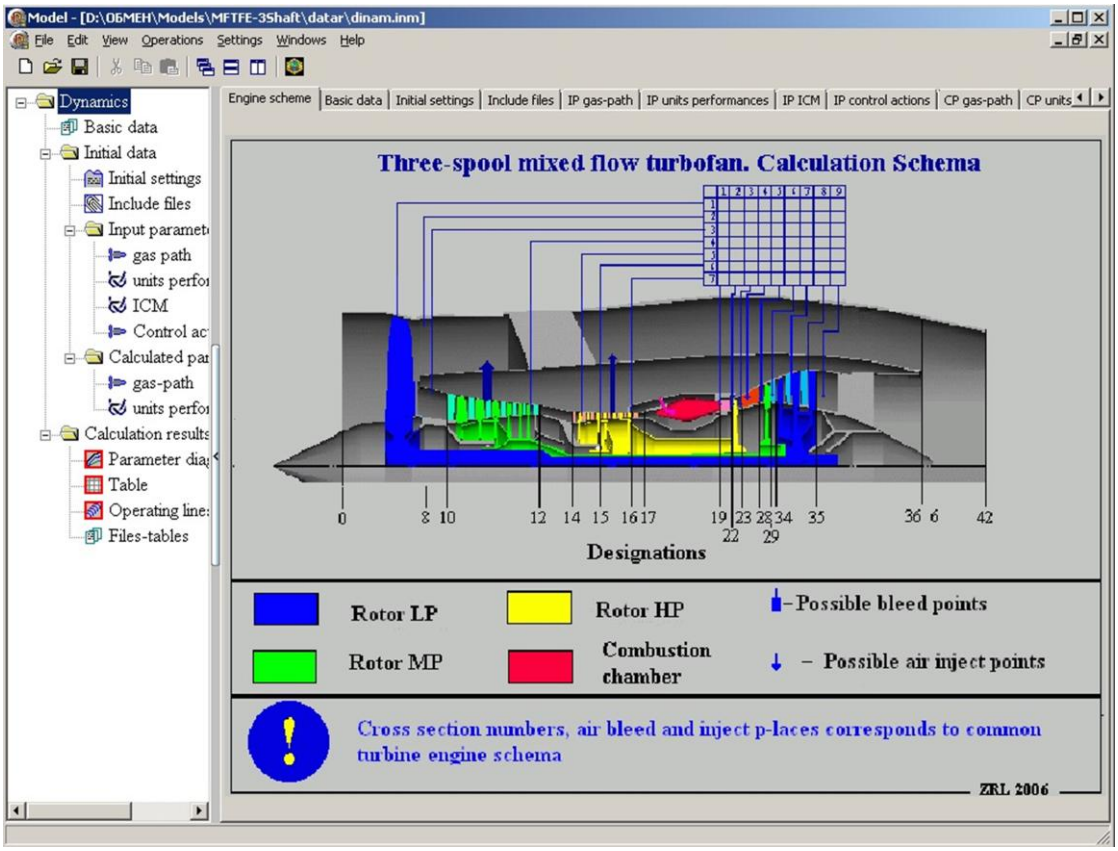


Figure 6.2 – Main window of the project

If there is a necessity in creating new file with input data, User can load already existing file and save it with another name, clicking icon or main window menu item ***File⇒Save as***. Then appears dialogue window, where User can save currently open project into the file with another name.

Besides, it is possible to create a new file with input data by the following main menu item ***File⇒New*** (Ctrl+N) or by clicking icon at control panel of the main window. Then dialogue window without additional information appears to set input data (fig. 6.2).

6.1.1 Configuring a calculating mode of an engine

Configuring a calculating mode is done by control elements of main menu ***Initial data ⇒ Initial settings*** (fig. 6.3). This menu item consists of seven tabs with boxes for inputting information.

Tab "***Study***" (fig. 6.3) allows User to choose (put checkmark) any of five variants of calculation:

– "***Dynamics***" means that at next starting program will calculate transient process that is characterized by its parameters, chosen according to given

next description of gas-path parameters, and parameters, which determine operating line of transient process.

- ***"Static performance map"*** means that at next starting program will calculate gas-path parameters at steady modes. Choosing this variant of calculation User can calculate throttle, velocity, altitude and climatic performances.
- ***"Gas-path straight-through calculation"*** means that at next starting program will calculate first given point and display information about residuals and independent numerical values. This calculation variant allows checking the correctness of set independent variables at program debugging phase. If residuals are too big, that means that independent variables differ a lot from those, which are proper for the simultion.
- ***"Identification"*** means that at next starting program will run the interface for identification of model parameters according to results of tests.
- ***"LDM coefficients calculation"*** means that at next starting program will start the subroutine for LDM coefficients calculation.

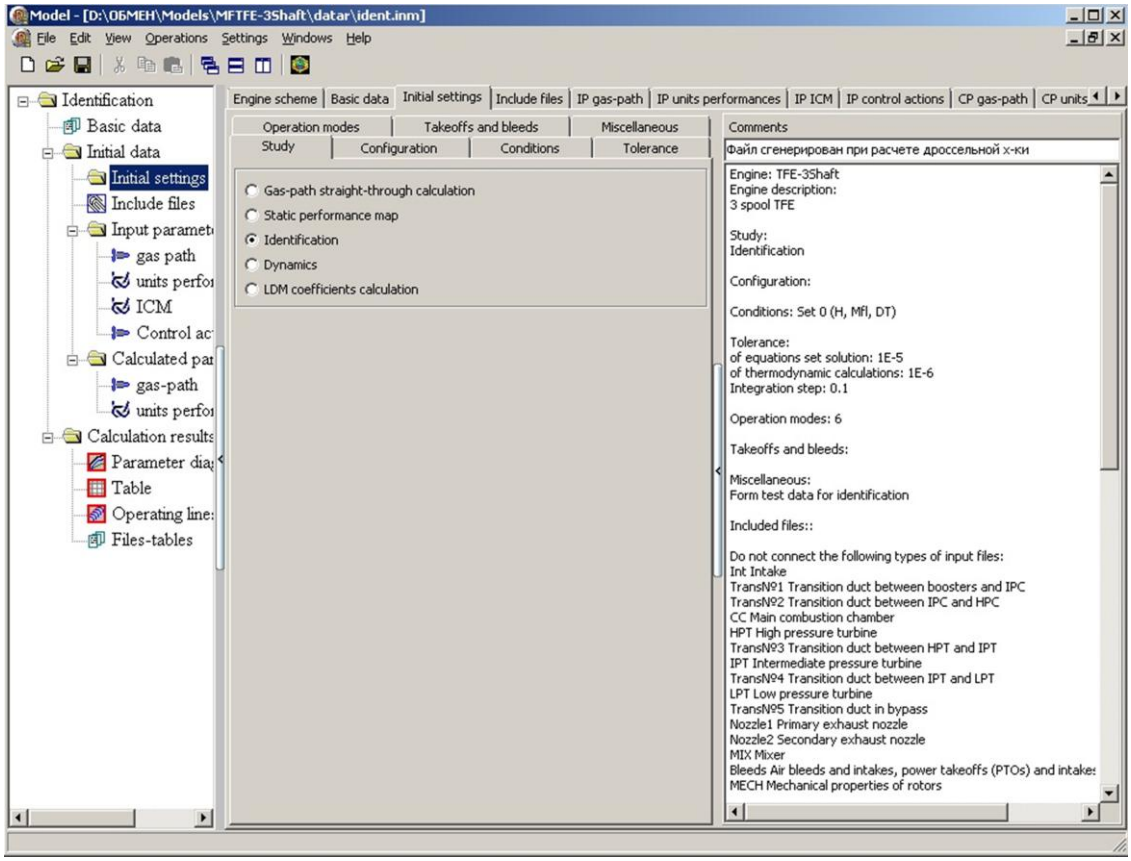


Figure 6.3 – Menu item "Initial settings"

Universal mathematical models do not need engine types to be configured, that is why only one configuration is present in "***Configuration"*** tab (fig. 6.4):

In tab ***"Configuration"*** User can choose the analysis type:

- without automatic control system (ACS) – program simulates the engine operation without limiters according to given parameters;

– with automatic control system (ACS) – program simulates the engine operation taking into account the active limiters. Engine operating mode is set by the position of control stick.

Tab ***"Miscellaneous"*** has no functions in this version.

"Modes" (fig. 6.5) is the tab, where User can set information about number of benchmark points to calculated parameters of transient process or about number of points to calculate throttle performance.

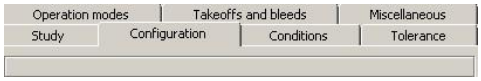


Figure 6.4 - Tab "Configuration"

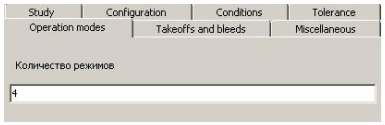


Figure 6.5 - Tab "Operation modes"

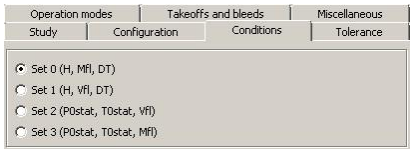


Figure 6.6 - Tab "Conditions"

To configure airflow conditions at engine inlet (for more information see subsection 1.1.2.4), User must go to "Conditions" tab (fig. 6.6) and put the checkmark in front of one of four present sets:

– H, M, ΔT;
– H, v_f, ΔT;
– P_H^*, T_H^*, v_f;
– P_H^*, T_H^*, M.

"Tolerance" tab (fig. 6.7) has numerical parameters of calculation:

– tolerance of equations set solving;
– tolerance of thermodynamic calculations;
– integration step, which is actually a discretization step, used to calculate transient process (c).

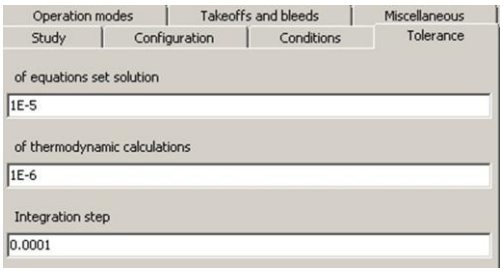


Figure 6.7 - Tab «Tolerance»

To edit mentioned above parameters, User must place cursor in corresponding box and click twice by LMB; then, using keyboard, change values of parameters.

All done changes are immediately displayed on the right (fig. 6.3). In comment box User may type any text information about the calculation.

6.1.2 Editing parameters of units performances

Next step after configuring information about engine operating mode is specifying information about units of an engine. This procedure is carried out in menu item ***Initial data⇒Included files*** (fig. 6.8), or in tab ***Included files***.

Each engine unit is represented with its performance, which is in its own file. For convenience, program displays not all files with performances, but only files with performances of chosen unit. Unit specifying is done by pull-down menu in the bottom of

central window; clicking appears, list of units (fig. 6.8). To choose the unit, which parameters need to be changed (or reviewed), it is necessary to place the cursor and to click by LMB. List of files with performances of this unit appears (fig. 6.10 shows an example of file list with intake performances). Number of files can be arbitrary, but only one file with performances of unit can be included to project by putting the checkmark in front of selected file.

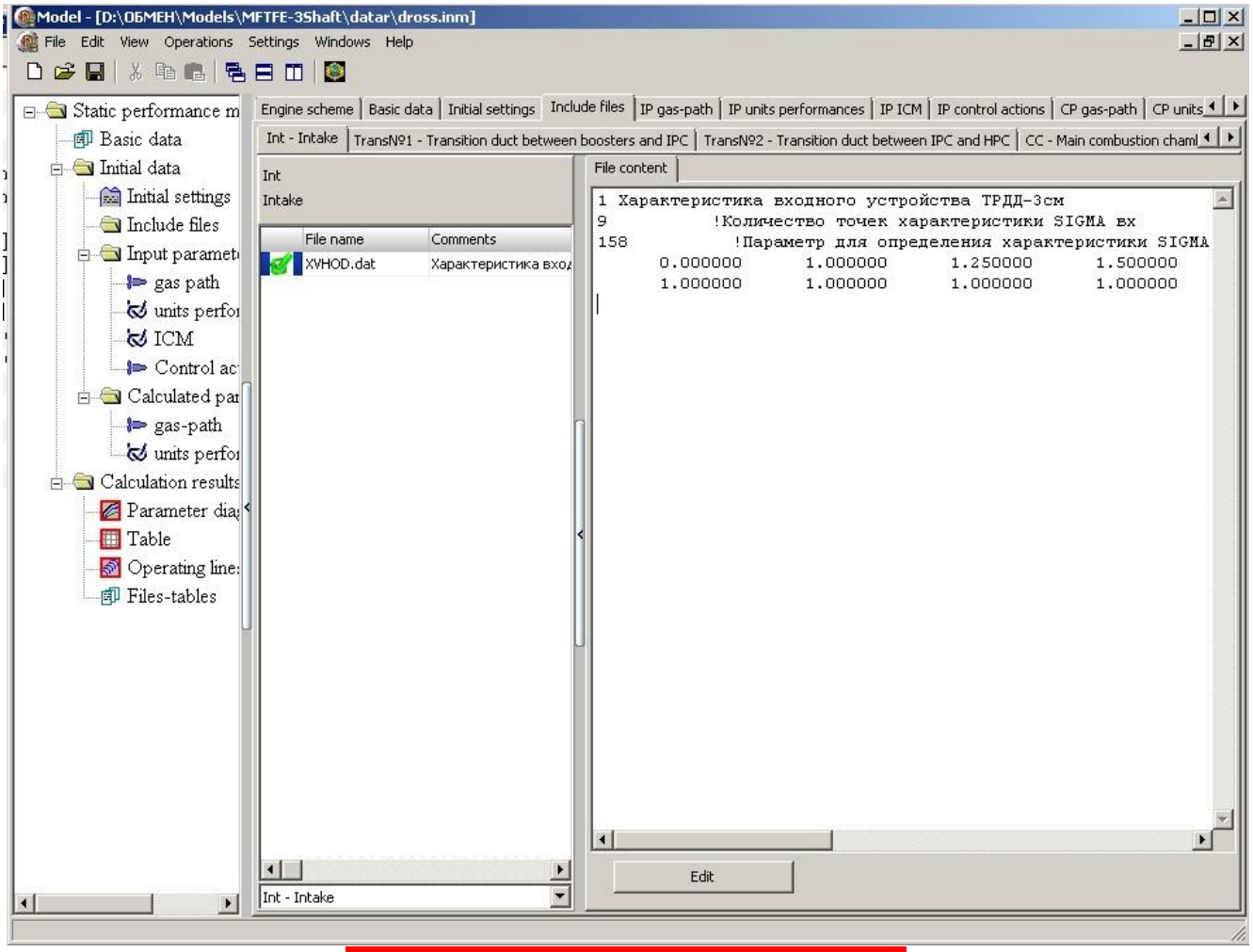


Figure 6.8 – Menu item «Include files »

If User creates new project, which is not based on existing one, then list of included files must be checked (fig. 6.8) and only one file must be chosen in each item of menu. Switching between include files can be done by tabs, which are present above the window with chosen file content. Tabs can be scrolled by control elements .



Figure 6.9 –Message informing user about not full set of include files

Each project type must comprise the definite number of files. This number corresponds to the list of included files. Number shows file belonging to this group of engine units.

In case when file is absent in any group because of any reason, User will see the information massage (fig. 6.9) about this when will start calculation. (see subsection 6.1.7)

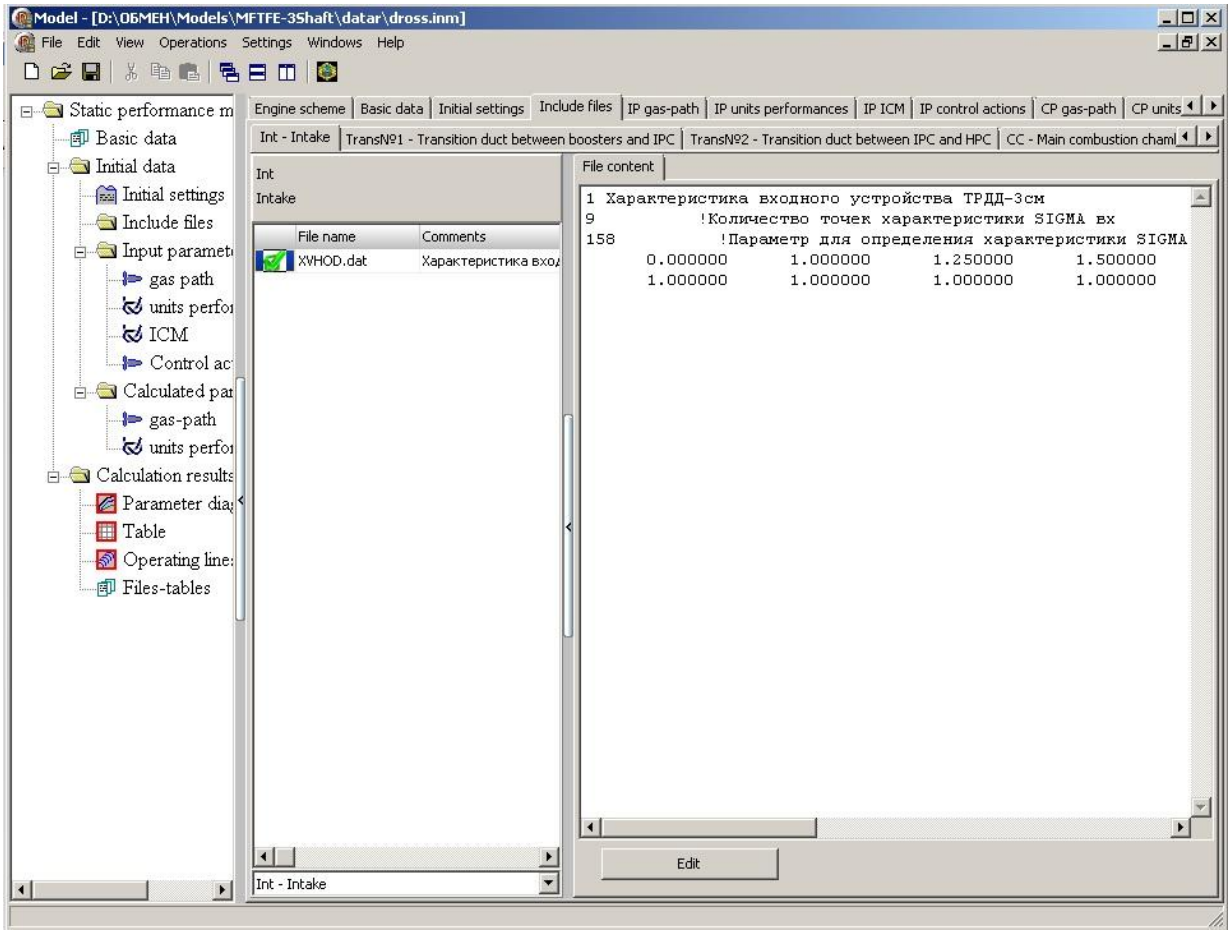


Figure 6.10 – Choosing performance of engine unit

All missing files are shown in the right part of information window (menu item ***Initial data⇒Initial settings*** fig. 6.3) in the list ***"Following types of input files are not included"*** (fig. 6.11).. The list is under list ***"Included files"***.

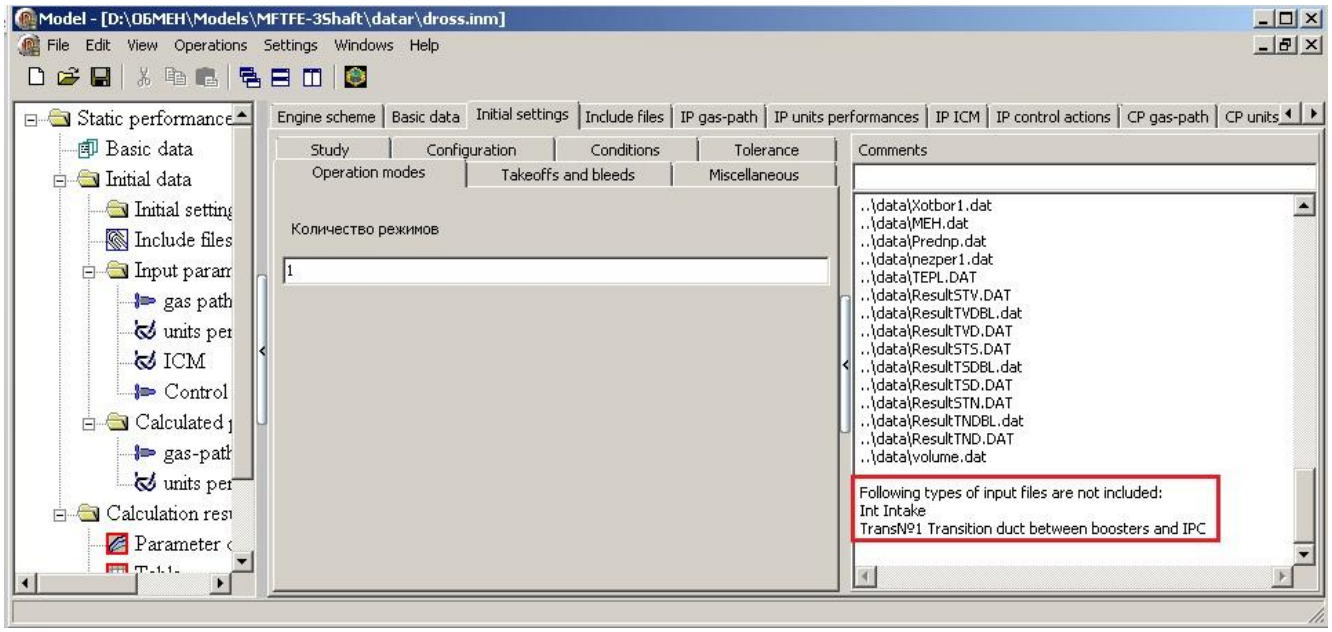


Figure 6.11 – The list of files that are not included in the project

Table 6.1 - The list of files, which are included to the model

Component name	NO of file
Intake – performance of intake	1
KND2 - fan (secondary flow part**)**	3
KND1 - fan (primary flow part)	4
Trans NO1 – Transition duct between boosters and variable bleeding valve VBV	8
Trans №2 – Transition duct between boosters and HPC	11
HPC(ALF) performance of HPC at set value of α_{IGV} (there are three performances, which are stratified by the α_{IGV})	1210 1220 1230
CC - main combustion chamber	14
HPT – high pressure turbine	15
Trans №3 – Transition duct between HPT and LPT	16
LPT1 – low pressure turbine, 1st stage	17
LPT – low pressure turbine	19
Trans №4 – transition duct between fan and variable bleeding valve VBV in the bypass	18
Trans№5 - transition duct between variable bleeding valve VBV and secondary nozzle discharge	20
Nozzle1 – primary nozzle	21
Nozzle2 – secondary nozzle	22
Bleeds – air bleeds and taps, power take-offs and supplies for different active clearance control programs (three programs)	28 281 282
MECH – mechanical properties of rotors, mechanization	30
NEZP – file with initial approximations of parameters	31
LIMIT – limits of independent variables	32
ACS – constants of control laws	40
TEPL – heat exchange parameters	41
TRANSSTV – transient performance of HPT casing	42
TRANSBLV - transient performance of HPT blade	43
TRANSDV - transient performance of HPT disk	44
TRANSSTN - transient performance of LPT casing	48
TRANSBLN - transient performance of LPT blade	49
TRANSDN - transient performance of LPT disk	50
VOLUME – Characterisitc volumes	51
VBV – calculated performace of variable bleeding valve VBV	1500

Content of chosen file is displayed in tabular format in the right window.

To edit files with performances it is necessary to press the button ***"Edit"***. Each file type has its own editor.

6.1.2.1 Editing parameters of intake

After choosing intake performance (fig. 6.10), and setting checkbox in front of one of files in the corresponding group (fig. 6.8), User will find text information about intake performance in the right box.

Button ***"Edit"*** calls the editor of the performance (fig. 6.12).

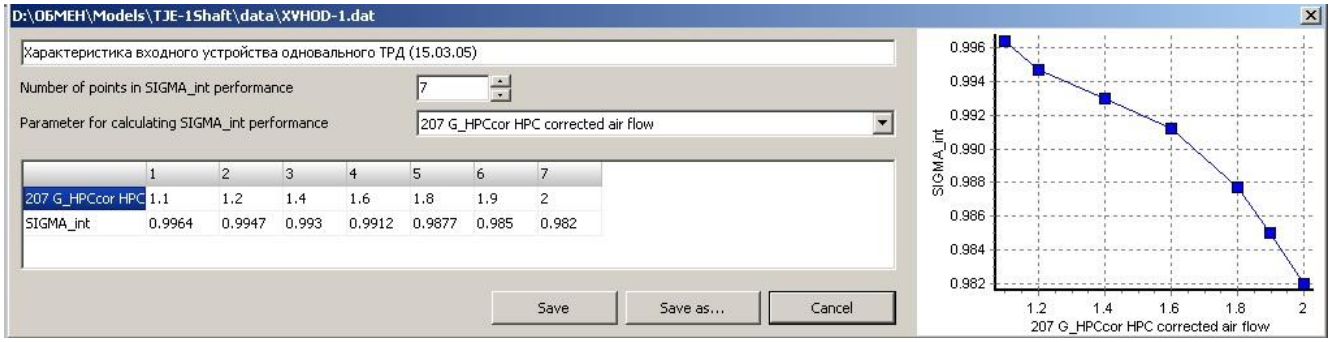


Figure 6.12– Editor of intake performance

User has an opportunity to add any comment in the upper part of the window. Next User must enter information about number of points that form the performance. Pushing up arrow ▲ once increases number of points by one, and conversely, pushing down arrow ▼ decreases number of points by one. According to number of points program forms the table (number of points corresponds to number of columns in the table) to input the data.

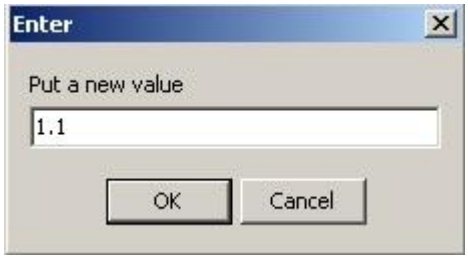


Figure 6.13– Menu for inputting a value



Figure 6.14– Menu for box editing

Choosing the parameter, that will be an argument of the performance σ_{in} can be done by pull-down menu. It consists of list of parameters according to general list (Appendix E). The most commonly used arguments are Mach number (M_{fl}) and corrected airflow at engine inlet ($G_{cor\ in}$).

There is a box for information entering in central part of the window. There are two ways to edit the values.

Way 1: move cursor to the cell, which needs to be changed, and click twice by LMB. This will reveal the input menu (fig. 6.13). After entering the value User must confirm the value by pressing ***«OK»*** or cancel the value by pressing ***«Cancel»*** (input menu will close without saving).

Way 2: is based on using the RMB. User must move cursor to the cell, which needs to be changed, and click once by LMB. This cell

will change color to blue (see fig. 6.12). Then, after pushing RMB, menu to edit value in the editing appears (fig. 6.14). It consists of four items:

- *Copy* – to copy the value from chosen cell into clipboard (hot key is Ctrl+C).
- *Paste* – to past the value from clipboard to chosen cell (hot key is Ctrl+V).
- *Paste from Excel* (hot key is Ctrl+Alt+V) – allows pasting the table from *Excel*. User must move pointer to cell, which is first to paste the data, and click once by LMB. This cell must change its color to blue. Then, clicking RMB, reveals the menu with item *«Paste from Excel»*. Values can be pasted by lines or by array. If copied array has more than two rows, then only first two rows, starting from blue cell will be copied from clipboard (in the fig. 6.10 this row is named 207 $G_{HPC\ cor}$). The rest rows will be cropped. If User have chosen the cell from the second row to be the starting cell (this cell has the name *Sigmain* in the fig. 6.10), then only first row will be pasted from the clipboard. If number of columns exceeds previously given number of points in the performance, then number of columns is increased and this number is adopted as a new number of columns instead of previously set. Number of points must be within the limits from 2 to 100. If data inputting happens by lines, and number of values in the line exceeds previously set number of points in the performance then, similar to array, number of columns will be increased automatically. But as opposed to array all cells in the line without inserted values will be assigned zeroes. This zeroes can be chosen by User before saving.
- *Delete* (hot key Del) – to delete value from marked cell.

The graph representing the data in the table will automatically appear in the right part of the editor.

The effect technical state of gas path makes on engine parameters needs to be determined when developing the ACS of GTE and alogithms for gas-path diagnostics. The degradation reveals itself in a form of deviations of gas-path geometrical parameters. But considered above model does not addresses the geometrical parameters, but parameters of components. They are calculated according to performances of components. Hence, given problem can be solved by consistent deforming the performances of components that will imitate the change of geometrical parameters.

The performances are generally functions of one or two variables. Let`s consider the initial performance $\nu_0 = f(y_1, y_2)$. All possible degradation effects can be simulated by the following deformations: parallel displacement of relative values of performance and turning the performance around the axes that intersect the performance. The axes are parallel to the reference axis. So, for example, the 2D performance in the parameterized form is the following:

$$\nu = \nu_0(1 + a\frac{y_1 - y_{10}}{\Delta y_1} + b\frac{y_2 - y_{20}}{\Delta y_2} + c) \tag{6.1}$$

where ν_0 - the value of parameter calculated by the initial performance; y_1, y_2 - current value of parameters, which are arguments of performance; y_{10}, y_{20} are reference values

of parameters that correspond to the point, which is the center around which turning is performed; $\Delta y_1, \Delta y_2$ are ranges of parameters variation in the performance; a, b, c are parameters of the considered component performance. The full list of state parameters is presented in Appendix G.

User can input the values of y_{10}, y_{20} and $\Delta y_1, \Delta y_2$ in the editor. The performance of intake is the function of one variable (Mach number at engine inlet). Hence, User must use only coefficients affecting the Mach number (i.e. *b*, see Appendix G). That is why only fields to input M_{n0} and ΔM_n become active. If the values in these fields (see fig. 6.45) differ from zeroes, then the performance will be recalculated by the formula 6.1. To show the new performance User must put the checkmark in the field "Shoe the corrected performance". Then two performances will be shown. The first one will be initial and the second one – corrected according to state parameters. For example, there is a corrected performance (coefficients are $b = 0.1$, $b = 0$, $M_{fl0} = 0.5$, $\Delta M_{fl} = 1$.) of the intake shown in fig. 6.15.

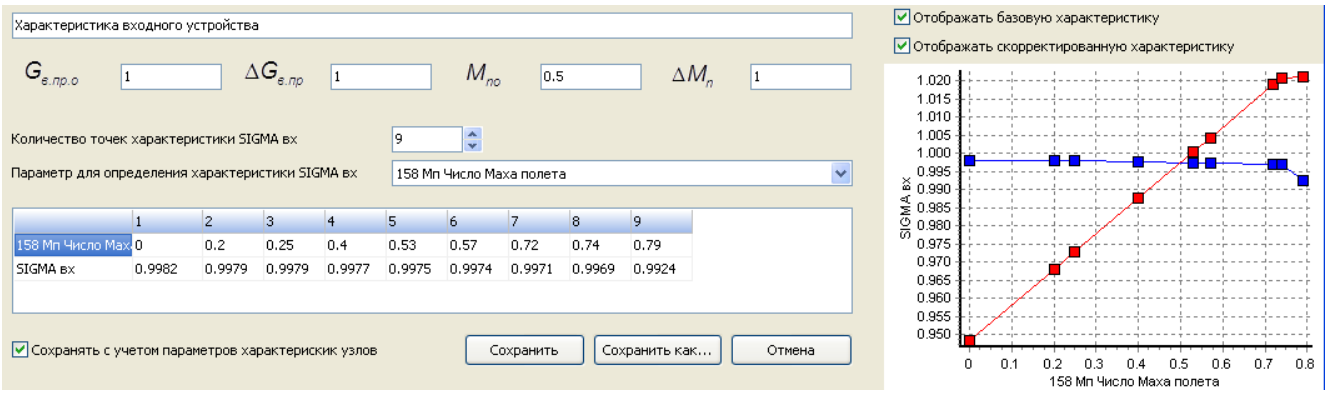


Figure 6.15– Editor of the intake performance with two performances shown

After editing the file User can save it with the same name by clicking the icon ***"Save"***, or with another name by clicking the icon ***"Save as…"***. After this appears dialogue window, where User can name the file (box ***File name***) and save it (button ***"Save"***). Default location for saving the file is the current folder with performances of engine. The saved file will contain the set performance in tabular format and also y_{10}, y_{20} and $\Delta y_1, \Delta y_2$, which were inputted in corresponding fields.

If the field "Save with parameters of component performances" is ticked, then the performance recalculated by formula 6.1 will be saved in file. There will appear the comment "corrected performance" in the file.

Important: If the corrected performance was saved then, when including it to the project User must set zeroes in the corresponding fields (fig. 6.44) for inputting state parameters (fields with coefficients). Otherwise the program will correct the performance twice.

Content of file with intake performances is presented in Appendix A1. Calculating σ_{int} is performed by linear interpolation between adjacent points.

6.1.2.2 Editing performances of compressor

Performances of HPC are stratified by IGVs installation angle (HPC performance has three positions of IGV). The performance of LPC consists of secondary flow performance (fan performance) and primary flow performance (fan performance and boosters performance together).

Performance of each spool is a dependence of π_C^* and η_C^* from spool inlet corrected rotational speed $n_{C\ cor} = n_C \sqrt{\frac{T_{ACS}}{T_{in\ C}^*}}$ and inlet corrected airflow rate $G_{a\ cor} = G_a \sqrt{\frac{T_{in\ C}^*}{T_{ACS}}} \frac{p_{ACS}}{p_{in\ C}^*}$, where: $T_{in\ C}^*$ is the temperature at spool inlet; $p_{in\ C}^*$ is the pressure at spool inlet; $n_{C\ cor}$, n_C are inlet corrected and real rotor rotational speeds, $G_{a\ cor}$, G_a are inlet corrected and real airflows at spool inlet. So, the performance of the compressor is of the following form:

$$\pi_C^* = f\left(n_{C\ cor}, G_{a\ cor}\right);\ \eta_C^* = f\left(n_{C\ cor}, G_{a\ cor}\right)$$

This performance (fig. 6.16) is presented as a set of constant speed lines, which correspond to constant values of $n_{C\ cor}$. Universal method to parameterize one pair of these lines is distinguishing characteristic points (fig. 6.16) and describing them with corresponding π_C^*, $G_{C\ cor}$ and η_C^*.

Current engine operating mode is a point on compressor performance map, and the set of steady operating modes form operating line. So, compressor performance is expressed by two bivariate functions. That is why, to calculate all 4 parameters, which characterize compressor operating mode, it is necessary to specify any two of them as arguments to find two others from performances.

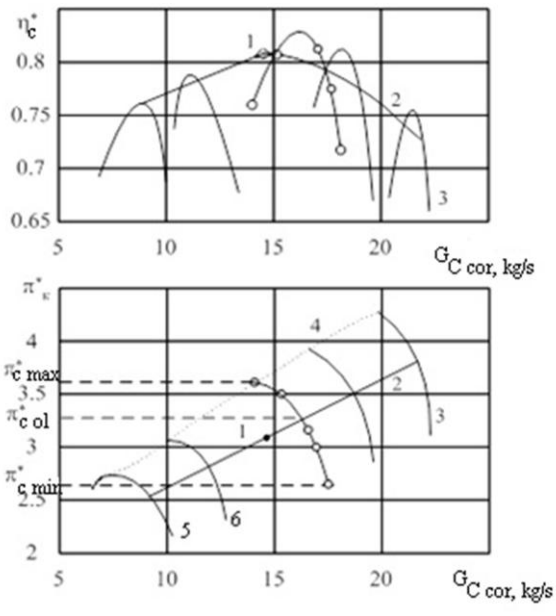


Figure 6.16 – Performance of compressor spool:
1 – Point of current mode;
2 – operating line;
3 – lines $n_{C\ cor} = const$;
4 – stability margin;
5 – low-head line;
6 – high-head line.

On the face of it, $n_{C\ cor}$ and $G_{C\ cor}$ suit better as arguments. But it is necessary to take into account the feature of performance shown in fig. 6.16: first argument $n_{C\ cor}$ has corresponding variation range of the second argument $G_{C\ cor}$.

As it was considered in paragraph 1, iterative procedure is used for engine parameters calculating, during which input parameters for calculating each unit, may vary in wide range and be mutually non-consistent. Specified mismatch between $n_{C\ cor}$ and $G_{a\ cor}$ may cause given

value of the second argument breaking variation range of the first argument.

To cope with this problem compressor performance map uses relative position of working point at constant corrected speed line (fig. 6.17) as the second argument of the correlations:

$$Z_C = \frac{\pi_C^* - \pi_{C\,min}^*}{\pi_{C\,max}^* - \pi_{C\,min}^*} \text{ or } Z_C = \frac{G_C - G_{C\,min}}{G_{C\,max} - G_{C\,min}}.$$

The used equation for calculating relative position of working point at constant corrected speed line Z_C depends on form of constant speed lines. If all constant speed lines in the performance map are high-pressure, i.e. small change of $G_{C\,cor}$ results in significant change of π_C^* (almost vertical lines, fig. 6.16, line 3), then Z_C must be determined according to π_C^*. If constant speed lines have inflection by π_C^* meant that single value of π_C^* corresponds to two values of $G_{C\,cor}$, fig. 6.16 line 5), then Z_C must be determined according to $G_{C\,cor}$.

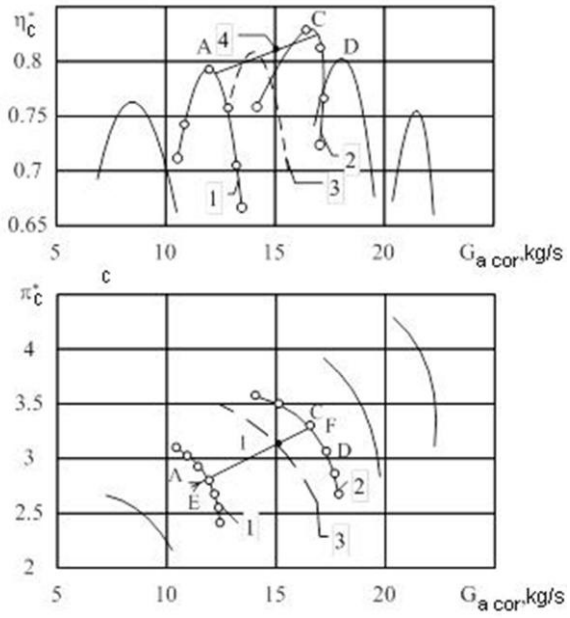


Figure 6.17 – Calculating parameters in given point of the performance map of the compressor:
1 – line $n_{C\,cor\,i} = const;$
2 – line $n_{C\,cor\,i+1} = const;$
3 – lines $n_{C\,cor} = const;$
4 – design point.

For constant corrected speed lines this parameter has fixed variation range: from 0 to 1.

The simplest algorithm for $G_{C\,cor}$, π_C^* and η_C^* calculating, according to given $n_{C\,cor}$ and Z_C is based on linear interpolation. $n_{C\,cor\,i}$ and $n_{C\,cor\,i+1}$ tabular values are initially calculated to meet the condition $n_{C\,cor\,i} \le n_{C\,cor} \le n_{C\,cor\,i+1}$ (fig. 6.17). Then, points A, B, C, D must be calculated on each corresponding constant speed line, which are given in the table of compressor performances (in a form of constant speed lines set). These points must meet next requirements

$$Z_B \le Z_C \le Z_A \; u \; Z_D \le Z_C \le Z_C.$$

Program interpolates constant speed lines to find points ***E*** and ***F*** according to known coordinates of points ***A***, ***B***, ***C*** and ***D***. Then corresponding values of parameters in points ***E*** and ***F*** are calculated.

$$G_{C\,cor\,E} = G_{C\,cor\,A} + \frac{Z_{C\,A} - Z_C}{Z_{C\,A} - Z_{C\,B}}\left(G_{C\,cor\,B} - G_{C\,cor\,A}\right);$$

$$\pi^*_{C\,E} = \pi^*_{C\,A} - \frac{Z_{C\,A} - Z_C}{Z_{C\,A} - Z_{C\,B}}\left[\pi^*_{C\,A} - \pi^*_{C\,B}\right]; \eta^*_{C\,E} = \eta^*_{C\,A} + \frac{Z_{C\,A} - Z_C}{Z_{C\,A} - Z_{C\,B}}\left[\eta^*_{C\,B} - \eta^*_{C\,A}\right];$$

$$G_{C\,cor\,F} = G_{C\,cor\,C} + \frac{Z_{C\,C} - Z_C}{Z_{C\,C} - Z_{C\,D}}\left(G_{C\,cor\,D} - G_{C\,cor\,C}\right);$$

$$\pi^*_{C\,F} = \pi^*_{C\,C} - \frac{Z_{C\,C} - Z_C}{Z_{C\,C} - Z_{C\,D}}\left[\pi^*_{C\,A} - \pi^*_{C\,D}\right]; \eta^*_{C\,F} = \eta^*_{C\,C} + \frac{Z_{C\,C} - Z_C}{Z_{C\,C} - Z_{C\,D}}\left[\eta^*_{C\,C} - \eta^*_{C\,D}\right].$$

And finally, program linearly interpolates between points ***E*** and ***F*** according to $n_{C\,cor}$, to get unknown parameters in modeled point:

$$G_{C\,cor} = G_{C\,cor\,E} + \frac{n_{C\,cor} - n_{C\,cor\,i}}{n_{C\,cor\,i+1} - n_{C\,cor\,i}}\left(G_{C\,cor\,F} - G_{C\,cor\,E}\right);$$

$$\pi^*_C = \pi^*_{C\,E} + \frac{n_{C\,cor} - n_{C\,cor\,i}}{n_{C\,cor\,i+1} - n_{C\,cor\,i}}\left(\pi^*_{C\,cor\,F} - \pi^*_{C\,cor\,E}\right);$$

$$\eta^*_C = \eta^*_{C\,E} + \frac{n_{C\,cor} - n_{C\,cor\,i}}{n_{C\,cor\,i+1} - n_{C\,cor\,i}}\left(\eta^*_{C\,cor\,F} - \eta^*_{C\,cor\,E}\right);$$

One more group of parameters, effecting compressor performance map, is installation angles of guide vanes (GV). Turning mechanism is common for all movable GVs of each spool of compressor. That is why, installation angles of GVs are coordinated and usually their influence on compressor performance map is taken into account by single parameter, which is installation angle of inlet guide vanes α_{IGV}. Important assumption of this method is that α_{GV} of stages correspond to "standard" values, which are used in ACS designing for "standard" engine and determining the performances of "standard" engine as a representative of engine family.

The reference performance (base) performance was calculated according to design IGV control law (next VGV control law). The argument of the law is HPC inlet corrected rotational speed $n_{cor\ HPC}$ (see table 6.2). This information is inputted in the editor "Mechanical properties and mechanization", "VSV" tab (see fig. 6.18).

Table 6.2 – Design VSV installation angle

$n_{cor\ HPC}$, rpm	4011	5000	5798	7000	8074	9000	9364	9500	10031
VSV_0(°)	38	38	35.58	34	33.29	32.08	31.52	30.94	28.3
$n_{cor\ HPC}$, rpm	10250	10759	11100	11271	11725	12036	12131	12250	12400
VSV_0(°)	27	23.08	20	18.09	11.6	3.7	2.7	1.8	1
$n_{cor\ HPC}$, rpm	12500	13000							
VSV_0(°)	0.7	0							

This model implies the method of taking into account the change of α_{IGV}:

- The position of blades at steady modes is controlled according to the law $VSV = \mathrm{f}\left(n_{HP}, T^*_{in\ \mathrm{HPC}}\right)$ (see Table 6.3). The control program is set in the editor "Mechanical properties and mechanization" (see fig. 6.18). The algorithm to input the data is similar to considered above algorithm for the intake, see subsec. 6.1.2.1;
- At transient modes (acceleration or deceleration) the position of blade is controlled according to the law $VSV = \mathrm{f}\left(n_{HP}, T^*_{in\ \mathrm{HPC}}\right)$ (see Table 6.4). The editor is shown in fig. 6.18, where T^*_{in} – total temperature at engine inlet; $T^*_{in\ \mathrm{HPC}}$ – total temperature at HPC inlet, n_{HP} – HP rotor rotational speed, $n_{cor\ \mathrm{HPC}\,0}$ – engine inlet corrected HP rotor rotational speed.

Table 6.3 –VSV control program at steady modes

$n_{\mathrm{cor\ HPC}}$, rpm	4011	5000	5798	7000	8074	9000	9364	9500	10031
VSV(°)	38	38	35.58	34	33.29	32.08	31.52	30.94	28.3
$n_{\mathrm{cor\ HPC}}$, rpm	10250	10759	11100	11271	11725	12036	12131	12250	12400
VSV(°)	27	23.08	20	18.09	11.6	3.7	2.7	1.8	1
$n_{\mathrm{cor\ HPC}}$, rpm	12500	13000							
VSV(°)	0.7	0							

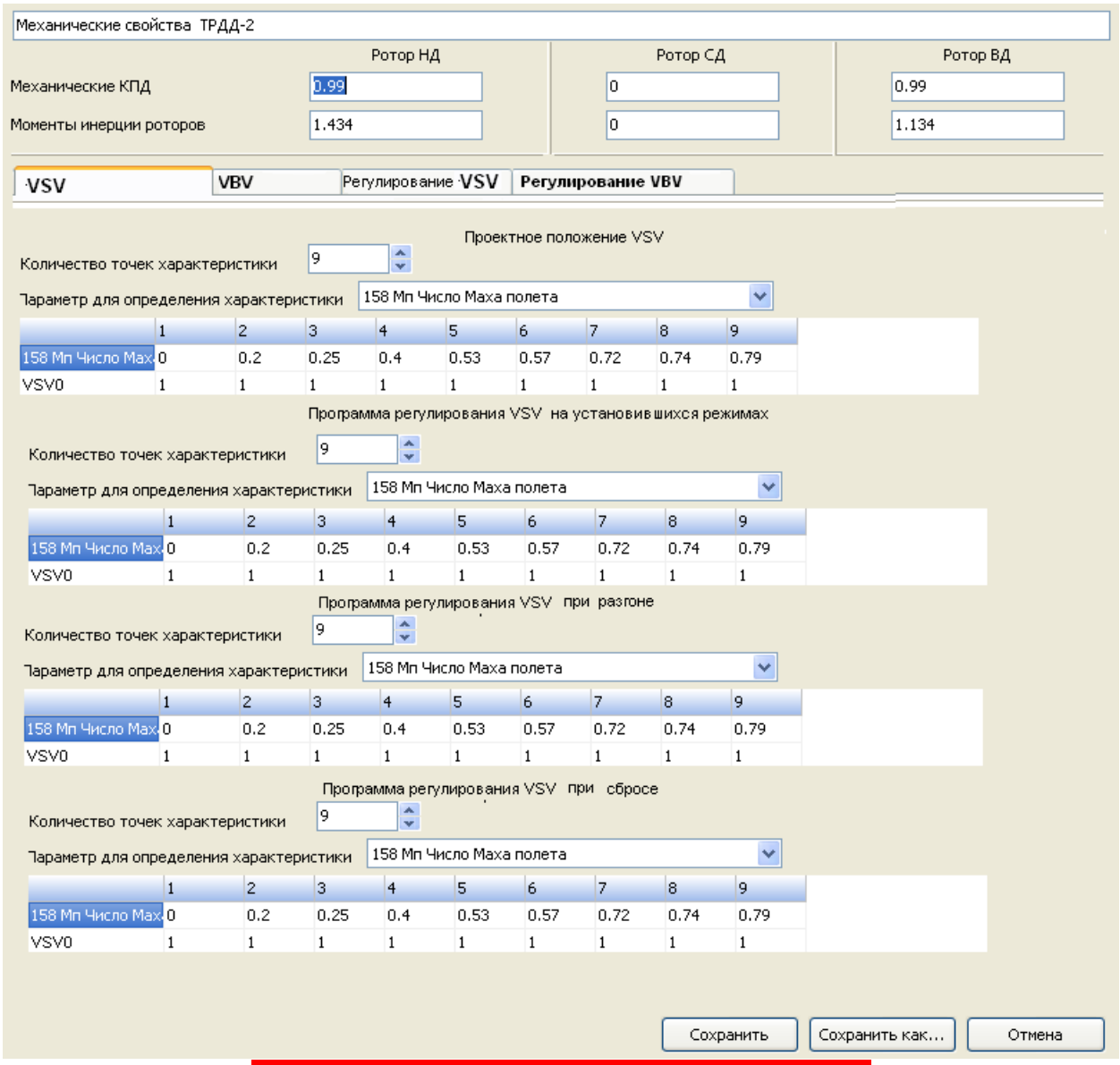


Figure 6.18 – Editor of VSV control programs

Table 6.4 - VSV control program at trasient modes (acceleration and deceleration)

$n_{cor\ HPC}$, rpm	4056.33	4490.4	4984.344	5059.18	6002.17	7843.23
VSV(°)	38.0	38.0	38.0	38.0	35.58	33.57
$n_{cor\ HPC}$, rpm	8920.93	9699.26	9878.88	10657.22	11360.71	11974.4
VSV(°)	32.85	31.52	30.12	25.8	21.2	16.0
$n_{cor\ HPC}$, rpm	12707.83	13171.84	13516.104	13905.27	14459.09	14848.26
VSV(°)	9.5	6.0	3.8	2.5	1.4	0.5
$n_{cor\ HPC}$, rpm	14968.0					

VSV(°)	0.0					

To determine the correct VSV(°) control program, it is necessary to add condition that switches between programs for steady and transient modes.

There are two variants of these conditions. They depend on engine configuration (see subsec. 6.1.1).

– If "without limiters (ACS)" configuration is active, then time derivative of rotational speed ($\dot{n}_{HP}$) is used to switch between control programs. If $\dot{n}_{HP} > f(n_2)$ (the relation $f(n_2)$ is presented in Table 6.5) then VSV position is determined according to control program at transient modes (acceleration), if $\dot{n}_{HP} < 0,6 \cdot f(n_2)$ then VSV position is determined according to control program at steady modes. If $\dot{n}_{HP} < -f(n_2)$ then VSV position is determined according to control program at transient modes (deceleration), if $\dot{n}_{HP} > -0,6 \cdot f(n_2)$ then VSV position is determined according to control program at steady modes;

– If "with limiters (ACS)" configuration is active, then the difference between set and current rotational speed of low pressure rotor is used to switch between control programs. If $n_{LP\,\mathrm{giv}} - n_{LP} > 255.3$ rpm then VSV position is determined according to control program at transient modes (acceleration), if $n_{LP\,\mathrm{giv}} - n_{LP} < 102.12$ rpm then VSV position is determined according to control program at steady modes. If $n_{LP\,\mathrm{giv}} - n_{LP} < -255.3$ rpm then VSV position is determined according to control program at transient modes (deceleration), if $n_{LP\,\mathrm{giv}} - n_{LP} > -102.12$ rpm then VSV position is determined according to control program at steady modes.

Table 6.5 – The relation $f(n_2)$

n_2, rpm	65%	70%	75%	80%	85%	90%	95%	100%
f(n_2) (%)	0.00675	0.009	0.01125	0.0144	0.0189	0.02925	0.04725	0.05

This information is inputted in the editor *"Mechanical properties and mechanization"*, *"VSV"* tab (see fig. 6.19).

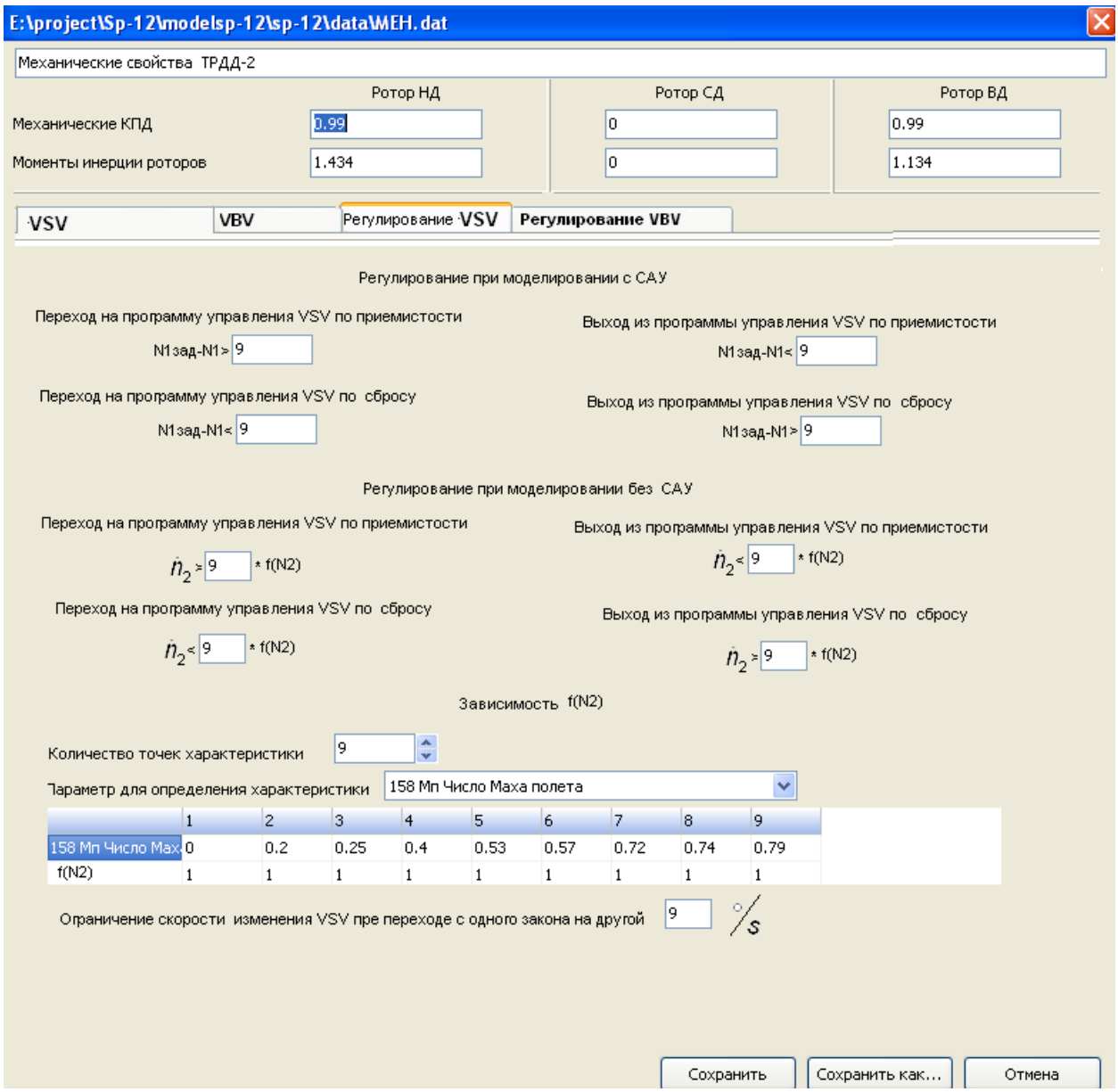


Figure 6.19 – Tab VSV control

User also has an opportunity to limit the speed of VSV position changing when switching from control law at transient mode to control law at stead mode (see fig. 6.19,, field «Limit the VSV change»).

In this case the program determines the deviation of current position (VSV) from design one (VSV_0) $\Delta VSV = VSV - VSV_0$. On this basis, program determines the HPC performance to be used in the analysis. As it was mentioned, there are three of them ($\Delta VSV = +3^\circ$, $\Delta VSV = 0^\circ$ and $\Delta VSV = -3^\circ$).

Inputting parameters can be done in the following way. User must press ***Edit*** in the box with performances of compressor (fig. 6.20). After this appears menu to input points of performances (fig. 6.21).

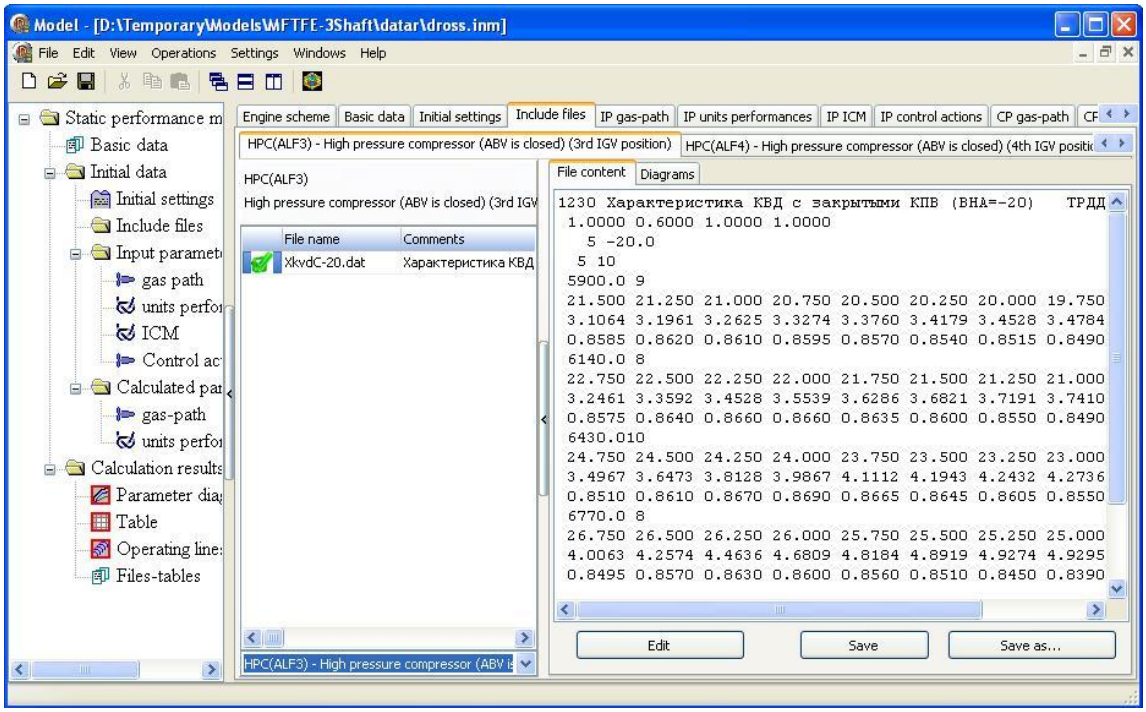


Figure 6.20 – Box for displaying performances of compressor

There are boxes in the upper part of the window to add comments to current performance.

To shift the performance in one of the arguments, without manual recalculating of each point at constant rotational speed, special adjustment factors are used. Adjustment factors are calculated as follows:

$$K_n = \frac{n}{n_{perf}}; \quad K_G = \frac{G}{G_{perf}}; \quad K_\pi = \frac{\pi}{\pi_{perf}}; \quad K_\eta = \frac{\eta}{\eta_{perf}}.$$

Box *ALFAigv* must have information about devation of IGV installation angle from design one, for which performance of compressor is inputted. User must remember that value of *ALFAigv* must increase from file to file, i.e. *ALFAigv* in the file HPC(ALF1) must be lower than in file HPC(ALF2), and so on. The performances of other spools are inputted in the same way.

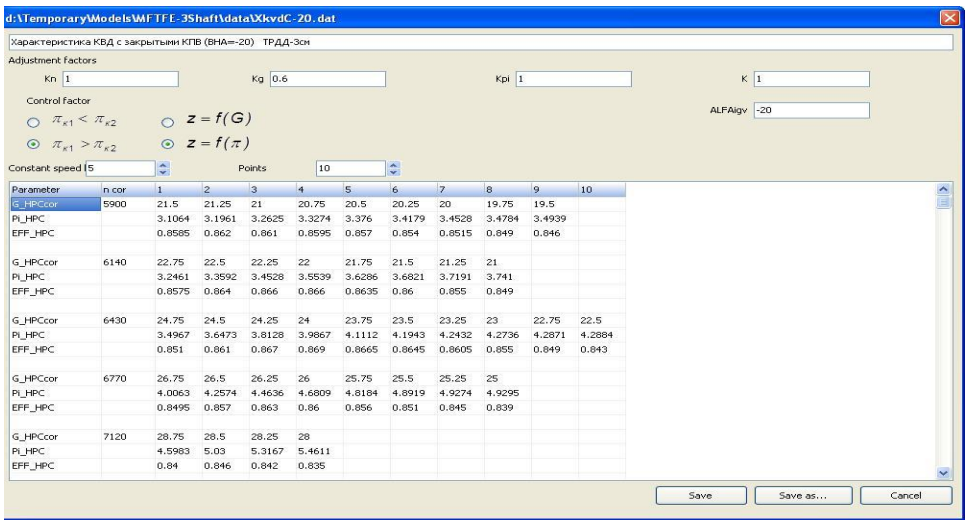


Figure 6.21 – Menu to input and edit performances of compressor

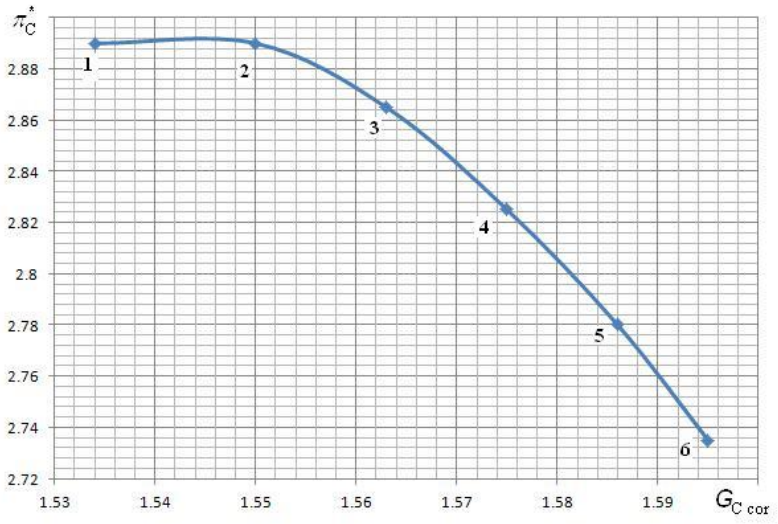


Figure 6.22 – Constant rotational speed line

To explain performances inputting in the box ***"Control coefficient"*** let`s consider an example of constant rotational speed line of compressor in the fig. 6.22.

In case table is filled in the following sequence p1-p2-p3-p4 (points according to fig. 6.22) and so on, then points must be arranged in pressure ratio (π_C^*) decreasing order. Therefore User must switch radio button to $\pi_{C1} > \pi_{C2}$ (in this case control coefficient will be negative, see Appendix A2). In case inputting of inputting points in pressure ratio increasing order (from point 6 to point 1), then User must switch radio button to $\pi_{C1} < \pi_{C2}$ (control coefficient will be positive).

If the performance has inflections by π_C^* (fig. 6.22 – value in point 2 is bigger that in points 1 and 3) then User must switch radio button to $z = f(G)$ (control coefficient becomes equal to 10), else to radio button $z = f(\pi)$ (control coefficient becomes equal to 5). Resuming all mentioned above, control coefficient may have the following values:

- -10 – if $z = f(G)$ and $\pi_{C1} < \pi_{C2}$;
- 10 – if $z = f(G)$ and $\pi_{C1} > \pi_{C2}$;
- -5 – if $z = f(\pi)$ and $\pi_{C1} < \pi_{C2}$;
- 5 – if $z = f(\pi)$ and $\pi_{C1} > \pi_{C2}$.

Attention. Values of control coefficient (5 and 10) for all performances of the spool must be the same

In the box ***"Constant speed lines"*** User must specify the number of constant speed lines at compressor performance map (maximum number of lines is 30). Pushing the button increases number of constant speed lines by one, which means adding three empty lines to input points of constant speed line $G_{i\ cor}$, Π_i, EFF_i (in the program index ***i*** will be substituted by the name of the spool). Number of cells in each row is determined by the value, given in the box ***"Points"***. In this box User must specify the number of points in the constant speed line with maximum number of points. It is important to know that not all constant speed lines must have the number of points given in the box ***"Points"*** (it is not necessary to input the same number of points for all constant speed lines). Data editing can be done in the same ways as were mentioned in subsection 6.1.2.1. If data is inputted from Excel and number of points exceeds number, specified in the box ***"Points"***, then number of columns in the table is automatically increased up to needed number. Program automatically recalculates number of columns and changes the value in the box ***"Points"***. All data is logged in the editing area in the line (up to 100 values).

User has an opportunity to input the values of y_{10}, y_{20} and $\Delta y_1, \Delta y_2$ for each performance. The description of these parameters is given in Appendix G). The algorithm of coefficients usage is similar to the one, presented in subsection 6.1.2.1. To visualize the corrected performance User must save the corrected performance by ticking the field "Save with parameters of component performances".

After editing the file User can save it with the same name by clicking the icon ***"Save"***, or with another name by clicking the icon ***"Save as…".*** When saving the file, all data is written in the order, specified in Appendix A.2.

Compressor performances can be represent in tabular and graphic form. To switch to graphic representation User must choose the tab ***"Diagrams"*** in menu item ***"Include files"*** to show performances in graphic form (fig. 6.23). To choose necessary performance it is necessary to put checkmark in front of its name.

There are two variants to represent performances – lines and lines with markers. There is a legend with values of corrected rotational speeds $n_{C\ cor}$ (these rotational speeds correspond to constant speed lines displayed in the diagram) above the diagram. It is possible to show all constant speed lines or some of them by putting and removing the checkmarks in front of needed values of $n_{C\ cor}$.

Program has an opportunity to zoom certain fragment of the performance. To zoom the fragment User must put the cursor in the upper left corner of zoomed fragment and holding LMB displace the cursor to the bottom right corner. During this operation appears the rectangle, and all inside this rectangle will be scaled to full screen of the corresponding performance. If User scales performance $\pi_C^* = f\left(G_{C\ cor}\right)$ (the lower diagram in the fig. 6.23), then performance $\eta_C^* = f\left(G_{C\ cor}\right)$ (upper diagram) is scaled only along $G_{C\ cor}$ axis. The scale of another axis (η_C^*) stays the same. If there is necessary to scale η_C^*, then the mentioned above procedure must be applied to the performance $\eta_C^* = f\left(G_{C\ cor}\right)$.

To cancel scaling User must put the cursor anywhere in the diagram and holding LMB displace the cursor up and then leftward, and finally release LMB. Diagrams will get back to initial state.

If faulty values were discovered during performance visualizing (fig. 6.24), then there is an opportunity to correct them. To correct the faulty value User must put the cursor in the faulty point at compressor performance map and holding LMB displace it to the needed position. Then to save the changes it is necessary to choose the tab ***"File content"*** and press the button ***"Save"*** (to save changes in current file), or ***"Save as"*** (to save changes in the file with new name).To cancel changes User must click file name (after this all done changes will be canceled and the performance will get back to its initial state).

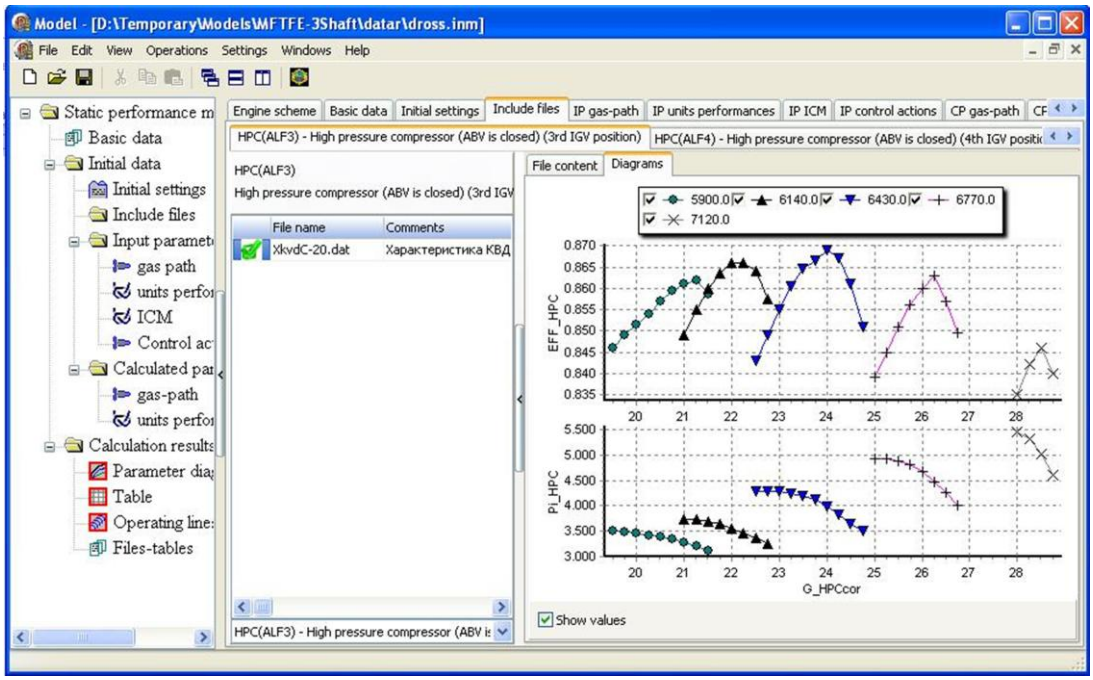


Figure 6.23 – Tab to review the performances of compressors

The chosen performance can be printed in tabular format or in a form of diagram. To print the performance in tabular format, it is necessary to switch the performance representation to text format (fig. 6.20), and to print the performance in a form of diagram, it is necessary to switch the performance representation to diagram format (fig. 6.23) and press menu item ***File⇒Print***. Program uses default printer specified in system settings. To change printer settings click ***File⇒Print settings*** (this command will call printer set-

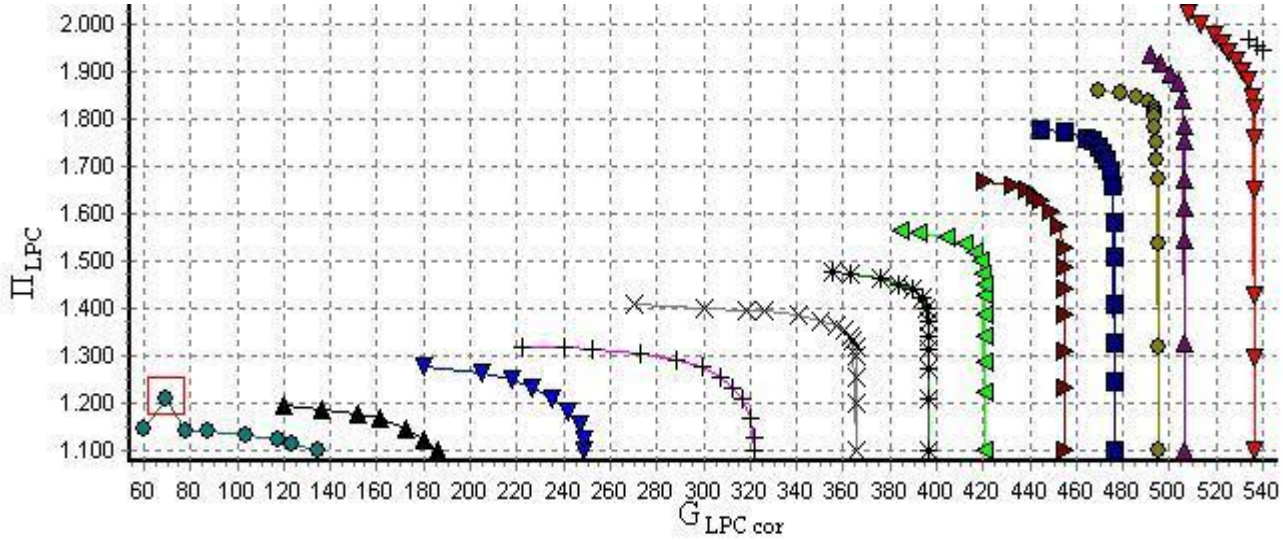


Figure 6.24 – Performance of compressor with faulty value

tings).

6.1.2.3 Editing parameters of transition ducts

Transition duct is characterized by losses of total pressure in the transition duct σ_{trans}. The editor is the same for all transition ducts. That is why this subsection considers editing of only one performance, and all other performances are edited in analogical way.

Choosing performance of transition duct in menu ***"Include files"*** and pressing the button ***"Edit"*** calls the editor of transition duct performance (fig. 6.25).

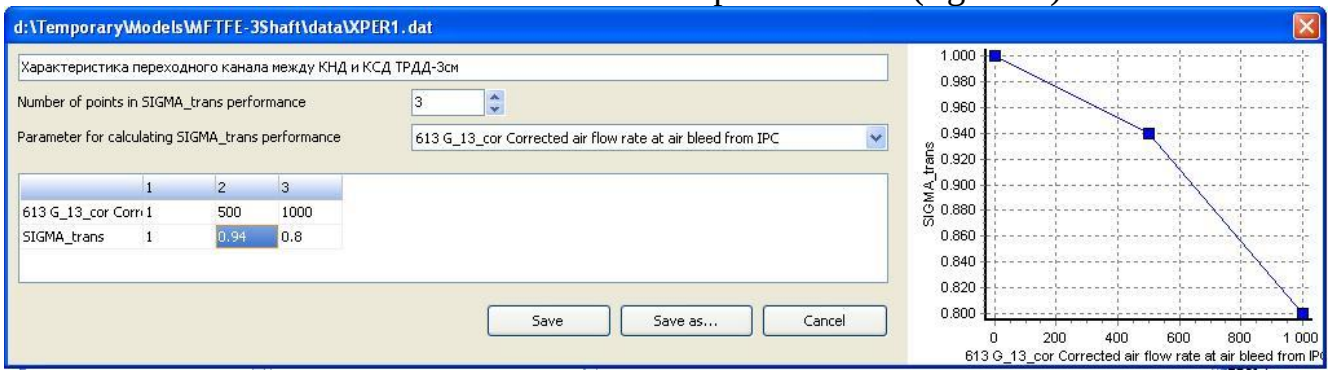


Figure 6.25 – Editor of transition duct performance

Editor has the similar form to the editor of intake (subsection 6.1.2.1). User may write any comments in the upper part of the window. Then it is necessary to specify

number of points and number of argument in the performance σ_{trans}, and input array of values (maximum is 100 values).

Performance of transition duct is saved in the file, which has the format described in Appendix A.4. Value of parameter is calculated by linear interpolation of adjacent values of performance.

6.1.2.4 Editing parameters of combustion chamber

Editor of CC performances consists of three blocks (fig. 6.26):

- block for inputting losses in combustion chamber σ_{CC};
- block for inputting combustion efficiency η_{CC};
- block for inputting parameters of fuel.

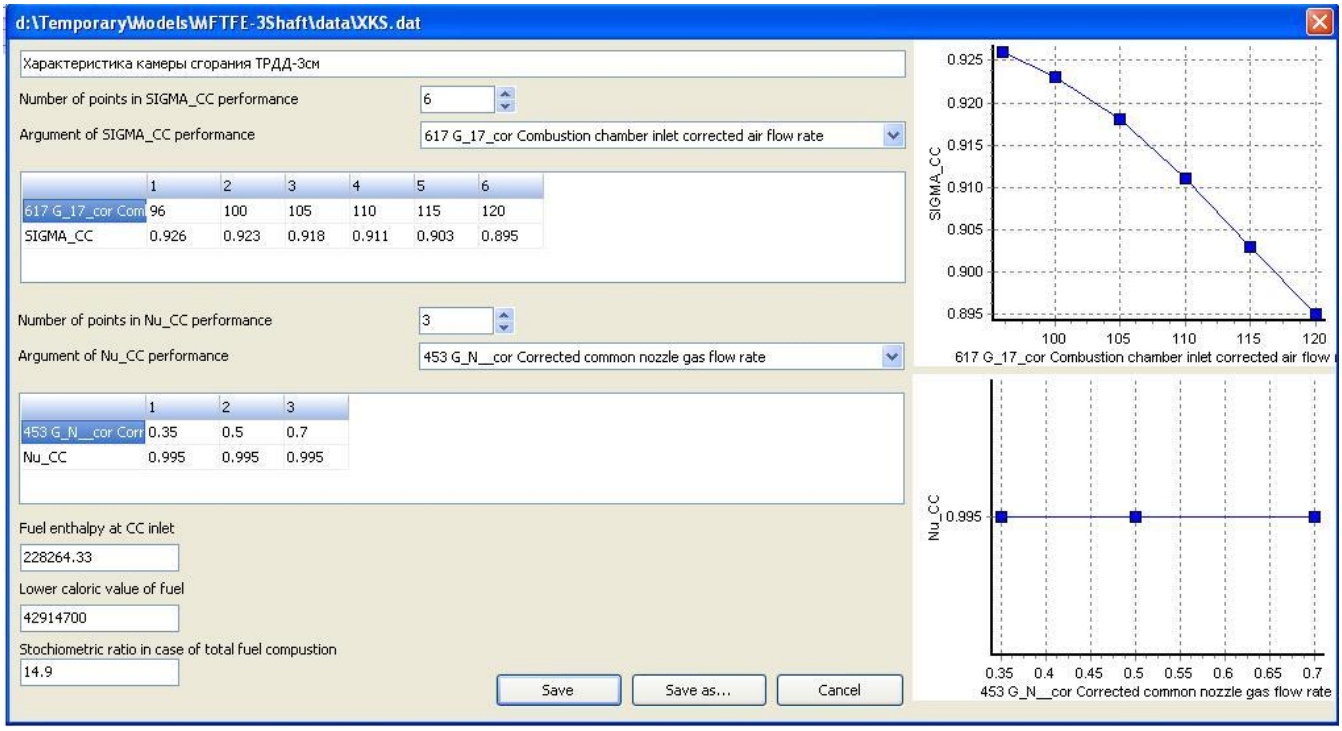


Figure 6.26 – Editor of CC performances

Methods of editing were described above. Number of points in the performances σ_{CC}, η_{CC} must be lower than 100 points

User must also input the following parameters: enthalpy of fuel at CC inlet i_o (subsection 1.1.2.9), lower caloric value of the fuel H_u, ratio between mass flow and fuel consumption in case of complete fuel burning L_o.

File format of CC performance is presented in Appendix A.5.

6.1.2.5 Editing parameters of turbine

Performance of each turbine spool is the dependence of gas flow rate though turbine nozzle vanes $A_T = \frac{G_g \sqrt{T^*_{in\ T}}}{p^*_{in\ T}}$ and efficiency η^*_T from pressure ratio of turbine π^*_T and corrected rotational speed $n_{T\ cor} = \frac{n}{\sqrt{T^*_{in\ T}}}$:

$$A_T = f\left(\pi^*_{T\ cor}, \pi^*_T\right); \eta^*_T = f\left(n_{T\ cor}.\pi^*_T\right),$$

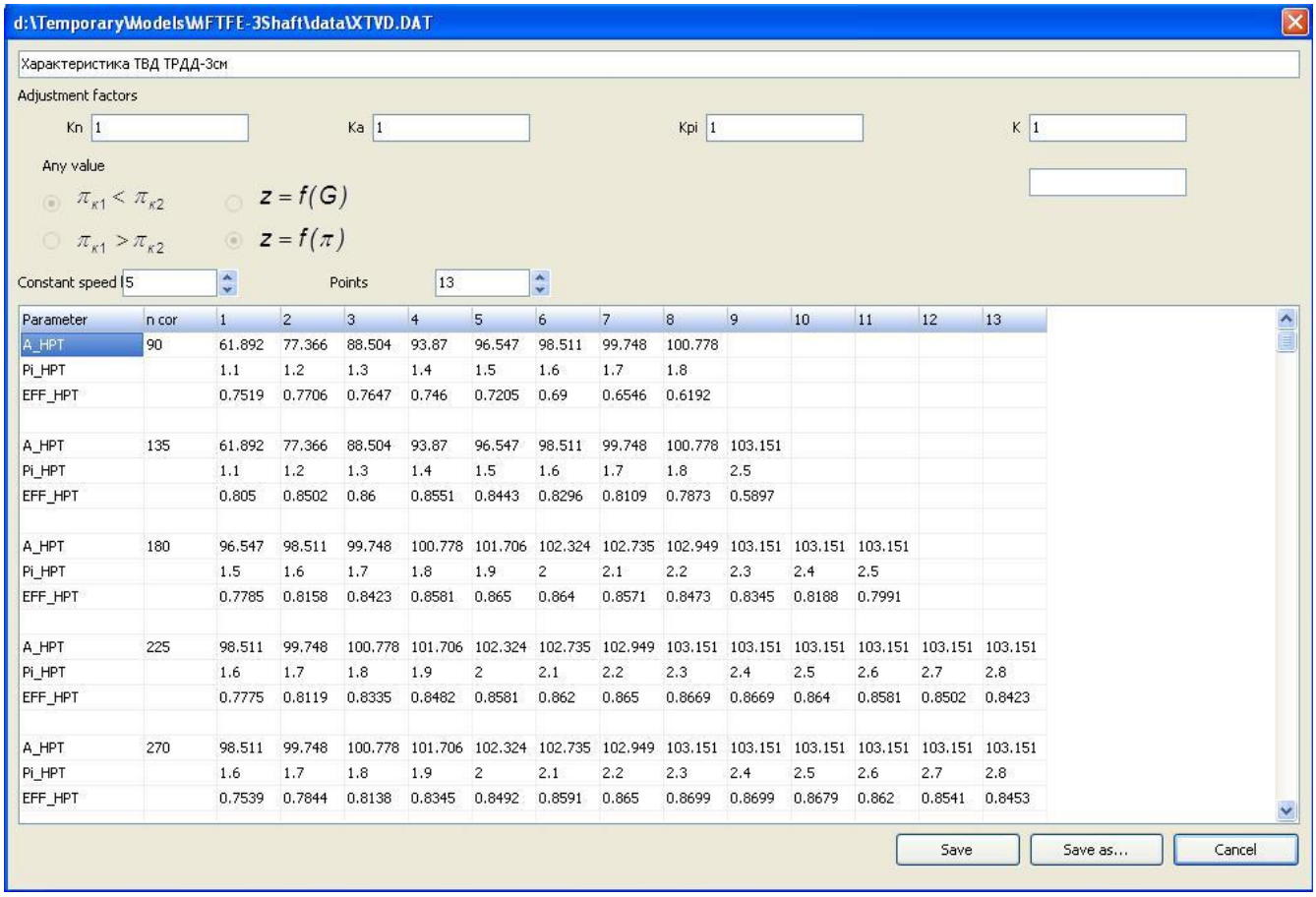
where G_g is gas flow rate at turbine spool inlet; $T^*_{in\ T}$, $p^*_{in\ T}$ are temperature and pressure at turbine spool inlet, $n_{T\ cor}$ is a corrected rotational speed of turbine rotor.

This performance is expressed as set of lines of $A_T = f\left(\pi^*_T\right)$ and $\eta^*_T = f\left(\pi^*_T\right)$, which correspond to constant corrected rotational speed $n_{T\ cor}$. Current engine operating mode is expressed by the point in the performance map, and set of steady modes forms operating line of the engine.

So, performance of turbine is described by two bivariate functions. That is why, to calculate all 4 parameters, which characterize turbine operation, it is necessary to specify any two of them as arguments to find two others from performance.

Sloping performances of turbine and significant variation range of π^*_T for each constant speed line $n_{T\ cor} = const$, allow using standard linear interpolation for bivariate function. In this case, arguments are π^*_T and $n_{T\ cor}$.

Performance editing can be done same to compressor performance in two ways – in tabular and in graphic forms Tabular editor is expressed in fig. 6.27.



Parameter	n cor	1	2	3	4	5	6	7	8	9	10	11	12	13
A_HPT	90	61.892	77.366	88.504	93.87	96.547	98.511	99.748	100.778					
Pi_HPT		1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8					
EFF_HPT		0.7519	0.7706	0.7647	0.746	0.7205	0.69	0.6546	0.6192					
A_HPT	135	61.892	77.366	88.504	93.87	96.547	98.511	99.748	100.778	103.151				
Pi_HPT		1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	2.5				
EFF_HPT		0.805	0.8502	0.86	0.8551	0.8443	0.8296	0.8109	0.7873	0.5897				
A_HPT	180	96.547	98.511	99.748	100.778	101.706	102.324	102.735	102.949	103.151	103.151	103.151		
Pi_HPT		1.5	1.6	1.7	1.8	1.9	2	2.1	2.2	2.3	2.4	2.5		
EFF_HPT		0.7785	0.8158	0.8423	0.8581	0.865	0.864	0.8571	0.8473	0.8345	0.8188	0.7991		
A_HPT	225	98.511	99.748	100.778	101.706	102.324	102.735	102.949	103.151	103.151	103.151	103.151	103.151	103.151
Pi_HPT		1.6	1.7	1.8	1.9	2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8
EFF_HPT		0.7775	0.8119	0.8335	0.8482	0.8581	0.862	0.865	0.8669	0.8669	0.864	0.8581	0.8502	0.8423
A_HPT	270	98.511	99.748	100.778	101.706	102.324	102.735	102.949	103.151	103.151	103.151	103.151	103.151	103.151
Pi_HPT		1.6	1.7	1.8	1.9	2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8
EFF_HPT		0.7539	0.7844	0.8138	0.8345	0.8492	0.8591	0.865	0.8699	0.8699	0.8679	0.862	0.8541	0.8453

Figure 6.27 – Editor of turbine performance

Adjustment factors are calculated as follows:

$$K_n = \frac{n}{n_{char}}; \quad K_A = \frac{A}{A_{char}}; \quad K_\pi = \frac{\pi}{\pi_{char}}; \quad K_\eta = \frac{\eta}{\eta_{char}}.$$

Format of the file with turbine performances is shown in Appendix A.6.

6.1.2.6 Editing parameters of nozzles

Performances of primary, secondary nozzles have common structure and consist of three parameters:

- pressure recovery σ_N;
- flow coefficient μ_N;
- velocity factor φ_C.

Each from these parameters may depend on its own arguments (there can be up to 100 points in the dependence). Main window of the editor is shown in fig. 6.28. The way of performances editing is similar to that, which was described in subsection 6.1.2.1.

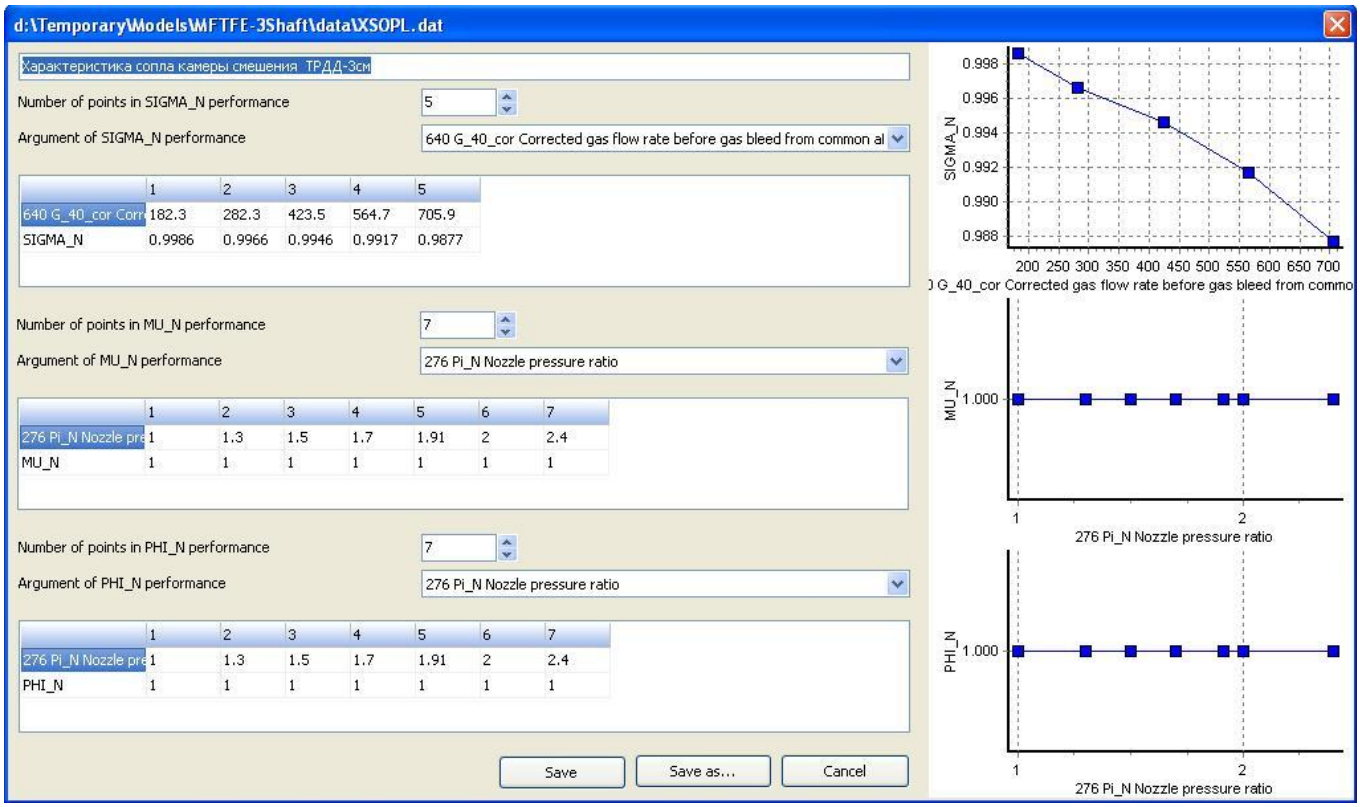


Figure 6.28 – Editor of nozzle performance

Format of the file with nozzle performances is shown in Appendix A.7.

6.1.2.7 Editing geometrical parameters of engine

Main parameters for inputting are areas of nozzles, i.e. gas-path area at primary nozzle discharge (F NI) and secondary nozzle discharge (F NII), gas-path area of primary and secondary flows at station with VBV, area at CC inlet, gas-path areas at stations with thermocouples (at boosters discharge and at first stage of LPC discharge).

Main window of the editor is shown in fig. 6.29.

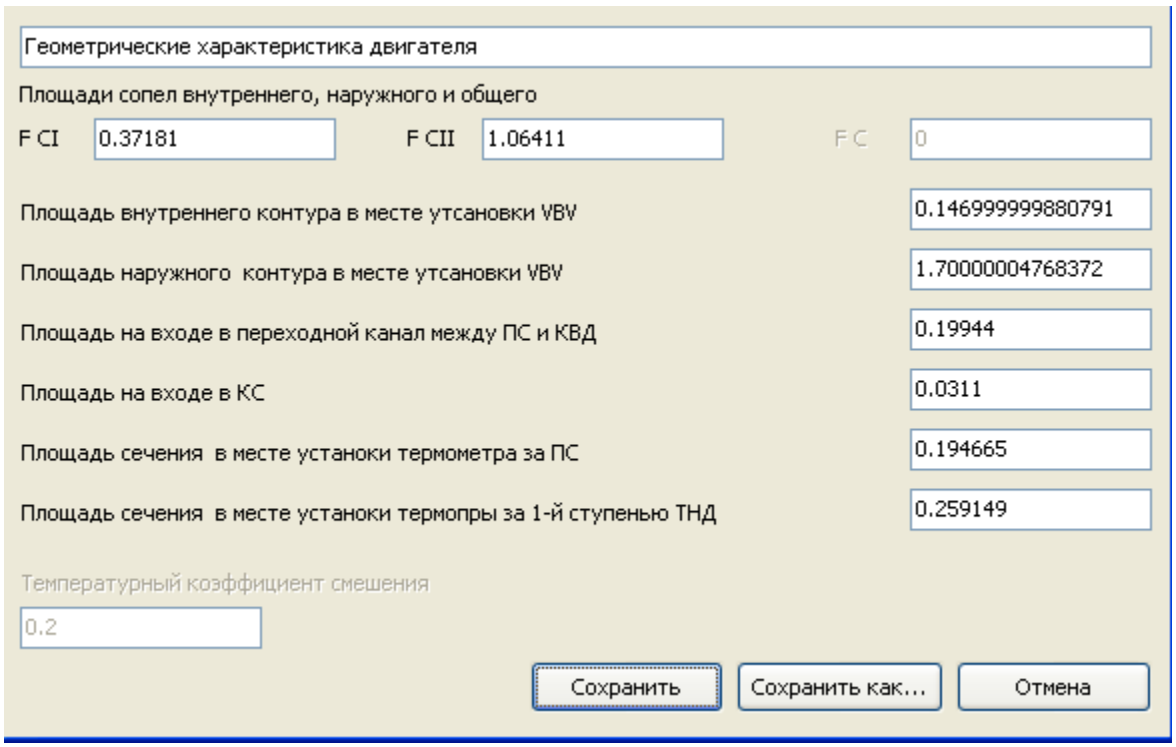


Figure 6.29 –Editor of geometrical parameters of mixer

File format is presented in Appendix A.8.

6.1.2.8 Editing parameters of air bleeds and taps, power take-offs and supplies

As it was mentioned in subsection 1.1.1, structure of gas-path of each particular engine is based on generic scheme of gas-path by decomposing this scheme. Numbers of stations of air bleeds and taps, power take-offs and supplies persist. On this basis it becomes clear that editing different engine types requires editing of definite cells of array. To facilitate user`s operation, only parameters that make sense for this engine type are available for editing (fig. 6.27). Other cells in the file will be set zeroes.

Relative air bleeds are calculated as follows:

$$\bar{G}_i = \frac{G_{bleed\ i}}{G_j},$$

where G_j is mass flow at inlet of jth spool of compressor, relative to which mass flow of bleed air is calculated.

Mass flow of supplied air is calculated same to bleed one. All bleeds are negative and taps are positive.

Here User may set bleeds, which are not correlated with engine operating mode. Stations, from which air is bleed, are determined by coefficients k_π and k_η (see subsection 1.1.2.13). There is an option to set the mode-independent power take-offs from rotors $\boldsymbol{W}$.

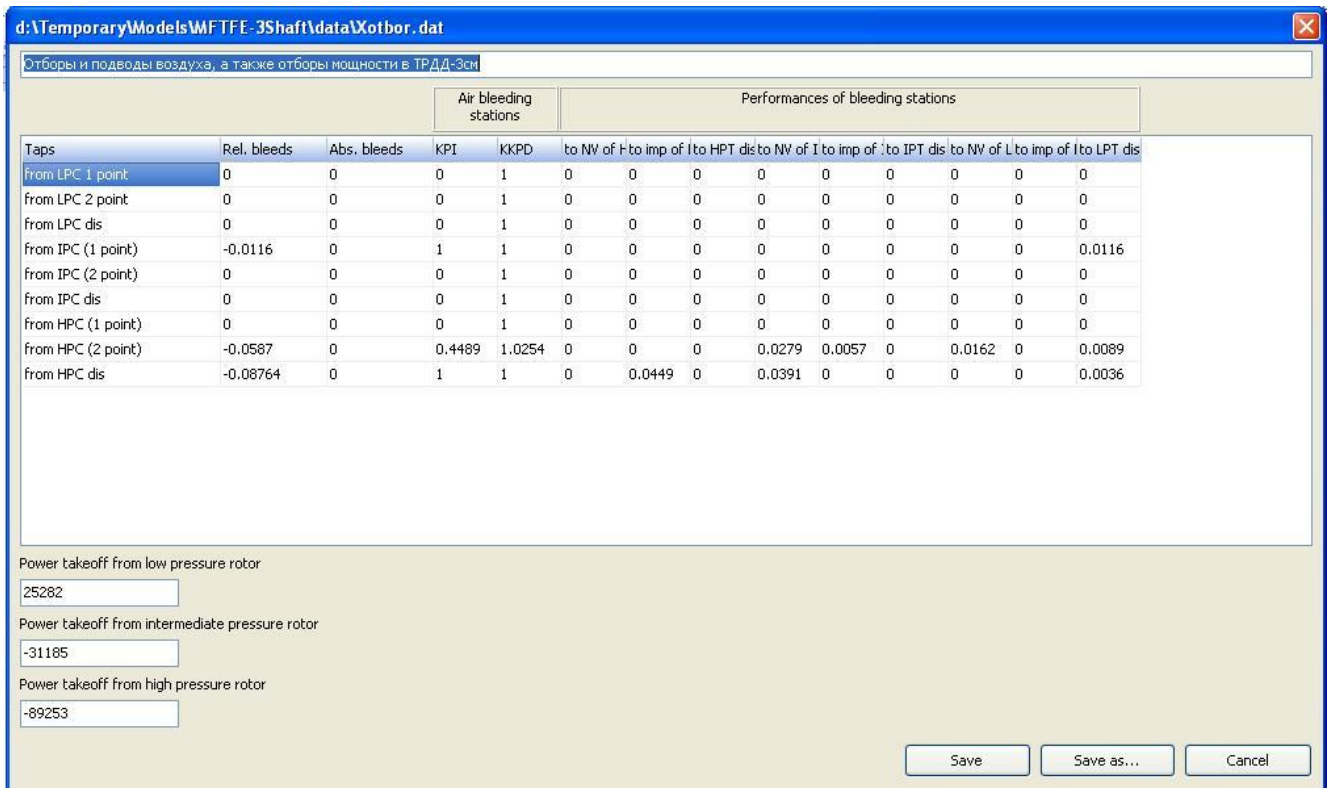


Figure 6.30 – Editor for setting parameters of air bleeds and taps for active clearance control system (Air is bled from the 5th stage of HPC)

There are three ways to bleed air for active clearance control system, therefore, User must input the parameters for each of them. That is why, three performances must be set in the project (see Table 6.1).

Except parameters of bleeds and taps User must input the coefficients that are used to recalculate the parameters of air in the control cavities of the casing. Laws of active clearance control system of HPT are presented in Tables 6.6 - 6.8. Rotational speed of high pressure rotor (n_{HP}) is the argument of law.

This information is inputted in the editor "Mechanical properties and mechanization", tab "Air bleeds control" see fig. 6.31. First program is when air is bled from the fifth stage, second program – from the ninth stage, third – from both fift and ninth stages.

Table 6.6 – Programs that control air bleeds and taps for active clearance control system of HPT at steady modes.

n_{HP}, rpm	<12070	$12070 \le n_{HP}$ <12570	$12570 \le n_{HP}$ <13570	$13570 \le n_{HP}$ <14070	$n_{HP} \ge 14070$
Bleeding station	9th stage	5th and 9th stages	5th stage	5th and 9th stages	9th stage

Table 6.7 - Programs that control air bleeds and taps for active clearance control system of HPT during acceleration.

n_{HP}, rpm	<12070	$12070 \le n_{HP}$ <12570	$12570 \le n_{HP}$ <13570	$13570 \le n_{HP}$ <14070	$n_{HP} \ge 14070$
Bleeding	9th	5th and 9th stages	5th stage	5th and 9th stages	9th stage

station	stage				

Table 6.8 - Programs that control air bleeds and taps for active clearance control system of HPT during deceleration.

n_{HP}, rpm	<11930	11930 ≤ n_{HP} <12430	12430≤ n_{HP} <13430	13430≤ n_{HP} <13930	n_{HP} ≥13930
Bleeding station	9th stage	5th and 9th stages	5th stage	5th and 9th stages	9th stage

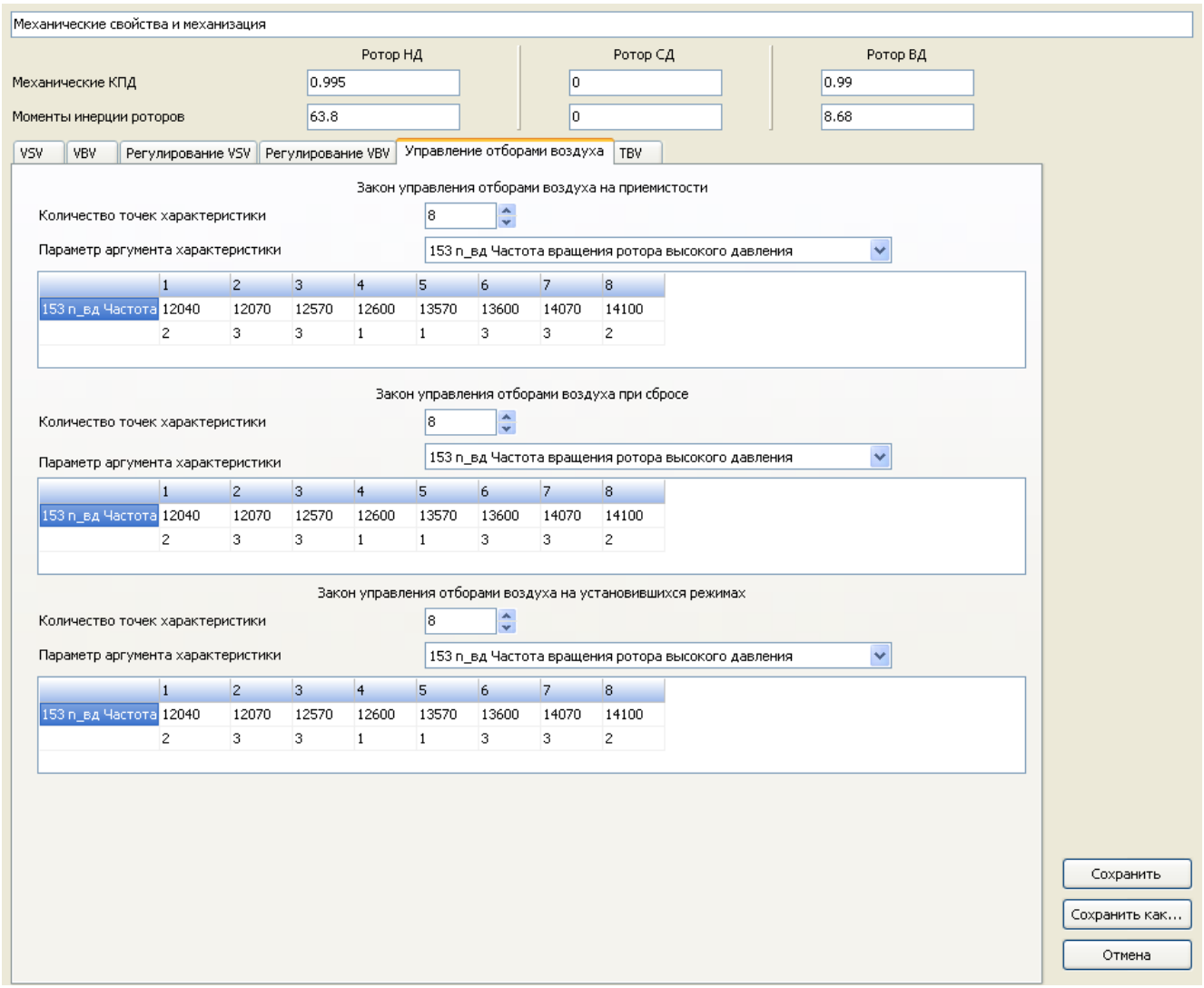


Рисунок 6.31 – Tab “Air bleeding control” in the editor “Mechanical properties and mechanization”

Format of the file with performances of air bleeds and taps, power take-offs and supplies is shown in Appendix A.9.

6.1.2.9 Editing mechanical properties and mechanization of compressor

Editor of parameters (fig. 6.32) consists of one block and six taps:

- mechanical properties of rotors: mechanical losses η_{mech} and inertia of jth rotor J_i;
- tab "VSV" gives an opportunity to input the information about reference (base) position of VSV, about control programs at steady and transient modes (see subsec. 6.1.2.2);
- tab "VSV control". Here User can input the information that is used to determine the control of VSV at transient and steady modes (see subsec. 6.1.2.2);
- "VBV" tab. Here User can input the information about control programs at transient and steady operational modes (see subsec. 6.1.2.2);
- "Air bleeding control" tab. Here User can input the information about air bleeding control at transient and steady operational modes;
- "ABV" tap. Here User can input the information about air bleeding valve, which is mounted at HPC discharge (next ABV) and about control program of this valve (see subsec. 6.1.2.2).

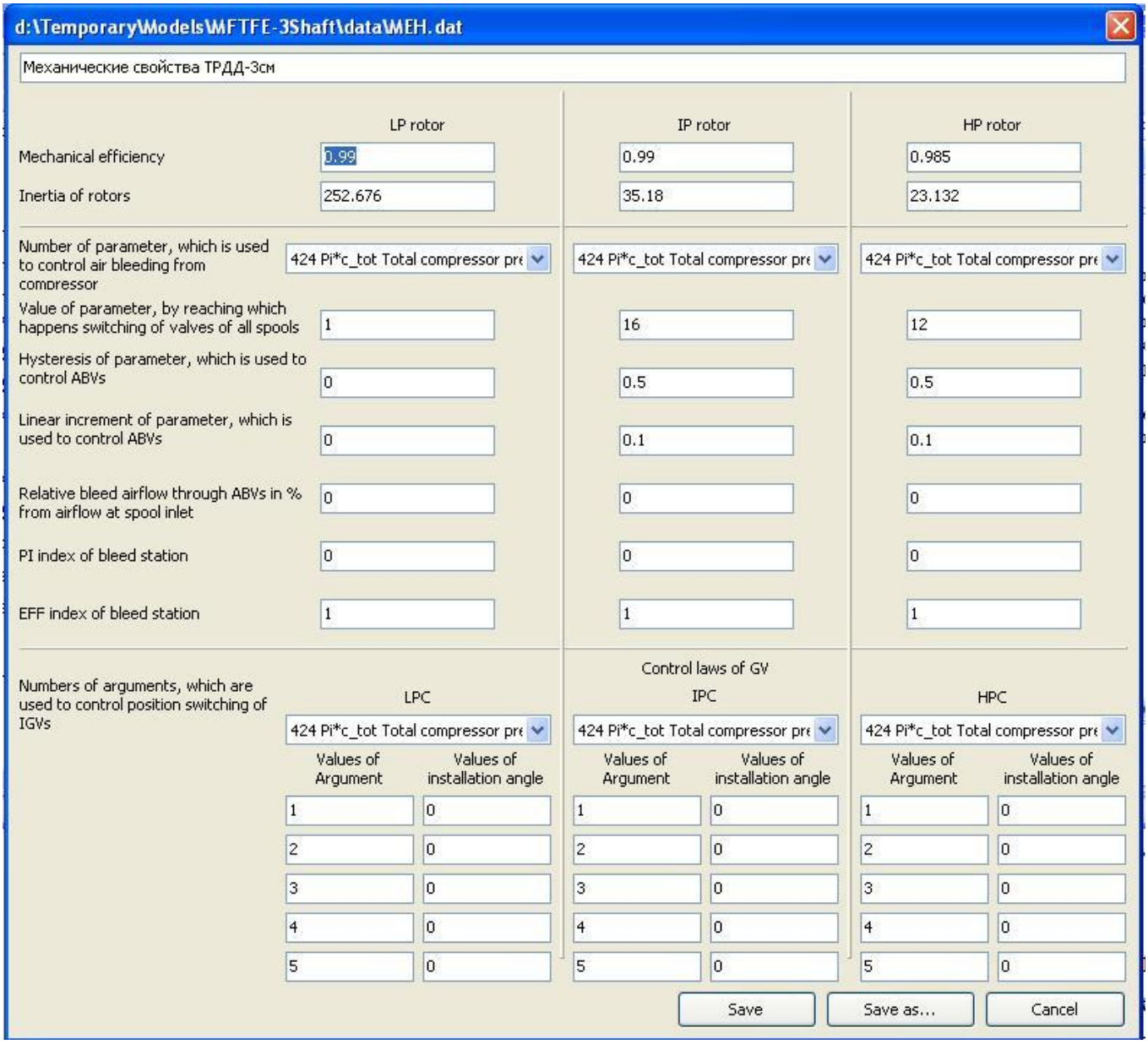


Figure 6.32 – Editor of mechanical properties and mechanization of compressor of single spool TJE.

Format of the file with mechanical properties and mechanization is shown in Appendix A.10.

6.1.2.10 Editing initial approximations and variation limits of independent variables.

User may call the editor of initial approximations and variation limits of independent variables (fig. 6.33) by pressing the button "Edit" in the tab ***"Independent variables (initial approximations)"*** or ***"Independent variables limits"***. These files are edited jointly in one editor to let User control the value of initial approximations and their variation limits.

There are boxes for comments in the upper part of the editor. Left box is used to write comment in the file with initial approximations, and right one –file with variation limits.

There is a list of parameters in central part of the window. The description of each parameter consists of the following columns:

- *key* – number according to generic list;
- *designation* – abridged form of parameter designation;
- *name* – deciphering of parameter;
- *Initial value* – cell to input initial value of parameter;
- *minimum* – cell to input minimum value;
- *maximum* – cell to input maximum value.

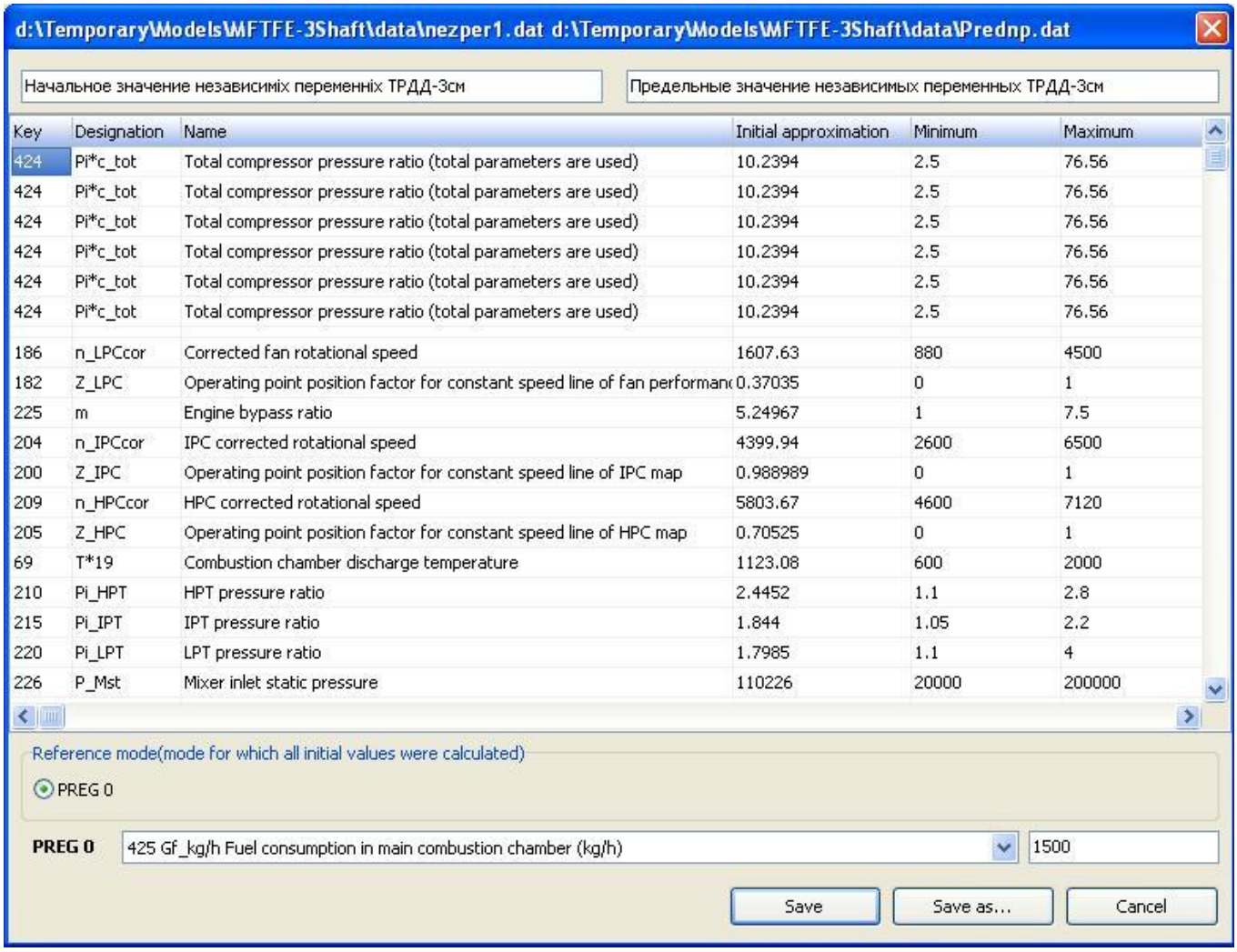


Figure 6.33 –Editor used for inputting initial approximations and variation limits of independent variables

User cannot change the list of parameters. User can only set numerical values in boxes ***"Initial value"***, ***"Minimum"***, ***"Maximum"***.

The list of parameters for SP12 turbofan is shown in Tables 6.9

Table 6.9 – List of independent variable

	Designation	Description of parameter
1	n_LPCcor	corrected rotational speed of LPC rotor, rpm;
2	Zlpc1	factor that determines the postion of operating point at constant speed line of LPC map (primary flow);
3	Zlpc2	factor that determines the postion of operating point at constant speed line of LPC map (secondary flow);
4	Zhpc	factor that determines the postion of operating point at constant speed line of HPC map;
5	n_HPCcor	corrected rotational speed of HPC rotor, rpm;
6	T*19	temperature at CC discharge, K;
7	Пhpt	HPT pressure ratio;
8	Пlpt	LPT pressure ratio;
9	G_19	gas flow at CC discharge, kg/s;
10	GfCC	fuel flow in CC, kg/s;
11	T*37	temperature of primary airflow at primary nozzle inlet, K;
12	G_37	airflow through core duct at primary nozzle inlet, kg/s;
13	T*6	temperature of secondary airflow at secondary nozzle inlet, K;
14	G_6	airflow through bypass duct at secondary nozzle inlet, kg/s;
15	TSTTVD	temperature of HPT stator, K;
16	TLTVD	temperature of HPT blade, K;
17	TDTVD	temperature of HPT disk, K.
18	TSTTND	temperature of LPT stator, K;
19	TLTND	temperature of LPT blade, K;
20	TDTND	temperature of LPT disk, K;
21	Пlpt1	pressure ratio of the first stage of LPT;

Variation range of independent variables can be randomly chosen. But its strongly recommended to choose variation range being guided by physical sense of parameter. Also it is necessary to take into account engine type and all possible operating conditions. Some recommendations for choosing variation limits are given below.

Temperature at combustion chamber discharge (T*19). For example variation range can be from 100 K to 10000K (or another), but this makes no physical sense, because temperature at combustion chamber discharge cannot be 100 K at all and 10000 K is impossible to reach in any engine type. So to short operating range it is reasonable to set this range within the limits 500 K…2000 K.

Corrected rotational speed of compressor. Variation limits of this parameter are determined by value of minimum constant speed line (bottom limit, which User must input in the box ***"Minimum"***) and value of maximum constant speed line (superior limit, which User must input in the box ***"Maximum"***).

Pressure ratio of the turbine. They are set according to performances of turbine spools. It is necessary to find minimum and maximum pressure ratios among all values given in the performance map.

Operating point position factor for constant speed line at compressor performance map. Minimum value is always "0", and maximum – 1.

Corrected rotational speed of turbine. Variation limits are set analogically to corrected rotational speed of compressor.

Gas flow. Depends on engine type and its application purposes. To make an initial approximation User can use performance of low pressure compressor. It is recommended to determine minimum and maximum corrected airflow from performance. Then calculate real limiting values by formulas for correcting parameters, taking into account minimum and maximum possible ambient temperatures and pressures at engine inlet.

Temperature of SE. Choosing variation range of this parameter depends on engine type (see the example of choosing gas flow temperature at combustion chamber discharge).

Then User must set the value of mode parameter, for which initial approximations were chosen. Any parameter from present in pull-down can be accepted as mode parameter. Engines GTE-1 and GTE-2 need two mode parameters. All changes will be saved in marked files (menu item ***«Initial data»*** → ***«Include files»***) when pressing button ***"Save"***.

When pressing button ***"Save as"*** first to appear is menu to enter name of file with initial approximations, and then – menu to enter name of file with variation limits.

Format of the file with initial approximations of independent variables is shown in Appendix A.11, and file with variation limits of independent variables is shown in Appendix A.12. File content depends of engine type.

6.1.2.11 Editing parameters of heat exchange

To set parameters of heat exchange (fig. 6.34) it is necessary to go to ***"Initial data"*** → ***"Include files"*** →Tab ***"Parameters of heat exchange"***.

Number of cells, available for editing, depends on number of turbine spools. User must set masses and specific heat capacities of structural elements (SE) (stator, disk and blade), which are present in the formula (1.34).

To describe heat flows:

- from gas to SE (equation (1.35)): program needs heat-transfer coefficient ($\boldsymbol{ALFA_i}$) and area ($\boldsymbol{F_i}$) of surfaces, which are flown by hot gas;
- between blades and disk of turbine (equation (1.19)): program needs contact heat transfer coefficient ($\boldsymbol{K_{lock\ i}}$) and area of contact between blade and disk ($F_{lock\ i}$);
- from SE to cooling air (equation (1.35)): program needs heat-transfer coefficient ($\boldsymbol{ALFA_{i\ cool}}$) and area ($\boldsymbol{F_{i\ cool}}$) of surfaces, which are flown by cooling air, and temperature of cooling air.

Format of the file with parameters of heat exchange is shown in Appendix A.13.

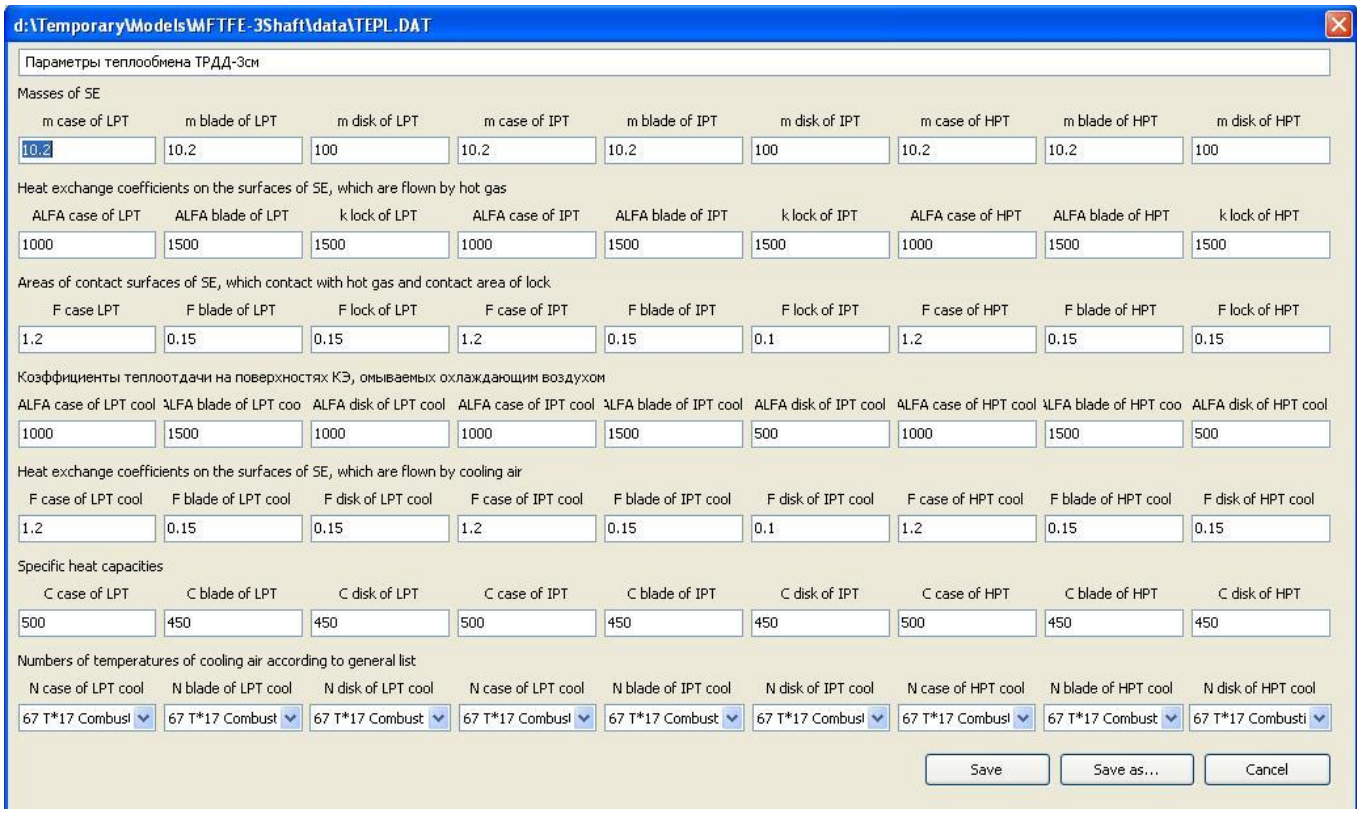


Figure 6.34 – Editor of heat exchange parameters

6.1.2.12 Editing files with transient performances of SE

This editor (fig. 6.35) is used to input transient performances of checkpoints radal displacements ***S***, caused by single stepped impact referring base value T_{ref} (see eq. (**Ошибка! Источник ссылки не найден.**)). Value of reference temperature must be inputted in the corresponding box.

There is a difference between transient performances of blade and stator (disk). Editor of stator and disk have no boxes ***"Influence coefficient"*** and ***"Blade length"***. These boxes are present only in the editor of blade transient performances. The sense of these parameters becomes clear from equation (1.38).

Setting transient performance consists of filling array of time and array of displacements of two checkpoints (see fig. 1.5).

Increasing value in the box "Number of points in array" increases of decreases number of data slices. Next User must manually input time and displacements of two points.

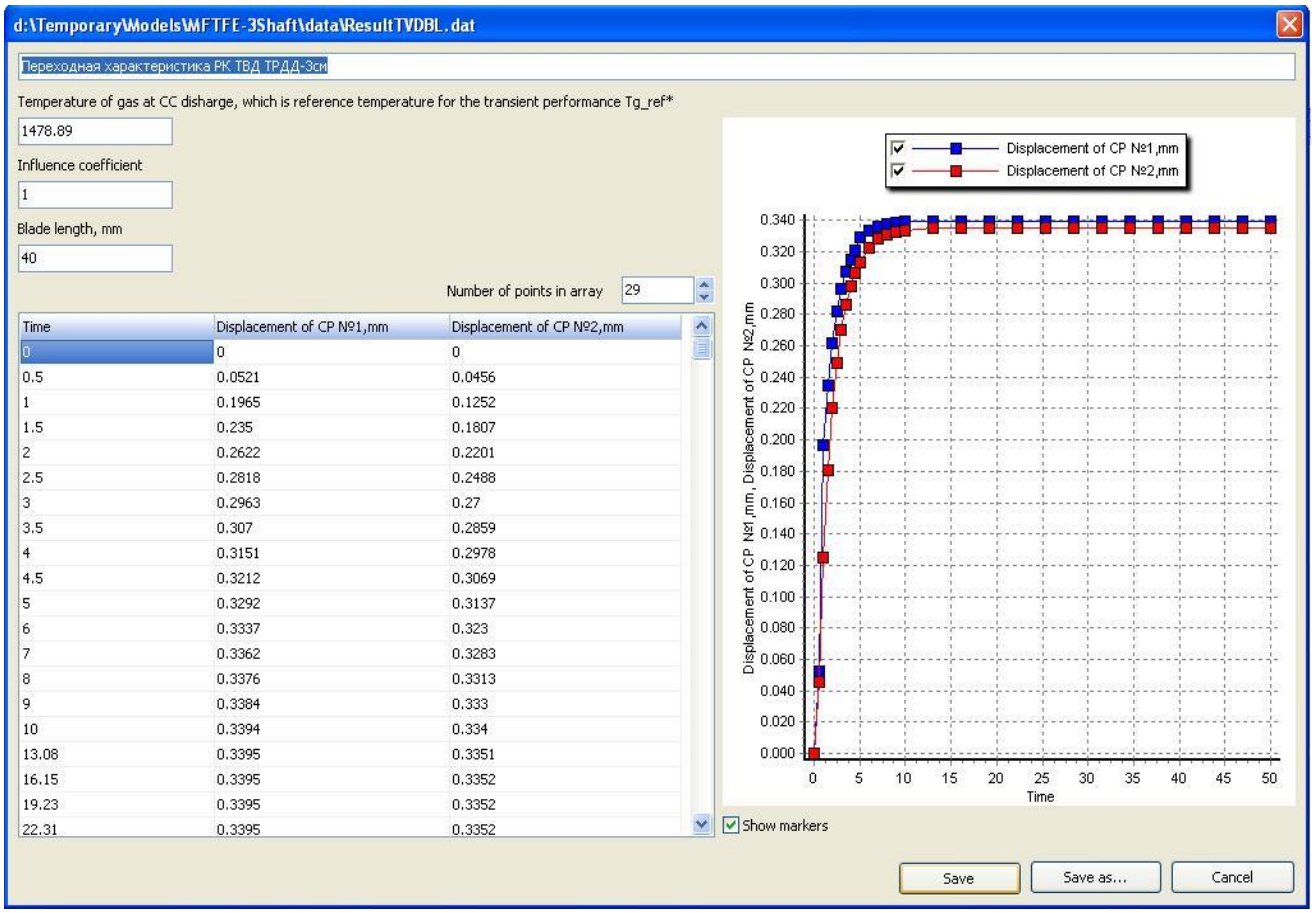


Figure 6.35 – Editor of transient performance of turbine impeller blade.

There is an opportunity to input the data from Excel (by RMB) by ***"Input from Excel"***. The insert has maximum 3 columns and 1000 lines. If the insert has more columns or lines, then they will be automatically cut off.

Format of the file with transient performances of working blade is shown in Appendix A.14, with transient performances of stator and disk – in Appendix A.15.

6.1.2.13 Editing parameters of characteristic volumes

Main parameters, which must be set are characteristic volumes, i.e. CC volume (V_{CC}), volume of transition duct in common nozzle (V_N),volume of transition duct in primary nozzle ($V_{N\,I}$) and volume of transition duct in secondary nozzle ($V_{N\,II}$) (see subsection 1.3.2).

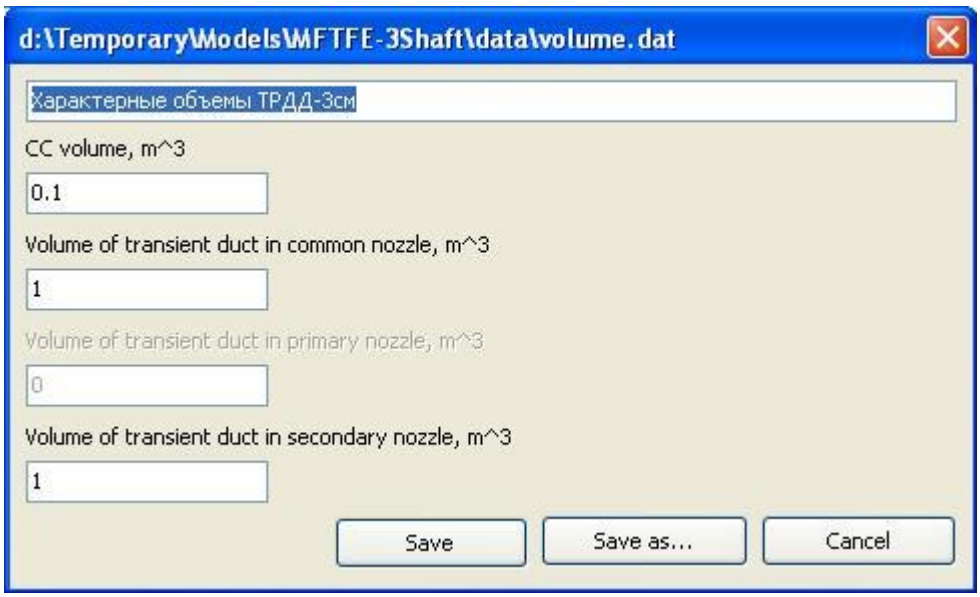


Figure 6.36 – Editor of characteristic volumes of an engine

Format of the file is shown in Appendix A.16.

6.1.2.14 Editing the performance of the airflow through throttle valve (VBV)

Airflow through throttle valve (airflow through the valve means the airflow related to airflow at intermediate casing inlet $\overline{G}_{bleed}$) is determined as a ratio between static pressures at intermediate casing inlet and VBV discharge (π). It also depends on valve position (the angle of VBV, α). This performance is stratificated according to Mach number of primary flow at the station where throttle valve is mounted. The general view of the performance for the single Mach number is presented in fig. 6.37. The editor is presented in fig. 6.38.

Using the up/down arrows in the ***"Performance number"*** field User can choose the needed number of performances, which corresponds to number of constant Mach number curves of VBV performance. Then radio-buttons appear in the box "***Current performance***".

Next User must switch the radio-button and input the following information for each radio-button:

- number of constant throttle valve position curves (use the up/down arrows);
- maximum number of points of each constant throttle valve position curve (use the up/down arrows);
- Mach number for which current performance was inputted.

Important: Mach number must permanently increase from performance to performance.

For each constant throttle valve position curve User must input π in the field PI, $\overline{G}_{bleed}$ – airflow through valve, α – throttle valve position (must be inputted in the similar way to compressor performance, see subsec. . 6.1.2.2).

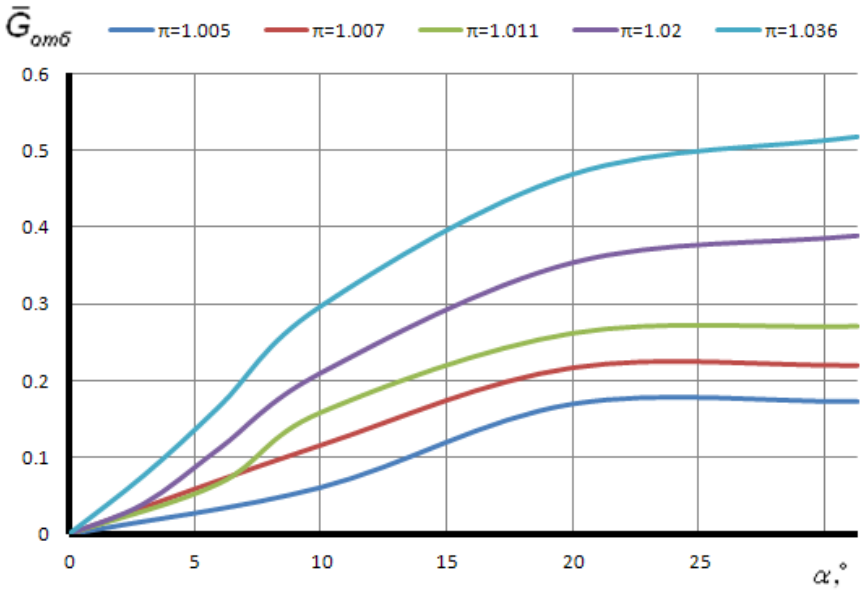


Figure 6.37 – The performance of throttle valve for M=0,1

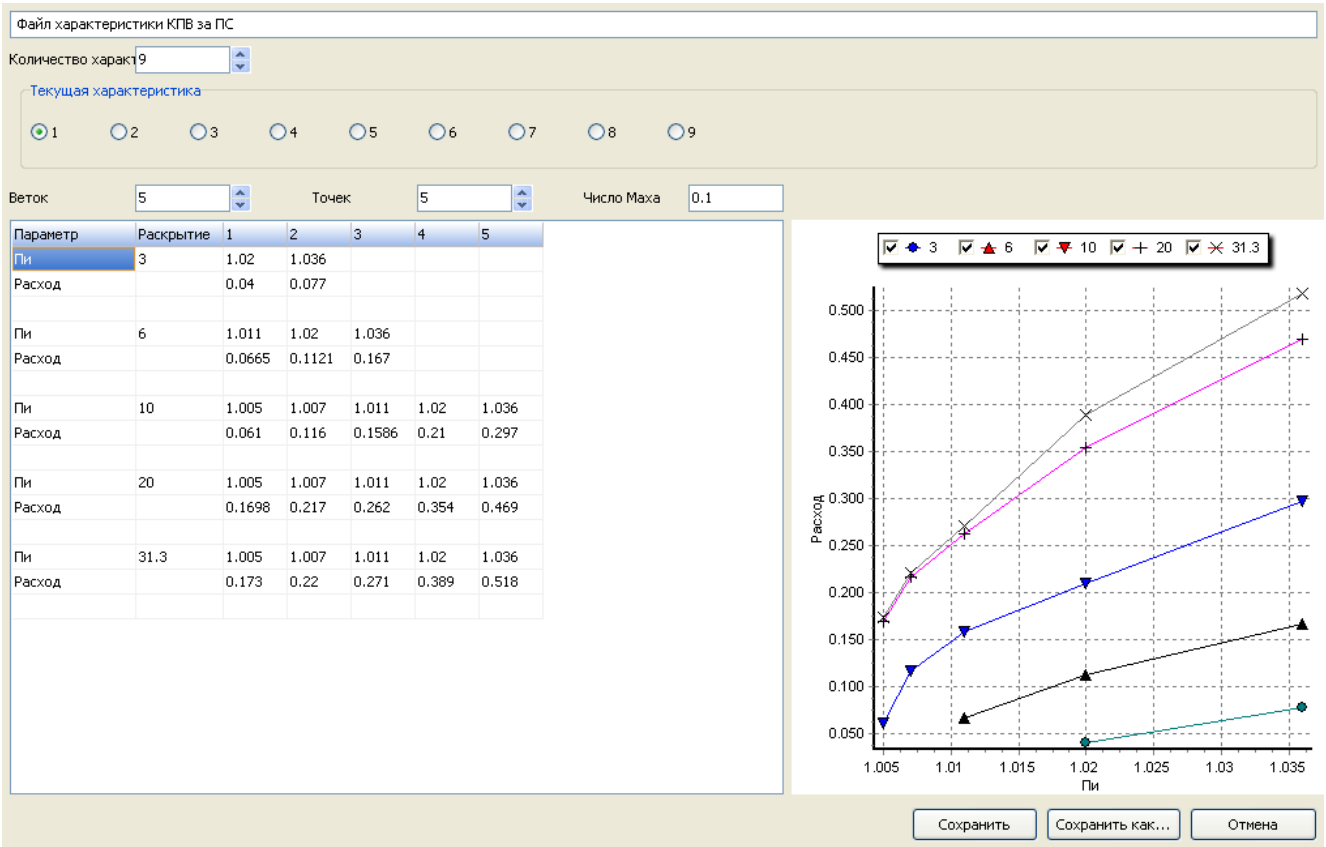


Figure 6.38 – The editor of throttle valve performance

6.1.3 Editing mode parameters

Tab ***"Input parameters → Gas path"*** (fig. 6.39) shows:

- mode parameters of an engine;
- flight conditions;
- time in base points (this parameter is valid only for dynamics and makes no sense for steady modes analysis);
- Additional input parameters (such as additional power take-offs);
- Additional dynamic factors.

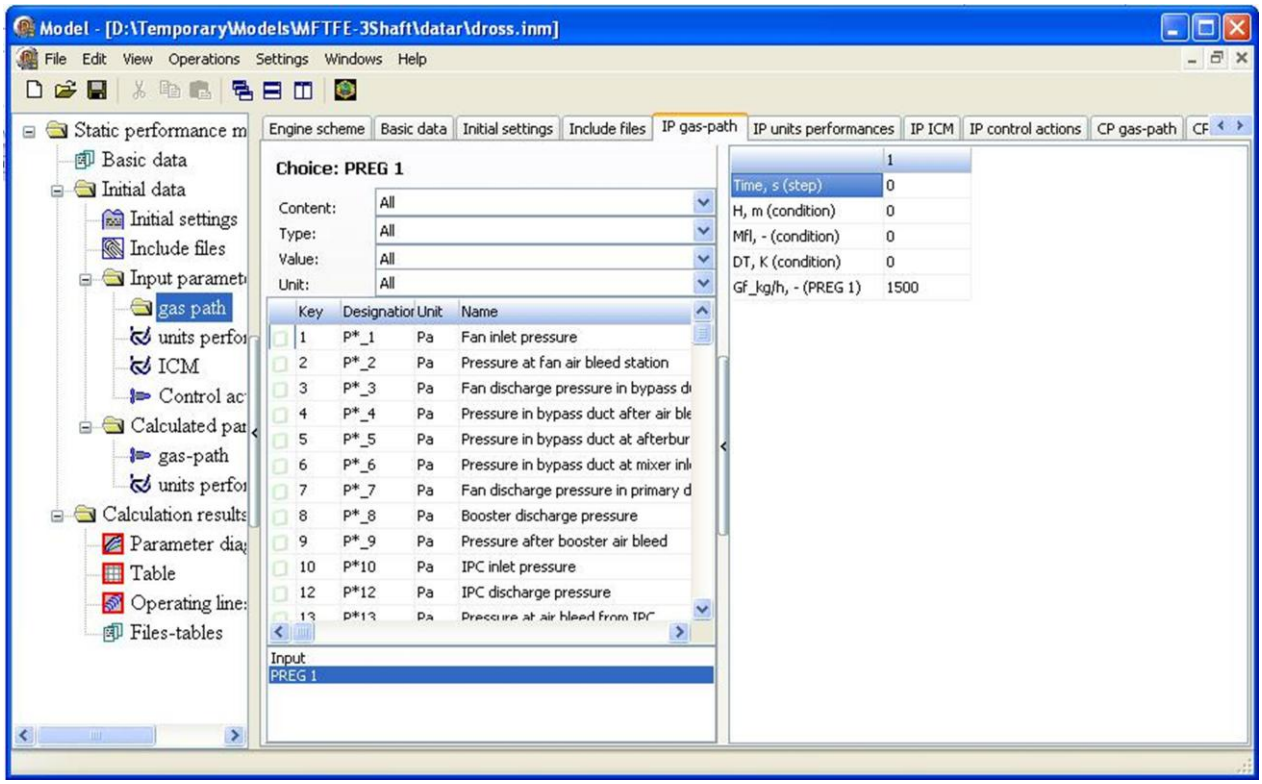


Figure 6.39 - Tab «Mode parameters»

This tab consists of three blocks:

1. Choosing parameters that will determine operating modes, and additional parameters.
2. Input of flight parameters and mode parameters;
3. Choosing additional dynamic factors (only for studies ***«Dynamics»*** and ***«Calculating coefficients of QLDM»***.

6.1.3.1 Choosing the list of operating mode parameters

As it was previously mentioned, number of operating mode parameters equals to number of freedom degrees in the engine, that is why, to determine engine operating mode for SP12 turbofan, numerical values of one parameter must be set.

Choosing control parameters is done in the block ***"Choice"***.

To choose control parameter, user must click icon ***PREG1*** by LMB. The inscription will appear above parameters ***Choice: PREG1***. After this, user must choose the parameter from general list by putting the checkmark in the box in front of parameter`s designation by clicking twice with LMB (chosen parameter will be marked by ✔). Only one parameter can be simultaneously chosen in one box for choosing operating mode parameter. If one control parameter is already chosen and User tries to choose one more parameter, then next warning appears (fig. 6.41).

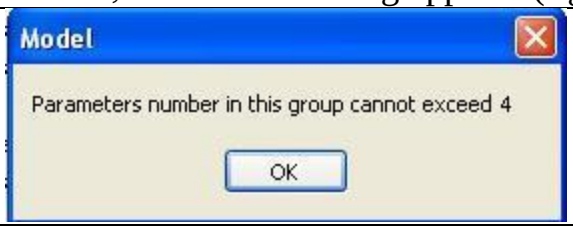


Figure 6.40 – Warning that informs about exiding number of input parameters to characterize engine operating mode



Figure 6.41 – Warning that informs about exiding number of input parameters to characterize engine operating mode

Input parameters are chosen in the same way. Maximum number of parameters is four. If number of parameters exceeds 4 then the warning shown in fig. 6.40 appears.

Program has an option to filter parameters for simplifying their searching. Parameters can be filtered:

- *by content of the list*:
  - ***"All"*** means that all parameters of the list will be displayed;
  - ***"Chosen"*** means that only parameters which were chosen for this part will be displayed;
- *by type*:
  - ***"All"***;
  - ***"Main parameters"*** means that only real-time controlled parameters of engine will be displayed (G_f n_{LP} , n_{LP}, etc.);
  - ***"Units performances"*** means that only parameters, which were calculated from performances will be displayed;
  - ***"Initial values"*** means that only parameters, which are constantly chosen for this engine will be displayed (η_T , σ_{int}, etc.);
  - ***"Bleeds and taps, takeoffs and supplies";***
  - ***"Signs"***
  - ***"Specification needed"*** means that parameters, which do not belong to any of previous groups and need specification from user, will be displayed.
  - ***"Others"***
- *by values:*
  - ***"All"***;
  - ***"Dimensionless factors"***;
  - ***"Mass flows"***;
  - ***"Rotational speeds"***;
  - ***"Pressures"***;
  - ***"Temperatures"***;

- ***"Powers"***;
- ***"Others"***;

– *by the unit, this parameter belongs to.*

All this filters work according to Boolean expression "AND".

6.1.3.2 Inputting operating mode parameters and flight conditions

After choosing parameters of operating mode and set of parameters to describe ambient conditions, (see subsection 6.1.3.1) user must specify their numerical values.

There is a table in the right block. Lines of this table are parameters and columns – are operating modes. Therefore, all parameters of one mode are input in one column. The method to set number of operating mode (Tab ***"Modes")*** is described in paragraph 6.1.1.

Program has different ways to input information about operating mode parameters and flight conditions. To input value of parameter, it is necessary to move cursor to name of parameter (parameter, which need to be changed) and click twice by LMB. Then appears the window for data inputting (fig. 6.42).

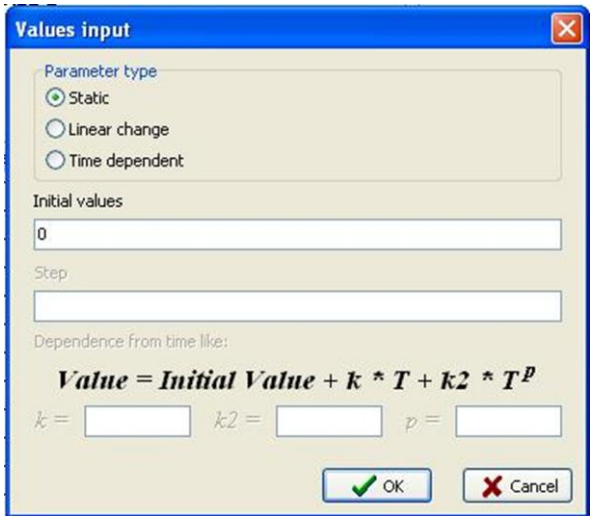


Figure 6.42 – Menu for inputting information

If to choose ***"Static"*** in the box ***"Parameter type"*** and set the value in the box ***"Initial data"***, then this value will not change from mode to mode.

If to choose ***"Linear change"***, then the box ***"Step"*** becomes active. Consequently, values of parameter at different modes will be calculated as follows:

$$Value = Initial\ value + step \cdot (N_{mode} - 1),$$

where N_{mode} is a mode number (number of column to input the data).

If to choose ***"Time dependent"*** then value will be calculated by formula:

$$Value = Initial\ value + k \cdot T + k2 \cdot T^{p}$$

where k, $k2$ and p are user defined constants, T is the value of parameter, which was set in box ***"Time"*** (fig. 6.39).

6.1.3.3 Choosing additional dynamic factors

As it was marked in subsection 1.3, there are some types of dynamic factors. Main factor is inertia of rotor. The impact of other dynamic factors is less, but they increase number of differential equations and time of calculation. That is why, when it became clear, that particular factor is inexpedient, it must be excluded from calculation by removing the checkmark oppose it in box ***"Additional dynamic factors"***. This box is active only for ***"Dynamics"*** and ***"LDM coefficients calculation".***

6.1.4 Inputting performances of units

Changing of technical state parameters of gas-path influence engine parameters. The level of this influence need to be assumed in development of gas-path diagnostic algorithms. These changes are implemented by deviations of gas-path geometrical parameters (areas of gas-path stations, installation angles of blades, characteristic sizes of micro relief on surfaces, etc.). But considered above model does work with geometrical parameters, it works with parameters of units, calculated by their performances maps. This is caused by level of considered gas-path model, in which internal units structure is neglected. That is why model can solve given task only by simulating changes in the geometrical parameters with deformations of corresponding performances of units. There is no necessity in considering big number of performance deformations because number of regularly measured parameters in gas-path is little and because measuring tolerance is low. Besides, most geometrical parameters cause same deformations of units performances, and correspondingly engine parameters.

Performances of units are usually expressed in a form of one- or two parametric function. Let`s consider initial performance in a following form $\nu_o = f(y_1, y_2)$. Taking into account previous notations, program provides the next deformations of performances: parallel shift of performance relative values and pivot turns around crossing axes, which are parallel to axes of coordinates.

To provide the possibility of changing values and directions these shifts, there is necessary to introduce parameters (next, parameters of units performances). Thus let`s obtain parameterized performances of units. For example, parameterized form of mentioned above 2-D performance has the following form:

$$\nu = \nu_0 \left(1 + a \frac{y_1 - y_{1_0}}{\Delta y_1} + b \frac{y_2 - y_{2_0}}{\Delta y_2} + c \right), \qquad (6.2)$$

where ν_0 is a value of unit parameter calculated by initial performance; y_1, y_2 are current values of performance arguments; $(y_1)_0$, $(y_2)_0$ are reference values of parameters, which correspond to chosen reference point of unit performance (performance turns round these points); Δy_1, Δy_2 are variation ranges parameters in the performance maps (they are used to normalize the deviations of current point referring to $(\mathbf{y_1})_0$, $(\mathbf{y_2})_0$); a, b, c are parameters of the considered performance of unit. Setting order of $(\mathbf{y_1})_0$, $(\mathbf{y_2})_0$ and Δy_1 and Δy_2 are described insubsec. 6.1.2.1.

When parameter a changes then initial performance of unit turns in plane $y_1 0 \nu$ (where "0" is coordinates origin) round point with coordinates $\left(y_{1_0}, y_{2_0}, \nu\left(y_{1_0}, y_{2_0}\right)\right)$. When parameter b changes then analogical turning in plane $y_2 0 \nu$ happens. When parameter c changes then initial performance is shifted along 0ν axis and shift is proportional to initial value (ν_0).

Let's consider parameterization of turbine performance as an example. Basic performance $\eta^*_{T\,ref} = f\left(\pi^*_T, \bar{n}_T\right)$ is shown in fig. 6.43. Real performance η^*_T will differ because of difference in gas-path geometrical parameters. As a result of previous considerations the following correlation between η^*_T and $\eta^*_{T\,ref}$ is proposed by introducing parameters θ_1, θ_2 and θ_3:

$$\eta^*_T = \eta^*_{T\,ref}\left(1 + \theta_1 + \theta_2 \frac{\pi^*_T - \pi^*_{T\,0}}{\Delta\pi^*_T} + \theta_3 \frac{n_{T\,cor} - n_{T\,cor\,0}}{\Delta n_{T\,cor}}\right), \tag{6.3}$$

where $\left(\pi^*_T\right)_0$ and $\left(\overline{n_T}\right)_0$ are randomly chosen reference values of arguments; $\Delta\pi^*_T$, $\Delta\bar{n}_T$ are variation ranges of arguments; θ_1 is a parameter, which describes constant relative changing of η^*_T in all definitional domain; θ_2 is a parameter, which describes turning of initial performance round line $\eta^*_T = f\left(\left(\pi^*_T\right)_{cor\ 0}, \bar{n}_T\right)$ in all definitional domain; θ_3 is a parameter, which describes turning of initial performance round line $\eta^*_T = f\left(\left(\pi^*_T\right)_0, \left(\bar{n}_T\right)_{cor\ 0}\right)$ in all definitional domain.

Basic 1 and real 3 performances at $\bar{n}_T = const$ are shown in fig. 6.44. Differences between them are described by parameters θ_1 (constant relative shift of line 1 to position 2) and θ_2 (line 2 turning to position 3) at $\theta_3 = 0$.

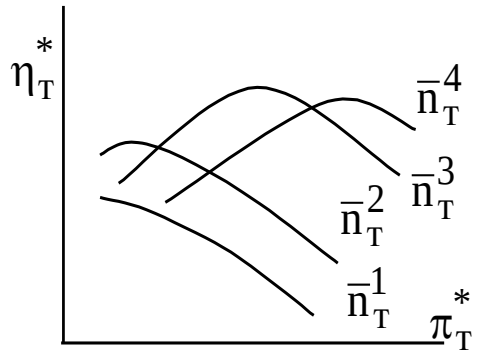


Figure 6.43- Reference performance of turbine

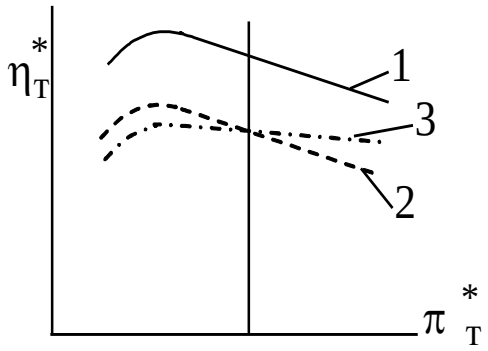


Figure 6.44 – Reference and real performances of turbine

In accordance with represented concept of engine universal mathematical model, all performances of units included into gas-path generic scheme are parameterized and parameters of units performances are sorted in accordance with Appendix G.

Designating parameters of units performances as $\vec{\theta}$, an engine static model may be represented as follows:

$$\vec{H}\left(\vec{Q},\ \vec{U}_{fl},\ \vec{U}_{om},\ \vec{\theta}\right) = 0; \tag{6.4}$$

$$\text{or } \vec{Y} = \vec{F}\left(\vec{U}_{fl}, \vec{U}_{om}, \vec{\theta}\right). \quad (6.5)$$

After choosing modes, user must set shifts of performances from base position (base values) in menu item ***Static performance=>Initial data=>Input parameters=>Unit performances*** (see fig. 6.45, marked region).

Note: menu item "ICM" in this version of program is not activated.

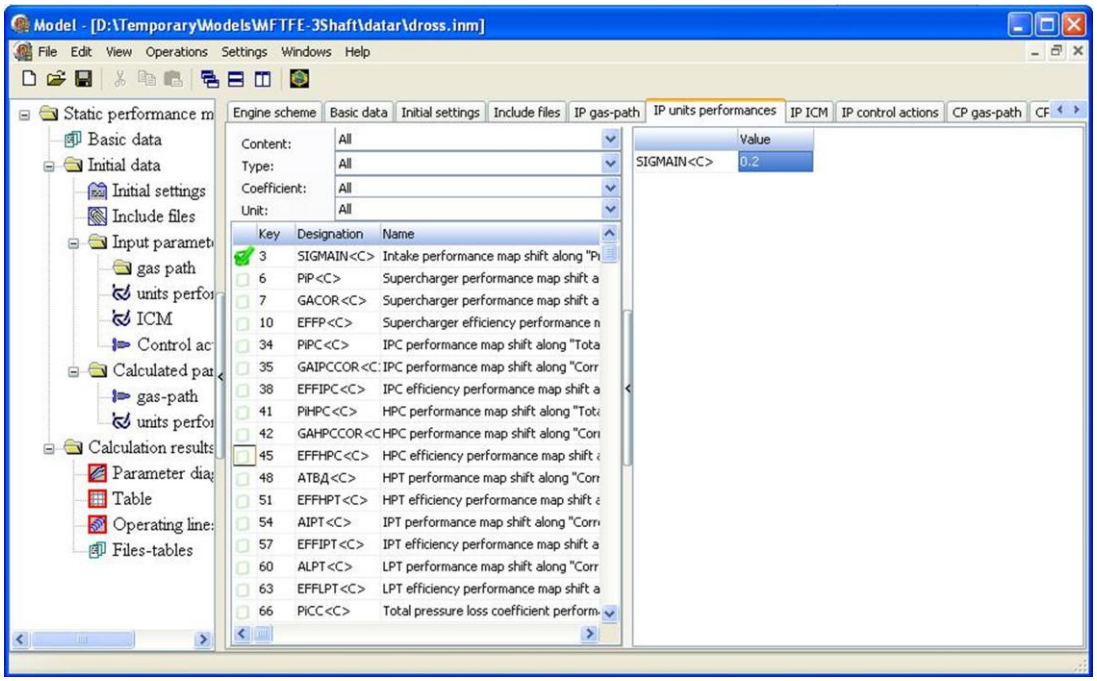


Figure 6.45 – Window for inputting state parameters

6.1.5 Inputting control action

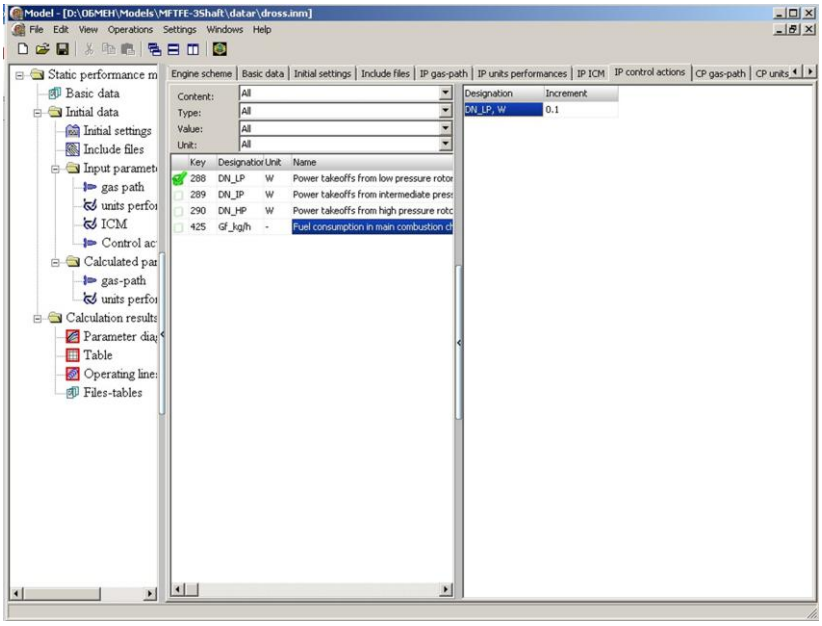


Figure 6.46 – Window for changing control actions

This tab serves for configuring LDM calculation. Here User may choose content of control vector $\Delta\bar{U}$ (see equations (1.100), (1.101)) and variation ranges of parameters that comprise this vector to calculate elements of matrixes B and D (fig. 6.46).

6.1.6 Inputting list of calculated parameters

After inputting initial data, User must choose parameters, which need to be calculated (fig. 6.47). This menu item has the same filters as were described in subsection 6.1.3.1.

User has an option to choose all parameters from general list by clicking twice by LMB in the box ***"*"***.Repeating previous command will deselect all parameters from general list. Each single parameter User can choose by clicking twice LMB on each parameter.

Note: menu item "Calculated parameters" ⇒ "Units performances" is not active in this version of the program.

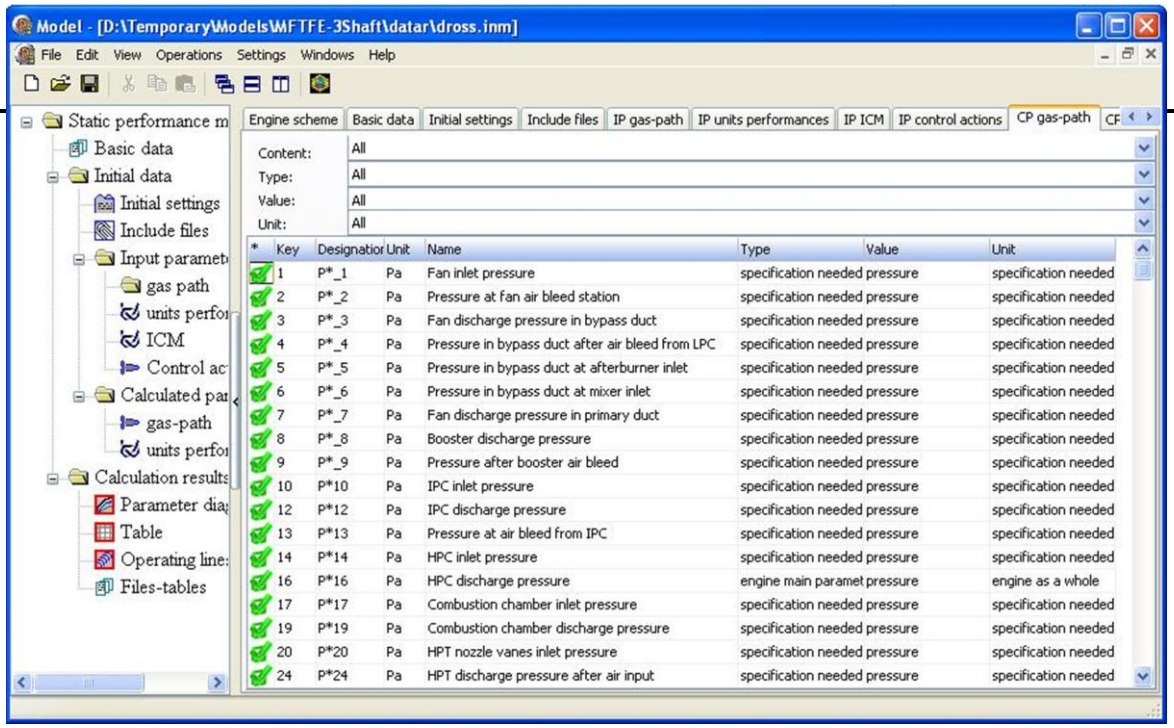
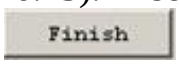


Figure 6.47 – Dialogue window for choosing gas-path parameters that need to be calculated

6.1.7 Starting calculation

 icon starts the calculation, according to chosen algorithm. Then appears dialogue window, where User can check entered information. Window has information about numbers of operating mode parameters, their numbers in general list and values at given operating modes. Also, calculation tolerance, tolerance of equations set solution and maximum number of iterations are stated (fig. 6.45). If set parameters are correct, user must click  icon, otherwise – click  icon and correct values, which were set incorrectly (subsections 6.1.1-6.1.6) and then run program again.

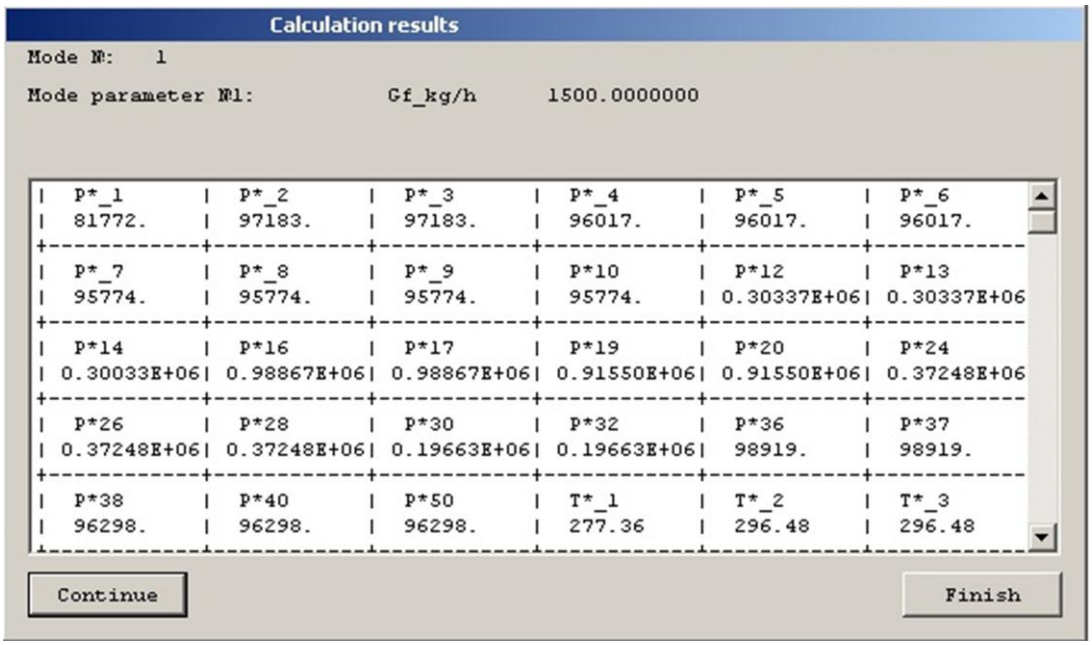


Figure 6.48 – Window to display iterim results of calculation

During throttle performance calculation, User has an opportunity to control calculation proceeding at each operating mode. To make program show intermediate results User must put checkmark in the box in front of ***"Display interim results***". Herewith, after each point calculation interim results will be displayed as it is shown in fig. 6.46.

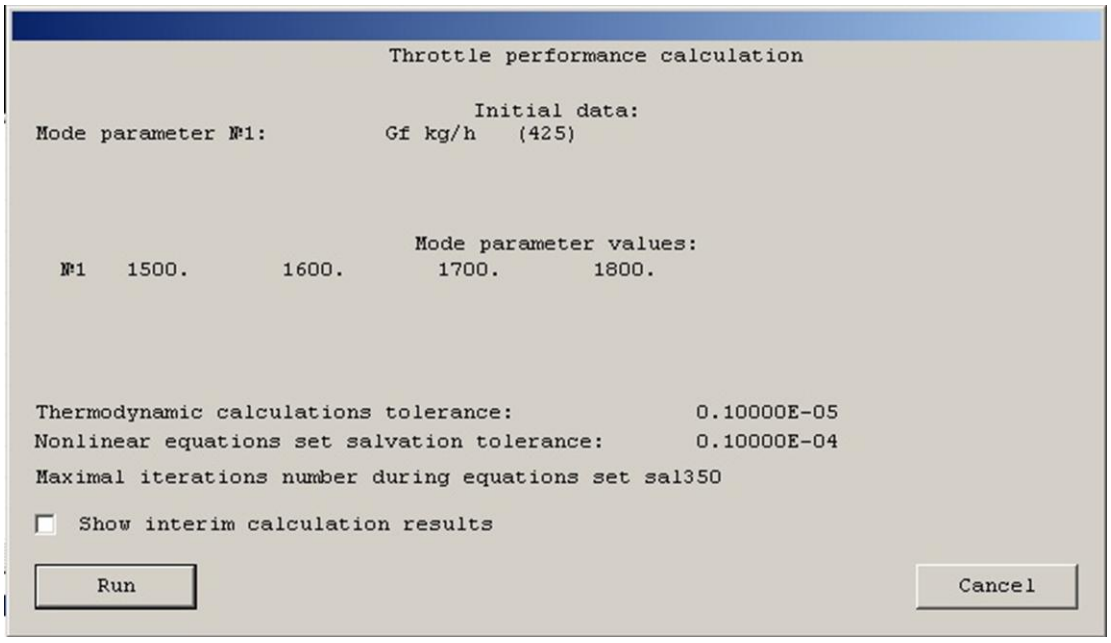


Figure 6.49 – Dialogue window to check the inputted information

If calculation result satisfies user, then user must click Run to run the calculation for next point (fig. 6.46). Otherwise, user must click Cancel and terminate the

calculation. All successfully calculated operating modes will be outputted to the file and will be available for browsing.

Some mistakes may occur during calculation. Program displays corresponding messages. Description of these messages is shown in Appendix D.

6.2 Displaying of calculation results

In case of successful calculation, the results are outputted to the files GRAFIX.DAT and PARAM.DAT (see their appointment in subsection 3.2)

Results can be browsed in tabular or graphical form.

6.2.1 Browsing calculation results in graphical form

To visualize calculation results, User must go to menu item ***"Calculation results" ⇒ "Parameter diagrams"*** (fig. 6.50).

This menu item also consists of two blocks:

1. block for choosing parameters;
2. block for visualizing parameters.

In the first block user can choose any calculated parameters.

It is necessary to choose the parameter, which will be used for abscissa axis. For this, User must choose needed parameter from list of parameters for ***X-axis***. This parameter will be used as an argument to form correlation for other parameter dependences, chosen from ***Y-axis*** list.

There is an opportunity to define parameter value in each point. To do this, user must put the checkmark in the box ***"Show markers"*** and ***"Show values"*** (fig. 6.50).

If there is a necessity in comparing some close parameters in common scale, then User must put the checkmark in the box ***"In common scale"***. Putting checkmark in front of ***"Active graph"*** activates the scale of this parameter (Y axis). Parts of diagram can be scaled as it was described in subsection 6.1.2.2.

Diagrams can be printed using next menu item ***"File" ⇒"Print"***. Colors of printed diagram will be inverted. Also there is a possibility to print the scaled part.

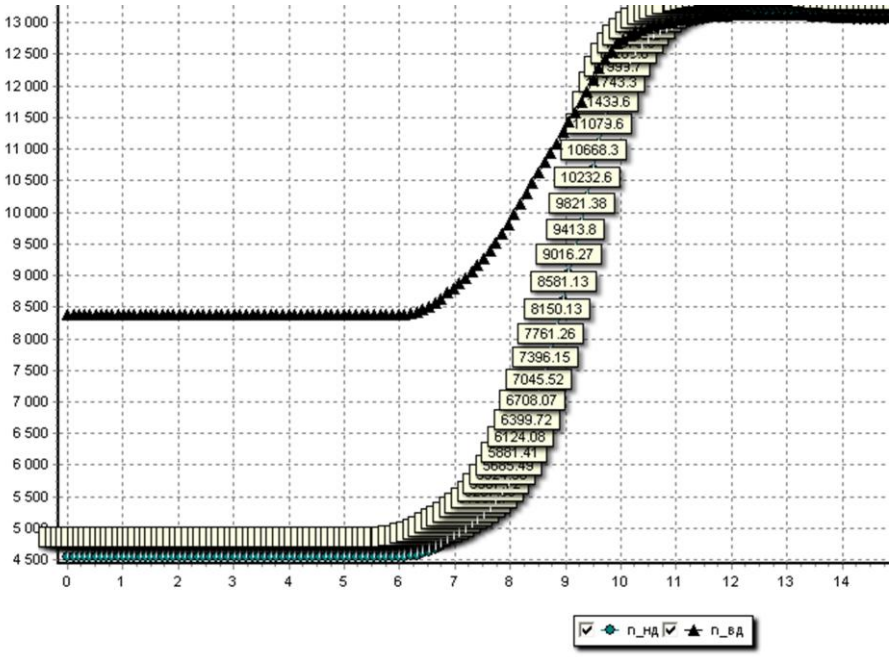


Figure 6.50 – Diagram with markers and values

6.2.2 Browsing results in tabular form and outputting results to file

Browsing results can be done in menu item ***"Calculation results" ⇒ "Table"*** (fig. 6.52).

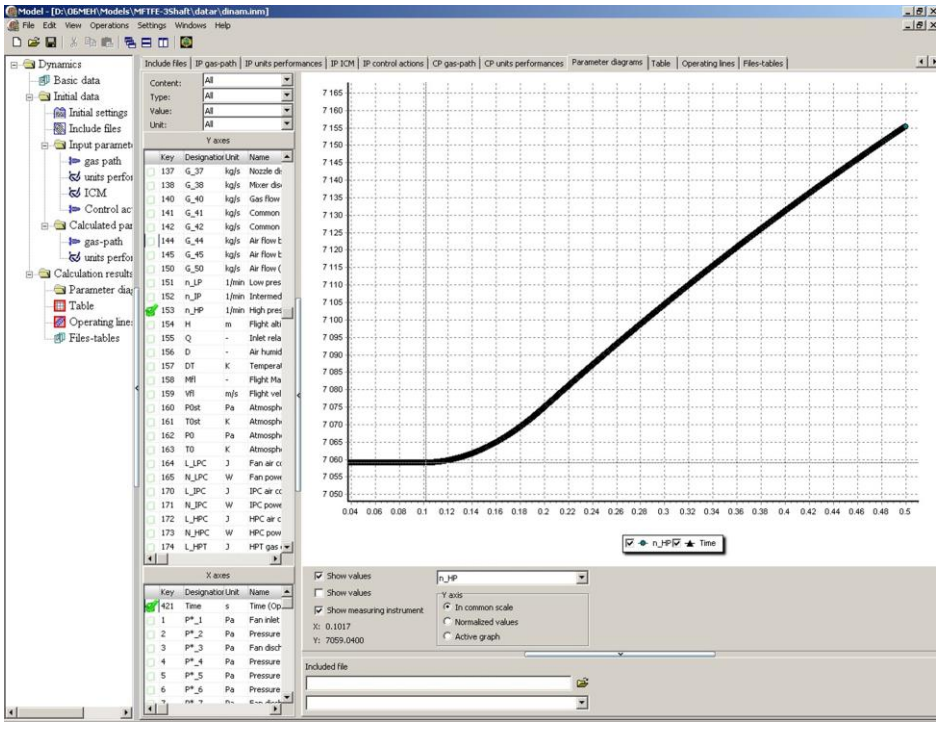


Figure 6.51 – Dialogue window to output calculation results in graphical form.

This menu item consists of two parts:

1) formatting the table;
2) misualizing chosen information.

Initially, all parameters and modes are marked. But user can edit content of table with calculation results, which need to be outputted to file-report. Any parameter (operating mode) can be added or removed from parameters (mode) list by double click on parameter or operation mode names. Marking and unmarking of all parameters and modes is done by clicking twice above all boxes with checkmarks (same to described in subsection 6.1.6).

File content can be chosen in menu item ***"Settings"⇒ "Result settings"***. Then appears dialogue window to set table format (fig. 6.53). These window consists of three inserts.

Results can be saved in three formats: text format, HTML format and full report. Each format has an option to choose the order of results organizing inside the file, meant an option to output results in columns or in lines. To configure this option User must mark or unmark the box ***"Transpose"***.

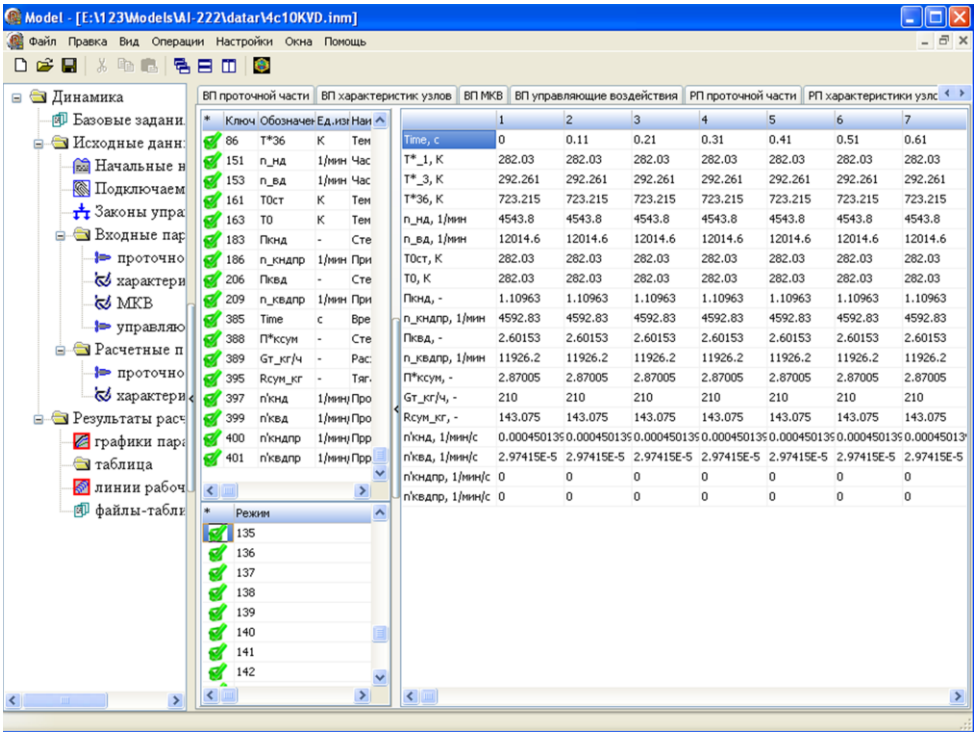


Figure 6.52 – Dialogue window to form tables with calculation results

There is an opportunity to add information about initial data (number of calculation points, list of included files, tolerances, etc.) in the file. This is done by choosing the corresponding item.

Last general setting is the calculation tolerance (number of figures after the point).

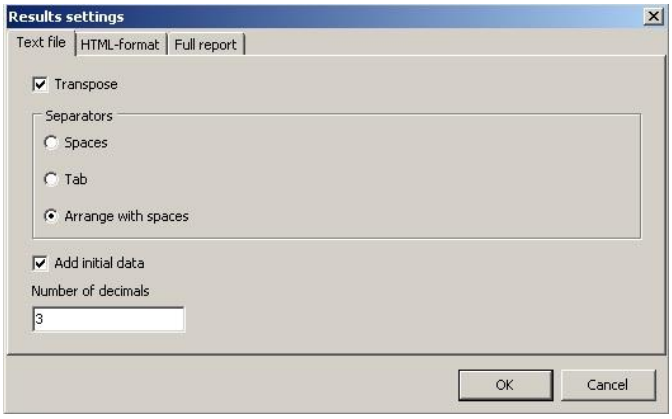


Figure 6.53 – Dialogue window to configure the table for outputting calculation results

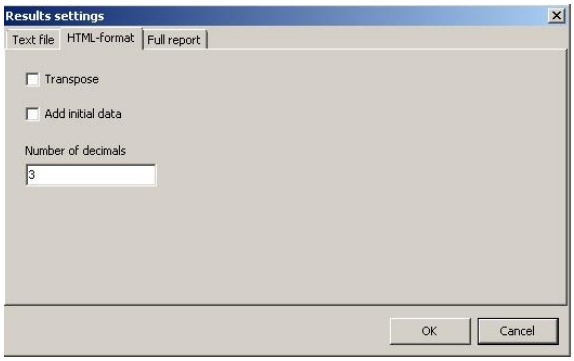


Figure 6.54 – Dialogue window to save calculation results

After configuring corresponding output files and choosing necessary modes and parameters in menu item ***"Calculation results" => "Table"***, the results can be saved by the following menu item ***"Operations" => "Save the table with results"***. Then appears dialogue window, where User can set format of output file and path for saving (fig. 6.54).

6.2.3 Browsing calculation results on performances of compressor and turbine (operating line)

To browse operating line in compressor and turbine performances maps, it is necessary to use menu item ***"Calculation results" ⇒"Engine operating line".*** There will appear the box for diagrams outputting and list of files, using which engine operating line can be browsed. These files are compressor and turbine performances. After choosing one of the performances (it is necessary to put checkmark in the box in front of name of the performance) it will appear in the box for diagrams (fig. 6.55). Browsing diagrams, scaling and printing is done in the mentioned above way.

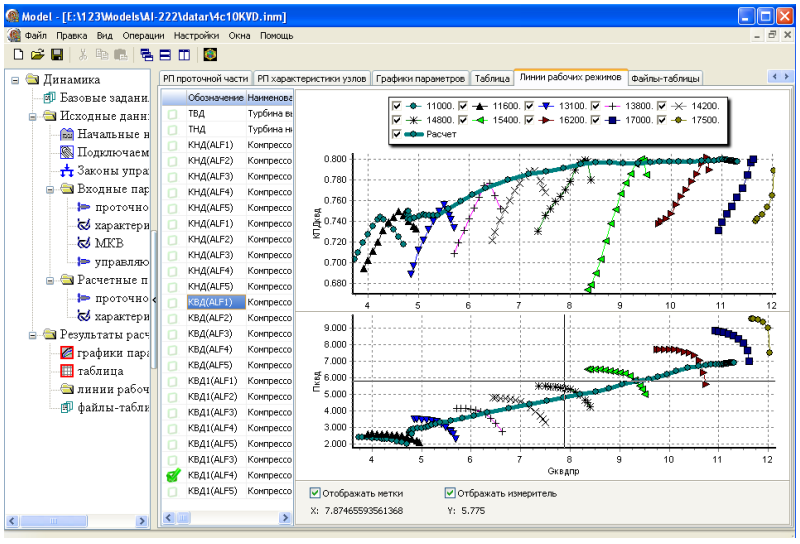


Figure 6.55 –Dialogue window to visualize operating line

at compressor performance map

6.3 Configuring studies

Depending on chosen study (menu item ***Initial data⇒Initial settings⇒Studies*** (subsection 6.1.1) there are some unique features, which will be considered next.

6.3.1 Static performance

Choosing ***"Static performance"*** in menu item ***"Initial data" ⇒"Initial settings"*** (fig. 6.3) allows performing the following calculations:

- throttle performance of an engine;
- climatic performance of an engine;
- velocity performance of an engine;
- altitude performance of an engine.

6.3.1.1 Throttle performance

Throttle performance is a dependence of basic parameters of engine operation (first of all its power, thrust, specific fuel consumption) from main control parameter (it characterizes the intensity of engine operation) at constant ambient conditions.

To get the throttle performance, User must follow the considering next algorithm:

1. It is better to choose already customized project to make the calculation (see subsection 6.1), or otherwise it will be necessary to include all files to a new project.
2. Choose ***"Static performance"*** in the tab ***"Studies"*** (fig. 6.3).
3. Choose the one of sets to describe ambient conditions in the tab ***"Conditions"*** (fig. 6.6).
4. Set tolerances of solving equations set and thermodynamic calculations (fig. 6.7). Integration step may be set 0.
5. Set number of points in the throttle performance (fig. 6.5).
6. Choose the mode parameter (see subsection 6.1.3.1) in the tab ***"Input parameters" ⇒ "Gas-path"***(fig. 6.39) and set its value for each point of throttle performance (see subsection 6.1.3.2).
7. Choose the list of calculated parameters (see subsection 6.1.6).
8. Run the calculation (see subsection 6.1.7).
9. Results are available in graphical (subsection 6.2.1) or in tabular form. Also there is an option to output calculation results to the file (subsection 6.2.2).

6.3.1.2 Climatic performance

Climatic performance is a dependence of thrust and other parameters of engine from ambient pressure p_0 and temperature T_0^* in case engine operating mode is the same and $H = const$, $V_{fl} = const$, where H is flight altitude, V_{fl} is flight velocity.

To get the climatic performance, User must follow the considering next algorithm:

1. It is better to choose already customized project to make the calculation (see subsection 6.1), or otherwise it will be necessary to include all files to a new project.
2. Choose ***"Static performance"*** in the tab ***"Studies"*** (fig. 6.3).
3. Choose the one of sets to describe ambient conditions in the tab ***"Conditions"*** (fig. 6.6).
4. Set tolerances of solving equations set and thermodynamic calculations (fig. 6.7). Integration step may be set 0.
5. Set number of points in the climatic performance (fig. 6.5).
6. Choose the mode parameter (see subsection 6.1.3.1) in the tab ***"Input parameters"*** $\Rightarrow$ ***"Gas-path"***(fig. 6.39) and set it constant at all modes (see subsection 6.1.3.2). Then set the parameter, which determines climatic conditions (e.g. ΔT_0 in the ***Set 0*** or ***Set 1***, or p_0 and T_0 in the ***Set 2*** or ***Set 3***).
7. Choose the list of calculated parameters (see subsection 6.1.6).
8. Run the calculation (see subsection 6.1.7).
9. Results are available in graphical (subsection 6.2.1) or in tabular form. Also there is an option to output calculation results to the file (subsection 6.2.2).

6.3.1.3 Velocity performance

Velocity performance is the dependence of parameters of GTE from flight velocity (Mach number) at given control program.

Algorithm of initial data inputting and analysis running is the same as for climatic performance (see subsection 6.3.1.2), except item 6. In this menu item User must specify the velocity change (for ***Set 1*** of ***Set 2***) and Mach number (for ***Set 0*** and ***Set 3***).

6.3.1.4 Altitude performance

Altitude performance is dependence of parameters of GTE from altitude at given control program.

Algorithm of initial data inputting and analysis running is the same as for climatic performance (see subsection 6.3.1.2), except item 3. In this item User must choose only ***Set 0*** or ***Set 1***.

6.3.2 Identification

As it was mentioned in subsection 6.1.1, when running "***Identification"*** there will appear an interface to identify parameters of model by testing results. First of all User must choose file with customized project (see subsection 6.1). Next it is necessary to choose ***"Identification"*** in menu ***"Studies"*** (see subsection 6.1.1). And finally User must run the calculation as it was described in subsection 6.1.7, which will call an interface for customizing the input data for identification. The description of the interface is given in the summary "Software for identification of gas turbine engines using thermodynamic models" (file ***Identification.pdf***).

6.3.3 Dynamics

Dynamic model simulates non-steady (transient) operating modes of an engine. Dynamic operating modes of the engine (its units and assemblies) are modeled by differential equations, which are presented in their dynamic performances (subsection 1.3).

1. It is better to choose already customized project to make the calculation (see subsection 6.1), or otherwise it will be necessary to include all files to a new project.

2. Choose ***"Dynamics"*** in the tab ***"Studies"*** (fig. 6.3).

3. Choose the one of sets to describe ambient conditions in the tab ***"Conditions"*** (fig. 6.6).

4. Set tolerances of solving equations set and thermodynamic calculations (fig. 6.7). Opposite to other studies, when running the analysis of dynamic perfomances is necessary to set ***"Integration step"***. The bigger is the step, the faster will be the calculation of transient process. But sometimes big step may cause the poor robustness when solving set of equations 1.80 - 1.89. In this case the integration step must be lessen.

5. Set number of datum points to set the changing of control parameter (fig. 6.5).

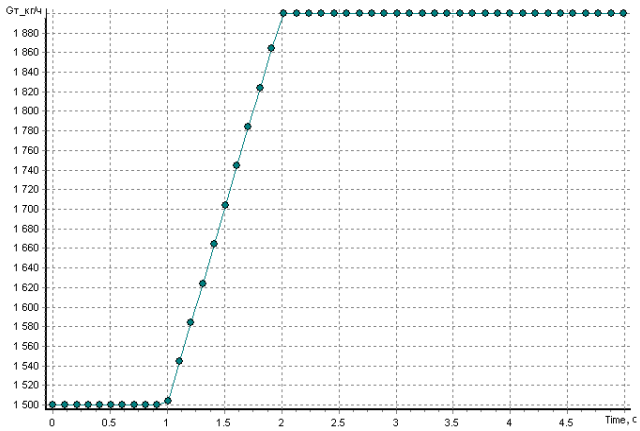


Figure 6.56 – Changing of control parameter in time

	1	2	3	4
Time, s (step)	0	0.1	0.2	0.5
H, m (condition)	0	0	0	0
Mfl, - (condition)	0	0	0	0
DT, K (condition)	0	0	0	0
Gf_kg/h, - (PREG 1)	1500	1500	1900	1900

Figure 6.57 – An example of control parameters customizing

6. Choose the mode parameter (see subsection 6.1.3.1) in the tab ***"Input parameters" → "Gas-path"***(fig. 6.36) and set it values at datum points (see subsection 6.1.3.2). Set parameter, which determines climatic conditions in all datum points. Opposite to static mode, ***"Time"*** is important to be set correctly. To understand what do datum points mean, let`s consider fig. 6.56. There 4 characteristic points in the figure. From the beginning to first second fuel consumption does not change and equals to 1500 kg/h. Starting from 1 second and to 2 second fuel consumption linearly changes from 1500 kg/h to 1900 kg/h. Then, starting from the 2 second and up to the end, i.e. up to 5th second, fuel consumption stays constant and equals to 1900 kg/h. Ambient conditions do not change during the whole process. That is why ***"Input parameters" → "Gas-path"*** will look like it is shown in the fig. 6.57.

7. Besides inputting mode parameters, ambient conditions and time in datum points, this study has an opportunity to choose the set of differential equations (dynamic factors). To choose the list of dynamic factors User must go to ***"Input parameter" →"Gas-path"*** →box ***"Add. Dynamic factors"*** (the example of list for TFE-3mix is shown in fig. 6.58). Default options take into account only inertia of rotors (equations 1.64 - 1.65). All other differential equations are equal to zero. Marking the box ***"Volume of CC"*** (by LMB double click), the set of differential equations will be complemented with equations 1.66 and 1.69. When marking the box ***"Volume of transition duct, arranged in bypass"***, the set of differential equations will be complemented with equations 1.67 and 1.70. When marking the box ***"Nozzle volume"***, the set of differential equations will be complemented with equations 1.68 and 1.71. When marking the box ***"Heat exchange in LPT"***, the set of differential equations will be complemented with equations 1.75 - 1.77. Generally, integration step for compressible working substance should not exceed 0,001s.

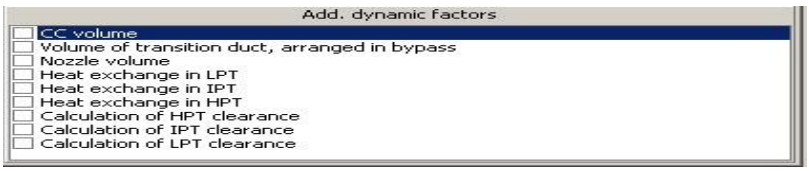


Figure 6.58 – An example of choosing the additional dynamic factors.

8. Choose the list of calculated parameters (see subsection 6.1.6).
9. Run the calculation (see subsection 6.1.7).
10. Results are available in graphical (subsection 6.2.1) or in tabular form. Also there is an option to output calculation results to the file (subsection 6.2.2).

6.3.4 Calculating QLDM coefficients

This study allows calculating coefficients in lineal equations, by deviating control parameters from steady mode and recording deviations of working process parameters, i.e. calculating coefficients of matrixes A, B, C, D which enter in the equations 1.100 and 1.101.

It is better to choose already customized project to make the calculation (see subsection 6.1), or otherwise it will be necessary to include all files to a new project.

1. Choose ***"LDM coefficients calculation"*** in the tab ***"Studies"*** (fig. 6.3).

2. Choose the one of sets to describe ambient conditions in the tab ***"Conditions"*** (fig. 6.6).

3. Set tolerances of solving equations set and thermodynamic calculations (fig. 6.7). Integration step may be set 0, because it is not involved in the analysis.

4. Set number of points that correspond to number of steady modes.

5. Choose the mode parameter (see subsection 6.1.3.1) in the tab ***"Input parameters"*** → ***"Gas-path"***(fig. 6.36) and set its value at all modes (see subsection 6.1.3.2). Also set the value of parameter, which determines climatic conditions. Box ***"Time"*** can be left empty.

6. Besides inputting mode parameters and ambient conditions in datum points, User must set the list of dynamic factors to determine the steady state (state vector 1.100). To choose the list of dynamic factors User must go to ***"Input parameter" →"Gas-path" →box "Add. Dynamic factors"*** (see the ***Dynamics***). Default state vector consist of rotational speeds of rotors. When marking the box ***"Volume of CC"*** (by LMB double click), then state vector will be complemented with parameters m_{CC} and $T^*_{dis\ \mathrm{CC}}$. When marking the box ***"Volume of transition duct, arranged in bypass"***, then state vector will be complemented with parameters $m_{N\ \mathrm{II}}$ and $T^*_{dis\ \mathrm{N\ II}}$. When marking the box "***Nozzle volume"***, then state vector will be complemented with parameters m_N and $T^*_{dis\ \mathrm{N}}$. When marking the box ***"Heat exchange in HPT",*** then state vector will be complemented with parameters $T_{\mathrm{ST\ HPT}}$, $T_{\mathrm{BL\ HPT}}$ and $T_{\mathrm{DISK\ HPT}}$; ***"Heat exchange in IPT"*** – $T_{\mathrm{ST\ IPT}}$, $T_{BL\ IPT}$ and $T_{DISK\ IPT}$; ***"Heat exchange in LPT"*** – $T_{ST\ LPT}$, $T_{BL\ LPT}$ and $T_{DISK\ LPT}$.

7. Choose the composition of vector of observations (see. 1.101) in menu item ***"Calculated parameters" → "Gas-path"*** (see subsection 6.1.6).

8. Choose the composition of control vector (vector $\Delta\vec{U}$ in equations 1.100 and 1.101) in menu item ***"Input parameters" → "Control actions"***. Composition of vector is limited by fuel consumption, power take-offs and air bleeds from available for bleeding stations.

9. Then specify the increment of one parameter. The increment is specified in parts from value of parameter at steady mode, which in its turn is determined by parameters, given in item 5.

10. Run the calculation (see subsection 6.1.7).

11. Calculation results can be browsed in tabular format (Calculation results →***Files-Tables***). Choose the file LDMPAR.DAT in central part of the window. Next

appears the table with coefficients of matrix A, B, C, D in the right part of the window (fig. 6.59). File has the following format:

1-st line – number of mode.

2-nd line – number of mode parameter and its value (here is the parameter, which have been chosen in item 5, see fig. 6.59).

Next follows matrixes A and B. Number of lines on matrixes A and B depends on what was chosen in box ***"Add. Dynamic factors"***. If all boxes in ***"Add. Dynamic factors***" were left empty, then there will be number of lines, which corresponds to number of rotors. In case of TFE-3mix, there will be three rotors. Minimum number of lines will always equal to three for this engine type. If to choose the parameter that takes into account that fact that gas is compressed inside the volume, then each considered volume will increase the size of matrixes A and B by two parameters for each (m_i - mass, stored in ith volume, T_i^* - temperature at volume discharge). In to choose heat exchange in any spool of the turbine, then number of parameters will increase by 3 per each spool ($T_{ST\,i}$, $T_{BL\,i}$, $T_{D\,i}$ are temperature of stator, blade and disk respectively). Number of columns of matrix A corresponds to number of lines (i.e. matrix A is square). Number of columns of matrix B depends on number of parameters, which were chosen in item 8. In considered example (fig. 6.59), we have chosen only one additional dynamic factor (heat exchange in HPT) and only one control parameter (fuel consumption), that is why matrix A is 6×6 (three values of rotational speeds + n_{HP}, T_{IP} T_{LP}) and matrix B is 6×1.

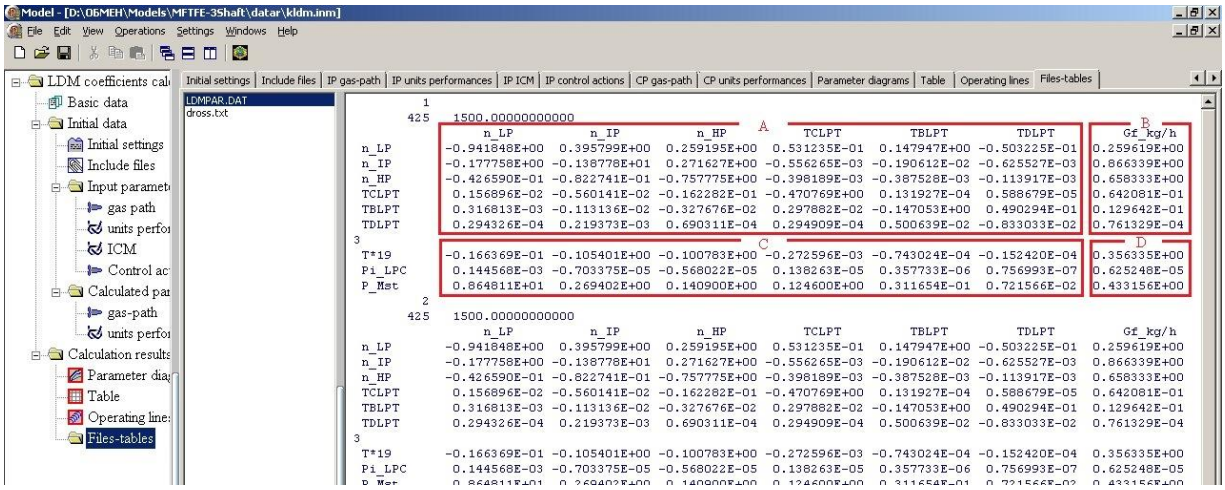


Figure 6.59 – Outputting results of QLDM coefficients calculation

There is the number, which specifies number of parameters in vector of observation (they were chosen in item 7) next after matrices A and B. Then follow coefficients of matrixes C and D. In our example size of matrix C is 3×6 and size of matrix D is 3×1.

If User have chosen more than one mode, then matrixes C and D are followed by number of next mode and the same information as for the first mode (sey fig. 6.59).

7 UNINSTALLING SOFTWARE

To uninstall the software go to ***"Start" ⇒ "Program files" ⇒ "Models" ⇒"Uninstall Models".***

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APPENDIXES

APPENDIX A
Format of files with performances of units

A.1 Intake performance

№	Parameter designation
1.	Number of file and comment
2.	Number of points in σ_{in} performance. Values, which are needed to parametrize the performance: – reference value of the first argument y_1; – variation range of the first argument Δy_1; – reference value of the second argument y_2; – variation range of the second argument Δy_2;
3.	Number of an argument according to generic list of arguments to calculate σ_{in} performance
4.	Value of an argument in σ_{in} performance
5.	Value of σ_{in} (or k factor)

A.2 Compressor spool performance (LPC2 – the secondary flow part of the fan, LPC1 – the primary flow part of the fan with booster stages and HPC)

№		Parameter designation
1.		Number of file and comment
2.		Adjustment factors Kn, KG, Kπ, Kη
3.		– control index (it may take values -5, 5, -10 or 10 depending on the way constant speed lines of compressor throttle performance and Z parameter were specified); – value of α_{IGV}; – reference value of the first argument y1; – variation range of the first argument Δy1; – reference value of the second argument y2; – variation range of the second argument Δy_2;
4.		Number of constant speed lines, maximum number of points at one constant speed line
5.		current constant speed line n_{cor}, number of points at current constant speed line
1-st constant speed line	6.	$G_{a\ cor\ C}$
	7.	π_C
	8.	η_C
	9.	$G_{a\ cor\ C}$
	10.	π_C
	11.	η_C
Lines 5 – 11 must be repeated for the rest constant speed lines of the performance		

A.3 Transition duct performance

№	Parameter designation
1.	Number of file and comment
2.	Number of points in σ_{trans} performance
3.	Number of argument in σ_{trans} performance
4.	Value of argument
5.	Value of σ_{trans}

A.4 Performance of combustion chamber

№	Parameter designation
1.	Number of file and comment
2.	Number of points in σ_{CC} performance Values, which are needed to parametrize the performance: – reference value of the first argument y_1; – variation range of the first argument Δy_1; – reference value of the second argument y_2; – variation range of the second argument Δy_2;
3.	Number of argument in σ_{CC} performance
4.	Array of argument values
5.	Array of σ_{CC}
6.	Number of points in η_{CC} performance
7.	Number of argument in η_{CC} performance
8.	Array of argument values in η_{CC} performance
9.	Array of η_{CC}
10.	Parameters of fuel: – enthalpy of fuel at CC inlet i_o, – lower calorific value of the fuel H_u; – stoichiometric ratio L_o

A.5 Turbine performance

№		Parameter designation
1.		Number of file and comment
2.		Adjustment factors K_n, K_A, $K\pi$, $K\eta$
3.		Any value
4.		– control index (it may take values -5, 5, -10 or 10 depending on the way constant speed lines of turine throttle performance and Z parameter were specified); Values, which are needed to parametrize the performance: – reference value of the first argument y1; – variation range of the first argument $\Delta y1$; – reference value of the second argument y2; – variation range of the second argument Δy_2;
5.		Number of constant speed lines, maximum number of points that form constant speed line
6.		Value of n_{cor} at current constant speed line, number of points at current constant speed line
1-st constant speed line	7.	A_T
	8.	π_T
	9.	η_T
	10.	A_T
	11.	π_T
	12.	η_T
Lines 6 – 15 must be repeated for the rest constant speed lines of the performance		

A.6 Nozzle performances

№	Parameter designation	Data format
1.	Number of file and comment	
2.	Number of points in σ_N performance	
3.	Number of argument in σ_N performance	
4.	Array of argument values of σ_N performance	
5.	Array of σ_N	
6.	Number of points in μ_N performance	
7.	Number of argument in μ_N performance	
8.	Array of argument values of μ_N performance	
9.	Array of μ_N	
10.	Number of points in ϕ_N performance	
11.	Number of argument in ϕ_N performance	
12.	Array of argument values of ϕ_N performance	
13.	Array of ϕ_N	

A.7 Geometrical performances of the engine

№	Parameter designation
1.	number of file and comment
2.	Primary nozzle area Secondary nozzle area Any value
3.	Any value
4.	The area of core duct at the station where VBV is mounted
5.	The area of bypass at the station where VBV is mounted
6.	The area at the inlet of the transition duct between boosters and HPC
7.	CC inlet area
8.	That area of the gas-path at the station where thermometer is mounted (after boosters)
9.	The area of core duct at the station where thermocouple is mounted (after the first stage of HPT)
10.	

A.8 Performances of air bleeds and taps, power takeoffs and supplies.

№	Parameter designation
1.	number of file and comment
2.	comment
3.	Parameters of air bleeding form LPC: – relative air bleed; – additional absolute bleed, kg/s; – factors K_π, K_η; – relative tap to NV of HPT; – relative tap to IMP of HPT; – relative tap to HPT discharge; – relative tap to NV of LPT; – relative tap to IMP of LPT; – relative tap to LPT discharge;
4.	Parameters of air bleeding form boosters discharge: – relative air bleed; – additional absolute bleed, kg/s; – factors K_π, K_η; – relative tap to NV of HPT; – relative tap to IMP of HPT; – relative tap to HPT discharge; – relative tap to NV of LPT; – relative tap to IMP of LPT; – relative tap to LPT discharge;
5.	Parameters of air bleeding form the first stage of HPC:

№	Parameter designation
	– relative air bleed; – additional absolute bleed, kg/s; – factors $К_{\pi}$, $К_{\eta}$; – relative tap to NV of HPT; – relative tap to IMP of HPT; – relative tap to HPT discharge; – relative tap to NV of LPT; – relative tap to IMP of LPT; – relative tap to LPT discharge;
6.	Parameters of air bleeding form the second stage of HPC: – relative air bleed; – additional absolute bleed, kg/s; – factors $К_{\pi}$, $К_{\eta}$; – relative tap to NV of HPT; – relative tap to IMP of HPT; – relative tap to HPT discharge; – relative tap to NV of LPT; – relative tap to IMP of LPT; – relative tap to LPT discharge;
7.	Parameters of air bleeding form HPC discharge: – relative air bleed; – additional absolute bleed, kg/s; – factors $К_{\pi}$, $К_{\eta}$; – relative tap to NV of HPT; – relative tap to IMP of HPT; – relative tap to HPT discharge; – relative tap to NV of LPT; – relative tap to IMP of LPT; – relative tap to LPT discharge;
8.	Absolute power take-off from HP rotor, W. Any value. Absolute power take-off from LP rotor, W.
9.	Coefficients to determine the parameters of air for active clearance control purposes

A.9 Mechanical properties and compressor mechanization

№	Parameter designation
1.	number of file and comment
2.	Mechanical efficiency of LPR Any value Mechanical efficiency of HPR
3.	Inertia moment of LPR

№	Parameter designation
	Any value Inertia moment of HPR
4.	Number of points of the reference (design) VSV position performance
5.	Argument of the reference (design) VSV position performance
6.	Array of argument values of the the reference (design) VSV position performance
7.	Array of VSV values of the the reference (design) VSV position performance
8.	Number of points of the VSV position performance to run the control at steady modes
9.	Argument of the VSV position performance to run the control at steady modes
10.	Array of arguments of the VSV position performance to run the control at steady modes
11.	Array of VSV values of the VSV position performance to run the control at steady modes
12.	Number of points of the VSV position performance to run the control during acceleration
13.	Argument of the VSV position performance to run the control during acceleration
14.	Array of arguments of the VSV position performance to run the control during acceleration
15.	Array of VSV values of the VSV position performance to run the control during acceleration
16.	Number of points of the VSV position performance to run the control during deceleration
17.	Argument of the VSV position performance to run the control during deceleration
18.	Array of arguments of the VSV position performance to run the control during deceleration
19.	Array of VSV values of the VSV position performance to run the control during deceleration
20.	Indicator that determines the switching to the VSV control program for accelerations to model the engine with ACS. Indicator that determines the switching out of the VSV control program for accelerations to model the engine with ACS.
21.	Indicator that determines the switching to the VSV control program for decelerations to model the engine with ACS. Indicator that determines the switching out of the VSV control program for decelerations to model the engine with ACS.
22.	Indicator that determines the switching to the VSV control program for accelerations to model the engine without ACS. Indicator that determines the switching out of the VSV control pro-

№	Parameter designation
	gram for accelerations to model the engine without ACS.
23.	Indicator that determines the switching to the VSV control program for decelerations to model the engine without ACS. Indicator that determines the switching out of the VSV control program for decelerations to model the engine without ACS.
24.	Number of points of the performance f(N2) that determines the VSV control mode
25.	Number of argument of the performance f(N2) that determines the VSV control mode
26.	Array of arguments f(N2) that determines the VSV control mode
27.	Array of function values f(N2) that determines the VSV control mode
28.	Maximum repositioning speed of VSV when transferring from one performance to another
29.	Number of points of the VBV control performance at steady modes
30.	Number of argument of the VBV control performance at steady modes
31.	Array of arguments of the VBV control performance at steady modes
32.	Array of position values of the VBV control performance at steady modes
33.	Number of points of the VBV control performance during acceleration
34.	Number of argument of the VBV control performance during acceleration
35.	Array of arguments of the VBV control performance during acceleration
36.	Array of position values of the VBV control performance during acceleration
37.	Number of points of the VBV control performance during deceleration
38.	Number of argument of the VBV control performance during deceleration
39.	Array of arguments of the VBV control performance during deceleration
40.	Array of position values of the VBV control performance during deceleration
41.	Indicator that determines the switching to the VBV control program for accelerations to model the engine with ACS. Indicator that determines the switching out of the VBV control program for accelerations to model the engine with ACS.
42.	Indicator that determines the switching to the VBV control program for decelerations to model the engine with ACS. Indicator that determines the switching out of the VBV control program for decelerations to model the engine with ACS.
43.	Indicator that determines the switching to the VBV control program for accelerations to model the engine without ACS. Indicator that determines the switching out of the VBV control pro-

№	Parameter designation
	gram for accelerations to model the engine without ACS.
44.	Indicator that determines the switching to the VBV control program for decelerations to model the engine without ACS. Indicator that determines the switching out of the VBV control program for decelerations to model the engine without ACS.
45.	Number of points of the performance f(N2) that determines the VBV control mode
46.	Number of argument of the performance f(N2) that determines the VBV control mode
47.	Array of arguments f(N2) that determines the VBV control mode
48.	Array of function values f(N2) that determines the VBV control mode
49.	Maximum repositioning speed of VBV when transferring from one performance to another
50.	Number of points of control performance that determines air bleeding during accelerations
51.	Number of argument of control performance that determines air bleeding during accelerations
52.	Array of arguments of control performance that determines air bleeding during accelerations
53.	Array of indicators of control performance that determines air bleeding during accelerations
54.	Number of points of control performance that determines air bleeding during decelerations
55.	Number of argument of control performance that determines air bleeding during decelerations
56.	Array of arguments of control performance that determines air bleeding during decelerations
57.	Array of indicators of control performance that determines air bleeding during decelerations
58.	Number of points of control performance that determines air bleeding at steady modes
59.	Number of argument of control performance that determines air bleeding at steady modes
	Array of arguments of control performance that determines air bleeding at steady modes
60.	Array of indicators of control performance that determines air bleeding at steady modes
61.	Number of points of TBV valve performance
62.	Number of argument of TBV valve performance
63.	Array of arguments of TBV valve performance
64.	Array of TBV positions
65.	Number of points of TBV control performance
66.	Number of argument of TBV control performance

№	Parameter designation
67.	Array of arguments of TBV control performance
68.	Array of TBV valve position indicators (open/closed)

A.10 Initial values of independent variables

№	Parameter designation
1.	number of file and comment
2.	$n_{LPC\ cor}$ – corrected roational speed of LPC
3.	$Z_{LPC\ 1}$ – factor that determines operating point position at constant speed line in performance map of core part of fan and boosters
4.	$Z_{LPC\ 2}$ – factor that determines operating point position at constant speed line in performance map of bypass part of fan
5.	Z_{HPC} – factor that determines operating point position at constant speed line in HPC performance map
6.	$n_{HPC\ cor}$ – corrected roational speed of HPC
7.	T^*_{19} – temperature at CC discharge
8.	π_{HPT} – pressure ratio of HPT
9.	π_{LPT} – pressure ratio of LPT
10.	G_{19} – gas flow at CC discharge
11.	G_f – fuel flow (fuel consumption)
12.	T^*_{37} – temperature at primary nozzle discharge
13.	G_{37} - gas flow at primary nozzle discharge
14.	T^*_6 - temperature at secondary nozzle discharge
15.	G_6 – gas flow at secondary nozzle discharge
16.	$T_{st\ HPT}$ - temperature of HPT casing
17.	$T_{bl\ HPT}$ - temperature of HPT blade
18.	$T_{d\ HPT}$ - temperature of HPT disk
19.	Any value
20.	Any value
21.	$T_{d\ LPT}$ - temperature of LPT disk
22.	$П_{LPT\ 1}$ – pressure ratio of the first stage of LPT

A.11 Variation limits of independent variables

№	Parameter designation
1.	number of file and comment
2.	Max and min value of $n_{КНДпр}$
3.	Max and min value of $Z_{КНД\ 1}$
4.	Max and min value of $Z_{КНД\ 2}$
5.	Max and min value of $Z_{КВД}$
6.	Max and min value of $n_{КВД\ пр}$
7.	Max and min value of T^*_{19}
8.	Max and min value of $П_{ТВД}$

№	Parameter designation
9.	Max and min value of $П_{ТНД}$ –
10.	Max and min value of G_{19}
11.	Max and min value of G_T
12.	Max and min value of T^*_{37}
13.	Max and min value of G_{37}
14.	Max and min value of T^*_6
15.	Max and min value of G_6
16.	Max and min value of $T_{ст\ ТВД}$
17.	Max and min value of $T_{л\ ТВД}$
18.	Max and min value of $T_{д\ ТВД}$
19.	Any value
20.	Any value
21.	Max and min value of $T_{д\ ТНД}$
22.	Max and min value of $П_{ТНД\ 1}$

A.12 Parameters of heat exchange

№	Parameter designation
1.	number of file and comment
2.	Masses of SE: – $m_{ST\ LPT}$, – $m_{BL\ LPT}$, – $m_{D\ LPT}$, – Any value, – Any value, – Any value, – $m_{ST\ HPT}$, – $m_{BL\ HPT}$, – $m_{D\ HPT}$
3.	Heat-transfer coefficients at SE surfaces, which are streamed by hot gas and heat-transfer coefficients of heat resistance: – $\alpha_{ST\ LPT}$, – $\alpha_{BL\ LPT}$, – $k_{LOCK\ LPT}$, – Any value, – Any value, – Any value, – $\alpha_{ST\ HPT}$, – $\alpha_{BL\ HPT}$, – $k_{LOCK\ HPT}$

№	Parameter designation
4.	Areas of SE surfaces including lock area, streamed by hot gases: – $F_{ST\ LPT}$, – $F_{BL\ LPT}$, – $F_{LOCK\ LPT}$, – Any value – Any value – Any value – $F_{ST\ HPT}$, – $F_{BL\ HPT}$, – $F_{LOCK\ HPT}$
5.	Areas of SE surfaces, streamed by cooling air: – $\alpha_{ST\ LPT\ cool}$, – $\alpha_{BL\ LPT\ cool}$, – $\alpha_{D\ LPT\ cool}$, – Any value, – Any value, – Any value, – $\alpha_{ST\ HPT\ cool}$, – $\alpha_{BL\ HPT\ cool}$, – $\alpha_{D\ HPT\ cool}$.
6.	Areas of SE surfaces, streamed by cooling air: – $F_{ST\ LPT\ cool}$, – $F_{BL\ LPT\ cool}$, – $F_{D\ LPT\ cool}$, – Any value, – Any value, – Any value, – $F_{ST\ HPT\ cool}$, – $F_{BL\ HPT\ cool}$, – $F_{D\ HPT\ cool}$.
7.	Heat capacities of structural elements – $c_{ST\ LPT}$, – $c_{BL\ LPT}$, – $c_{D\ LPT}$, – Any value, – Any value, – Any value,

№	Parameter designation
	– $c_{ST\ HPT}$, – $c_{BL\ HPT}$, – $c_{D\ HPT}$.
8.	Numbers of cooling temperatures according to general list: – $N_{ST\ LPT\ cool}$, – $N_{BL\ LPT\ cool}$, – $N_{D\ LPT\ cool}$, – Any value, – Any value, – Any value, – $N_{ST\ HPT\ cool}$, – $N_{BL\ HPT\ cool}$, – $N_{D\ HPT\ cool}$.

A.13 Transient performances of HPT casing (stator)

№	Parameter designation
1.	number of file and comment
2.	Basic values of parameters: - gas temperature at CC discharge $T^*_{G\ ref}$, - pressure at HPC discharge $p^*_{HPC\ ref}$, - rotational speed $n_{HP\ ref}$, - temperature of cooling air $T^*_{cool\ ref}$, - pressure of cooling air $p^*_{cool\ ref}$, - airflow of cooling air $G_{cool\ ref}$
3.	Dimentions of table with radial displacements: – number of heating temperature values $N(T^*_G)$, – number of values $N(K_{\alpha\ cool})$, – number of cooling temperature values $N(T^*_{cool})$, – number of values $N(K_{\alpha\ gas})$
4.	Array of heating temperatures
5.	Array of $K_{\alpha\ cool}$
6.	Value of cooling temperature for the first data set
7.	Value of $K_{\alpha\ cool}$ for the first data table
8.	Absolute radial displacements of the first table of the first data block. They correspond to first $K_{\alpha\ cool}$ and corresponding values of heating temperature
9.	Absolute radial displacements of the first table of the first data block.

№	Parameter designation
	They correspond to second $K_{\alpha\ cool}$ and corresponding values of heating temperature
	…
10.	Absolute radial displacements of the first table of the first data block. They correspond to last $K_{\alpha\ cool}$ and corresponding values of heating temperature
11.	Rows 7 – 10 are repeated for the second table of the first data block
12.	Rows 7 – 10 are repeated for the last table of the first data block
	Rows 6 – 12 are repeated for the second and the following data blocks. Number of blocks corresponds to number of cooling temperatures, which was set in the third line
	Dimentions of table with coefficients that describe the dynamics of radial clearance during heating: - number of $N_{exp\ heat}$ exponents, which approximate the transient performance; - number of argument values (coefficietnts $K_{\alpha\ heat}$)
	The first power gain
	…
	The last power gain (number $N_{exp\ heat}$)
	The first response time
	…
	The last response time (number $N_{exp\ heat}$)
	Dimentions of table with coefficients that describe the dynamics of radial clearance during cooling: - number of $N_{exp\ heat}$ exponents, which approximate the transient performance; - number of argument values (coefficietnts $K_{\alpha\ heat}$)
	The first power gain
	…
	The last power gain (number $N_{exp\ heat}$)
	The first response time
	…
	The last response time (number $N_{exp\ heat}$)

A.14 Transient performances of HPT blade

№	Parameter designation
1.	Number of file and comment
2.	– reference (base) value of gas temperature at CC discharge $T^*_{G\ ref}$, – influence coefficient, – blade length, mm
3.	Number of points of static performance (blade elongation perfor-

№	Parameter designation
	mance)
4.	Number of arguments (relations $T_G^* / T_{G\ ref}^*$) of static performance
5.	Array of static blade elongations
6.	Dimentions of table with coefficients that describe the dynamics of radial clearance during heating: number of $N_{exp\ heat}$ exponents, which approximate the transient performance; - number of argument values (coefficietnts $K_{\alpha\ heat}$)
7.	The first power gain
8.	…
9.	The last power gain (number $N_{exp\ heat}$)
10.	The first response time
11.	…
12.	The last response time (number $N_{exp\ heat}$)

A.15 Transient performances of HPT disk

№	Parameter designation
1.	Number of file and comment
2.	Basic values of parameters: - gas temperature at CC discharge $T_{G\ ref}^*$, – temperature of cooling air $T_{cool\ ref}^*$, - pressure at HPC discharge $p_{HPC\ ref}^*$, - rotational speed $n_{HP\ ref}$,
3.	Dimentions of table with radial displacements: – number of values $N(K_t)$, – number of values $N(K_{\alpha\ cool})$ – number of cooling temperature values $N(T_{cool}^*)$, –
4.	Array of K_t
5.	Array of $K_{\alpha\ cool}$
6.	T_{cool}^* for the first data table
7.	Absolute radial displacements of the first data table. They correspond to first $K_{\alpha\ cool}$ and corresponding values of K_t
8.	Absolute radial displacements of the first data table. They correspond to second $K_{\alpha\ cool}$ and corresponding values of K_t
9.	…
10.	Absolute radial displacements of the first data table. They correspond to las $K_{\alpha\ cool}$ and corresponding values of K_t
11.	Rows 6 – 10 are repeated for the second and the following data

№	Parameter designation
	blocks. Number of blocks corresponds to number of cooling temperatures, which was set in the third line
12.	Dimentions of table with coefficients that describe the dynamics of radial clearance during heating: number of $N_{exp\ heat}$ exponents, which approximate the transient performance; - number of argument values (coefficietnts $K_{\alpha\ heat}$)
13.	The first power gain
14.	…
15.	The last power gain (number $N_{exp\ heat}$)
16.	The first response time
17.	…
18.	The last response time (number $N_{exp\ heat}$)

A.16 Transient performances of LPT casing radial displacements

№	Parameter designation
1.	Number of file and comment
2.	Reference (base) value of gas temperature at CC discharge $T^*_{G\ ref}$. The performace was calculated for this temperature.
3.	Parameters of the 1st point: – time; – radial displacement of the first checkpoint; – radial displacement of the second checkpoint;
4.	…
5.	Parameters of the Nth point

A.17 Transient performances of LPT blade radial displacements

№	Parameter designation
1.	Number of file and comment
2.	– reference (means that the performance was calculated for this temperature) value of gas temperature at CC discharge $T^*_{G\ ref}$, – influence coefficient, – blade length, mm
3.	Parameters of the 1st point: – time; – radial displacement of the first checkpoint; radial displacement of the second checkpoint;
4.	…
5.	Parameters of the Nth point

A.18 Transient performances of LPT disk radial displacements

№	Parameter designation
1.	Number of file and comment
2.	Reference (base) value of gas temperature at CC discharge $T^*_{G\,ref}$. The performace was calculated for this temperature.
3.	Parameters of the 1st point: – time; – radial displacement of the first checkpoint; – radial displacement of the second checkpoint;
4.	…
5.	Parameters of the Nth point

A.19 Characteristic volumes

№	Parameter designation
1.	Number of file and comment
2.	Volumes of: – combustion chamber; – any value; – transition duct of primary nozzle; – transition duct of secondary nozzle;

A.20 Characteristic volumes

№	Parameter designation
1.	Number of file and comment
2.	Number of performances (for different Mach numbers)
3.	Parameters of the first performance: - number of constant constant throttle valve position curves, - maximum number of points at single constant throttle valve position curve; - Mach number
4.	Parameters of the first constant throttle valve position curve: - constant for this curve valve position; - number of points of this curve
5.	Number of π_{VSV} for the curve
6.	Array of airflow through the valve for the curve
7.	Rows 4-6 are repeated for the rest constant throttle valve position curve of the performance
8.	Rows 3–6 are repeated for the second and rest performances. Number of the performances is set in the second row.

APPENDIX B
Description of subroutines used for engine simulation

ATMOS

APPLICABILITY

Calculating parameters of the Standard Atmosphere.

CALLING

CALL ATMOS(H,DT,T0ST,P0ST).

DESCRIPTION OF PARAMETERS

H – altitude (input parameter), *m*.

DT – deviation of from a standard temperature (input parameter), *K*;

T0ST – static temperature (output parameter), *K*.

P0ST – static pressure (output parameter), *Pa*.

METHOD:

The method is described in subsection 1.1.2.4 of this Manual.

ATMOSX

APPLICATION

Subroutine for calculating parameters at engine inlet.

CALLING

CALL ATMOSX(TYPE,PAR1, PAR1, PAR1)

DESCRIPTION OF PARAMETERS

TYPE – type of atmospheric conditions inputting (input parameter).

PAR1 – first parameters of atmospheric conditions group (input parameter).

PAR2 – second parameters of atmospheric conditions group (input parameter).

PAR3 – third parameters of atmospheric conditions group (input parameter).

NOTES

Parameters must strongly correspond to the way of atmospheric conditions inputting. Values of TYPE and corresponding to it groups of input parameters are:

0 – *H, m;* M_{fl}*;* ΔT_H*, K;*

1 – *H, m;* V_{fl}*, m/s;* ΔT_H*, K;*

2 – P_0*, Pa;* T_0*, K;* V_{fl}*, m/s;*

3 – P_0*, Pa;* T_0*, K;* M_{fl}*;*

COMMON – BLOCKS

GAZ, VHOD

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

ATMOS, CPFT, TFI.

METHOD:

A method is described in subsection 1.1.2.4 of this Manual (equations 1.3 - 1.5).

BCN

APPLICATION

A function that calculates correction coefficient for rotational speed (square root from ratio of temperatures).

CALLING

BCN (T)

DESCRIPTION OF PARAMETERS

T – temperature at station of rotational speed correction (input parameter), *K*.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

No.

METHOD:

$$BCN = \sqrt{\frac{T}{T_{ACS}}}\text{, where } T_{ACS} = 288{,}15\ K$$

BG

APPLICATION

Function to calculate the factor in equation of corrected airflow calculation.

CALLING

BG (T,P)

DESCRIPTION OF PARAMETERS

T – temperature at the station of corrected airflow (input parameter), *K*.

P – pressure at the station of corrected airflow (input parameter), *Pa*.

METHOD:

$$BG = \frac{p_{ACS}}{p} \cdot \sqrt{\frac{T}{T_{ACS}}}\text{, where } T_{ACS} = 288{,}15\ K,\ p_{ACS} = 101300\ Pa.$$

CPFT

APPLICATION

Subroutine to calculate physical properties of working substance according to given temperature and composition.

CALLING

CALL CPFT(Q,D,T,I,S,CP)

DESCRIPTION OF PARAMETERS

Q – relative combustion products content of fossil fuel (input parameter);

D – moisture content (input parameter);

T – temperature (input parameter) *K*;

I – specific enthalpy (output parameter) *J/kg*;

S – specific entropy (output parameter) *J/kg*;

CP – specific heat capacity (output parameter) *[J/kg]*.

NOTES

Values of Q and D must be within the limits [0,00001, 1,00001], otherwise they are assigned “0”.

COMMON – BLOCKS

GAZ

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

SITTER.

METHOD:

A method is described in subsection 1.1.2.1 of this Manual.

CRIT

APPLICATION

Subroutine to calculate critical parameters of working substance at given temperature and composition.

CALLING

CALL CRIT(Q,D,TVHS,FI,TCR,PICR,VCR)

DESCRIPTION OF PARAMETERS

Q – relative fuel flow (input parameter);

D – moisture content (input parameter);

T – total temperature at station of parameters calculating (input parameter), *K*;

FI – velocity factor (input parameter);

TCR – total temperature of the flow at critical conditions (input parameter), *K*;

PICR – critical pressure ratio;

VCR – flow velocity at critical conditions, *m/s*.

NOTES

Value FI must be within the limits [0, 1], otherwise to it a value is assigned “1”.

COMMON – BLOCKS

GAZ,

EPSTER,

SIGSFI.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT ,SITTER, TFI

METHOD:

A method is described in subsection 1.1.2.5 of this Manual.

DHPCG

APPLICATION

Subroutine to solve a set of first order simple differential equations with given initial approximations. The set of equations has the following form:

$$\begin{cases} y_1' = \dfrac{\partial y_1}{\partial x} = f_1(x, y_1, y_2, \ldots, y_{n\ dim}); \\ y_2' = \dfrac{\partial y_2}{\partial x} = f_1(x, y_1, y_2, \ldots, y_{n\ dim}); \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ y_{n\ dim}' = \dfrac{\partial y_1}{\partial x} = f_1(x, y_1, y_2, \ldots, y_{n\ dim}) \end{cases}$$

CALLING

CALL DHPCG(PRMT,Y,DERY,NDIM,IHLF,FCT,OUTP,AUX)

Parameters FCT and OUTP must be declared as EXTERNAL

DESCRIPTION OF PARAMETERS

PRMT is a 5th or higher order vector of double-precision input and output data. This vector comprises the integration limits and acceptable calculation error; this vector also provides the interaction of output (formed according to User requirements) and DHPCG subroutines. The subroutine DHPCG doesn't change components of PRMT, except PRMT(5).

INPUT parameters of PRMT.

PRMT(1) – lower limit of integration limits.

PRMT(2) - upper limit of integration limits.

PRMT(3) – initial step of integration.

PRMT(4) – acceptable error of calculations.

If an absolute error exceeds PRMT(4), then an integration step equals to half of initial step. The integration step is doubled if it is less than PRMT(4)/50.The User can change PRMT(4) by output subroutine.

PRMT(5) – is not input parameter. Subroutine DHPCG initially assigns PRMT(5)=0.

If User wishes to stop the subroutine DHPCG working in any point within interval of integration, he should assign PRMT (5) non-zero value by the subroutine OUTP.

PRMT vector may have more components if its dimension exceeds 5. But DHPCG subroutine addresses only first five elements and doesn't require and change the rest. Nevertheless, they can be used to transfer output data from subroutine OUTP to main program (where from DHPCG was called).

Y is a vector of double precision input values (is not saved). Next Y becomes vector of dependent variables calculated in intermediate points X (X is internal variable, used as an argument for integration, e.g. time).

DERY is a vector of double precision input weight coefficients of errors (is not saved). The sum of its components must be equal to "1". Next DERY becomes vector of derivatives of functions in intermediate points X.

NDIM is an input parameter, which sets number of equations in the set.

IHLF is an output parameter that specifies number of halvings of initial integration step. If IHLF becomes more than 10, the subroutine DHPCG passes the control to the main program with error message IHLF=11. The error messages IHLF=12 or IHLF=13 accordingly arise in cases when PRMT (3)=0 or SIGN(PRMT (3)) ≠SIGN(PRMT(2)-PRMT(1)).

FCT – name of applied external subroutine. It evaluates right parts of equations in DERY by given X and Y. The list of arguments must be X, Y, DERY. The subroutine must save X and Y.

OUT is a name of external subroutine, used for outputting. It makes processing of output parameters. A list of its parameters is the next: X, Y, DERY, IHLF, NDIM, PRMT.

Any of these parameters should not be changed by the subroutine OUTP. If PRMT(5) assigns non-zero value, the subroutine DHPCG passes the control to the main program.

AUX is a double precision accessory array with 16 lines and NDIM columns.

NOTES

Calculation terminates and control is passed to the main program in cases when:

1) achieving needed tolerance requires more than 10 halvings of initial integration step (error message IHLF=11);
2) the initial integration step equals to zero or has incorrect sign (error message IHLF=12 or IHLF=13);
3) calculations within limits are completed;
4) OUTP has assigned non-zero value to PRMT (5).

COMMON – BLOCKS

No.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

External subroutines FCT(X,Y,DERY) and OUTP(X, Y, DERY, IHLF, NDIM, PRMT) should be formed by User.

METHOD:

The calculations are executed by formulas of modified Hamming prediction-correction method. This is the fourth-order method. It takes the solutions in four prior points to calculate of new vector Y of dependent variables. To set initial integration step and to calculate mentioned above in first three points the fourth-order Runge-Kutt method (offered by Ralston) is used

The subroutine DHPCG automatically regulates an integration step by halving or doubling it during calculation process. To use OUTP capabilities as full as possible, it must be tuned by User.

DINAM

APPLICATION

Main program to calculate transient modes of an engine at law of control parameter in time.

CALLING

PROGRAM DINAM

COMMON – BLOCKS

BALANS, CTRLD, DOPPAR, DPARUN, ENERG, EPSTER, ETAMEH, FILE, HARAKT, IDNTFC, ITMP, KAMERA, OBOROT, OTBORM, PARAM, PREDEL, PRINT, REGIM, SAU, STAT, SWITCH, TEPLO, TXT, UPRAVL, VHOD.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

ATMOSX, DHPCG, INOUTD, NACHBL, PDD2S (EXTERNAL), PRDIN (EXTERNAL), DINDD2 (EXTERNAL), POINTN, PRNTF, RSUMN.

DINDD2

APPLICATION

Subroutine that calculates right parts of differential equations that describe engine dynamics.

CALLING

CALL DINDD2(TIME,N,F)

DESCRIPTION OF PARAMETERS

TIME is a double precision value of parameter for which the integration is carried out;

N is a double precision input vector of dependent variables calculated in intermediate point TIME.

F is a double precision vector of derivatives of functions in intermediate point TIME.

COMMON – BLOCKS

BALANS, DINROT, DOPPAR, DPARUN, ENERG, ETAMEH, GAZ, INER, KAMERA, OTBORM, PARAM, REGIM, SPR, TERMO, UPRAVL, VHOD.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

PARREG, RSUMN.

DROSS

APPLICATION

Main program that calculates static parameters of an engine.

CALLING

PROGRAM DROSS

COMMON – BLOCKS

CONTRL, DOPPAR, EPSTER, FILE, FOREXEC, GLBUPR, HARAKT, IDNTFC, ITMP, OTBORM, PARAM, PRINT, REGIM, SAU, TXT, UPRAVL, VHOD.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

ATMOSX, IERWIN, INOUTD, NACHBL, PDD2S, POINTN, PRNTF, PRNTW, RSUMN.

METHOD

See subsection 1.2 of this Manual.

DSIMQ

APPLICATION

Subroutine for double-precision solving the set of non-linear algebraic equations (AX+B) by the method of exclusions.

CALLING

CALL DSIMQ(A,B,N,KS)

DESCRIPTION OF PARAMETERS

A is a double precision input matrix of system coefficients (NxN order); varies at course of calculation;

B is a vector of absolute term of equations set (length equals to N). It is substituted by vector X when transferring from the subroutine.

N – number of equations

KS is an error code (0 – solution succeeded;1 – for specific equations set).

NOTES

Matrix A must be general. If matrix is singular then solution makes no sense.

METHOD

Method of exclusion with choosing of principal element is used for solving. Each iteration has the procedure of rows shift, which prevents division by zero or by very small element. Each ith unknown value is calculated by N iterations during direct solution. The rest unknown elements are calculated by successive replacements during inverse solution.

Calculated results are put into vector B: first unknown is B(1), second – B(2) and so on. When principal element is enough accurate ($tol = 0.995 \cdot 10^{-16}$), then matrix transforms to singular and KS is set 1.

ETAG

APPLICATION

Function that calculates combustion efficiency of combustion chamber.

CALLING

ETAG

COMMON – BLOCKS

AIDENT, KSDAT, PRRG

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT, STATPAR

FIS {FIS1, FIS2}

APPLICATION

Function that calculates velocities ratio λ of nozzle.

CALLING

FIS – for common nozzle;

FIS1 – for primary nozzle;

FIS2 – for secondary nozzle;

COMMON – BLOCKS

AIDENT, SDAT {S1DAT, S2DAT}, SOPL

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT

GG1

APPLICATION

Subroutine that calculates parameters of a single-spool gas-generator.

CALLING

CALL DGG1(NVP)

DESCRIPTION OF PARAMETERS

NVP – number of varied parameter (is set when calculating the Jacobi matrix of equation set. The set describes the joint operation of engine components).

NOTES

NVP is needed for speed up the throughflow engine analysis.

If NVP>7, then calculation starts from HPT, skipping former components during straight through analysis of the engine.

If NVP>6, then calculation starts from combustion chamber, skipping former components during straight through analysis of the engine.

Else, calculation starts from HPC.

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, ENERG, HARAKT, KAMERA, OBOROT, OTBOLD, OTBORV, PARAM, PAROTB, POTERI, PRRG, TEPLO, TERMO, VHOD

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

BG, BCN, CPFT, DSUM, ETAG, KOMPR, SIGKS, TAU, TFI, TURBIN, XKVD, XTVD

GG2

APPLICATION

Subroutine that calculates parameters of a two-spool gas-generator.

CALLING

CALL DGG2(NVP)

DESCRIPTION OF PARAMETERS

NVP – number of varied parameter (is set when calculating the Jacobi matrix of equation set. The set describes the joint operation of engine components).

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, ENERG, HARAKT, KAMERA, OBOROT, OTBOLD, OTBORV, PARAM, PAROTB, POTERI, PRRG, TEPLO, TERMO, VHOD

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

BG, BCN, CPFT, DSUM, ETAG, KOMPR, SIGKS, TAU, TFI, TURBIN, XKVD, XTVD

GG3

APPLICATION

Subroutine that calculates parameters of three-spool gas-generator.

CALLING

CALL GG3(NVP)

DESCRIPTION OF PARAMETERS

NVP – number of varied parameter (is set when calculating the Jacobi matrix of equation set. The set describes the joint operation of engine components).

NOTES

NVP was used in previous versions of the program.

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, DVHKON, ENERG, HARAKT, KAMERA, OBOROT, OTBORM, OTBORV, PARAM, PAROTB, POTERI, PRRG, TERMO, VHOD, VVR, XRPDAT

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

BCN, BG, CPFT, DSUM, GG2, HARRP, KOMPR, TAU, TFI, TURBIN, SIGP4, SIGP5, SIGVH, XKND, XTND.

HARRP

APPLICATION

Subroutine that calculates parameters of flow splitter.

CALLING

CALL HARRP

COMMON – BLOCKS

DVHKON, XRPDAT

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT

IERWIN

APPLICATION

Subroutine to show windows with error messages when solving set of non-linear equations.

CALLING

CALL IERWIN(I,NEV,NEZP)

DESCRIPTION OF PARAMETERS

I – input value of error code.

NEV –input vector of double precision independent variables.

NEZP - input vector of double precision residuals.

COMMON – BLOCKS

GLBUPR, UPRAVL

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

ABDX, WHAT

INIST1

APPLICATION

Subroutine to initialize the performances of units.

CALLING

CALL INIST1

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, DVHKON, ETAMEH, FILE, FKSDAT, HARAKT, INER, KAMERA, KSDAT, NEZPD, OBOROT, OTBOLD, OTBORV, PARAM, PAROTB, PER2DAT, PER3DAT, PER5DAT, PEREOUSK, PREDEL, S1DAT, S2DAT, SAU, SAUPI, SDAT, SMESHN, SOPL, UPRAVL, VHODDAT, VVR, XKN1DT, XKNDAT, XKNDCDT, XKV1DT, XKVDAT, XKVDCDT, XRPDAT, XTNDAT, XTVDAT.

INOUTD

APPLICATION

Program to initialize the given study for further calculations and forming input file to carry out identification.

CALLING

INOUTD(IODAT,FDATA,MNPAR1,MNUZL,MNUID,MNUPAR,YIZM,W, EPSR,KREG,KPAR1,KUZL,KUID,KUPAR,ENV1,ENV2,ENV3,ENV4,ENV 5,ENV6,ENV7,ENV,ENV9, ENV10,ENV11,ENV12,TYPEIN,YR,YRF)

DESCRIPTION OF PARAMETERS

IODAT – parameter that determines reading (0) or generating input file for identification (1) (input parameter).

FDATA – name of input file (input parameter).

MNPAR1 – array of parameters numbers (size of array is [KPAR1]) to be printed (output parameter).

MNUZL – array of units numbers (size of array is [KUZL]), which needs correction (output parameter).

MNUID – array of units numbers (size of array is [KUID]) necessary for identification (output parameter).

MNUPAR – array of numbers of external control actions (size of array is [KUPAR]) (output parameter).

YIZM – array of measured parameters with a dimension (size of array is [KPAR1, KREG]) (output parameter).

W – array of weight coefficients of measured parameters (size of array is [KPAR1]) (output parameter).

EPSR – solution tolerance of equations set (output parameter).

KREG – a number of modes (input parameter).

KPAR1 – a number of calculated parameters (input parameter).

KUZL – amount of units which needs to be corrected (input parameter).

KUID – amount of units which needs identification (input parameter).

KUPAR – amount of external control actions (input parameter).

ENV1 – array of modes numbers or array of a time for a transient characteristic (size of array is [KPAR1]) (output parameter).

ENV2 – array that shows the value of first parameter from group of external conditions at different modes (size of array is [KPAR1]) (output parameter).

ENV3 – array that shows the value of second parameter from group of external conditions at different modes (size of array is [KPAR1]) (output parameter).

ENV4 – array that shows the value of third parameter from group of external conditions at different modes (size of array is [KPAR1]) (output parameter).

ENV5 – array that shows the value of first parameter of operating mode at different modes (size of array is [KPAR1]) (output parameter).
ENV6 – array that shows the value of second parameter of operating mode at different modes (size of array is [KPAR1]) (output parameter).
ENV7 – array that shows the value of third parameter of operating mode at different modes (size of array is [KPAR1]) (output parameter).
ENV8 – array that shows the value of fourth parameter of operating mode at different modes (size of array is [KPAR1]) (output parameter).
ENV9 - array that shows the value of first additional input control action at different modes (size of array is [KPAR1]) (output parameter).
ENV10 - array that shows the value of second additional input control action at different modes (size of array is [KPAR1]) (output parameter).
ENV10 - array that shows the value of third additional input control action at different modes (size of array is [KPAR1]) (output parameter).
ENV11 - array that shows the value of fourth additional input control action at different modes (size of array is [KPAR1]) (output parameter).
TYPEIN – the way ambient conditions are set (input parameter).
YR – array of values of parameters calculated by identification procedure (size of array is [KPAR1, KREG]) (input parameter).
YRF - array of real values of parameters calculated by identification procedure (size of array is [KPAR1, KREG]) (input parameter).

NOTES

IODAT=1 means that all parameters are input parameters.

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, DVHKON, ETAMEH, FILE, FKSDAT, HARAKT, INER, KAMERA, KSDAT, NEZPD, OBOROT, OTBOLD, OTBORV, PARAM, PAROTB, PER2DAT, PER3DAT, PER5DAT, PEREOUSK, PREDEL, S1DAT, S2DAT, SAU, SAUPI, SDAT, SMESHN, SOPL, UPRAVL, VHODDAT, VVR, XKN1DT, XKNDAT, XKNDCDT, XKV1DT, XKVDAT, XKVDCDT, XRPDAT, XTNDAT, XTVDAT.

INTCN

APPLICATION

Double precision linear interpolation of B, C, D functions for given arguments P and A according to given tables (AX, BX, CX), (AX, BX, DX).

CALLING

CALL INTCN(VX, V, P, B, C, D, PRM, NA, NOMAX, IL, LIML, IH, LIMH, BX, CX, DX, NCODE, ISEARC)

DESCRIPTION OF PARAMETERS

VX – double precision input array of the first argument.
V – double precision input value of the first argument.
P – double precision input value of the second argument.
B – double precision interpolated value of the first function.
C – double precision interpolated value of the second function.
D – double precision interpolated value of the third function.

PRM – double precision input coefficient to recalculate function.
NA – a number of AX correlations.
NOMAX – maximum number of points of correlation.
IL – number of a first argument, the closest one to V that meets the requirement VX[IL]<V (input parameter).
LIML – number of significant (non-zero) points in the IL column of BX, CX, DX arrays (input parameter).
IH – number of a first argument, the closest one to V that meets the requirement VX[IH]>V (input parameter)..
LIMH – number of significant (non-zero) points in the IH column of BX, CX, DX arrays (input parameter).
BX – double precision input array of the first function values (array size is NA×NOMAX).
CX – double precision input array of the second function values (array size is NA×NOMAX).
DX – double precision input array of the third function values (array size is NA×NOMAX).
NCODE –value of error parameter.
NCODE=0, if VX(min)<V<VX(max);
NCODE=10, if V<VX(min);
NCODE=20, if V>VX(NA);
ISEARC – logical sign: FALSE – if function VX grows, TRUE - if function VX decays.

NOTES

Values of arguments of arrays VX, BX, CX, DX.

METHOD

See subsection 6.1.2.2 of this manual.

INTERF

APPLICATION

Subroutine to initialize the dialogue window for outputting information previously specified as input data.

CALLING

CALL INTERF

COMMON – BLOCKS

BALANS, CTRLD, EPSTER, GLBUPR, REGIM, TXT, UPRAVL.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

ABDX, WHAT

KAMSM

APPLICATION

Subroutine for calculating parameters of working substance at mixer discharge.

CALLING

CALL KAMSM(Q1 ,D1 ,T1 ,P1 ,G1 ,V1 ,F1 ,
Q2 ,D2 ,T2 ,P2 ,G2 ,V2 ,F2 ,
QSM,DSM,TSM,PSM,GSM,VSM,FSM,PSMST)

DESCRIPTION OF PARAMETERS

Q1, Q2 – double precision input values of relative fuel flow at core and bypass ducts discharges.

D1, D2 – double precision input values of relative moisture content at core and bypass ducts discharges.

T1, T2 - double precision input values of total temperatures at core and bypass ducts discharges.

P1, P2 - double precision input values of total pressures at core and bypass ducts discharges.

G1, G2 - double precision input values of mass flows at core and bypass ducts discharges.

V1, V2 - double precision output values of airflow velocities at core and bypass ducts discharges.

F1, F2 - double precision input values of cross-sectional areas at core and bypass ducts discharges.

QSM– double precision output value of relative fuel flow at mixer discharge

DSM – double precision output value of moisture content at mixer discharge.

TSM – double precision output value of total temperature at mixer discharge.

PSM – double precision output value of total pressure at mixer discharge.

GSM - double precision output value of mass flow of working substance at mixer discharge.

VSM - double precision output value of gas flow velocity at mixer discharge.

FSM - double precision output value of cross-sectional area at mixer discharge.

PSMST - double precision output value of static pressure at mixer discharge.

COMMON – BLOCKS

EPSTER, GAZ

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, SITHAR, TFI

METHOD

See item 1.1.2.7 of this Manual.

KOMPR

APPLICATION

Subroutine that calculates parameters of airflow at the compressor discharge.

CALLING

CALL KOMPR(Q,D,TVHK,TVIHK,PIK,LK,KPDK,NUM)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel at compressor inlet.

D – double precision input value of relative moisture content at compressor inlet.

TVHK – double precision input value of total temperature at compressor inlet.

TVIHK - double precision output value of total temperature at compressor discharge.

PIK - double precision input value of compressor pressure ratio.

LK - double precision output value of work done by compressor.

KPDK - double precision input value of compressor efficiency.

NUM – number of compressor spool used for displaying messages about compressor spools:

21 – LPC, 22 – IPC, 23 - HPC.

NOTES

No.

COMMON – BLOCKS

GAZ.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, TFI, TFS, SITTER

METHOD

See item 1.1.2.7 of this Manual.

LAM

APPLICATION

Function that calculates relative velocity factor of working substance.

CALLING

LAM(Q,D,T,P,G,F)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel flow at compressor inlet.

D - double precision input value of relative moisture content at compressor inlet.

T - double precision input value of total temperature at design station.

P - double precision input value of total pressure at given station.

G - double precision input value of mass flow of working substance at given station.

F - double precision input value of a cross-sectional area of given station.

COMMON – BLOCKS

EPSTER, GAZ

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, CRIT, TFI

METHOD

See subsection 1.1.2.6 of this Manual.

LINTX

APPLICATION

Subroutine for interpolation of function of single variable.

CALLING

CALL LINTX(X,Y,AX,AY,N,IER)

DESCRIPTION OF PARAMETERS

X – double precision value of input variable.

Y – double precision calculated value of function.

AX – double precision input array of variable values.

AY – double precision input array of function values.

N – size of AX and AY arrays.

IER – result value of error:

IER=0 – if AX(1)<X<AX(N)

IER=1 – if X>AX(N)

IER=-1 – if X<AX(1)

NOTES

A table (AX, AY) is a function of single variable, which is filled in order of input argument increasing.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

SEAR.

METHOD

Interpolation is performed by the equation of linear interpolation:

$$f(x) = f_k + (f_{k+1} - f_k)\frac{x - x_k}{x_{k+1} - x_k},$$

where

$$f_k = f(x_k), x \in [x_k, x_{k+1}], \quad k = 1,2,..n-1$$

LINXYT

APPLICATION

Subroutine to interpolate turbine performances.

CALLING

CALL LINXYT(X,Y,Z,AX,N,AY,M,MM,AZ,IERX,IERY)

DESCRIPTION OF PARAMETERS

X – double precision input value of the first variable.

Y – double precision input value of the second variable.

Z – double precision value of function.

AX – double precision input array of the first variable values (size of array is N).

N – size of AX array.

AY – double precision input array of the second variable values.

M – maximum number of points in a table (AY,AZ).

MM – array that indicates number of function variables tha correspond to each value of argument.

AZ - double precision input array of a function values.

IERX – result value of error caused by the first argument.

IERX=0 – if AX(1)<X<AX(N)

IERX=1 – if X>AX(N)

IER=-1 – if X<AX(1)

IERY – result value of error caused by the first argument.

NOTES

A table (AX, AY) is a function of single variable, which is filled in order of input arguments increasing.

COMMON – BLOCKS

No.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

SEAR

METHOD

Standard methof of linear interpolation for function of two variables.

MUS (MUS1, MUS2)

APPLICATION

Function that calculates flow factor of common nozzle.

CALLING

MUS – for common nozzle

MUS1 – for primary nozzle

MUS2 – for secondary nozzle

COMMON – BLOCKS

AIDENT, SDAT (S1DAT, S2DAT), SOPL

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT

NACHBL

APPLICATION

Common block to start different studies.

CALLING

CALL NACHBL (KNEV,NEZPER,SG,MAXIT,IER)

DESCRIPTION OF PARAMETERS

KNEV – number elements in array NEZPER.

NEZPER – double precision output vector of input values of independent variables.

GS – double precision output value of independent variables variation

MAXIT – output value of maximum number of iterations when calculating by a Newton method.

IER – error code

COMMON – BLOCKS

DOPPAR, DVHKON, HARAKT, KAMERA, OBOROT, NEZPD, PARAM, PREDEL, SMESHN, UPRAVL.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

INIDD2, TEXTIN.

PARREG

APPLICATION

Subroutine to calculate external and mode parameters at current value of parameter used as argument in integration process.

CALLING

CALL PARREG (TIME)

DESCRIPTION OF PARAMETERS

TIME – double precision input parameter, used as argument in integration process.

COMMON – BLOCKS

CTRLD, DOPPAR, EPSTER, OTBORM, PARAM, REGIM, SAU, SPR, TXT, UPRAVL.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

ATMOSX, LINTX

PARUTND

{**PARUTSD, PARUTSD, PARU_BL_TND, PARU_BL_TSD, PARU_BL_TVD, PARU_ST_TND, PARU_ST_TSD, PARU_ST_TVD}**

APPLICATION

Subroutines to calculate displacements of check points.

CALLING

CALL PARUTND (time,T,U1,U2) – to calculate displacements of LPT disk

CALL PARUTSD (time,T,U1,U2) – to calculate displacements of IPT disk

CALL PARUTVD (time,T,U1,U2) – to calculate displacements of HPT disk

CALL PARU_BL_TND (time,T,U1,U2) – to calculate displacements of LPT impeller

CALL PARU_BL_TSD (time,T,U1,U2) – to calculate displacements of IPT impeller

CALL PARU_BL_TVD (time,T,U1,U2) – to calculate displacements of HPT impeller

CALL PARU_ST_TND (time,T,U1,U2) – to calculate displacements of LPT stator

CALL PARU_ST_TSD (time,T,U1,U2) – to calculate displacements of IPT stator

CALL PARU_ST_TVD (time,T,U1,U2) – to calculate displacements of HPT stator

DESCRIPTION OF PARAMETERS

TIME – double precision input parameter, used as argument in integration process.

T – double precision value of temperature at current moment TIME.

U1 – double precision value of 1st check point displacement of structural element

U2- double precision value of 2nd check point displacement of structural element

COMMON – BLOCKS

PARPERHAR.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX

METHOD

Displacements are calculated by the equation

Ошибка! Источник ссылки не найден.

PDD2S

APPLICATION

Subroutine to provide joint proceeding of universal subroutine that solves the set of non-linear algebraic equations by Newton method and subroutine that calculates residuals.

CALLING

CALL PDD2S(NEZPER,NEV,KNEV,NU,IER)

DESCRIPTION OF PARAMETERS

NEZPER – double precision input array of independent variables (size of array is KNEV).

NEV – double precision output array of residuals (size of array is KNEV).

KNEV – size of arrays.

NU – input value of a number of independent variable, which is varied when Jacobi matrix calculating (in other calculations NU=0)

IER – exit code of calculation.

COMMON – BLOCKS

DINROT, DOPPAR, DPARUN, DVHKON, ENERG, ETAMEH, HARAKT, KAMERA, NEZPD, OBOROT, OTBORM, PARAM, POTERI, PREDEL, REGIM, SMESHN, SOPL, SPR, TERMO, UPRAVL, VHOD.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

RDD2S

POINT

APPLICATION

Function that forms value of thermodynamic parameter by its number according to general array.

CALLING

CALL POINT(IND)

DESCRIPTION OF PARAMETERS

IND – number of parameter in a general array.

NOTES

IND must be within the limits [1, 650].

METHOD

Knowing the number of parameter from general array, subroutine assigns its value to variable POINT.

POINTN

APPLICATION

Subroutine that assigns given value to thermodynamic parameter by its number in a general array.

CALLING

CALL POINTN(IND,ZNAC)

DESCRIPTION OF PARAMETERS

IND – number of parameter in a general array.

ZNAC – double precision value which must be assigned to parameter.

NOTES

No.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

No.

METHOD

Knowing the number of parameter from general array, subroutine selects parameter and assigns it the value of parameter ZNAC.

PRDIN

APPLICATION

Subroutine that outputs results of solution of a set of first order ordinary differential equations according to given initial approximations.

CALLING

CALL PRDIN (TIME,N,F,IEH,KVAL,PRMT)

DESCRIPTION OF PARAMETERS

TIME – double precision input parameter, used as argument in integration process.

N – double precision input vector of effect variables at given value of the parameter, used as argument in integration process.

F – double precision input vector of functions derivatives at given value of the parameter, used as argument in integration process.

IEH – input parameter that specifies number of equations.

PRMT – double precision input vector (see description of the subroutine DHPCG).

NOTES

No input parameter can be changed, except PRMT.

COMMON – BLOCKS

DOPPAR, HARAKT, KAMERA, OBOROT, PARAM, PRINT, SAU, STAT, SWITCH, TEPLO.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

POINT, SAUNUM

PRNTF

APPLICATION

Subroutine for printing results of thermodynamic calculations and identification Each of parameters is printed with its set name and measuring units.

CALLING

CALL PRNTF (MNPAR,NREG,KPAR,IREG,ITEST,KREG,YIZM)

DESCRIPTION OF PARAMETERS

MNPAR – input array containing numbers of parameters to be printed (size of array is KPAR).
NREG – number of mode parameter.
KPAR – number of elements in array MNPAR.
IREG – number of a mode.
ITEST – forming and outputting test values to make identification.
0 – don't print, 1 - print.
KREG – number of modes.
YIZM – double precision array of calculated parameters for further using as input parameters of testing procedure during identification (size of array is [KPAR,KREG]).

COMMON – BLOCKS

TXT

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

POINT.

PRNTW

APPLICATION

Subroutine that forms dialogue window for outputting intermediate results of calculation (see fig. 6.48).

CALLING

CALL PRNTW(MNPAR,NEZPER,NEV,KNEV,KPAR,IRR)

DESCRIPTION OF PARAMETERS

MNPAR – input array containing numbers of parameters to be printed (size of array is KPAR).
NEZPER – input array of independent variables (size of array is KNEV).
NEV - input array of residuals (size of array is KNEV).
KNEV – size of arrays NEZPER and NEV.
KPAR – number of elements in array MNPAR.
IRR – input error code.

COMMON – BLOCKS

ABDX, GLBUPR, REGIM

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

POINT, WHAT.

PRPKND(PRPKSD, PRPKVD)

APPLICATION

Subroutine, used to calculate the mass flow of air, bleed form each spool of compressor.

CALLING

CALL PRPKND (GP,GVPR) – to calculate the bleed from low pressure compressor.

CALL PRPKSD (GP,GVPR) – to calculate the bleed from intermediate pressure compressor.

CALL PRPKVD (GP,GVPR) – to calculate the bleed from high pressure compressor.

DESCRIPTION OF PARAMETERS

GP – airflow bleed through ABV.

GVPR – corrected airflow at spool inlet.

COMMON – BLOCKS

PARAM, PEREPUSK.FOR

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

BG.

METHOD

The method is described in subsection 6.1.2.9.

RDD2S

APPLICATION

Subroutine that carries out direct thermodynamic calculation of two-spool turbofan engine with mixing of primary and secondary flows.

CALLING

CALL RDD2S(NVP)

DESCRIPTION OF PARAMETERS

NVP – number of varied parameter (is set during numerical calculating of Jacobi matrix of the set of equations, which expresses the conditions of units` joint operation).

COMMON –BLOCKS

DOPPAR, DPARUN, DVHKON, ENERG, ETAMEH, GAZ, INER, KAMERA, OBOROT, OTBORM, PARAM, PAROTB, POTERI, PRRG, REGIM, SAU, SMESHN, SOPL, TERMO, UPRAVL, VHOD, VVR.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, DGG2, ETAFF, FIS, FIS1, FIS2, KAMSGF, KAMSM, LAM, MUS, MUS1, MUS2, RDPAR, SAUPAR, SIGGP, SIGS, SIGS1, SIGS2, SIGTP, SOPLO, SOPLOP, TFI.

RDPAR

APPLICATION

Subroutine that calculates additional parameters which were not calculated during engine straight-through calculation.

CALLING

CALL RDPAR

COMMON – BLOCKS

DOPPAR, DPARUN, ENERG, ETAMEH, GAZ, HARAKT, INER, KAMERA, OBOROT, OTBORM, PARAM, SOPL, TERMO, VHOD.

RP5

APPLICATION

Subroutine for calculating airflow parameters in transient duct of secondary flow.

CALLING

CALL RP5

COMMON – BLOCKS

PARAM, POTERI, PRRG, TERMO, VHOD.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

SIGP5.

RSUMN

APPLICATION

Subroutine that solves set of non-linear algebraic equations by Newton method.

CALLING

CALL RSUMN(PS,HP,ZP,KNZP,VAR,DB,C,PRECES,MAXIT,SG,IER)

DESCRIPTION OF PARAMETERS

PS – a name of a procedure that calculates residuals (EXTERNAL).

HP – double precision input vector of independent variables with a dimension KNZP. KNZP may vary during the caluclation.

ZP – double precision output vector of residuals (size of vector is KNZP), this parameter can be changed during calculations.

KNZP – number of equations.

VAR – double precision output vector of independent variables (size of vector is KNZP). KNZP may vary during the caluclation.

DB – double precision output Jacobi matrix (sizes of matrix are [KNZP,KNZP]).

C – vector of intermediate results of residuals calculating when varying independent variables.

PRECES – double precision input parameter that expresses the tolerance of solution of algebraic equations set.

MAXIT – input parameter that limits maximum number of iterations in Newton method.

SG - double precision input value that limits varying of independent variables.

IER – result error code:

IER=0 – successful solution;

IER=-2 – break because the subroutine reached the limit by number of iterations;

IER=±1 – degenerated Jacobi matrix.

IER=2 – break because of residuals became not stable.

NOTES

A system of linear equations for calculating corrective factors is formed in every step. This system is checked on degeneration. If the system is degenerated, then solving procedure emergently breaks.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

DSIMQ,

PS(VAR,C,KNZP,NU,IER) must be formed by User.

SEAR

APPLICATION

Subroutine for searching the interval with XX value in the table AA. Subroutine calculates the index MM that satisfies the requirement:

$$XX \in \left[AA(MM), AA(MM+1) \right]$$

CALLING

CALL SEAR(XX,AA,NN,MM,IER)

DESCRIPTION OF PARAMETERS

XX – double precision input value which variation range in under calculation.

AA – double precision input array of values.

NN – size of array AA.

MM – output index of lower range limit.

IER – calculated error code:

IER=0 – successful calculation.

IER=1 – $XX > AA(NN)$

IER=1 – $XX < AA(1)$

NOTES

Table (AA) is a function of single variable. It is filled in argument value increasing order. Forming an error code subroutine at the same time changes value XX.

If $XX > AA(MM)$ then $XX = AA(NN)$

If $XX > AA(1)$ then $XX = AA(1)$

METHOD

Index of lower range limit is obtained by running down values *AA(I)* and *AA(I+1)* where *I=1,...,NN-1* and checks the condition $XX \in \left[AA(I), AA(I+1) \right]$. If condition is met then MM=I.

SEARCH

APPLICATION

Subroutine for linear interpolation of double precision functions B, C, D for given values of arguments P and A according to given tables (AX,BX,CX), (AX,BX,DX).

CALLING

CALL SEARCH(P, A, B, C, D, AX, NA, BX, CX, DX, NO, NOMAX, NCODE, ISEARC)

DESCRIPTION OF PARAMETERS

P – double precision value of the first argument.

A – double precision value of the second argument.

B – double precision interpolated value of the first function.

C – double precision interpolated value of the second function.

D – double precision interpolated value of the third function.

AX – double precision input array with values of a second argument.

NA – size of array AX.

BX – double precision input array with values of the first function (size of array is NA*NOMAX).

CX – double precision input array with values of the second function (size of array is NA*NOMAX).

DX – double precision input array with values of the third function (size of array is NA*NOMAX).

NO – double precision input array of a amounts of table values for each value of a second argument A

NOMAX – maximum value of array NO.

NCODE – result error code:

NCODE=0 – if A(1)<A<A(NA);

NCODE=1 – if A<A(1);

NCODE=2 – if A>A(NA).

ISEARC – bitwise operator that indicates current variable of interest: FALSE – value of function B is determined according to the first argument P; TRUE – value of function C is determined according to the first argument P;

NOTES

Table (AX, BX, CX, DX) is a function of single variable. The table is filled in variables increasing order.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

INTCN

METHOD

Subroutine makes calculations by one of two proposed schemes. The type of calculating scheme is specified by parameter ISEARC.

If ISEARC=TRUE then it is assumed that

$$P = \left.\frac{C - C_{min}}{C_{max} - C_{min}}\right|_a = const \text{ (scheme N1).}$$

If ISEARC=FALSE then it is assumed that

$$P = \left.\frac{B - B_{min}}{B_{max} - B_{min}}\right|_a = const \text{ (scheme N2).}$$

Using second scheme allows excluding an ambiguity when interpolating the table [AX, BX,CX]. If A in this table are constant and there are segments with non-uniform change of CX. In all other instances it's recommended to use the first scheme.

Calculation begins from determining the values IL and IH, which match the requirements $(IH = IL + 1)$ and $AX(IL) < A < AX(IH)$. Then in chosen segments of tables $AX(IL)$ and $AX(IH)$ according to given value of P, program calculates that have B value, recalculated according to parameter P- BL and BH , and surrounding points, which are given in the input table with values $BX(IL,KL), BX(IL,KH)$ and $BX(IH,JL), BX(IH,JH)$, which in their turn must meet requirements $KH = KL + 1$ $JH = JL + 1$, $BX(IL,KL) < BL < BX(IL,KH)$, $BX(IH,JL) < BH < BX(IH,JH)$.

Calculation of required values B, C and D is performed by the next equations:

$$PRM = \frac{(A - AX[IL])}{(AX[IH] - AX[IL])};$$

$$PRH = \frac{(BX[IL,KL] - BH)}{(BX[IH,JH] - BX[IH,JL])};$$

$$PRL = \frac{(BX[IL,KL] - BH)}{(BX[IL,KH] - BX[IL,KL])};$$

$$CH = CX(IH,JL) - PRH*(CX(IH,JH) - CX(IH,JL));$$

$$CL = CX(IL,KL) - PRL*(CX(IL,KH) - CX(IL,KL));$$

$$DH = DX(IH,JL) - PRH*(DX(IH,JH) - DX(IH,JL));$$

$$DL = DX(IL,KL) - PRL*(DX(IL,KH) - DX(IL,KL));$$

$$B = BL + PRM*(BH - BL);$$

$$C = CL + PRM*(CH - CL);$$

$$D = DL + PRM*(DH - DL).$$

SIGKS

APPLICATION

Subroutine to calculate loses of total pressure in CC.

CALLING

SIGKS

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, PRRG, KSDAT, VHOD, GAZ, TERMO, PARAM.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, STATPAR

SIGP1 {SIGP2, SIGP3, SIGP4, SIGP5}

APPLICATION

Subroutines to calculate losses of total pressure in transition ducts.

CALLING

SIGP1 (SIGP2, SIGP3, SIGP4, SIGP5)

COMMON – BLOCKS

AIDENT, PER1DAT (PER2DAT, PER3DAT, PER4DAT, PER5DAT), PRRG.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT, STATPAR

SIGS {SIGS1, SIGS2}

APPLICATION

Subroutines to calculate loses of total pressure in nozzles.

CALLING

SIGS {SIGS1, SIGS2}

COMMON – BLOCKS

AIDENT, PRRG, SDAT (S1DAT, S2DAT)

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT, STATPAR.

SIGVH

APPLICATION

Subroutine that calculates heat loses of total pressure at engine inlet.

CALLING

SIGVH

COMMON – BLOCKS

AIDENT, PRRG, VHODDAT.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, POINT, STATPAR.

SITHAR

APPLICATION

Subroutine that forms dialogue windows to inform User about errors, which occurred when having been calculating parameters by performances.

CALLING

SITHAR(I,A)

DESCRIPTION OF PARAMETERS

I – number of performance, which had errors when have been calculating parameters of this performance.

A – input array of parameters calculated by performance.

NOTES

COMMON – BLOCKS

ABDX, GLBUPR.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

TABLE, WHAT

METHOD

SITTER

APPLICATION

Subroutine that forms dialogue windows to inform User about errors, which had occurred during thermodynamic calculation.

CALLING

SITTER(I,b)

DESCRIPTION OF PARAMETERS

I –number of problem.

A – parameter, which was the reason of the problem.

COMMON – BLOCKS

ABDX, GLBUPR.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

WHAT

SOPLO12

APPLICATION

Subroutine to calculate parameters of working substance and needed area of critical station.

CALLING

SOPLO(Q,D,TVH,FI,MU,PVH,FS,GS,PH,PVIH,VS,LAMS,PIS)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel flow at jet nozzle inlet.

D – double precision input value of relative humidity at jet nozzle inlet.

TVH - double precision input value of total temperature at jet nozzle inlet.

FI – double precision input value of velocity factor of jet nozzle.

MU – double precision input value of mass flow factor through jet nozzle.

PVH – double precision input value of total pressure at jet nozzle inlet.

FS – double precision output value of needed area of jet nozzle.

GS – double precision input value of mass flow of working substance at jet nozzle inlet.

PH – double precision input value of total pressure at jet nozzle discharge. (atmospheric pressure).

PVIH – double precision input value of static pressure at jet nozzle discharge.

VS – double precision output value of jet stream velocity at jet nozzle discharge.

LAMS – double precision output value of velocities ratio of gas (λ) at jet nozzle discharge.

PIS – double precision output value of nozzle pressure ratio.

COMMON – BLOCKS

GAZ.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, CRIT, SITTER, TFI, TFS.

METHOD

See subsection 1.1.2.10 of this Manual.

SQINTP

APPLICATION

Subroutine for quadratic interpolation of single-variable function.

CALLING

CALL SQINTP(X,Y,N,A,F)

DESCRIPTION OF PARAMETERS

X – double precision input value of argument.

Y – double precision calculated value of function.

N – sizes of arrays AX and AY.

A – double precision input array of argument values.

F – double precision input array of function values.

NOTES

Table (AX, AY) is a function of single variable, which values are sorted in variable increasing order.

METHOD

Standard method of squared interpolation

STATPAR

APPLICATION

Subroutine to calculate static pressures and temperatures.

CALLING

STATPAR (Q,D,TVH,PVH,FS,GS,TVih,PVIH,VS,MH)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel flow.

D – double precision input value of relative moisture content.

TVH – double precision input value of total temperature, K.

PVH – double precision input value of total pressure, Pa.

FS – double precision output value of needed area of jet nozzle, m^2.

GS – double precision input value of mass flow at jet nozzle inlet, kg/s.

TVIH – double precision output value of static temperature, K.

PVIH – double precision output value of static pressure, Pa.

VS – double precision output value of jet stream velocity at nozzle discharge, m/s.

MH – double precision output value of Mach number.

COMMON – BLOCKS

GAZ

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, LAM, TFS.

TABLE

APPLICATION

Subroutine to assign text information for forming dialogue window with information about errors in thermodynamic calculations.

CALLING

CALL TABLE(I,STR)

DESCRIPTION OF PARAMETERS

I – number of performance, parameters of which had resulted in error.

STR – output array with text information about this unit.

TAU

APPLICATION

Subroutine that calculates temperature rise factor at arbitrary station of compressor.

CALLING

CALL TAU(NUM,KPD,PI,TT)

DESCRIPTION OF PARAMETERS

NUM – number of air bleed station according to generic scheme.

KPD –double precision input value of compressor spool efficiency.

PI –double precision input value of compressor spool pressure ratio.

TVHK – double precision input value of temperature at compressor spool inlet.

TT – temperature rise factor at arbitrary intermediate compressor station.

COMMON – BLOCKS

AIDENT, HARAKT, PAROTB.

METHOD

See subsection 1.1.2.13 of this Manual.

$$TT = \frac{T^*_{bleed}}{T^*_{in\ C}}$$

TEXTIN

APPLICATION

Subroutine for filling with TEXT and TEXTMKV by string values. These parameters are used for printing.

CALLING

CALL TEXTIN

NOTES

This subroutine must be initialized before calling the output.

COMMON – BLOCKS

FILE, TXT, TXTMKV.

TFI

APPLICATION

Subroutine for physical properties calculation according to given enthalpy.

CALLING

CALL TFI(Q,D,T,I,S)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel flow.

D – double precision input value of humidity.

T – double precision output value of total temperature, *K*.

I – double precision input value of enthalpy, *J/kg*.

S – double precision output value of entropy *J/kg*.

COMMON – BLOCKS

EPSTER

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT.

METHOD

The method is described in subsection 1.1.2.2 of this Manual.

TFS

APPLICATION

Subroutine for physical properties calculation according to given entropy.

CALLING

CALL TFS (Q,D,T,I,S)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel flow.

D – double precision input value of humidity.

T – double precision output value of total temperature, *K*.

I – double precision output value of enthalpy, *J/kg*.

S – double precision input value of entropy, *J/kg*.

COMMON – BLOCKS

EPSTER

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT.

METHOD

The method is described in subsection 1.1.2.31.1.2.2 of this Manual.

TURBIN

APPLICATION

Subroutine that calculates parameter of air flow at turbine discharge.

CALLING

CALL TURBIN(Q,D,TVHT,TVIHT,PIT,LT,KPDT,NUM)

DESCRIPTION OF PARAMETERS

Q – double precision input value of relative fuel flow at turbine inlet.

D – double precision input value of relative humidity at turbine inlet.

TVHT – double precision input value of total temperature at turbine inlet.

TVIHT – double precision output value of total temperature at turbine discharge.

PIT – double precision output value of turbine pressure ratio.

LT – double precision output value of specific work done by turbine.

KPDT – double precision input value of turbine efficiency.

NUM – number of spool used in forming messages about this turbine spool.

27 – LPT, 28 – IPT, 29 – HPT.

COMMON – BLOCKS

GAZ.

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

CPFT, TFI, TFS, SITTER

METHOD

See subsection 1.1.2.8 of this Manual.

XKND
{XKVD, XKSD}

APPLICATION

Subroutine to calculate parameters of compressor at given operating point on compressor performance map.

CALLING

CALL XKND (XKVD, XKSD)

COMMON – BLOCKS

AIDENT, DOPPAR, DPARUN, HARAKT, OBOROT, PARAM, PEREPUSK, XKNDAT (XKVDAT, XKSDAT), XKNDCDT (XKVDCDT, XKSDCDT).

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINTX, SEARCH, SITHAR, SQINTP

METHOD

See item 6.1.2.2 of this Manual.

XTND
{XTVD, XTSD}

APPLICATION

Subroutine to calculate parameters of turbine at given operating point on turbine performance map.

CALLING

CALL XTND (XTVD, XTSD)

COMMON – BLOCKS

AIDENT, HARAKT, PARAM, XTNDAT (XTVDAT, XTSDAT).

USED SUBROUTINES AND SUBROUTINE-FUNCTIONS

LINXYT, SITHAR.

METHOD

See subsection 6.1.2.5 of this Manual.

APPENDIX D
Error messages, that may appear during calculations

Error because of exceeding maximum number of iterations

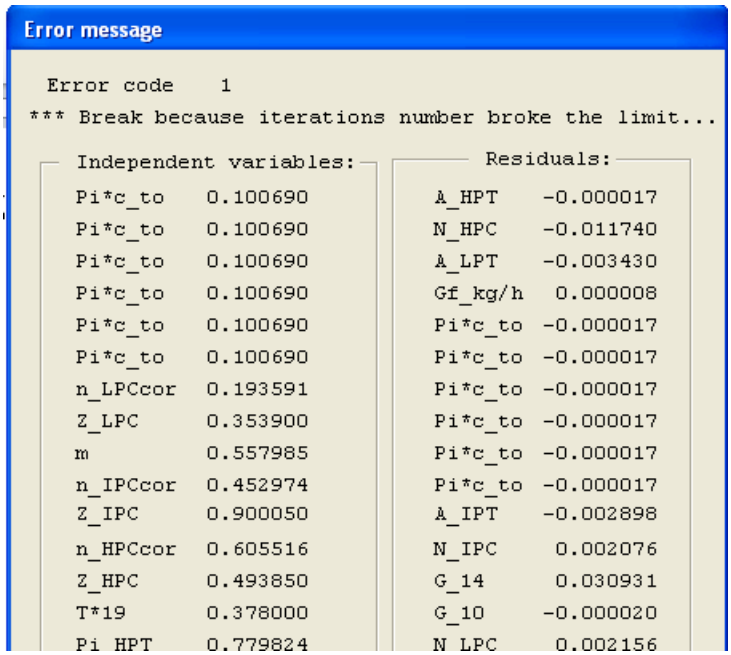


Fig. D.1 – Error message

This message is formed by subroutine, which solves the set of non-linear equations by Newton method (RSUMN). It appears when the solution was not obtained for 350. One of possible reasons is that given operating mode is very close to valves switching operating mode, and $\Delta\Pi_{ABV\ L}$ in compressor performance map was set little.

Degenerate Jacobi matrix

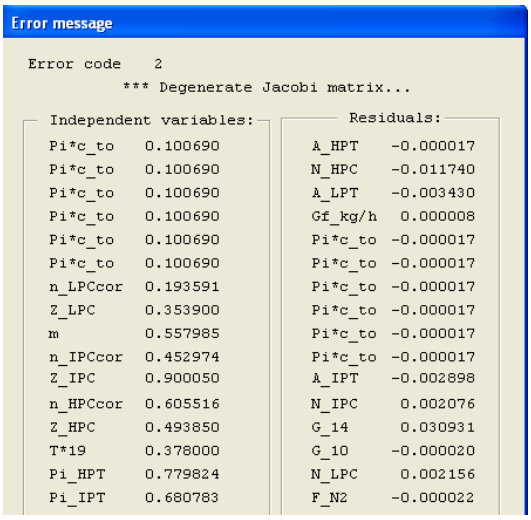


Fig. D.2 – Error message

This message is formed by subroutine, which solves the set of non-linear equations by Newton method (RSUMN). It appears during the calculation when one from

independent variables does not affect the residuals. Let`s consider booster stages as an example for this situation. In case we model TFE-2 without boosters we must set the conventional performance (pressure ratio of these stages must be equal to 1 for all airflows; this is necessary to prevent pressure rise in this stages or power consuming, produced by the corresponding turbine). If User will set $z = f(\pi)$ in the editor of compressor performances (see subsection 6.1.2.2), then independent variable z_B will not result in changing of any residual; hence there will be a line in the Jacobi matrix consisting of zeros. This problem can be avoided by choosing $z = f(G)$ in the editor of compressor performances (see subsection 6.1.2.2). Then changing of z_B will result in changing of residual by airflow between primary flow of LPC and booster stages, and residual by airflow between booster stages and HPC inlet. So, matrix will not become singular.

Independent variables have overran the limit

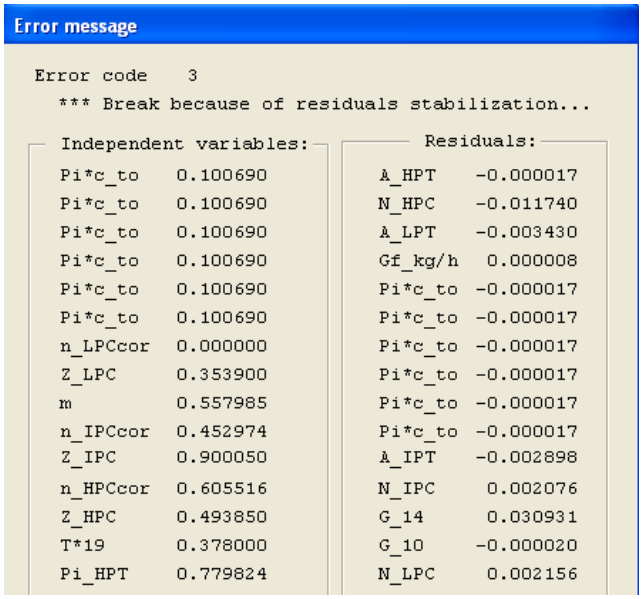


Fig. D.3 – Error message

This message is formed by subroutine, which solves the set of non-linear equations by Newton method (RSUMN). It appears when during the calculation one of independent variables have overran its limit, which was set in file with limitations (see subsection . 6.1.2.10). Then parameter, which have reached bottom limit will be equal to 0, and parameter, which have reached top limit will be equal to 1. This information will be displayed in error message (fig. D.3) in the box ***"Independent variables"***. This situation can be avoided by broaden the range of corresponding parameter (by increasing the top limit and decreasing the bottom limit) in the editor (see 6.1.2.10). Limitations, which were set according to performances of corresponding unit generally need changing of performances themselves. Only after changing the performances User can change the limitations. Changing of performances must be done ONLY by highly qualified specialists of corresponding department.

Overrunning the limit

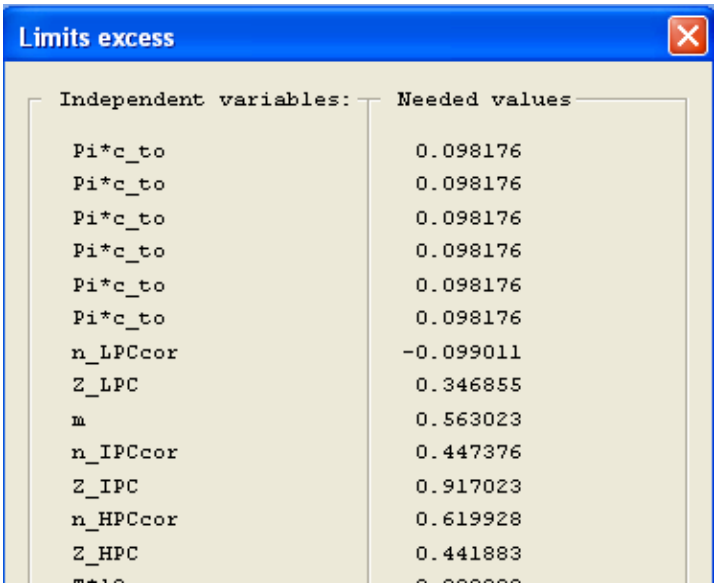


Fig. D.4 – Error message

This error appears together with message, shown in fig. D.3. The error message shows the list of independent variables and their needed values.

Important! The content of error messages with information about independent variables depends on engine type.

Qt value in CPFT program is invalid

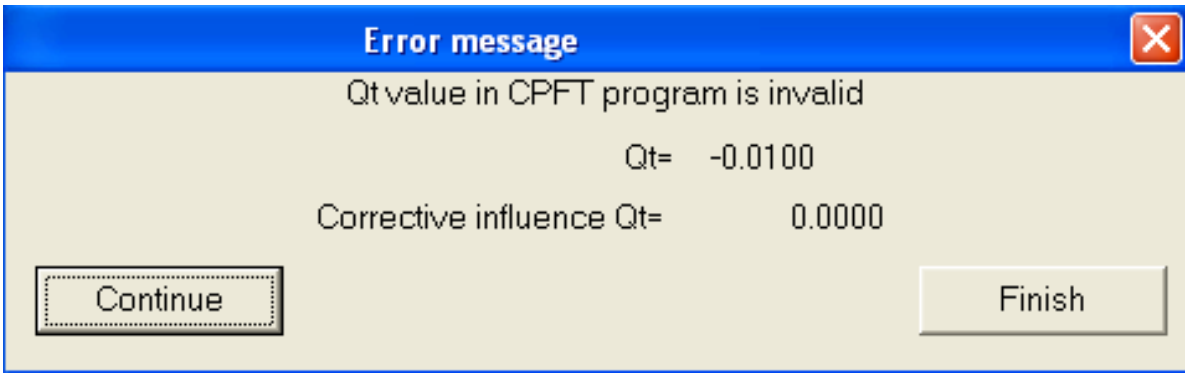


Fig. D.5 – Error message (Q is not correct to be inputted to CPFT)

This message is formed by program SITTER in case when Qt (relative content of combustion products in hydrocarbon fuel) in the program CPFT overruns the limits [$-1 \cdot 10^{-5}$, 1,00001]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 0 (this value is shown in line ***"Corrective influence"***).

D value in CPFT program is invalid

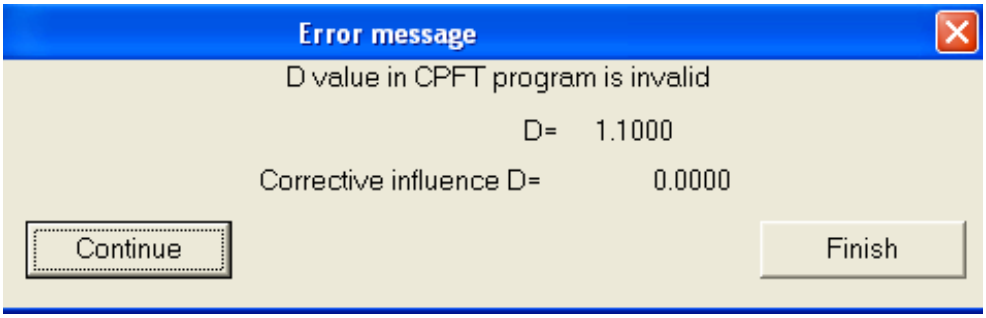


Fig. D.6 – Error message (D is not correct to be inputted to CPFT)

This message is formed by program SITTER in case when D (humidity) in the program CPFT overruns the limits [$-1 \cdot 10^{-5}$, 1,00001]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 0 (this value is shown in line ***"Corrective influence"***).

Fi value in CRIT program is invalid

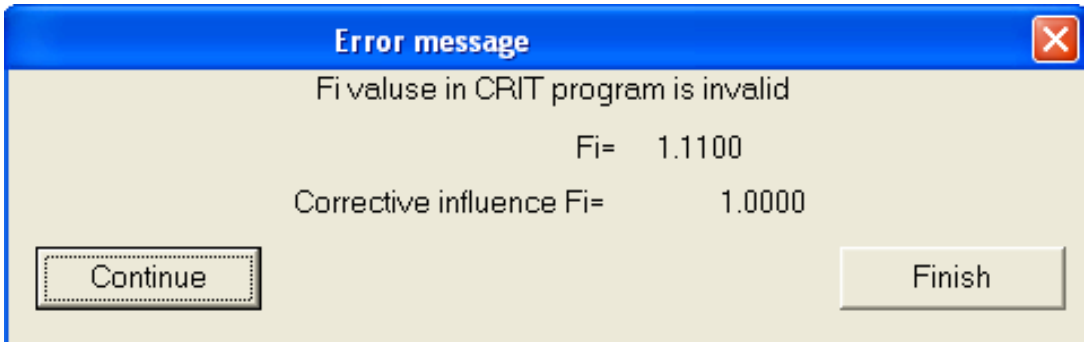


Fig. D.7 – Error message (velocity factor was incorrectly set in CRIT)

This message is formed by program SITTER in case when Fi (velocity factor) in the program CRIT overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

T at combustion chamber discharge is lower than T at its inlet

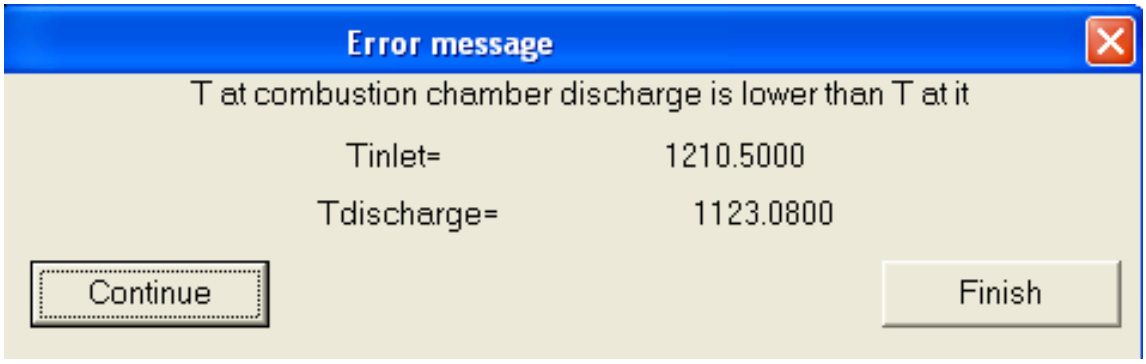


Fig. D.8 – Error message (TIT temperature was incorrectly set)

This message is formed by program SITTER in case when temperature at CC inlet at arbitrary iteration is higher than temperature at CC discharge. Error message shows current values of temperatures. If this message have appeared then program makes the following corrective influence: $T_{dis\ CC} = T_{in\ CC} + 100$

ETAKS value in program for combustion chamber calculation is invalid

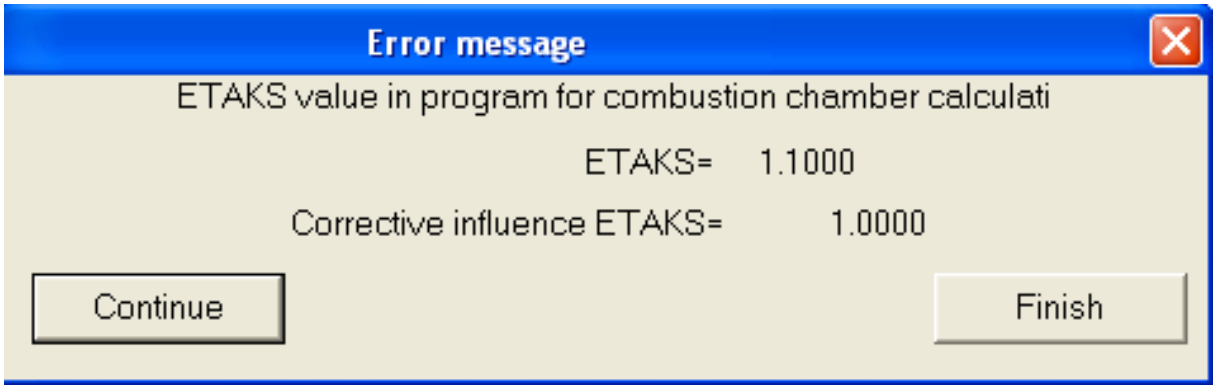


Fig. D.9 – Error message (combustion efficiency was incorrectly set)

This message is formed by program SITTER in case when ETAKS (combustion efficiency) in the program for combustion chamber calculation overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

"Uninflow" to nozzle mode

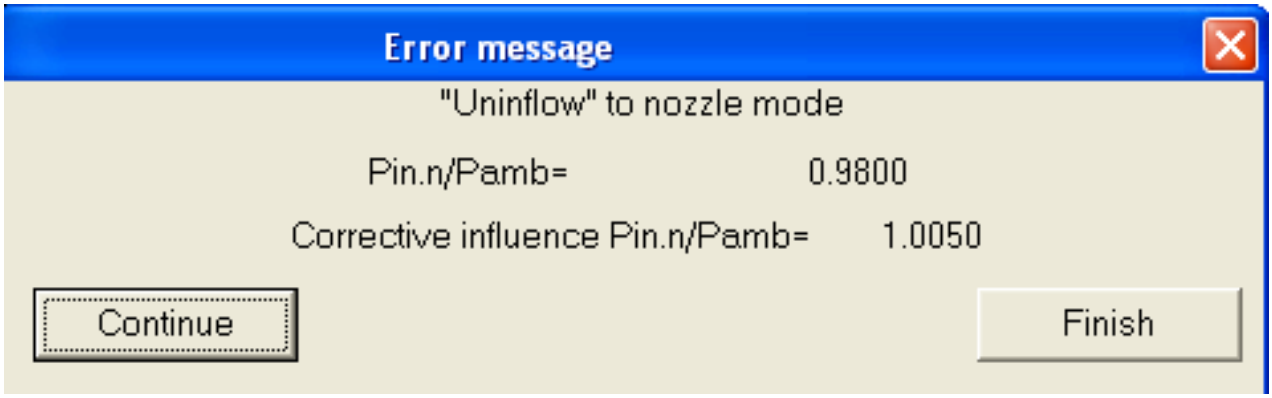


Fig. D.10 – Error message (wrong nozzle parameters)

This message is formed by program SITTER in case when ratio of total pressure to static pressure at nozzle discharge is less than 1,005. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1,005 (this value is shown in line ***"Corrective influence"***).

Pi_comp value in XKND program is invalid

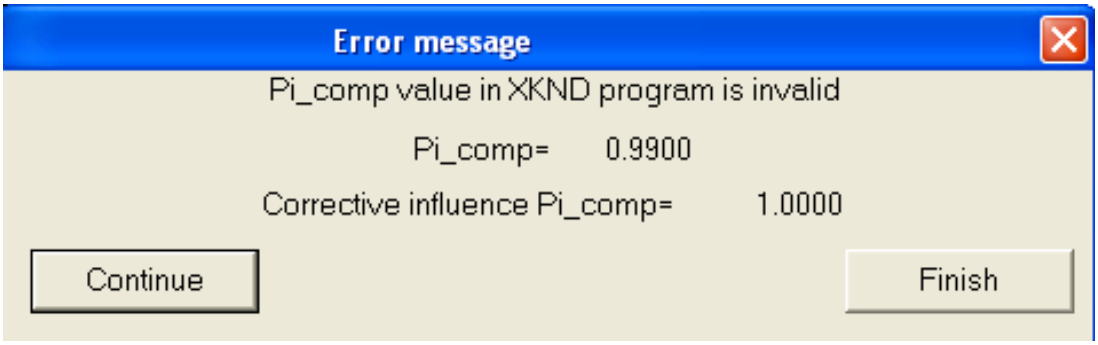


Fig. D.11 – Error message (pressure ratio of LPC was incorrectly set)

This message is formed by program SITTER in case when Pi_comp (LPC pressure ratio) in the program XKND becomes less than 1. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

Pi_comp value in XKSD program is invalid

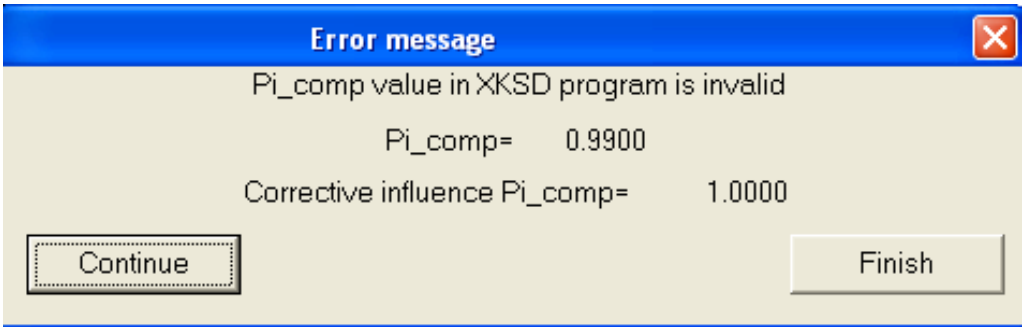


Fig. D.12 – Error message (pressure ratio of IPC was incorrectly set)

This message is formed by program SITTER in case when Pi_comp (IPC pressure ratio) in the program XKSD becomes less than 1. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

PiK value in XKVD program is invalid

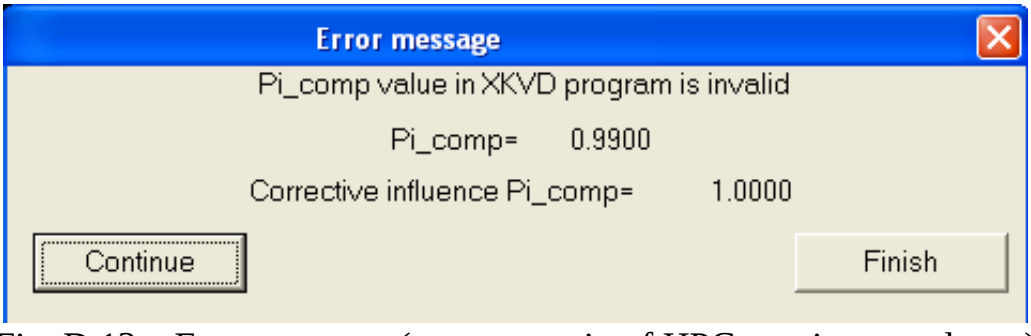


Fig. D.13 – Error message (pressure ratio of HPC was incorrectly set)

This message is formed by program SITTER in case when Pi_comp (HPC pressure ratio) in the program XKVD becomes less than 1. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

KPDK value in XKND program is invalid

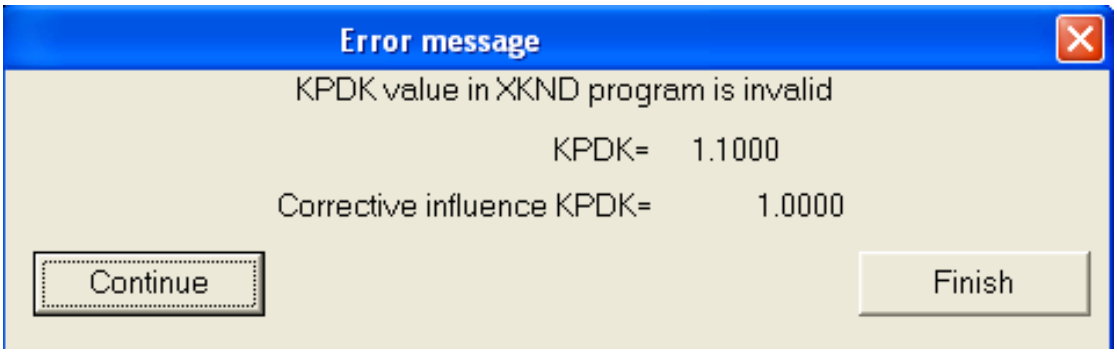


Fig. D.14 – Error message (efficiency of LPC was incorrectly set)

This message is formed by program SITTER in case when KPDK (efficiency of compressor spool) in the program XKND (the performance of low pressure compressor) overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

KPDK value in XKSD program is invalid

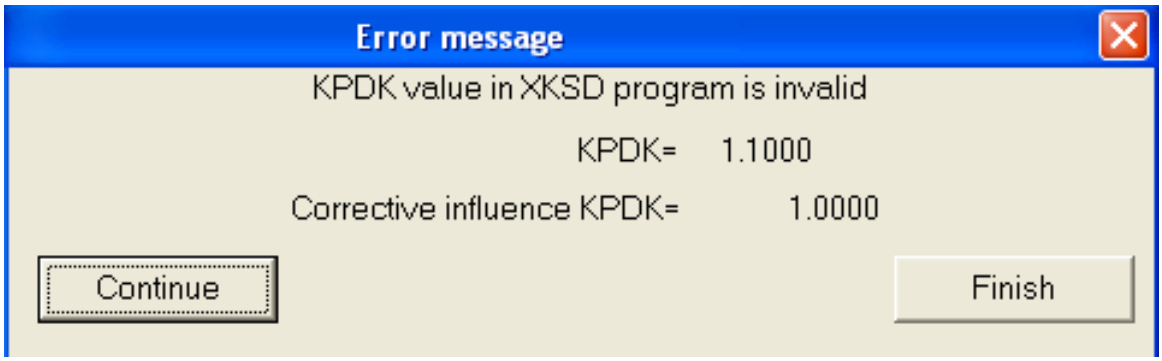


Fig. D.15 – Error message (efficiency of IPC was incorrectly set)

This message is formed by program SITTER in case when KPDK (efficiency of compressor spool) in the program XKSD (the performance of intermediate pressure compressor) overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

KPDK value in XKVD program is invalid

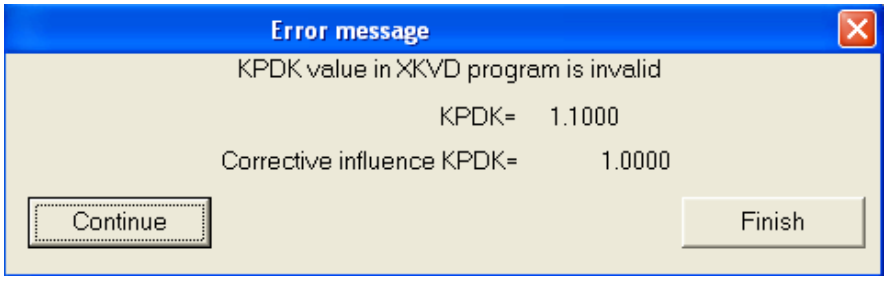


Fig. D.16 – Error message (efficiency of HPC was incorrectly set)

This message is formed by program SITTER in case when KPDK (efficiency of compressor spool) in the program XKVD (the performance of high pressure compressor) overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

Pi_turb value in XTND program is invalid

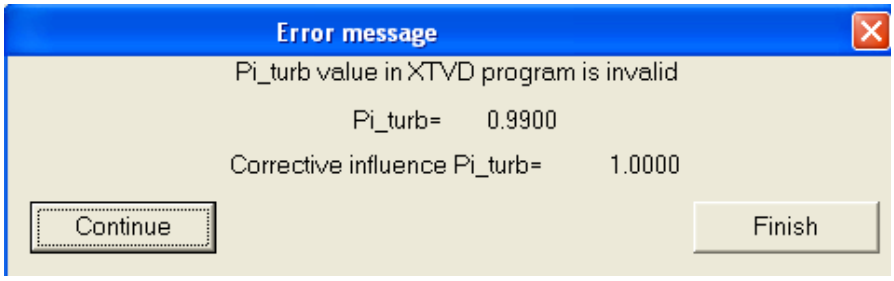


Fig. D.17 – Error message (pressure ratio of LPT was incorrectly set)

This message is formed by program SITTER in case when Pi_turb (pressure ratio of turbine spool) in the program XTND (the performance of high pressure turbine) have become less than 1. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

Pi_turb value in XTSD program is invalid

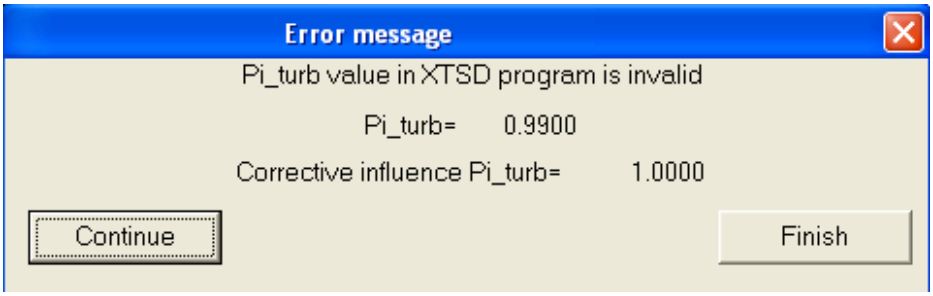


Fig. D.18 – Error message (pressure ratio of IPT was incorrectly set)

This message is formed by program SITTER in case when Pi_turb (pressure ratio of turbine spool) in the program XTSD (the performance of intermediate pressure turbine) have become less than 1. The message shows current value of parameter. If pa-

rameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

Pi_turb value in XTVD program is invalid

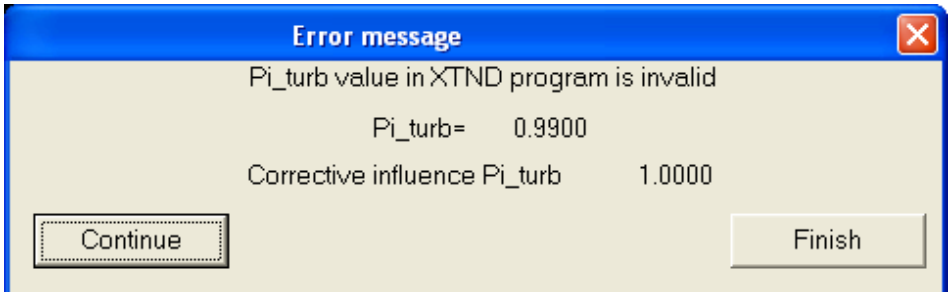


Fig. D.19 – Error message (pressure ratio of HPT was incorrectly set)

This message is formed by program SITTER in case when Pi_turb (pressure ratio of turbine spool) in the program XTVD (the performance of high pressure turbine) have become less than 1. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

KPDT value in XTND program is invalid

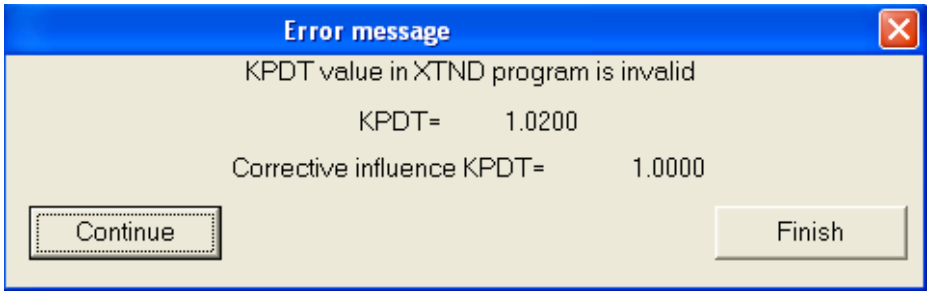


Fig. D.20 – Error message (efficiency of LPT was incorrectly set)

This message is formed by program SITTER in case when KPDT (efficiency of spool) in the program XTND (the performance of low pressure turbine) overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

KPDT value in XTSD program is invalid

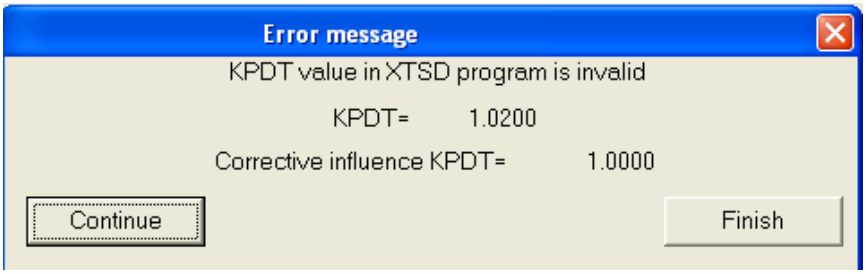


Fig. D.21 – Error message (efficiency of IPT was incorrectly set)

This message is formed by program SITTER in case when KPDT (efficiency of spool) in the program XTSD (the performance of intermediate pressure turbine) overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

KPDT value in XTVD program is invalid

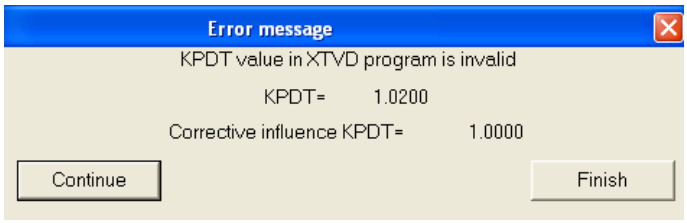


Fig. D.22 – Error message (efficiency of HPT was incorrectly set)

This message is formed by program SITTER in case when KPDT (efficiency of spool) in the program XTVD (the performance of high pressure turbine) overruns the limits [0, 1]. The message shows current value of parameter. If parameter have overran the limit then it is assigned 1 (this value is shown in line ***"Corrective influence"***).

Calculation point is outside LPC performances map

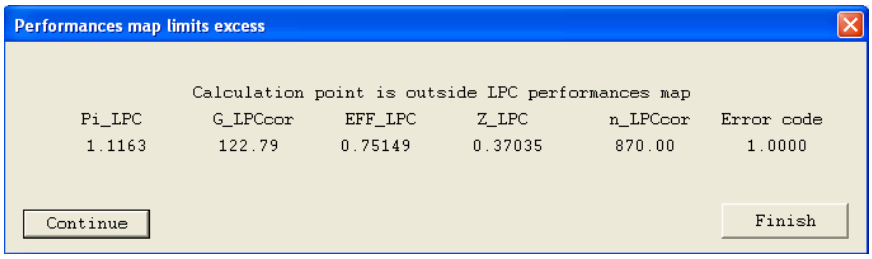


Fig. D.23 – Error message (operating point is outside the LPC performance)

The message is formed by subroutine SITHAR (error in subroutine XKND) in case when operating point is outside the compressor spool performance. In this message program shows parameters of operating point in the performance map (Pi_lpc – pressure ratio of LPC, G_lpccor – corrected airflow, EFFlpc – efficiency of compressor spool, Zlpc - parameter of operating point position in the performance map, n_lpccor – corrected rotational speed), and also ***Error code*** (1 – current rotational speed is lower than minimum rotational speed in the performance, 2– current rotational speed is higher than maximum rotational speed in the performance).

Calculation point is outside booster performances map

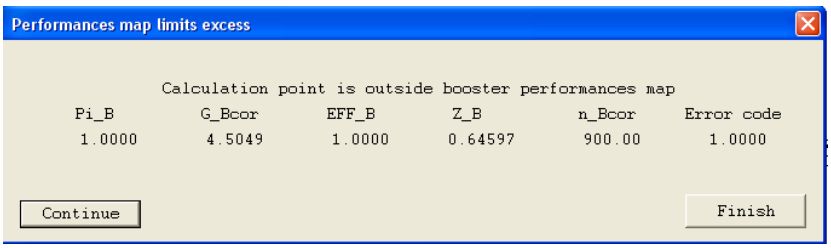


Fig. D.24 – Error message (operating point is outside the performance of boosters)

The message is formed by subroutine SITHAR, in case of error in subroutine XPS (the performance of boosters). This error appears when operating point is outside the booster stages performance . In this message program shows parameters of operating point in the performance map (Pi_b – pressure ratio of booster, G_bcor – corrected airflow, EFFb – efficiency of booster, Zb - parameter of operating point position in the performance map, n_bcor – corrected rotational speed), and also ***Error code*** (1 – current rotational speed is lower than minimum rotational speed in the performance, 2– current rotational speed is higher than maximum rotational speed in the performance).

Calculation point is outside IPC performances map

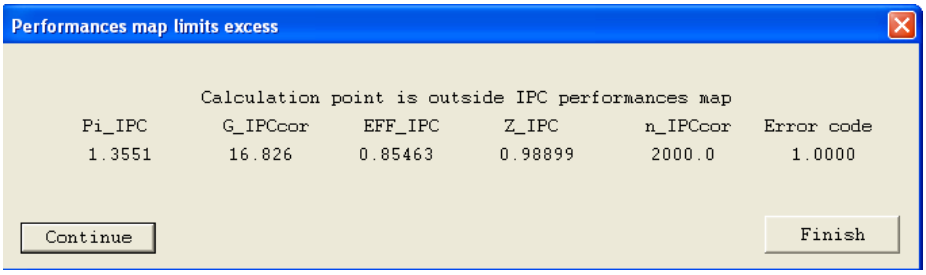


Fig. D.25 – Error message (operating point is outside the IPC performance)

The message is formed by subroutine SITHAR (error in subroutine XKSD) in case when operating point is outside the compressor spool performance. In this message program shows parameters of operating point in the performance map (Pi_ipc – pressure ratio of IPC, G_ipccor – corrected airflow, EFFipc – efficiency of compressor spool, Zipc - parameter of operating point position in the performance map, n_ipccor – corrected rotational speed), and also ***Error code*** (1 – current rotational speed is lower than minimum rotational speed in the performance, 2– current rotational speed is higher than maximum rotational speed in the performance).

Calculation point is outside HPC performances map

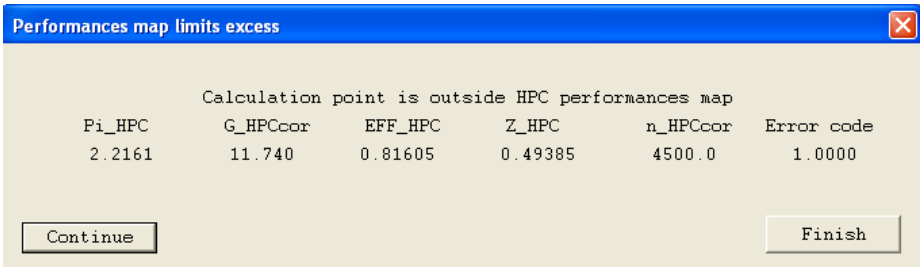


Fig. D.26 – Error message (operating point is outside the HPC performance)

The message is formed by subroutine SITHAR (error in subroutine XKVD) in case when operating point is outside the compressor spool performance. In this message program shows parameters of operating point in the performance map (Pi_hpc – pressure ratio of IPC, G_hpccor – corrected airflow, EFFhpc – efficiency of compressor spool, Zhpc - parameter of operating point position in the performance map, n_hpccor – corrected rotational speed), and also ***Error code*** (1 – current rotational speed is lower than minimum rotational speed in the performance, 2– current rotational speed is higher than maximum rotational speed in the performance).

Calculation point is outside HPT performances map

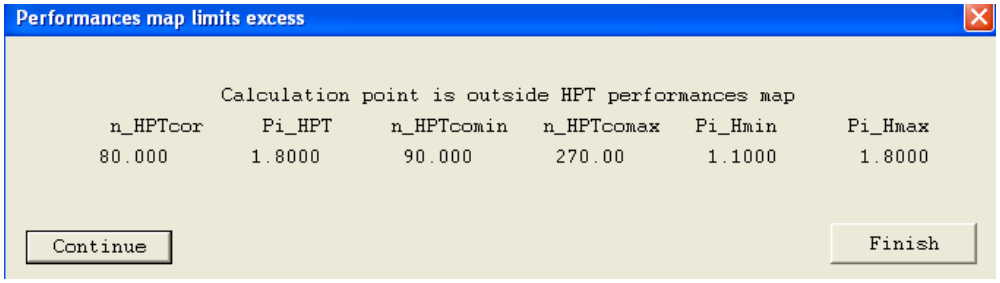


Fig. D.27 – Error message (operating point is outside the HPT performance)

The message is formed by subroutine SITHAR (error in subroutine XTVD) in case when one of parameters: n_hptcor (corrected rotational speed) or Pi_hpt (pressure ratio) overruns the limits [n_hptmin, n_hptmax] or [Пhptmin, Пhptmax].

Calculation point is outside IPT performances map

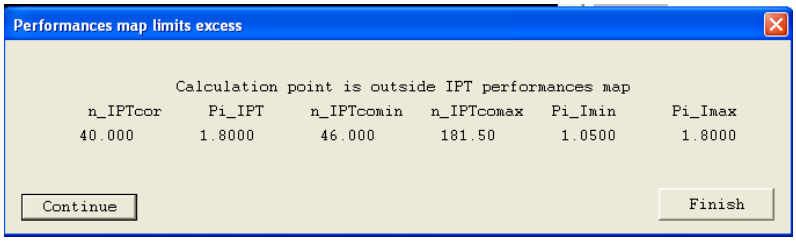


Fig. D.28 – Error message (operating point is outside the IPT performance)

The message is formed by subroutine SITHAR (error in subroutine XTSD) in case when one of parameters: n_iptcor (corrected rotational speed) or Pi_ipt (pressure ratio) overruns the limits [n_iptmin, n_iptmax] or [Пiptmin, Пiptmax].

Calculation point is outside LPT performances map

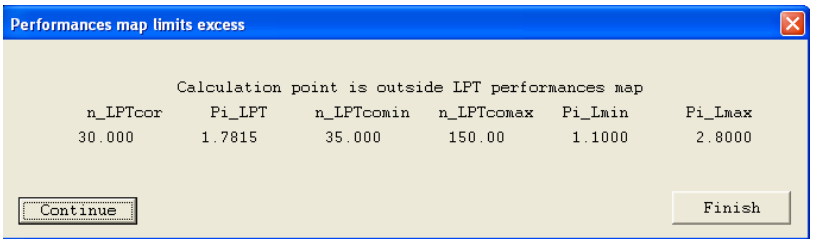


Fig. D.29 – Error message (operating point is outside the LPT performance)

The message is formed by subroutine SITHAR (error in subroutine XTND) in case when one of parameters: n_lptcor (corrected rotational speed) or Pi_lpt (pressure ratio) overruns the limits [n_lptmin, n_lptmax] or [Πlptmin, Πlptmax].

APPENDIX E
General list of gas-path parameters

№	Identifier	Designation	Name of parameter
1	2	3	4
1.	P(1)	P*_1	Fan inlet pressure
2.	P(2)	P*_2	Pressure at fan air bleed station
3.	P(3)	P*_3	Fan discharge pressure in bypass duct
4.	P(4)	P*_4	Pressure in bypass duct after air bleed from LPC
5.	P(5)	P*_5	Pressure in bypass duct at afterburner inlet
6.	P(6)	P*_6	Pressure in bypass duct at mixer inlet
7.	P(7)	P*_7	Fan discharge pressure in primary duct
8.	P(8)	P*_8	Booster discharge pressure
9.	P(9)	P*_9	Pressure after booster air bleed
10.	P(10)	P*10	IPC inlet pressure
11.	P(11)	P*11	Pressure at air bleed from IPC
12.	P(12)	P*12	IPC discharge pressure
13.	P(13)	P*13	Pressure at air bleed from IPC
14.	P(14)	P*14	HPC inlet pressure
15.	P(15)	P*15	Pressure at 2nd air bleed from HPC
16.	P(16)	P*16	HPC discharge pressure
17.	P(17)	P*17	Combustion chamber inlet pressure
18.	P(18)	P*18	Pressure at gas bleed from combustion chamber
19.	P(19)	P*19	Combustion chamber discharge pressure
20.	P(20)	P*20	HPT nozzle vanes inlet pressure
21.	P(21)	P*21	HPT nozzle vanes discharge pressure
22.	P(22)	P*22	HPT impeller inlet pressure
23.	P(23)	P*23	HPT impeller discharge pressure
24.	P(24)	P*24	HPT discharge pressure after air input
25.	P(25)	P*25	Pressure at transition duct between HPT and IPT discharge
26.	P(26)	P*26	IPT nozzle vanes inlet pressure
27.	P(27)	P*27	IPT nozzle vanes discharge pressure
28.	P(28)	P*28	IPT impeller inlet pressure
29.	P(29)	P*29	IPT impeller discharge pressure
30.	P(30)	P*30	IPT discharge pressure after air input
31.	P(31)	P*31	Pressure at transition duct between IPT and LPT discharge
32.	P(32)	P*32	LPT nozzle vanes inlet pressure
33.	P(33)	P*33	LPT nozzle vanes discharge pressure
34.	P(34)	P*34	LPT impeller inlet pressure
35.	P(35)	P*35	LPT impeller discharge pressure
36.	P(36)	P*36	LPT discharge pressure after air input
37.	P(37)	P*37	Nozzle discharge pressure
38.	P(38)	P*38	Mixer discharge pressure
39.	P(39)	P*39	Common afterburner inlet pressure
40.	P(40)	P*40	Pressure before gas bleed from common afterburner
41.	P(41)	P*41	Common nozzle inlet pressure

Continuation of Appendix E

1	2	3	4
42.	P(42)	P*42	Common nozzle discharge pressure
43.	P(43)	P*43	Pressure at 1st air bleed from HPC
44.	P(44)	P*44	Pressure before HPC ABV
45.	P(45)	P*45	Pressure before booster ABV
46.	P(46)	P*46	Pressure (reserve)
47.	P(47)	P*47	Pressure (reserve)
48.	P(48)	P*48	Pressure (reserve)
49.	P(49)	P*49	Pressure (reserve)
50.	P(50)	P*50	Pressure (reserve)
51.	T(1)	T*_1	Fan inlet temperature
52.	T(2)	T*_2	Temperature at air bleed from fan station
53.	T(3)	T*_3	Fan discharge temperature in bypass duct
54.	T(4)	T*_4	Temperature in bypass duct after air bleed from LPC
55.	T(5)	T*_5	Temperature in bypass duct at afterburner inlet
56.	T(6)	T*_6	Temperature in bypass duct at mixer inlet
57.	T(7)	T*_7	Fan discharge pressure in primary duct
58.	T(8)	T*_8	Booster discharge temperature
59.	T(9)	T*_9	Temperature after booster air bleed
60.	T(10)	T*10	IPC inlet temperature
61.	T(11)	T*11	Temperature at air bleed from IPC
62.	T(12)	T*12	IPC inlet discharge temperature
63.	T(13)	T*13	Temperature at air bleed from IPC
64.	T(14)	T*14	HPC inlet temperature
65.	T(15)	T*15	Temperature at 2nd air bleed from HPC
66.	T(16)	T*16	HPC discharge temperature
67.	T(17)	T*17	Combustion chamber inlet temperature
68.	T(18)	T*18	Temperature at gas bleed from combustion chamber
69.	T(19)	T*19	Combustion chamber discharge temperature
70.	T(20)	T*20	HPT nozzle vanes inlet temperature
71.	T(21)	T*21	HPT nozzle vanes discharge temperature
72.	T(22)	T*22	HPT impeller inlet temperature
73.	T(23)	T*23	HPT impeller discharge temperature
74.	T(24)	T*24	HPT discharge temperature after air input
75.	T(25)	T*25	Temperature at transition duct between HPT and IPT discharge
76.	T(26)	T*26	IPT nozzle vanes inlet temperature
77.	T(27)	T*27	IPT nozzle vanes discharge temperature
78.	T(28)	T*28	IPT impeller inlet temperature
79.	T(29)	T*29	IPT impeller discharge temperature
80.	T(30)	T*30	IPT discharge temperature after air input
81.	T(31)	T*31	Temperature at transition duct between IPT and LPT discharge
82.	T(32)	T*32	LPT nozzle vanes inlet temperature
83.	T(33)	T*33	LPT nozzle vanes discharge temperature
84.	T(34)	T*34	LPT impeller inlet temperature
85.	T(35)	T*35	LPT impeller discharge temperature
86.	T(36)	T*36	LPT discharge temperature after air input

Continuation of Appendix E

1	2	3	4
87.	T(37)	T*37	Nozzle discharge temperature
88.	T(38)	T*38	Mixer discharge temperature
89.	T(39)	T*39	Common afterburner inlet temperature
90.	T(40)	T*40	Temperature before gas bleed from common afterburner
91.	T(41)	T*41	Common nozzle inlet temperature
92.	T(42)	T*42	Common nozzle discharge temperature
93.	T(43)	T*43	Temperature at 1st air bleed from HPC
94.	T(44)	T*44	Temperature before HPC ABV
95.	T(45)	T*45	Temperature before booster ABV
96.	T(46)	T*46	Temperature (reserve)
97.	T(47)	T*47	Temperature (reserve)
98.	T(48)	T*48	Temperature (reserve)
99.	T(49)	T*49	Temperature (reserve)
100.	T(50)	T*50	Temperature (reserve)
101.	G(1)	G__1	Fan inlet air flow rate
102.	G(2)	G__2	Air flow at air bleed from fan
103.	G(3)	G__3	Fan discharge air flow in bypass duct
104.	G(4)	G__4	Air flow in bypass duct after air bleed from LPC
105.	G(5)	G__5	Air flow in bypass duct at afterburner inlet
106.	G(6)	G__6	Air flow in bypass duct at mixer inlet
107.	G(7)	G__7	Fan discharge air flow in primary duct
108.	G(8)	G__8	Booster discharge air flow
109.	G(9)	G__9	Air flow after booster air bleed
110.	G(10)	G_10	IPC inlet air flow
111.	G(11)	G_11	Air flow at air bleed from IPC
112.	G(12)	G_12	IPC discharge air flow
113.	G(13)	G_13	Air flow at air bleed from IPC
114.	G(14)	G_14	HPC inlet air flow
115.	G(15)	G_15	Air flow at 2nd air bleed from HPC
116.	G(16)	G_16	HPC discharge air flow
117.	G(17)	G_17	Combustion chamber inlet air flow
118.	G(18)	G_18	Air flow at gas bleed from combustion chamber
119.	G(19)	G_19	Combustion chamber discharge gas flow
120.	G(20)	G_20	HPT nozzle vanes inlet gas flow
121.	G(21)	G_21	HPT nozzle vanes discharge gas flow
122.	G(22)	G_22	HPT impeller inlet gas flow
123.	G(23)	G_23	HPT impeller discharge gas flow
124.	G(24)	G_24	HPT discharge gas flow after air input
125.	G(25)	G_25	Gas flow at transition duct between HPT and IPT discharge
126.	G(26)	G_26	IPT nozzle vanes inlet gas flow
127.	G(27)	G_27	IPT nozzle vanes discharge gas flow
128.	G(28)	G_28	IPT impeller inlet gas flow
129.	G(29)	G_29	IPT impeller discharge gas flow
130.	G(30)	G_30	IPT discharge gas flow after air input
131.	G(31)	G_31	Gas flow at transition duct between IPT and LPT discharge
132.	G(32)	G_32	LPT nozzle vanes inlet gas flow
133.	G(33)	G_33	LPT nozzle vanes discharge gas flow
134.	G(34)	G_34	LPT impeller inlet gas flow

Continuation of Appendix E

1	2	3	4
135.	G(35)	G_35	LPT impeller discharge gas flow
136.	G(36)	G_36	LPT discharge gas flow after air input
137.	G(37)	G_37	Nozzle discharge gas flow
138.	G(38)	G_38	Mixer discharge gas flow
139.	G(39)	G_39	Common afterburner inlet gas flow
140.	G(40)	G_40	Gas flow before gas bleed from common afterburner
141.	G(41)	G_41	Common nozzle inlet gas flow
142.	G(42)	G_42	Common nozzle discharge gas flow
143.	G(43)	G_43	Air flow at 1st air bleed from HPC
144.	G(44)	G_44	Air flow before HPC ABV
145.	G(45)	G_45	Air flow before booster ABV
146.	G(46)	G_46	Air flow (reserve)
147.	G(47)	G_47	Air flow (reserve)
148.	G(48)	G_48	Air flow (reserve)
149.	G(49)	G_49	Air flow (reserve)
150.	G(50)	G_50	Air flow (reserve)
151.	NND	n_lp	Low pressure rotor rotational speed
152.	NSD	n_ip	Intermediate pressure rotor rotational speed
153.	NVD	n_hp	High pressure rotor rotational speed
154.	H	H	Flight altitude
155.	Q	Q	Inlet relative fuel consumption
156.	D	D	Air humidity at engine inlet
157.	DT	DT	Temperature deviation from standard
158.	MP	Mfl	Flight Mach number
159.	VP	Vfl	Flight velocity
160.	P0ST	P0st	Atmospheric static pressure
161.	T0ST	T0st	Atmospheric static temperature
162.	P0	P0	Atmospheric stagnation pressure
163.	T0	T0	Atmospheric stagnation temperature
164.	LKND	Llpc	Fan air compression specific work
165.	MKND	Nlpc	Fan power
166.	LKND1	Llpc1	Fan primary flow air compression specific work
167.	MKND1	Nlpc1	LPC primary flow power
168.	LKND2	Llpc2	LPC bypass flow air compression specific work
169.	MKND2	Nlpc2	LPC bypass flow power
170.	LKSD	Llpc	IPC air compression specific work
171.	MKSD	Nipc	IPC power
172.	LKVD	Lhpc	HPC air compression specific work
173.	MKVD	Nhpc	HPC power
174.	LTVD	Lhpt	HPT gas expansion specific work
175.	MTVD	Nhpt	HPT power
176.	LTSD	Lipt	IPT gas expansion specific work
177.	MTSD	Nipt	IPT power
178.	LTND	Llpt	LPT gas expansion specific work
179.	MTND	Nlpt	LPT power
180.	LPS	Lb	Booster air compression specific work
181.	MPS	Nb	Booster power
182.	ZKND	Zlpc	Operating point position factor for constant speed line of fan

Continuation of Appendix E

1	2	3	4
			performance map
183.	PIKND	Πlpc*	Fan pressure ratio
184.	GKNDPR	Glpccor	Fan corrected air flow
185.	KPDKND	EFFlpc*	Fan efficiency
186.	NKNDPR	n_lpccor	Corrected fan rotational speed
187.	ZKND1	Zlpc1	Operating point position factor for constant speed line of fan performance map (primary duct)
188.	PIKND1	Πlpc1*	Fan pressure ratio (primary duct)
189.	GND1PR	Glpc1cor	Fan corrected air flow (primary duct)
190.	KPDND1	EFFlpc1*	Fan efficiency (primary duct)
191.	ZKND2	Zlpc2	Operating point position factor for constant speed line of fan performance map (bypass)
192.	PIKND2	Πlpc2*	Fan pressure ratio (bypass)
193.	GND2PR	Glpc2cor	Fan corrected air flow (bypass)
194.	KPDND2	EFFlpc2*	Fan efficiency (bypass)
195.	ZPS	Zipc	Operating point position factor for constant speed line of booster performance map
196.	PIPS	Πipc*	Booster pressure ratio
197.	GPSPR	Zb	Booster corrected air flow
198.	KPDPS	Πb*	Booster efficiency
199.	NPSPR	Gbcor	Booster corrected rotational speed
200.	ZKSD	EFFb*	Operating point position factor for constant speed line of IPC map
201.	PIKSD	n_bcor	IPC pressure ratio
202.	GKSDPR	Gipccor	IPC corrected air flow
203.	KPDKSD	EFFipc*	IPC efficiency
204.	NKSDPR	n_ipccor	IPC corrected rotational speed
205.	ZKVD	Zhpc	Operating point position factor for constant speed line of HPC map
206.	PIKVD	Πhpc*	HPC pressure ratio
207.	GKVDPR	Ghpccor	HPC corrected air flow
208.	KPDKVD	EFFhpc*	HPC efficiency
209.	NKVDPR	n_hpccor	HPC corrected rotational speed
210.	PITVD	Πhpt*	HPT pressure ratio
211.	ATVD	Ahpt	HPT corrected gas flow (according to HPT performance map)
212.	ATVDR	Ahpt cor	HPT corrected gas flow (calculated)
213.	KPDTVD	EFFhpt*	HPT efficiency
214.	NTVDPR	n_hptcor	HPT corrected rotational speed
215.	PITSD	Πipt*	IPT pressure ratio
216.	ATSD	Aipt	IPT corrected gas flow (according to IPT performance map)
217.	ATSDR	Aiptcor	IPT corrected gas flow (calculated)
218.	KPDTSD	EFFipt*	IPT efficiency
219.	NTSDPR	n_iptcor	IPT corrected rotational speed
220.	PITND	Πlpt*	LPT pressure ratio
221.	ATND	Alpt	LPT corrected gas flow (according to LPT performance map)
222.	ATNDR	Alptcor	LPT corrected gas flow (calculated)
223.	KPDTND	EFFlpt*	LPT efficiency

Continuation of Appendix E

1	2	3	4
224.	NTNDPR	n_lptcor	LPT corrected rotational speed
225.	MDV	m	Engine bypass ratio
226.	PKSMST	Pmixst	Mixer inlet static pressure
227.	GTKS	Gfcc	Main combustion chamber fuel consumption
228.	QTKS	Qfcc	Main combustion chamber relative fuel consumption
229.	ETGKS	ETAfcc	Main combustion chamber combustion efficiency
230.	ALFAKS	ALFAcc	Main combustion chamber excess air/fuel ratio
231.	GTFK	Gfaft	Afterburner fuel consumption
232.	QTFK	Qfaft	Afterburner relative fuel consumption
233.	ETGFK	ETAfcc	Afterburner combustion efficiency
234.	ALFKSR	ALFAcc	Total excess air/fuel ratio
235.	GTFK2	Gfaft2	Bypass duct afterburner fuel consumption
236.	QTFK2	Qfaft2	Bypass duct afterburner relative fuel consumption
237.	ETGFK2	ETAfaft2	Bypass duct afterburner combustion efficiency
238.	ALFFK2	ALFAaft2	Bypass duct afterburner excess air/fuel ratio
239.	SGMVH	SIGMAin	Intake pressure recovery factor
240.	SGMP1	SIGMAtrans1	Transition duct between LPC and IPC pressure recovery factor
241.	SGMP2	SIGMAtrans 2	Transition duct between IPC and HPC pressure recovery factor
242.	SGMP3	SIGMAtrans 3	Transition duct between HPT and IPT pressure recovery factor
243.	SGMP4	SIGMAtrans 4	Transition duct between IPT and LPT pressure recovery factor
244.	SGMP5	SIGMAtrans 5	Bypass pressure recovery factor
245.	SGMP6	SIGMAtrans 6	Transition duct between fan and boosters pressure recovery
246.	SGMKST	SIGMAcc_h	Combustion chamber pressure recovery factor (hydraulic)
247.	SGMKSG	SIGMAcc_h	Combustion chamber pressure recovery factor (heat)
248.	SGMKS	SIGMAcc	Combustion chamber pressure recovery factor (total)
249.	SGMFKG2	SIGMaft2_h	Bypass duct afterburner pressure recovery factor (hydraulic)
250.	SGMFKT2	SIGMaft2_t	Bypass duct afterburner pressure recovery factor (heat)
251.	SGMFKG1	SIGMaft_h	Afterburner pressure recovery factor (hydraulic)
252.	SGMFKT1	SIGMaft_t	Afterburner pressure recovery factor (heat)
253.	SGMFK2	SIGMAaft2	Bypass duct afterburner pressure recovery factor (total)
254.	SGMFK1	SIGMAaft	Afterburner pressure recovery factor (total)
255.	SGMS1	SIGMAn1	Primary nozzle pressure recovery
256.	SGMS2	SIGMAn2	Bypass nozzle pressure recovery factor
257.	SGMKSM	SIGMAmix	Mixer pressure recovery factor
258.	SGMST	SIGMAst	Pressure recovery factor in divergent duct after LPT
259.	SGMS	SIGMAn	Nozzle pressure recovery factor
260.	PIS1	Пn1	Primary nozzle pressure ratio
261.	PS1ST	Pn1st	Primary nozzle pressure discharge static pressure
262.	FS1	Fn1	Primary nozzle area (real)
263.	FS1R	Fn1des	Primary nozzle area (calculated)
264.	VS1	Vn1	Primary nozzle discharge velocity
265.	LAMS1	LAMn1	Primary nozzle discharge superficial velocity
266.	MUSPL1	MUn1	Primary nozzle flow factor
267.	FISPL1	Fi1	Primary nozzle velocity factor
268.	PIS2	Пn2	Bypass nozzle pressure ratio

Continuation of Appendix E

1	2	3	4
269.	PS2ST	Pn2st	Bypass nozzle discharge static pressure
270.	FS2	Fn2	Bypass nozzle area (real)
271.	FS2R	Fn2calc	Bypass nozzle area (calculated)
272.	VS2	Vn2	Bypass nozzle discharge velocity
273.	LAMS2	LAMn2	Bypass nozzle discharge superficial velocity
274.	MUSPL2	MUn2	Bypass nozzle flow factor
275.	FISPL2	FIn2	Bypass nozzle efficiency
276.	PIS	Пn	Nozzle pressure ratio
277.	PSST	Pnst	Nozzle discharge static pressure
278.	FS	Fn	Common nozzle area (real)
279.	FSR	Fncalc	Common nozzle area (calculated)
280.	VS	Vn	Common nozzle discharge velocity
281.	LAMS	LAMn	Common nozzle discharge superficial velocity
282.	MUSPL	MUn	Common nozzle flow factor
283.	FISPL	Fin	Common nozzle velocity factor
284.	KPDMND	EFFmech_LP	Low pressure rotor mechanical efficiency
285.	KPDMSD	EFFmech_IP	Intermediate pressure rotor mechanical efficiency
286.	KPDMVD	EFFmech_LP	High pressure rotor mechanical efficiency
287.	KPDRED	EFFmech_GB	Gearbox mechanical efficiency
288.	DMND	DN_LP	Power takeoffs from low pressure rotor
289.	DMSD	DN_IP	Power takeoffs from intermediate pressure rotor
290.	DMVD	DN_HP	Power takeoffs from high pressure rotor
291.	DGK(1)	DGc1	Relative bleeds from fan (1 point)
292.	DGK(2)	DGc2	Relative bleeds from fan (2 point)
293.	DGK(3)	DGc3	Relative bleeds from fan bypass part discharge
294.	DGK(4)	DGc4	Relative bleeds from boosters discharge
295.	DGK(5)	DGc5	Relative bleeds from IPC (1 point)
296.	DGK(6)	DGc6	Relative bleeds from IPC (2 point)
297.	DGK(7)	DGc7	Relative bleeds from IPC discharge
298.	DGK(8)	DGc8	Relative bleeds from HPC (1 point)
299.	DGK(9)	DGc9	Relative bleeds from HPC (2 point)
300.	DGK(10)	DGк10	Relative bleeds from HPC discharge
301.	DGKA(1)	DGcabs1	Absolute bleeds from fan (1 point)
302.	DGKA(2)	DGcabs2	Absolute bleeds from fan (2 point)
303.	DGKA(3)	DGcabs3	Absolute bleeds from fan bypass part discharge
304.	DGKA(4)	DGcabs4	Absolute bleeds from boosters discharge
305.	DGKA(5)	DGcabs5	Absolute bleeds from IPC (1 point)
306.	DGKA(6)	DGcabs6	Absolute bleeds from IPC (2 point)
307.	DGKA(7)	DGcabs7	Absolute bleeds from IPC discharge
308.	DGKA(8)	DGcabs8	Absolute bleeds from HPC (1 point)
309.	DGKA(9)	DGcabs9	Absolute bleeds from HPC (2 point)
310.	DGKA(10)	DGcabs10	Absolute bleeds from HPC discharge
311.	DGT(1,1)	DGt1_1	Relative taps of air, bled from fan (1 point), to nozzle vanes of HPT
312.	DGT(2,1)	DGt2_1	Relative taps of air, bled from fan (2 point), to nozzle vanes of HPT
313.	DGT(3,1)	DGt3_1	Relative taps of air, bled from fan discharge, to nozzle vanes of HPT
314.	DGT(4,1)	DGt4_1	Relative taps of air, bled from boosters discharge, to nozzle

Continuation of Appendix E

1	2	3	4
			vanes of HPT
315.	DGT(5,1)	DGt5_1	Relative taps of air, bled from IPC (1 point), to nozzle vanes of HPT
316.	DGT(6,1)	DGt6_1	Relative taps of air, bled from IPC (2 point), to nozzle vanes of HPT
317.	DGT(7,1)	DGt7_1	Relative taps of air, bled from IPC discharge, to nozzle vanes of HPT
318.	DGT(8,1)	DGt8_1	Relative taps of air, bled from HPC (1 point), to nozzle vanes of HPT
319.	DGT(9,1)	DGt9_1	Relative taps of air, bled from HPC (2 point), to nozzle vanes of HPT
320.	DGT(10,1)	DGt10_1	Relative taps of air, bled from HPC discharge, to nozzle vanes of HPT
321.	DGT(1,2)	DGt1_2	Relative taps of air, bled from fan (1 point), to impeller of HPT
322.	DGT(2,2)	DGt2_2	Relative taps of air, bled from fan (2 point), to impeller of HPT
323.	DGT(3,2)	DGt3_2	Relative taps of air, bled from fan discharge, to impeller of HPT
324.	DGT(4,2)	DGt4_2	Relative taps of air, bled from boosters discharge, to impeller of HPT
325.	DGT(5,2)	DGt5_2	Relative taps of air, bled from IPC (1 point), to impeller of HPT
326.	DGT(6,2)	DGt6_2	Relative taps of air, bled from IPC (2 point), to impeller of HPT
327.	DGT(7,2)	DGt7_2	Relative taps of air, bled from IPC discharge, to impeller of HPT
328.	DGT(8,2)	DGt8_2	Relative taps of air, bled from HPC (1 point), to impeller of HPT
329.	DGT(9,2)	DGt9_2	Relative taps of air, bled from HPC (2 point), to impeller of HPT
330.	DGT(10,2)	DGt10_2	Relative taps of air, bled from HPC discharge, to impeller of HPT
331.	DGT(1,3)	DGt1_3	Relative taps of air, bled from fan (1 point), to HPT discharge
332.	DGT(2,3)	DGt2_3	Relative taps of air, bled from fan (2 point), to HPT discharge
333.	DGT(3,3)	DGt3_3	Relative taps of air, bled from fan discharge, to HPT discharge
334.	DGT(4,3)	DGt4_3	Relative taps of air, bled from boosters discharge, to HPT discharge
335.	DGT(5,3)	DGt5_3	Relative taps of air, bled from IPC (1 point), to HPT discharge
336.	DGT(6,3)	DGt6_3	Relative taps of air, bled from IPC (2 point), to HPT discharge
337.	DGT(7,3)	DGt7_3	Relative taps of air, bled from IPC discharge, to HPT discharge
338.	DGT(8,3)	DGt8_3	Relative taps of air, bled from HPC (1 point), to HPT discharge

Continuation of Appendix E

1	2	3	4
339.	DGT(9,3)	DGt9_3	Relative taps of air, bled from HPC (2 point), to HPT discharge
340.	DGT(10,3)	DGt10_3	Relative taps of air, bled from HPC discharge, to HPT discharge
341.	DGT(1,4)	DGt1_4	Relative taps of air, bled from fan (1 point), to nozzle vanes of IPT
342.	DGT(2,4)	DGt2_4	Relative taps of air, bled from fan (2 point), to nozzle vanes of IPT
343.	DGT(3,4)	DGt3_4	Relative taps of air, bled from fan discharge, to nozzle vanes of IPT
344.	DGT(4,4)	DGt4_4	Relative taps of air, bled from boosters discharge, to nozzle vanes of IPT
345.	DGT(5,4)	DGt5_4	Relative taps of air, bled from IPC (1 point), to nozzle vanes of IPT
346.	DGT(6,4)	DGt6_4	Relative taps of air, bled from IPC (2 point), to nozzle vanes of IPT
347.	DGT(7,4)	DGt7_4	Relative taps of air, bled from IPC discharge, to nozzle vanes of IPT
348.	DGT(8,4)	DGt8_4	Relative taps of air, bled from HPC (1 point), to nozzle vanes of IPT
349.	DGT(9,4)	DGt9_4	Relative taps of air, bled from HPC (2 point), to nozzle vanes of IPT
350.	DGT(10,4)	DGt10_4	Relative taps of air, bled from HPC discharge, to nozzle vanes of IPT
351.	DGT(1,5)	DGt1_5	Relative taps of air, bled from fan (1 point), to impeller of IPT
352.	DGT(2,5)	DGt2_5	Relative taps of air, bled from fan (2 point), to impeller of IPT
353.	DGT(3,5)	DGt3_5	Relative taps of air, bled from fan discharge, to impeller of IPT
354.	DGT(4,5)	DGt4_5	Relative taps of air, bled from boosters discharge, to impeller of IPT
355.	DGT(5,5)	DGt5_5	Relative taps of air, bled from IPC (1 point), to impeller of IPT
356.	DGT(6,5)	DGt6_5	Relative taps of air, bled from IPC (2 point), to impeller of IPT
357.	DGT(7,5)	DGt7_5	Relative taps of air, bled from IPC discharge, to impeller of IPT
358.	DGT(8,5)	DGt8_5	Relative taps of air, bled from HPC (1 point), to impeller of IPT
359.	DGT(9,5)	DGt9_5	Relative taps of air, bled from HPC (2 point), to impeller of IPT
360.	DGT(10,5)	DGt10_5	Relative taps of air, bled from HPC discharge, to impeller of IPT
361.	DGT(1,6)	DGt1_6	Relative taps of air, bled from fan (1 point), to IPT discharge
362.	DGT(2,6)	DGt2_6	Relative taps of air, bled from fan (2 point), to IPT discharge
363.	DGT(3,6)	DGt3_6	Relative taps of air, bled from fan discharge, to IPT discharge
364.	DGT(4,6)	DGt4_6	Relative taps of air, bled from boosters discharge, to IPT

Continuation of Appendix E

1	2	3	4
			discharge
365.	DGT(5,6)	DGt5_6	Relative taps of air, bled from IPC (1 point), to IPT discharge
366.	DGT(6,6)	DGt6_6	Relative taps of air, bled from IPC (2 point), to IPT discharge
367.	DGT(7,6)	DGt7_6	Relative taps of air, bled from IPC discharge, to IPT discharge
368.	DGT(8,6)	DGt8_6	Relative taps of air, bled from HPC (1 point), to IPT discharge
369.	DGT(9,6)	DGt9_6	Relative taps of air, bled from HPC (2 point), to IPT discharge
370.	DGT(10,6)	DGt10_6	Relative taps of air, bled from HPC discharge, to IPT discharge
371.	DGT(1,7)	DGt1_7	Relative taps of air, bled from fan (1 point), to nozzle vanes of LPT
372.	DGT(2,7)	DGt2_7	Relative taps of air, bled from fan (2 point), to nozzle vanes of LPT
373.	DGT(3,7)	DGt3_7	Relative taps of air, bled from fan discharge, to nozzle vanes of LPT
374.	DGT(4,7)	DGt4_7	Relative taps of air, bled from boosters discharge, to nozzle vanes of LPT
375.	DGT(5,7)	DGt5_7	Relative taps of air, bled from IPC (1 point), to nozzle vanes of LPT
376.	DGT(6,7)	DGt6_7	Relative taps of air, bled from IPC (2 point), to nozzle vanes of LPT
377.	DGT(7,7)	DGt7_7	Relative taps of air, bled from IPC discharge, to nozzle vanes of LPT
378.	DGT(8,7)	DGt8_7	Relative taps of air, bled from HPC (1 point), to nozzle vanes of LPT
379.	DGT(9,7)	DGt9_7	Relative taps of air, bled from HPC (2 point), to nozzle vanes of LPT
380.	DGT(10,7)	DGt10_7	Relative taps of air, bled from HPC discharge, to nozzle vanes of LPT
381.	DGT(1,8)	DGt1_8	Relative taps of air, bled from fan (1 point), to impeller of LPT
382.	DGT(2,8)	DGt2_8	Relative taps of air, bled from fan (2 point), to impeller of LPT
383.	DGT(3,8)	DGt3_8	Relative taps of air, bled from fan discharge, to impeller of LPT
384.	DGT(4,8)	DGt4_8	Relative taps of air, bled from boosters discharge, to impeller of LPT
385.	DGT(5,8)	DGt5_8	Relative taps of air, bled from IPC (1 point), to impeller of LPT
386.	DGT(6,8)	DGt6_8	Relative taps of air, bled from IPC (2 point), to impeller of LPT
387.	DGT(7,8)	DGt7_8	Relative taps of air, bled from IPC discharge, to impeller of LPT
388.	DGT(8,8)	DGt8_8	Relative taps of air, bled from HPC (1 point), to impeller of LPT

Continuation of Appendix E

1	2	3	4
389.	DGT(9,8)	DGt9_8	Relative taps of air, bled from HPC (2 point), to impeller of LPT
390.	DGT(10,8)	DGt10_8	Relative taps of air, bled from HPC discharge, to impeller of LPT
391.	DGT(1,9)	DGt1_9	Relative taps of air, bled from fan (1 point), to LPT discharge
392.	DGT(2,9)	DGt2_9	Relative taps of air, bled from fan (2 point), to LPT discharge
393.	DGT(3,9)	DGt3_9	Relative taps of air, bled from fan discharge, to LPT discharge
394.	DGT(4,9)	DGt4_9	Relative taps of air, bled from boosters discharge, to LPT discharge
395.	DGT(5,9)	DGt5_9	Relative taps of air, bled from IPC (1 point), to LPT discharge
396.	DGT(6,9)	DGt6_9	Relative taps of air, bled from IPC (2 point), to LPT discharge
397.	DGT(7,9)	DGt7_9	Relative taps of air, bled from IPC discharge, to LPT discharge
398.	DGT(8,9)	DGt8_9	Relative taps of air, bled from HPC (1 point), to LPT discharge
399.	DGT(9,9)	DGt9_9	Relative taps of air, bled from HPC (2 point), to LPT discharge
400.	DGT(10,9)	DGt10_9	Relative taps of air, bled from HPC discharge, to LPT discharge
401.	DGT(1,10)	DGt1_10	Relative taps of air, bled from fan (1 point), to afterburner
402.	DGT(2,10)	DGt2_10	Relative taps of air, bled from fan (2 point), to afterburner
403.	DGT(3,10)	DGt3_10	Relative taps of air, bled from fan discharge, to afterburner
404.	DGT(4,10)	DGt4_10	Relative taps of air, bled from boosters discharge, to afterburner
405.	DGT(5,10)	DGt5_10	Relative taps of air, bled from IPC (1 point), to afterburner
406.	DGT(6,10)	DGt6_10	Relative taps of air, bled from IPC (2 point), to afterburner
407.	DGT(7,10)	DGt7_10	Relative taps of air, bled from IPC discharge, to afterburner
408.	DGT(8,10)	DGt8_10	Relative taps of air, bled from HPC (1 point), to afterburner
409.	DGT(9,10)	DGt9_10	Relative taps of air, bled from HPC (2 point), to afterburner
410.	DGT(10,10)	DGt10_10	Relative taps of air, bled from HPC discharge, to afterburner
411.	DGT(1,11)	DGt1_11	Relative taps of air, bled from fan (1 point), to common nozzle
412.	DGT(2,11)	DGt2_11	Relative taps of air, bled from fan (2 point), to common nozzle
413.	DGT(3,11)	DGt3_11	Relative taps of air, bled from fan discharge, to common nozzle
414.	DGT(4,11)	DGt4_11	Relative taps of air, bled from boosters discharge, to common nozzle
415.	DGT(5,11)	DGt5_11	Relative taps of air, bled from IPC (1 point), to common nozzle
416.	DGT(6,11)	DGt6_11	Relative taps of air, bled from IPC (2 point), to common nozzle
417.	DGT(7,11)	DGt7_11	Relative taps of air, bled from IPC discharge, to common nozzle

Continuation of Appendix E

1	2	3	4
418.	DGT(8,11)	DGt8_11	Relative taps of air, bled from HPC (1 point), to common nozzle
419.	DGT(9,11)	DGt9_11	Relative taps of air, bled from HPC (2 point), to common nozzle
420.	DGT(10,11)	DGt10_11	Relative taps of air, bled from HPC discharge, to common nozzle
421.	Time	Time	Time (Operation mode number in static analysis)
422.	ALRUD	Acl	Power control lever position
423.	PKVDST	Phpc_st	Static pressure at HPC discharge
424.	PKSM	Пctot	Total compressor pressure ratio
425.	GTCHAS	Gf_kg/h	Fuel consumption in main combustion chamber (kg/h)
426.	GTPR	Gf_ cor	Corrected to standard atmosphere fuel consumption
427.	CETEH	Ce	Specific fuel consumption, kg/(kgf·h)
428.	R1	Rn1	Primary nozzle thrust
429.	R2	Rn2	Secondary nozzle thrust
430.	RSUM	Rtot	Total thrust, N
431.	RSUMKG	Rtot_kg	Total thrust, kgf
432.	RUDEL	Rsp	Specific thrust
433.	NSKND	n'LPR	Time derivative of LP rotor rotational speed
434.	NSKSD	n'IPR	Time derivative of IP rotor rotational speed
435.	NSKVD	n'HPR	Time derivative of HP rotor rotational speed
436.	NSKNDP0	n'LPRcor	Corrected time derivative of LP rotor rotational speed
437.	NSKVDP0	n'HPRcor	Corrected time derivative of HP rotor rotational speed
438.	PKSST	Pic_tot	Total pressure ratio (experimental static pressure at HPC outlet/static pressure at engine inlet station)
439.	ISBROS	Sdecel	Deceleration indicator (mode parameter diminution)
440.	MKNDLS	N_LPC_hp	LPC power, hp
441.	MKSDLS	N_IPC_hp	IPC power, hp
442.	MKVDLS	N_HPC_hp	HPC power, hp
443.	MTNDLS	N_LPT_hp	LPT power, hp
444.	MTSDLS	N_IPT_hp	IPT power, hp
445.	MTVDLS	N_HPT_hp	HPT power, hp
446.	ATVTEH	A_HPT*	Corrected gas flow rate in HPT (technical system)
447.	ATSTEH	A_IPT*	Corrected gas flow rate in IPT (technical system)
448.	ATNTEH	A_LPT*	Corrected gas flow rate in LPT (technical system)
449.	IPRPPS	Srest_st	Restarting from booster indicator
450.	P16IZM	P_HPC_me	Measured pressure value at HPC discharge station
451.	GPRS2	G_N2_cor	Corrected secondary nozzle gas flow rate
452.	GPRS1	G_N1_cor	Corrected primary nozzle gas flow rate
453.	GPRSSM	G_N__cor	Corrected common nozzle gas flow rate
454.	LAMKSM	LAMmix	Mixer superficial velocity
455.	PS12ST	P_mix_st	Static pressure at mixing chamber inlet
456.	DKUKND	dKU_LPC	LPC stability factor
457.	DKUKSD	dKU_IPC	IPC stability factor
458.	DKUKVD	dKU_HPC	HPC stability factor
459.	DKUPS	dKU_B	Booster stability factor
460.	DKUSTN	dKundlrr	LPC stability factor relative to operating line
461.	DKUSTS	dKucdlrr	IPC stability factor relative to operating line
462.	DKUSTV	dKuvdlrr	HPC stability factor relative to operating line

Continuation of Appendix E

1	2	3	4
463.	DIACTR	Ncon_ACS	ACS active controller number
464.	KPIV	KPI_F	Fan pressure ratio correction adjustment
465.	KKPDV	KEFF_F	Fan efficiency correction adjustment
466.	PITSUM	Pi_T_tot	Total turbine pressure ratio
467.	IPRPKS	Sbleed_IPC	IPC bleed indicator
468.	IPRPKV	Sbleed_HPC	HPC bleed indicator
469.	IVNAKV	S_HPC_IGV	HPC IGV reposition indicator (needed for dynamics)
470.	VTP	V_TC	Velocity at temperature probe position
471.	TTERMP	T_TP	Temperature measured by temperature probe
472.	EPSEL	Eps_TC	Temperature probe time constant
473.	TIME3	Time3	Time increment Dt, s
474.	NSDPR0	N_IPcor0	IP rotor rotational speed corrected to engine inlet
475.	NVDPR0	N_HPcor0	HP rotor rotational speed corrected to engine inlet
476.	TTSPR0	T_IPcor0	Inlet corrected gas temperature over IP rotor
477.	TTNPR0	T_LPcor0	Inlet corrected gas temperature over LP rotor
478.	TGPR0	Tg_cor0	Inlet Corrected gas temperature over LP rotor
479.	NUMUCH	N_CM	Compressor mechanization segment number in static model
480.	MKRTND	Mtorq_FT	Free turbine torsion torque
481.	PKVDOT	P_HPC_abs	Excessive pressure at HPC discharge station
482.	PVHATM	Pin_atm	Ambient pressure, kg/sm^2 (technical system)
483.	TVHC	Tin_C	Ambient temperature, °C
484.	T19C	T19_C	Temperature at CC discharge, °C
485.	T29C	T29_C	Temperature, at impeller of IPC discharge, °C
486.	MTNDKW	N_FT_kW	Free turbine power, kW
487.	PKVTEH	P16tech	Pressure at HPC discharge, kgf
488.	KPDEF	EFF_ef	Effective efficiency
489.	DTHAR	dT_CC_rel	Relative temperature change in combustion chamber
490.	TKSM	DTsh	Shift coefficient
491.	SKOL	S0	Low pressure rotor slip
492.	SKOL1	S01	Rotors slip (ratio)
493.	PIKSUM	П*Cst	Total pressure ratio(calculated as Phpc.st/P*in)
494.	IPRPKN		Sign of air bleeding form LPC
		TJE-1	
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG41	MG_N	Mass of gas, stored in transition duct of nozzle
536.	TLTVD	TBHPT	HPT blade temperature
537.	TSTTVD	TCHPT	HPT case temperature
538.	TDTVD	TDHPT	HPT disk temperature
539.	DELTVD	delHPT	Dynamic clearance of HPT
		TJE-2	
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG41	MG_N	Mass of gas, stored in transition duct of nozzle

Continuation of Appendix E

1	2	3	4
536.	TLTVD	TBHPT	HPT blade temperature
537.	TSTTVD	TDHPT	HPT case temperature
538.	TDTVD	TDHPT	HPT disk temperature
539.	TLTSD	TBIPT	IPT blade temperature
540.	TSTTSD	TCIPT	IPT case temperature
541.	TDTSD	TDIPT	IPT disk temperature
542.	DELTVD	delHPT	Dynamic clearance of HPT
543.	DELTSD	delIPT	Dynamic clearance of IPT
		TFE-2	
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG37	MG_NI	Mass of gas, stored in transition duct of primary nozzle
536.	MG6	MG_NII	Mass of gas, stored in transition duct of secondary nozzle
537.	TLTVD	TBHPT	HPT blade temperature
538.	TSTTVD	TCHPT	HPT case temperature
539.	TDTVD	TDHPT	HPT disk temperature
540.	TLTND	TBLPT	LPT blade temperature
541.	TSTTND	TCLPT	LPT case temperature
542.	TDTND	TDLPT	LPT disk temperature
543.	DELTVD	delHPT	Dynamic clearance of HPT
544.	DELTND	delLPT	Dynamic clearance of LPT
		MTFE-2	
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG37	MG_NI	Mass of gas, stored in transition duct of primary nozzle
536.	MG6	MG_NII	Mass of gas, stored in transition duct of secondary nozzle
537.	MG41	MG_N	Mass of gas, stored in transition duct of common nozzle
538.	TLTVD	TBHPC	HPT blade temperature
539.	TSTTVD	TCHPC	HPT case temperature
540.	TDTVD	TDHPT	HPT disk temperature
541.	TLTND	TBLPT	LPT blade temperature
542.	TSTTND	TCLPT	LPT case temperature
543.	TDTND	TDLPT	LPT disk temperature
544.	DELTVD	delHPT	Dynamic clearance of HPT
545.	DELTND	delLPT	Dynamic clearance of LPT
		TFE-3	
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG37	MG_NI	Mass of gas, stored in transition duct of primary nozzle
536.	MG6	MG_NII	Mass of gas, stored in transition duct of secondary nozzle
537.	TLTVD	TBHPT	HPT blade temperature
538.	TSTTVD	TCHPT	HPT case temperature

Continuation of Appendix E

1	2	3	4
539.	TDTVD	TDHPT	HPT disk temperature
540.	TLTSD	TBIPT	IPT blade temperature
541.	TSTTSD	TCIPT	IPT case temperature
542.	TDTSD	TDIPT	IPT disk temperature
543.	TLTND	TBLPT	LPT blade temperature
544.	TSTTND	TCLPT	LPT casing temperature
545.	TDTND	TDLPT	LPT disk temperature
546.	DELTVD	delHPT	Dynamic clearance of HPT
547.	DELTVD	delIPT	Dynamic clearance of IPT
548.	DELTND	delLPT	Dynamic clearance of LPT
MTFE-3			
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG6	MG_NII	Mass of gas, stored in transition duct of primary nozzle
536.	MG41	MG_N	Mass of gas, stored in transition duct of secondary nozzle
537.	TLTVD	TBHPT	HPT blade temperature
538.	TSTTVD	TCHPT	HPT case temperature
539.	TDTVD	TDHPT	HPT disk temperature
540.	TLTSD	TBIPT	IPT blade temperature
541.	TSTTSD	TCIPT	IPT case temperature
542.	TDTSD	TDIPT	IPT disk temperature
543.	TLTND	TBLPT	LPT blade temperature
544.	TSTTND	TCLPT	LPT casing temperature
545.	TDTND	TDLPT	LPT disk temperature
546.	DELTVD	delHPT	Dynamic clearance of HPT
547.	DELTVD	delIPT	Dynamic clearance of IPT
548.	DELTND	delLPT	Dynamic clearance of LPT
TShE-1			
531.	ALVNAN	ALigv_l	Installation angle of LPC IGV
532.	ALVNAS	ALigv_i	Installation angle of IPC IGV
533.	ALVNAV	ALigv_h	Installation angle of HPC IGV
534.	MG19	m_CC	Stored in CC mass of gas
535.	MG41	MG_N	Mass of gas, stored in transition duct of nozzle
536.	TLTVD	TBHPT	HPT blade temperature
537.	TSTTVD	TCHPT	HPT case temperature
538.	TDTVD	TDHPT	HPT disk temperature
539.	TLTND	TBLPT	LPT blade temperature
540.	TSTTND	TCLPT	LPT case temperature
541.	TDTND	TDLPT	LPT disk temperature
542.	DELTVD	delHPT	Dynamic clearance of HPT
543.	DELTND	delLPT	Dynamic clearance of LPT
TShE-2			
544.	ALVNAN	ALigv_l	Installation angle of LPC IGV
545.	ALVNAS	ALigv_i	Installation angle of IPC IGV
546.	ALVNAV	ALigv_h	Installation angle of HPC IGV
547.	MG19	m_CC	Stored in CC mass of gas

Continuation of Appendix E

1	2	3	4
548.	MG41	MG_N	Mass of gas, stored in transition duct of nozzle
549.	TLTVD	TBHPT	HPT blade temperature
550.	TSTTVD	TCHPT	HPT case temperature
551.	TDTVD	TDHPT	HPT disk temperature
552.	TLTSD	TBIPT	IPT blade temperature
553.	TSTTSD	TCIPT	IPT case temperature
554.	TDTSD	TDIPT	IPT disk temperature
555.	TLTND	TBLPT	LPT blade temperature
556.	TSTTND	TCLPT	LPT casing temperature
557.	TDTND	TDLPT	LPT disk temperature
558.	DELTVD	delHPT	Dynamic clearance of HPT
559.	DELTVD	delIPT	Dynamic clearance of IPT
560.	DELTND	delLPT	Dynamic clearance of LPT
COMMON			
601.	RGPR(1)	G__1_cor	Fan inlet corrected gas flow rate
602.	RGPR(2)	G__2_cor	Corrected air flow rate at fan air bleed station
603.	RGPR(3)	G__3_cor	Fan discharge corrected gas flow rate in bypass duct
604.	RGPR(4)	G__4_cor	Corrected air flow rate after air bleed from LPC in bypass duct
605.	RGPR(5)	G__5_cor	Corrected gas flow rate in bypass duct at afterburner inlet
606.	RGPR(6)	G__6_cor	Corrected gas flow rate in bypass duct at mixer inlet
607.	RGPR(7)	G__7_cor	Fan discharge corrected air flow rate in primary duct
608.	RGPR(8)	G__8_cor	Booster corrected air flow rate
609.	RGPR(9)	G__9_cor	Corrected air flow rate after booster air bleed
610.	RGPR(10)	G__10_cor	IPC inlet corrected air flow rate
611.	RGPR(11)	G__11_cor	Corrected air flow rate at air bleed from IPC
612.	RGPR(12)	G__12_cor	IPC discharge corrected air flow rate
613.	RGPR(13)	G__13_cor	Corrected air flow rate at air bleed from IPC
614.	RGPR(14)	G__14_cor	HPC inlet corrected air flow rate
615.	RGPR(15)	G__15_cor	Corrected air flow rate at 2nd air bleed from HPC
616.	RGPR(16)	G__16_cor	Corrected air flow rate discharge pressure
617.	RGPR(17)	G__17_cor	Combustion chamber inlet corrected air flow rate
618.	RGPR(18)	G__18_cor	Corrected gas flow rate at gas bleed from combustion chamber
619.	RGPR(19)	G__19_cor	Combustion chamber discharge corrected gas flow rate
620.	RGPR(20)	G__20_cor	HPT nozzle vanes inlet corrected gas flow rate
621.	RGPR(21)	G__21_cor	HPT nozzle vanes discharge corrected gas flow rate
622.	RGPR(22)	G__22_cor	HPT impeller inlet corrected gas flow rate
623.	RGPR(23)	G__23_cor	HPT impeller discharge corrected gas flow rate
624.	RGPR(24)	G__24_cor	HPT discharge corrected gas flow rate after air input
625.	RGPR(25)	G__25_cor	Corrected gas flow rate at transition duct between HPT and IPT discharge
626.	RGPR(26)	G__26_cor	IPT nozzle vanes inlet corrected gas flow rate
627.	RGPR(27)	G__27_cor	IPT nozzle vanes discharge corrected gas flow rate
628.	RGPR(28)	G__28_cor	IPT impeller inlet corrected gas flow rate
629.	RGPR(29)	G__29_cor	IPT impeller discharge corrected gas flow rate
630.	RGPR(30)	G__30_cor	IPT discharge corrected gas flow rate after air input
631.	RGPR(31)	G__31_cor	Corrected gas flow rate at transition duct between IPT and LPT discharge

Continuation of Appendix E

1	2	3	4
632.	RGPR(32)	G__32_cor	LPT nozzle vanes inlet corrected gas flow rate
633.	RGPR(33)	G__33_cor	LPT nozzle vanes discharge corrected gas flow rate
634.	RGPR(34)	G__34_cor	LPT impeller inlet corrected gas flow rate
635.	RGPR(35)	G__35_cor	LPT impeller discharge corrected gas flow rate
636.	RGPR(36)	G__36_cor	LPT discharge corrected gas flow rate after air input
637.	RGPR(37)	G__37_cor	Nozzle discharge corrected gas flow rate
638.	RGPR(38)	G__38_cor	Mixer discharge corrected gas flow rate
639.	RGPR(39)	G__39_cor	Common afterburner corrected gas flow rate
640.	RGPR(40)	G__40_cor	Corrected gas flow rate before gas bleed from common afterburner
641.	RGPR(41)	G__41_cor	Common nozzle inlet corrected gas flow rate
642.	RGPR(42)	G__42_cor	Common nozzle discharge corrected gas flow rate
643.	RGPR(43)	G__43_cor	Corrected air flow rate at 1st air bleed from HPC
644.	RGPR(44)	G__44_cor	Corrected air flow rate before HPC ABV
645.	RGPR(45)	G__45_cor	Corrected air flow rate before booster ABV
646.	RGPR(46)	G__46_cor	Corrected air flow rate (reserve)
647.	RGPR(47)	G__47_cor	Corrected air flow rate (reserve)
648.	RGPR(48)	G__48_cor	Corrected air flow rate (reserve)
649.	RGPR(49)	G__49_cor	Corrected air flow rate (reserve)
650.	RGPR(50)	G__50_cor	Corrected air flow rate (reserve)

APPENDIX F
Description of main variables

Identifier	Name of parameter
1	2
AIDENT.FOR	
A(160)	Array of corrections to performances of units (see Appendix G), size of array is 160
AREG.FOR	
ALREG(160)	Array of control coefficients (see the description of identification $\alpha_{contr\ i}$), size of array is 160
BALANS.FOR	
HP(KNEV)	Double precision vector of independent variables, size of vector is KNEV. KNEV – number of equations depends on engine type (the list of equations is described in subsection 6.1.2.10)
ZP(KNEV)	Double precision vector of residuals, size of vector is KNEV. KNEV – number of equations depends on engine type (the list of equations is described in subsection 6.1.2.10)
VAR(KNEV)	Double precision vector of independent variables, size of vector is KNEV. KNEV – number of equations depends on engine type (the list of equations is described in subsection 6.1.2.10)
CM(KNEV)	Interim values of parameters that comprise vector of residuals. Residuals appear because of varying independent variables. Size of vector is KNEV. KNEV – number of equations depends on engine type (the list of equations is described in subsection 6.1.2.10)
DB(KNEV, KNEV)	Jacobian matrix, size of matrix is [KNEV, KNEV]. KNEV – number of equations depends on engine type (the list of equations is described in subsection 6.1.2.10)
EPSR	Solution tolerance of equations set
SG	Deviation of independent variables
MAXIT	Maximum number of iterations in Newton method
CONTRL.FOR	
CNTL(3)	Boolean parameter, which is used to control outputting of diagnostic information, CNTL(3)=.FALSE. – including tables of output SITHAR, CNTL(3)=.TRUE. – excluding tables of output SITHAR,
CTRLD.FOR	
KREG	Number of operating modes
TYPEIN	Identifier to storage the choice among sets of independent variables (see subsection 1.1.2.4)
ENV1(99)	Array of time values (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV2(99)	Array of first parameter that characterizes ambient conditions (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV3(99)	Array of second parameter that characterizes ambient conditions (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV4(99)	Array of third parameter that characterizes ambient conditions (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV5(99)	Array of first control parameter (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV6(99)	Array of second control parameter (inputting the values is described in subsection 6.1.3), maximum number is 99

Continuation of Appendix F

1	2
ENV7(99)	Array of third control parameter (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV8(99)	Array of fourth control parameter (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV9(99)	Array of first additional external action (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV10(99)	Array of second additional external action (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV11(99)	Array of third additional external action (inputting the values is described in subsection 6.1.3), maximum number is 99
ENV12(99)	Array of fourth additional external action (inputting the values is described in subsection 6.1.3), maximum number is 99
DINROT.FOR	
NNDDIN	LPR rotational speed, calculated by solving differential equation 1.64
NSDDIN	IPR rotational speed, calculated by solving differential equation **Ошибка! Источник ссылки не найден.**
NVDDIN	HPR rotational speed, calculated by solving differential equation 1.65
DINupr.FOR	
uprdin(10)	Sign that identifies using of dynamic factors, it may possess the values: 0 – there are no additional dynamic factors; 1 – additional factor is included. uprdin(1) – taking into account the compressibility of gas in CC; uprdin(2) – taking into account the compressibility of gas in common nozzle; uprdin(3) – taking into account the compressibility of gas in transient duct of secondary nozzle; uprdin(4) – taking into account the compressibility of gas in transient duct between turbine and nozzle discharge; uprdin(5) – taking into account heat exchange between gas and SE of HPT; uprdin(6) – taking into account heat exchange between gas and SE of IPT; uprdin(7) – taking into account heat exchange between gas and SE of LPT; uprdin(8) – taking into account the presence of clearance in HPT; uprdin(9) – taking into account the presence of clearance in IPT; uprdin(10) – taking into account the presence of clearance in LPT.
MG19DIN	Stored in CC mass of gas, which was calculated by solving the differential equation (1.66)
T19DIN	Temperature at CC discharge, which was calculated by solving the differential equation (1.69)
MG41DIN	Mass of gas, which was stored in transition duct of common nozzle; this mass was calculated by solving the differential equation (1.68)
T41DIN	Temperature at common nozzle discharge, which was calculated by solving the differential equation (1.71)
MG37DIN	Mass of gas, which was stored in transition duct of primary nozzle; this mass was calculated by solving the differential equation
T37DIN	Temperature at primary nozzle discharge, which was calculated by solving the differential equation
MG6DIN	Mass of gas, which was stored in transition duct of secondary nozzle; this mass was calculated by solving the differential equation (1.67)
T6DIN	Temperature at secondary nozzle discharge, which was calculated by solving the differential equation (1.70)
TSTTVDDIN	Temperature of HPT stator, which was calculated by solving the differential

1	2
	equation (1.72)

Continuation of Appendix F

1	2
TLTVDDIN	Temperature of HPT blade, which was calculated by solving the differential equation (1.73)
TDTVDDIN	Temperature of HPT disk, which was calculated by solving the differential equation (1.74)
KOFTVD	Coefficient that shows the impact of clearance in HPT on efficiency of HPT (see the equation (1.38))
LBLTVD	L:ength of HPT blade
TSTTSDDIN	Temperature of IPT stator, which was calculated by solving the differential equation (**Ошибка! Источник ссылки не найден.**)
TLTSDDIN	Temperature of IPT blade, which was calculated by solving the differential equation (**Ошибка! Источник ссылки не найден.**)
TDTSDDIN	Temperature of IPT disk, which was calculated by solving the differential equation (**Ошибка! Источник ссылки не найден.**)
KOFTSD	Coefficient that shows the impact of clearance in IPT on efficiency of IPT (see the equation (1.38))
LBLTSD	Length of IPT blade
TSTTNDDIN	Temperature of LPT stator, which was calculated by solving the differential equation (1.75)
TLTNDDIN	Temperature of LPT blade, which was calculated by solving the differential equation (1.76)
TDTNDDIN	Temperature of LPT disk, which was calculated by solving the differential equation (1.77)
KOFTND	Coefficient that shows the impact of clearance in LPT on efficiency of LPT (see the equation (1.38))
LBLTND	Length of LPT blade
VKS	Volume of CC
VC	Volume of transition duct in common nozzle
VC1	Volume of transition duct in primary nozzle
VC2	Volume of transition duct in secondary nozzle
DOPDINFACT.FOR	
MSTTND	Mass of LPT stator
MLTND	Mass of LPT blade
MDTND	Mass of LPT disk
MSTTSD	Mass of IPT stator
MLTSD	Mass of IPT blade
MDTSD	Mass of IPT disk
MSTTVD	Mass of HPT stator
MLTVD	Mass of HPT blade
MDTVD	Mas of HPT disk
ALSTTND	Heat transfer coefficient on the surface of LPT stator, which is flown by hot gas
ALLTND	Heat transfer coefficient on the surface of LPT blade, which is flown by hot gas
ALZTND	Heat transfer coefficient in lock part of LPT
ALSTTSD	Heat transfer coefficient on the surface of IPT stator, which is flown by hot gas
ALLTSD	Heat transfer coefficient on the surface of IPT blade, which is flown by hot gas

1	2
ALZTSD	Heat transfer coefficient in lock part of IPT
ALSTTVD	Heat transfer coefficient on the surface of HPT stator, which is flown by hot gas

Continuation of Appendix F

1	2
ALLTVD	Heat transfer coefficient on the surface of HPT blade, which is flown by hot gas
ALZTVD	Heat transfer coefficient in lock part of HPT
FSTTND	Surface of LPT stator, which is flown by hot gas
FLTND	Surface of LPT blade, which is flown by hot gas
FZTND	Surface of contact heat exchange in lock part of LPT
FSTTSD	Surface of IPT stator, which is flown by hot gas
FLTSD	Surface of IPT blade, which is flown by hot gas
FZTSD	Surface of contact heat exchange in lock part of IPT
FSTTVD	Surface of HPT stator, which is flown by hot gas
FLTVD	Surface of HPT blade, which is flown by hot gas
FZTVD	Surface of contact heat exchange in lock part of HPT
ALSTVTND	Heat transfer coefficient on the surface of LPT stator, which is flown by cooling air
ALLVTND	Heat transfer coefficient on the surface of LPT blade, which is flown by cooling air
ALZVTND	Heat transfer coefficient on the surface of LPT disk, which is flown by cooling air
ALSTVTSD	Heat transfer coefficient on the surface of IPT stator, which is flown by cooling air
ALLVTSD	Heat transfer coefficient on the surface of IPT blade, which is flown by cooling air
ALZVTSD	Heat transfer coefficient on the surface of IPT disk, which is flown by cooling air
ALSTVTVD	Heat transfer coefficient on the surface of HPT stator, which is flown by cooling air
ALLVTVD	Heat transfer coefficient on the surface of HPT blade, which is flown by cooling air
ALZVTVD	Heat transfer coefficient on the surface of HPT disk, which is flown by cooling air
FSTVTND	Surface of LPT stator, which is flown by cooling air
FLVTND	Surface of LPT blade, which is flown by cooling air
FZVTND	Surface of LPT disk, which is flown by cooling air
FSTVTSD	Surface of IPT stator, which is flown by cooling air
FLVTSD	Surface of IPT blade, which is flown by cooling air
FZVTSD	Surface of IPT disk, which is flown by cooling air
FSTVTVD	Surface of HPT stator, which is flown by cooling air
FLVTVD	Surface of HPT blade, which is flown by cooling air
FZVTVD	Surface of HPT disk, which is flown by cooling air
CSTTND	Specific heat of LPT stator
CLTND	Specific heat of LPT blade
CDTND	Specific heat of LPT disk
CSTTSD	Specific heat of IPT stator
CLTSD	Specific heat of IPT blade
CDTSD	Specific heat of IPT disk

1	2
CSTTVD	Specific heat of LPT stator
CLTVD	Specific heat of LPT blade
CDTVD	Specific heat of LPT disk

Continuation of Appendix F

1	2
NSTTND	Number (according to general list) that identifies the temperature of air, which is used for cooling stator of LPT
NLTND	Number (according to general list) that identifies the temperature of air, which is used for cooling blade of LPT
NDTND	Number (according to general list) that identifies the temperature of air, which is used for cooling disk of LPT
NSTTSD	Number (according to general list) that identifies the temperature of air, which is used for cooling stator of IPT
NLTSD	Number (according to general list) that identifies the temperature of air, which is used for cooling blade of IPT
NDTSD	Number (according to general list) that identifies the temperature of air, which is used for cooling disk of IPT
NSTTVD	Number (according to general list) that identifies the temperature of air, which is used for cooling stator of HPT
NLTVD	Number (according to general list) that identifies the temperature of air, which is used for cooling blade of HPT
NDTVD	Number (according to general list) that identifies the temperature of air, which is used for cooling disk of HPT
EPSTER.FOR	
EPST	Tolerance of gas-dynamic calculations
FILE.FOR	
VHOD	Text variable, that keeps the name of file with intake performance
KNDO(5)	Array of text variables, that keep names of files with LPC performances with open ABVs (five performances with different installation angles of IGV)
KNDC	Array of text variables, that keep names of files with LPC performances with closed ABVs (five performances with different installation angles of IGV)
RP	Text variable, that keeps the name of file with performance of flow splitter
PER1	Text variable, that keeps the name of file with performance of transition duct between LPC and IPC
PER5	Text variable, that keeps the name of file with performance of transition duct arranged in bypass
KSDO	array of text variables, that keep names of files with IPC performances with open ABVs (five performances with different installation angles of IGV)
KSDC	Array of text variables, that keep names of files with IPC performances with closed ABVs (five performances with different installation angles of IGV)
PER2	Text variable, that keeps the name of file with performance of transition duct between IPC and HPC
KVDO	Array of text variables, that keep names of files with HPC performances with open ABVs (five performances with different installation angles of IGV)
KVDC	Array of text variables, that keep names of files with HPC performances with closed ABVs (five performances with different installation angles of IGV)
KS	Text variable, that keeps the name of file with performance of combustion cham-

1	2
	ber
TVD	Text variable, that keeps the name of file with performance of HPT
PER3	Text variable, that keeps the name of file with performance of transition duct between HPT and IPT
TSD	Text variable, that keeps the name of file with performance of IPT
PER4	Text variable, that keeps the name of file with performance of transition duct between IPT and LPT
Continuation of Appendix F	
TND	Text variable, that keeps the name of file with performance of LPT
SOPL1	Text variable, that keeps the name of file with performance of primary nozzle
SOPL2	Text variable, that keeps the name of file with performance of secondary nozzle
KSM	Text variable, that keeps the name of file with performance of mixer
SOPL	Text variable, that keeps the name of file with performance of common nozzle
OTBOR	Text variable, that keeps the name of file with information about air bleeds and taps, power take-offs and supplies
MEH	Text variable, that keeps the name of file with mechanical properties
PREDEL	Text variable, that keeps the name of file with variation limits of independent variables
NEZP	Text variable, that keeps the name of file with initial approximations of independent variables
TEPL	Text variable, that keeps the name of file with parameters of heat exchange between SE and gas or cooling air
STTVD	Text variable, that keeps the name of file with transient performance of HPT stator
BLTVD	Text variable, that keeps the name of file with transient performance of HPT blade
DTVD	Text variable, that keeps the name of file with transient performance of HPT disk
STTSD	Text variable, that keeps the name of file with transient performance of IPT stator
BLTSD	Text variable, that keeps the name of file with transient performance of IPT blade
DTSD	Text variable, that keeps the name of file with transient performance of IPT disk
STTND	Text variable, that keeps the name of file with transient performance of LPT stator
BLTND	Text variable, that keeps the name of file with transient performance of LPT blade
DTND	Text variable, that keeps the name of file with transient performance of LPT disk
VOLUME	Text variable, that keeps the name of file with characteristic volumes
GAZ.FOR	
R	Gas constant
GLBUPR.FOR	
REGTEK	Number of current operating mode, which is transmitted to the subroutine that forms dialogue windows

Continuation of Appendix F

1	2
KOLR	Number of modes, which is transmitted to the subroutine that forms dialogue windows
MKITER	Maximum number of iterations, which is transmitted to the subroutine that forms dialogue windows
EPSRZ	Tolerance of equations set solution, which is transmitted to the subroutine that forms dialogue windows
ISTOP	Boolean identifier of pressing the button “Break”
IIFF	Boolean identifier, to output interim values in dialogue window
REGP1(9999)	Array of values of first mode parameter to output it in dialogue window; maximum number is 9999
REGP2(9999)	Array of values of second mode parameter to output it in dialogue window; maximum number is 9999
REGP3(9999)	Array of values of third mode parameter to output it in dialogue window; maximum number is 9999
REGP4(9999)	Array of values of fourth mode parameter to output it in dialogue window; maximum number is 9999
IDNTFC.FOR	
IDENTIFY	Sign, which identifies that files for identification have already been formed
INER.FOR	
JND	Inertia of low pressure rotor
JSD	Inertia of intermediate pressure rotor
JVD	Inertia of high pressure rotor
ITMP.FOR	
ITMP(4)	Array with numbers of additional parameter that describe external impact (up to 4 parameters)
KSDAT.FOR	
KTETAG	Number of points in dependence of CC combustion efficiency
KTSIG	Number of points in dependence of CC hydraulic efficiency
NARGET	Number of argument in general list (see Appendix E), which is used in dependence of CC combustion efficiency
NARGSG	Number of argument in general list (see Appendix E), which is used in dependence of CC hydraulic efficiency
SKSARGM(10)	Array of arguments for calculating losses of total pressure in CC
SIGKSM(10)	Array of losses of total pressure in CC
EKSARGM(10)	Array of arguments for calculating CC combustion efficiency
ETKSM(10)	Array of CC combustion efficiency
LDMDM.FOR	
DMNDI	Deviation part of power, which is produced by low pressure rotor, to calculate the coefficients of QLDM (matrixes A and C)
DMSDI	Deviation part of power, which is produced by intermediate pressure rotor, to calculate the coefficients of QLDM (matrixes A and C)
DMVDI	Deviation part of power, which is produced by high pressure rotor, to calculate the coefficients of QLDM (matrixes A and C)
DU	Part of control impact deviation to calculate the coefficients of QLDM (matrixes B and D)
KOLDF	Number of state equations (depends on engine type and number of additional dynamic factors)

Continuation of Appendix F

1	2
NEZP0.FOR	
NEZP0(KNEV)	Array with initial approximations of independent variables, which size is KNEV (KNEV depends on engine type)
OTBORM0.FOR	
DMND0	Constant power take-offs from low pressure rotor, which were set in MEH.DAT
DMSD0	Constant power take-offs from intermediate pressure rotor, which were set in MEH.DAT
DMVD0	Constant power take-offs from high pressure rotor, which were set in MEH.DAT
PARDATA.FOR	
TYPENG	Number of project according to list of engine
COLHAR	Number of files that were included to project
NUMPAR(4)	Number of parameters that describe ambient conditions
KPARDOP	Number of additional parameters for identification
MNPARDOP (600)	Array with numbers of additional parameters for identification
PARGV.FOR	
GVO(10)	Array with airflows, used as reference airflows for calculating the part of bleeding air
PARISH.FOR	
NREGISH	Number of first mode parameter, which initial approximations were set in file NEZPER.DAT
NREG1ISH	Number of second mode parameter, which initial approximations were set in file NEZPER.DAT
NREG2ISH	Number of third mode parameter, which initial approximations were set in file NEZPER.DAT
NREGDISH	Number of fourth mode parameter, which initial approximations were set in file NEZPER.DAT
NREGOLD	Initial number of first mode parameter, which was chosen by User
NREG2OLD	Initial number of second mode parameter, which was chosen by User
NREG1OLD	Initial number of third mode parameter, which was chosen by User
NREGDOLD	Initial number of fourth mode parameter, which was chosen by User
PREGISH	Value of first mode parameter, which initial approximations of independent variables were set in file NEZPER.DAT
PREG1ISH	Value of second mode parameter, which initial approximations of independent variables were set in file NEZPER.DAT
PREG2ISH	Value of third mode parameter, which initial approximations of independent variables were set in file NEZPER.DAT
PREGDISH	Value of fourth mode parameter, which initial approximations of independent variables were set in file NEZPER.DAT
PREGOLD	Initial value of first mode parameter, which was set by User
PREG2OLD	Initial value of second mode parameter, which was set by User

Continuation of Appendix F

1	2
PREG1OLD	Initial value of third mode parameter, which was set by User
PREGDOLD	Initial value of fourth mode parameter, which was set by User
PARMEH.FOR	
PERKND	Value of parameter that characterizes air bleeding through ABVs of LPC
PERKSD	Value of parameter that characterizes air bleeding through ABVs of IPC
PERKVD	Value of parameter that characterizes air bleeding through ABVs of HPC
VARVKND	Value of parameter that controls the position of IGV of LPC
VARVKSD	Value of parameter that controls the position of IGV of IPC
VARVKVD	Value of parameter that controls the position of IGV of HPC
PAROTB.FOR	
KPIOTB(10)	Array of k_π in ten bleeding stations
KKOTB(10)	Array of k_η in ten bleeding stations
PARPERHAR.FOR	
X1TVD(9999)	Array of first CP displacements in the performance of HPT disk
X2TVD(9999)	Array of second CP displacements in the performance of HPT disk
TTVD(9999)	Array of times in the performance of HPT disk
X1TSD(9999)	Array of first CP displacements in the performance of IPT disk
X2TSD(9999)	Array of second CP displacements in the performance of IPT disk
TTSD(9999)	Array of times in the performance of IPT disk
X1TND(9999)	Array of first CP displacements in the performance of LPT disk
X2TND(9999)	Array of second CP displacements in the performance of LPT disk
TTND(9999)	Array of times in the performance of LPT disk
NBASETVD	Temperature, which was used as reference temperature for getting transient performance of HPT disk
NBASESTV	Temperature, which was used as reference temperature for getting transient performance of HPT stator
NBASETVDBL	Temperature, which was used as reference temperature for getting transient performance of HPT blade
NBASETSD	Temperature, which was used as reference temperature for getting transient performance of IPT disk
NBASESTS	Temperature, which was used as reference temperature for getting transient performance of IPT stator
NBASETSDBL	Temperature, which was used as reference temperature for getting transient performance of IPT blade
NBASETND	Temperature, which was used as reference temperature for getting transient performance of LPT disk
NBASESTN	Temperature, which was used as reference temperature for getting transient performance of LPT stator
NBASETNDBL	Temperature, which was used as reference temperature for getting transient performance of LPT blade

Continuation of Appendix F

1	2
U10TVD	Displacement of HPT disk in the first checkpoint at initial steady mode
U20TVD	Displacement of HPT disk in the second checkpoint at initial steady mode
U10TSD	Displacement of IPT disk in the first checkpoint at initial steady mode
U20TSD	Displacement of IPT disk in the second checkpoint at initial steady mode
U10TND	Displacement of LPT disk in the first checkpoint at initial steady mode
U20TND	Displacement of LPT disk in the second checkpoint at initial steady mode
U10STV	Displacement of HPT stator in the first checkpoint at initial steady mode
U20STV	Displacement of HPT stator in the second checkpoint at initial steady mode
U10STS	Displacement of IPT stator in the first checkpoint at initial steady mode
U20STS	Displacement of IPT stator in the second checkpoint at initial steady mode
U10STN	Displacement of LPT stator in the first checkpoint at initial steady mode
U20STN	Displacement of LPT stator in the second checkpoint at initial steady mode
U10TVDBL	Displacement of HPT blade in the first checkpoint at initial steady mode
U20TVDBL	Displacement of HPT blade in the second checkpoint at initial steady mode
U10TSDBL	Displacement of IPT blade in the first checkpoint at initial steady mode
U20TSDBL	Displacement of IPT blade in the second checkpoint at initial steady mode
U10TNDBL	Displacement of LPT blade in the first checkpoint at initial steady mode
U20TNDBL	Displacement of LPT blade in the second checkpoint at initial steady mode
NVDOLDSTV (99999)	Value of parameter that characterizes displacement of HPT stator at the moment of the parameter jump
TIMEOLDSTV (99999)	Times when jumps of parameter, which characterizes the HPT stator displacement, have occurred.
NVDOLDSTS (99999)	Value of parameter that characterizes displacement of IPT stator at the moment of the parameter jump
TIMEOLDSTS (99999)	Times when jumps of parameter, which characterizes the IPT stator displacement, have occurred.
NVDOLDSTN (99999)	Value of parameter that characterizes displacement of LPT stator at the moment of the parameter jump
TIMEOLDSTN (99999)	Times when jumps of parameter, which characterizes the LPT stator displacement, have occurred.

Continuation of Appendix F

1	2
NVDOLDTVD (99999)	Value of parameter that characterizes displacement of HPT disk at the moment of the parameter jump
TIMEOLDTVD (99999)	Times when jumps of parameter, which characterizes the HPT disk displacement, have occurred.
NVDOLDTSD (99999)	Value of parameter that characterizes displacement of IPT disk at the moment of the parameter jump
TIMEOLDTSD (99999)	Times when jumps of parameter, which characterizes the IPT disk displacement, have occurred.
NVDOLDTND (99999)	Value of parameter that characterizes displacement of LPT disk at the moment of the parameter jump
TIMEOLDTND (99999)	Times when jumps of parameter, which characterizes the LPT disk displacement, have occurred.
NVDOLDTVDBL (99999)	Value of parameter that characterizes displacement of PT blade at the moment of the parameter jump
TIMEOLDTVDBL (99999)	Times when jumps of parameter, which characterizes the HPT blade displacement, have occurred.
NVDOLDTSDBL (99999)	Value of parameter that characterizes displacement of IPT blade at the moment of the parameter jump
TIMEOLDTSDBL (99999)	Times when jumps of parameter, which characterizes the IPT blade displacement, have occurred.
NVDOLDTNDBL (99999)	Value of parameter that characterizes displacement of LPT blade at the moment of the parameter jump
TIMEOLDTNDBL (99999)	Times when jumps of parameter, which characterizes the LPT blade displacement, have occurred.
T_ST_V(99999)	Array of times for transient performance of HPT stator
X1_S_TVD (99999)	Array of first CP displacements for transient performance of HPT stator
X2_S_TVD (99999)	Array of second CP displacements for transient performance of HPT stator
T_ST_S(99999)	Array of times for transient performance of IPT stator
X1_S_TSD (99999)	Array of first CP displacements for transient performance of IPT stator
X2_S_TSD (99999)	Array of second CP displacements for transient performance of IPT stator
T_ST_N(99999)	Array of times for transient performance of LPT stator
X1_S_TND (99999)	Array of first CP displacements for transient performance of LPT stator
X2_S_TND (99999)	Array of second CP displacements for transient performance of LPT stator
TTVDBL (99999)	Array of times for transient performance of HPT blade
X1TVDBL(99999)	Array of first CP displacements for transient performance of HPT blade
X2TVDBL(99999)	Array of second CP displacements for transient performance of HPT blade
TTSDBL (99999)	Array of times for transient performance of IPT blade
X1TSDBL(99999)	Array of first CP displacements for transient performance of IPT blade
X2TSDBL(99999)	Array of second CP displacements for transient performance of IPT blade

Continuation of Appendix F

1	2
TTNDBL (99999)	Array of times for transient performance of LPT blade
X1TNDBL(99999)	Array of first CP displacements for transient performance of LPT blade
X2TNDBL(99999)	Array of second CP displacements for transient performance of LPT blade
URTVDT1	Dynamic displacement of HPT disk in the first CP
URTVDT2	Dynamic displacement of HPT disk in the second CP
URTSDT1	Dynamic displacement of IPT disk in the first CP
URTSDT2	Dynamic displacement of IPT disk in the second CP
URTNDT1	Dynamic displacement of LPT disk in the first CP
URTNDT2	Dynamic displacement of LPT disk in the second CP
UBLTVDT1	Dynamic displacement of HPT blade in the first CP
UBLTVDT2	Dynamic displacement of HPT blade in the second CP
UBLTSDT1	Dynamic displacement of IPT blade in the first CP
UBLTSDT2	Dynamic displacement of IPT blade in the second CP
UBLTNDT1	Dynamic displacement of LPT blade in the first CP
UBLTNDT2	Dynamic displacement of LPT blade in the second CP
USTVDT1	Dynamic displacement of HPT stator in the first CP
USTVDT2	Dynamic displacement of HPT stator in the second CP
USTSDT1	Dynamic displacement of IPT stator in the first CP
USTSDT2	Dynamic displacement of IPT stator in the second CP
USTNDT1	Dynamic displacement of LPT stator in the first CP
USTNDT2	Dynamic displacement of LPT stator in the second CP
URTVDT1ST	Static displacement of HPT disk in the first CP
URTVDT2ST	Static displacement of HPT disk in the second CP
URTSDT1ST	Static displacement of IPT disk in the first CP
URTSDT2ST	Static displacement of IPT disk in the second CP
URTNDT1ST	Static displacement of LPT disk in the first CP
URTNDT2ST	Static displacement of LPT disk in the second CP
UBLTVDT1ST	Static displacement of HPT blade in the first CP
UBLTVDT2ST	Static displacement of HPT blade in the second CP
UBLTSDT1ST	Static displacement of IPT blade in the first CP
UBLTSDT2ST	Static displacement of IPT blade in the second CP
UBLTNDT1ST	Static displacement of LPT blade in the first CP
UBLTNDT2ST	Static displacement of LPT blade in the second CP
USTVDT1ST	Static displacement of HPT stator in the first CP
USTVDT2ST	Static displacement of HPT stator in the second CP
USTSDT1ST	Static displacement of IPT stator in the first CP
USTSDT2ST	Static displacement of IPT stator in the second CP
USTNDT1ST	Static displacement of LPT stator in the first CP
USTNDT2ST	Static displacement of LPT stator in the second CP
KOLTVD	Number of points in transient performance of HPT disk
PARITVD	Sign that identifies calculating initial position of HPT disk in the CP
PARISTTVD	Sign that identifies calculating initial position of HPT stator in the CP
METERTVD	Counter for jumps of control actions addition when calculating HPT disk
METERSTV	Counter for jumps of control actions addition when calculating HPT stator

Continuation of Appendix F

1	2
KOL_T_STV	Number of points in transient performance of HPT stator
KOLTVDBL	Number of points in transient performance of HPT blade
METERTVDBL	Counter for jumps of control actions addition when calculating displacements of HPT blade
PARITVDBL	Sign that identifies calculating initial position of HPT blade in the CP
KOLTSD	Number of points in transient performance of IPT disk
KOL_T_STS	Number of points in transient performance of IPT stator
KOLTSDBL	Number of points in transient performance of IPT blade
PARITSD	Sign that identifies calculating initial position of IPT disk in the CP
PARISTTSD	Sign that identifies calculating initial position of IPT stator in the CP
METERTSD	Counter for jumps of control actions addition when calculating IPT disk
METERSTS	Counter for jumps of control actions addition when calculating IPT stator
PARITSDBL	Sign that identifies calculating initial position of IPT blade in the CP
METERTSDBL	Counter for jumps of control actions addition when calculating displacements of IPT blade
KOLTND	Number of points in transient performance of IPT stator
KOL_T_STN	Number of points in transient performance of IPT disk
KOLTNDBL	Number of points in transient performance of IPT blade
PARITND	Sign that identifies calculating initial position of LPT disk in the CP
PARISTTND	Sign that identifies calculating initial position of LPT stator in the CP
METERTND	Counter for jumps of control actions addition when calculating LPT disk
METERSTN	Counter for jumps of control actions addition when calculating LPT stator
PARITNDBL	Sign that identifies calculating initial position of LPT blade in the CP
METERTNDBL	Counter for jumps of control actions addition when calculating displacements of LPT blade
PER1DAT.FOR	
KTPER1	Number of points in arrays for calculating pressure recovery in transient duct between LPC and IPC
NARGPER1	Number of argument, used to calculate pressure recovery in transient duct between LPC and IPC
PER1ARGM(10)	Array of arguments for calculating the pressure recovery in transient duct between LPC and IPC
SIGPER1M(10)	Array of pressure recoveries for calculating the pressure recovery in transient duct between LPC and IPC
PER21DAT.FOR	
KTPER21	Number of points in arrays for calculating pressure recovery in transient duct between IPC and HPC
NARGPER21	Number of argument, used to calculate pressure recovery in transient duct between IPC and HPC
PER21ARGM (10)	Array of arguments for calculating the pressure recovery in transient duct between IPC and HPC
SIGPER21M(10)	Array of pressure recoveries for calculating the pressure recovery in transient duct between IPC and HPC

Continuation of Appendix F

1	2
PER3DAT.FOR	
KTPER3	Number of points in arrays for calculating pressure recovery in transient duct between HPT and IPT
NARGPER3	Number of argument, used to calculate pressure recovery in transient duct between HPT and IPT
PER3ARGM(10)	Array of arguments for calculating the pressure recovery in transient duct between HPT and IPT
SIGPER3M(10)	Array of pressure recoveries for calculating the pressure recovery in transient duct between HPT and IPT
PER4DAT.FOR	
KTPER4	Number of points in arrays for calculating pressure recovery in transient duct between IPT and LPT
NARGPER4	Number of argument, used to calculate pressure recovery in transient duct between IPT and LPT
PER4ARGM(10)	Array of arguments for calculating the pressure recovery in transient duct between IPT and LPT
SIGPER4M(10)	Array of pressure recoveries for calculating the pressure recovery in transient duct between IPT and LPT
PER5DAT.FOR	
KTPER5	Number of points in arrays for calculating pressure recovery in transient duct arranged in bypass
NARGPER5	Number of argument, used to calculate pressure recovery in transient duct arranged in bypass
PER5ARGM(10)	Array of arguments for calculating the pressure recovery in transient duct arranged in bypass
SIGPER5M(10)	Array of pressure recoveries for calculating the pressure recovery in transient duct arranged in bypass
PEREPUSK.FOR	
PIPRPKND	Switching value of parameter, which is used to control position of LPC ABVs
DPIGSTKND	Hysteresis of parameter, which is used to control LPC ABVs opening.
DPILINKND	Linear increment of parameter, which is used to control LPC ABVs opening.
PIPRPKSD	Switching value of parameter, which is used to control position of IPC ABVs
DPIGSTKSD	Hysteresis of parameter, which is used to control IPC ABVs opening.
DPILINKSD	Linear increment of parameter, which is used to control IPC ABVs opening.
PIPRPKVD	Switching value of parameter, which is used to control position of HPC ABVs
DPIGSTKVD	Hysteresis of parameter, which is used to control HPC ABVs opening.
DPILINKVD	Linear increment of parameter, which is used to control HPC ABVs opening.
VNAKND(5)	Array of IGVs installation angles of LPC at given control program
VNAKSD(5)	Array of IGVs installation angles of IPC at given control program
VNAKVD(5)	Array of IGVs installation angles of HPC at given control program
AVNAKND(5)	Array of arguments for calculating IGVs installation angle of LPC at given control program
AVNAKSD(5)	Array of arguments for calculating IGVs installation angle of IPC at given control program
AVNAKVD(5)	Array of arguments for calculating IGVs installation angle of HPC at given control program

Continuation of Appendix F

1	2
NPRPKND	Number of parameter-argument, which is used for controlling ABVs of LPC
NPRPKSD	Number of parameter-argument, which is used for controlling ABVs of IPC
NPRPKVD	Number of parameter-argument, which is used for controlling ABVs of HPC
NVNAKND	Number of parameter-argument, which is used for controlling installation angle of PC IGV
NVNAKSD	Number of parameter-argument, which is used for controlling installation angle of IPC IGV
NVNAKVD	Number of parameter-argument, which is used for controlling installation angle of HPC IGV
KOLPERN	Part of airflow, which is bleed through ABVs of LPC; it was calculated referring the airflow at LPC inlet
KOLPERS	Part of airflow, which is bleed through ABVs of IPC; it was calculated referring the airflow at IPC inlet
KOLPERV	Part of airflow, which is bleed through ABVs of HPC; it was calculated referring the airflow at HPC inlet
KPIPERN	Ratio of pressure at bleeding station in LPC to pressure at LPC discharge
KPIPERS	Ratio of pressure at bleeding station in LPC to pressure at IPC discharge
KPIPERV	Ratio of pressure at bleeding station in HPC to pressure at HPC discharge
KKPDPERN	Ratio of efficiency η_1 to efficiency of LPC spool (η_1 is the efficiency of compressor part from LPC inlet to bleeding station of LPC)
KKPDPERS	Ratio of efficiency η_1 to efficiency of HPC spool (η_1 is the efficiency of compressor part from HPC inlet to bleeding station of HPC)
KKPDPERV	Ratio of efficiency η_1 to efficiency of IPC spool (η_1 is the efficiency of compressor part from IPC inlet to bleeding station of IPC)
PREDEL.FOR	
NVDPMX	Upper limit for varying corrected rotational speed of HPC (independent variable)
NVDPMN	Bottom limit for varying corrected rotational speed of HPC (independent variable)
TGMX	Upper limit for varying temperature at CC discharge (independent variable)
TGMN	Bottom limit for varying temperature at CC discharge (independent variable)
PIVDMX	Upper limit for varying pressure ratio of HPT (independent variable)
PIVDMN	Bottom limit for varying pressure ratio of HPT (independent variable)
NNDMX	Upper limit for varying corrected rotational speed of FT (independent variable)
NNDMN	Bottom limit for varying corrected rotational speed of FT (independent variable)
PINDMX	Upper limit for varying pressure ratio of LPT (independent variable)
PINDMN	Bottom limit for varying pressure ratio of LPT (independent variable)
GTKSMX	Upper limit for varying fuel flow in CC (independent variable)
GTKSMN	Bottom limit for varying fuel flow in CC (independent variable)

Continuation of Appendix F

1	2
PERKVDMN	Upper limit for varying independent variable, which is used to control ABVs of HPC
PERKVDMX	Bottom limit for varying independent variable, which is used to control ABVs of HPC
VARVKVDMN	Upper limit for varying independent variable, which is used to control IGV installation angles of HPC
VARVKVDMX	Bottom limit for varying independent variable, which is used to control IGV installation angles of HPC
NSDPMN	Upper limit for varying corrected rotational speed of IPC (independent variable)
NSDPMX,	Bottom limit for varying corrected rotational speed of IPC (independent variable)
PISDMN	Upper limit for varying pressure ratio of IPT (independent variable)
PISDMX	Bottom limit for varying pressure ratio of IPT (independent variable)
PERKSDMN	Upper limit for varying independent variable, which is used to control ABVs of IPC
PERKSDMX	Bottom limit for varying independent variable, which is used to control ABVs of IPC
VARVKSDMN	Upper limit for varying independent variable, which is used to control IGV installation angles of IPC
VARVKSDMX	Bottom limit for varying independent variable, which is used to control IGV installation angles of IPC
GTFKMX	Upper limit for varying fuel flow in afterburner (independent variable)
GTFKMN	Bottom limit for varying fuel flow in afterburner (independent variable)
PERKNDMN	Upper limit for varying independent variable, which is used to control ABVs of LPC
PERKNDMX	Bottom limit for varying independent variable, which is used to control ABVs of LPC
VARVKNDMX	Upper limit for varying independent variable, which is used to control LGV installation angles of LPC
VARVKNDMN	Bottom limit for varying independent variable, which is used to control LGV installation angles of LPC
NNDPMX	Upper limit for varying corrected rotational speed of LPC (independent variable)
NNDPMN	Bottom limit for varying corrected rotational speed of LPC (independent variable)
MDVMX	Upper limit for varying bypass ratio (independent variable)
MDVMN	Bottom limit for varying bypass ratio (independent variable)
PKSMMX	Upper limit for varying static pressure at mixer discharge (independent variable)
PKSMMN	Bottom limit for varying static pressure at mixer discharge (independent variable)
GGKMN	Upper limit for varying airflow at CC discharge (independent variable)

Continuation of Appendix F

1	2
GGKMX,	Bottom limit for varying airflow at CC discharge (independent variable)
T6MN	Upper limit for varying temperature at secondary nozzle discharge (independent variable)
T6MX	Bottom limit for varying temperature at secondary nozzle discharge (independent variable)
G6MN	Upper limit for varying gas flow at secondary nozzle discharge (independent variable)
G6MX	Bottom limit for varying gas flow at secondary nozzle discharge (independent variable)
T41MN	Upper limit for varying temperature at nozzle discharge (independent variable)
T41MX	Bottom limit for varying temperature at nozzle discharge (independent variable)
G41MN	Upper limit for varying gas flow at nozzle discharge (independent variable)
G41MX	Bottom limit for varying gas flow at nozzle discharge (independent variable)
T37MN	Upper limit for varying temperature at primary nozzle discharge (independent variable)
T37MX	Bottom limit for varying temperature at primary nozzle discharge (independent variable)
G37MN	Upper limit for varying gas flow at primary nozzle discharge (independent variable)
G37MX	Bottom limit for varying gas flow at primary nozzle discharge (independent variable)
TLTVDMX,	Upper limit for varying temperature of HPT blade (independent variable)
TLTVDMN	Bottom limit for varying temperature of HPT blade (independent variable)
TSTTVDMX	Upper limit for varying temperature of HPT stator (independent variable)
TSTTVDMN	Bottom limit for varying temperature of HPT stator (independent variable)
TDTVDMX	Upper limit for varying temperature of HPT disk (independent variable)
TDTVDMN	Bottom limit for varying temperature of HPT disk (independent variable)
TLTSDMX	Upper limit for varying temperature of IPT blade (independent variable)
TLTSDMN	Bottom limit for varying temperature of IPT blade (independent variable)
TSTTSDMX	Upper limit for varying temperature of IPT stator (independent variable)
TSTTSDMN	Bottom limit for varying temperature of IPT stator (independent variable)
TDTSDMX	Upper limit for varying temperature of IPT disk (independent variable)
TDTSDMN	Bottom limit for varying temperature of IPT disk (independent variable)
TLTNDMX	Upper limit for varying temperature of LPT blade (independent variable)
TLTNDMN	Bottom limit for varying temperature of LPT blade (independent variable)
TSTTNDMX	Upper limit for varying temperature of LPT stator (independent variable)
TSTTNDMN	Bottom limit for varying temperature of LPT stator (independent variable)
TDTNDMX	Upper limit for varying temperature of LPT disk (independent variable)
TDTNDMN	Bottom limit for varying temperature of LPT disk (independent variable)

Continuation of Appendix F

1	2
PRINT.FOR	
DTPR	Step of results outputting in dynamics calculation
TPRINT	Last time, at which calculation results were outputted into file for printing
KPAR	Number of parameters to output results for printing
MNPAR(600)	Array with numbers of parameters to output results for printing
NPRINT	Number of file to output calculation results for printing
REGIM.FOR	
NREG	Number of the first mode parameter
NREG1	Number of the second mode parameter
NREG2	Number of the third mode parameter
NREGD	Number of the fourth mode parameter
PREG	Given value of the first mode parameter
PREG1	Given value of the second mode parameter
PREG2	Given value of the third mode parameter
PREGD	Given value of the fourth mode parameter
S1DAT.FOR	
KTS1S	Number of points in dependence of pressure recovery in primary nozzle
KTS1FI	Number of points in dependence of velocity factor in primary nozzle
KTS1MU	Number of points in dependence of gas flow factor in primary nozzle
NARGS1S	Number of parameter, which is used as argument in dependence of pressure recovery in primary nozzle
NARGS1FI	Number of parameter, which is used as argument in dependence of velocity factor in primary nozzle
NARGS1MU	Number of parameter, which is used as argument in dependence of gas flow factor in primary nozzle
S1SARGM(10)	Array of arguments in dependence of pressure recovery in primary nozzle
S1FIARGM(10)	Array of arguments in dependence of velocity factor in primary nozzle
S1MUARGM(10)	Array of arguments in dependence of gas flow factor in primary nozzle
SIGS1M(10)	Array of values in dependence of pressure recovery in primary nozzle
FIS1M(10)	Array of values in dependence of velocity factor in primary nozzle
MUS1M(10)	Array of Values in dependence of gas flow factor in primary nozzle
S2DAT.FOR	
KTS2S	Number of points in dependence of pressure recovery in secondary nozzle
KTS2FI	Number of points in dependence of velocity factor in secondary nozzle
KTS2MU	Number of points in dependence of gas flow factor in secondary nozzle
NARGS2S	Number of parameter, which is used as argument in dependence of pressure recovery in secondary nozzle
NARGS2FI	Number of parameter, which is used as argument in dependence of velocity factor in secondary nozzle

Continuation of Appendix F

1	2
NARGS2MU	Number of parameter, which is used as argument in dependence of gas flow factor in secondary nozzle
S2SARGM(10)	Array of arguments in dependence of pressure recovery in secondary nozzle
S2FIARGM(10)	Array of arguments in dependence of velocity factor in secondary nozzle
S2MUARGM(10)	Array of arguments in dependence of gas flow factor in secondary nozzle
SIGS2M(10)	Array of values in dependence of pressure recovery in secondary nozzle
FIS2M(10)	Array of values in dependence of velocity factor in secondary nozzle
MUS2M(10)	Array of Values in dependence of gas flow factor in secondary nozzle
S2DAT.FOR	
KTSS	Number of points in dependence of pressure recovery in common nozzle
KTSFI	Number of points in dependence of velocity factor in common nozzle
KTSMU	Number of points in dependence of gas flow factor in common nozzle
NARGS	Number of parameter, which is used as argument in dependence of pressure recovery in common nozzle
NARGSFI	Number of parameter, which is used as argument in dependence of velocity factor in common nozzle
NARGSMU	Number of parameter, which is used as argument in dependence of gas flow factor in common nozzle
SSARGM(10)	Array of arguments in dependence of pressure recovery in common nozzle
SFIARGM(10)	Array of arguments in dependence of velocity factor in common nozzle
SMUARGM(10)	Array of arguments in dependence of gas flow factor in common nozzle
SIGSM(10)	Array of values in dependence of pressure recovery in common nozzle
FISM(10)	Array of values in dependence of velocity factor in common nozzle
MUSM(10)	Array of Values in dependence of gas flow factor in common nozzle
SIGSFI.FOR	
SIGFI	Losses of total pressure because of velocity factor
SPR.FOR	
SZ	Given derivative, in case when rotational speed of any rotor is a mode parameter (for dynamics)
SZ1	Given derivative, in case when rotational speed of any rotor is a second mode parameter (for dynamics)
SWITCH.FOR	
PVNAKV	Sign to identify current position of HPC IGV
PPRPKV	Sign to identify current position of HPC ABVs
JVNAKV	Newly formed sign to identify current position of HPC IGV
JPRPKV	Newly formed sign to identify current position of HPC ABVs
PVNAKS	Sign to identify current position of IPC IGV
PPRPKS	Sign to identify current position of IPC ABVs
JVNAKS	Newly formed sign to identify current position of IPC IGV
JPRPKS	Newly formed sign to identify current position of IPC ABVs
PVNAKN	Sign to identify current position of LPC IGV
PPRPKN	Sign to identify current position of LPC ABVs
JVNAKN	Newly formed sign to identify current position of LPC IGV
JPRPKN	Newly formed sign to identify current position of LPC ABVs

Continuation of Appendix F

1	2
TERMO.FOR	
I(50)	Array of enthalpies at fifty stations according to temperatures described in Appendix E
QT(19:50)	Array of relative combustion products content in gas (the array starts from 19th station, which is the station at CC discharge, and to 50th station.
TOPL.FOR	
I0	Enthalpy of fuel at CC inlet
HU	Lower caloric value of fuel
L0	Stoichiometric ratio in case of total combustion of fuel
TXT.FOR	
TP(1500)	Array of text variables, which are names of gas dynamic parameters used for printing
TXTMKV.FOR	
TM(150)	Array of text variables, which are names of state parameters used for printing
UPRAVL.FOR	
IJOB	Study code
ICONF	Configuration code
ITEST	Sign that test data to make an identification was formed
IVAR	The variable is not used
IPP	The variable is not used
IZAPAS(8)	Reserved variables, which are not used
VHOD.FOR	
KTVHOD	Number of points in dependence of pressure recovery at engine inlet
NARGVHOD	Number of parameter, used as argument in dependence of pressure recovery at engine inlet
VARGM(10)	Array of parameters, used as arguments in dependence of pressure recovery at engine inlet
SIGVHM(10)	Array of pressure recoveries at engine inlet
WINTXT.FOR	
tttt(100)	Array of text variables, used to form dialogue windows
XKNDCDT.FOR	
KVTKNC(5)	Number of constant speed lines in the LPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKNDC(5)	Maximum number of points in constant speed lines of LPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKNDCM(5,24)	Number of points in each constant speed line of LPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
NKNDPCM(5,24),	Corrected rotational speed of each constant speed line of LPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)

Continuation of Appendix F

1	2
PIKNDCM (5,10000)	Values of pressure ratio in the performance of LPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
GKNDPCM (5,10000)	Values of corrected air flow in the performance of LPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
KPDKNCM (5,10000)	Values of efficiencies in the performance of LPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
VNAKNDCM(5)	Values of IGV installation angles which are used for stratifying the performances of LPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KUKNDC	Sign that identifies that performance of LPC were already inputted
XKNDODT.FOR	
KVTKNO(5)	Number of constant speed lines in the LPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKNDO(5)	Maximum number of points in constant speed lines of LPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKNDOM (5,24)	Number of points in each constant speed line of LPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
NKNDPOM (5,24)	Corrected rotational speed of each constant speed line of LPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
PIKNDOM (5,10000)	Values of pressure ratio in the performance of LPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
GKNDPOM (5,10000)	Values of corrected air flow in the performance of LPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
KPDKNOM (5,10000)	Values of efficiencies in the performance of LPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
VNAKNDOM(5)	Values of IGV installation angles which are used for stratifying the performances of LPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KUKNDO	Sign that identifies that performance of LPC were already inputted
XKSDCDT.FOR	
KVTKSC(5)	Number of constant speed lines in the IPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)

Continuation of Appendix F

1	2
KTKSDC(5)	Maximum number of points in constant speed lines of IPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKSDCM(5,24)	Number of points in each constant speed line of IPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
NKSDPCM(5,24)	Corrected rotational speed of each constant speed line of IPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
PIKSDCM (5,10000)	Values of pressure ratio in the performance of IPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
GKSDPCM (5,10000)	Values of corrected air flow in the performance of IPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
KPDKSCM (5,10000)	Values of efficiencies in the performance of IPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
VNAKSDCM(5)	Values of IGV installation angles which are used for stratifying the performances of IPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KUKSDC	Sign that identifies that performance of IPC were already inputted
XKSDODT.FOR	
KVTKSO(5)	Number of constant speed lines in the IPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKSDO(5)	Maximum number of points in constant speed lines of IPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKSDOM(5,24)	Number of points in each constant speed line of IPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
NKSDPOM(5,24)	Corrected rotational speed of each constant speed line of IPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
PIKSDOM (5,10000)	Values of pressure ratio in the performance of IPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
GKSDPOM (5,10000)	Values of corrected air flow in the performance of IPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)

Continuation of Appendix F

1	2
KPDKSOM (5,10000)	Values of efficiencies in the performance of IPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
VNAKSDOM(5)	Values of IGV installation angles which are used for stratifying the performances of IPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KUKSDO	Sign that identifies that performance of IPC were already inputted
XKVDCDT.FOR	
KVTKVC(5)	Number of constant speed lines in the HPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKVDC(5)	Maximum number of points in constant speed lines of HPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKVDCM (5,24)	Number of points in each constant speed line of HPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
NKVDPCM (5,24)	Corrected rotational speed of each constant speed line of HPC performance with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
PIKVDCM (5,10000)	Values of pressure ratio in the performance of HPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
GKVDPCM (5,10000)	Values of corrected air flow in the performance of HPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
KPDKVCM (5,10000)	Values of efficiencies in the performance of HPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
VNAKVDCM(5)	Values of IGV installation angles which are used for stratifying the performances of HPC with closed valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KUKVDC	Sign that identifies that performance of HPC were already inputted
XKVDODT.FOR	
KVTKVO(5)	Number of constant speed lines in the HPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKVDO(5)	Maximum number of points in constant speed lines of HPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KTKVDOM (5,24)	Number of points in each constant speed line of HPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)

Continuation of Appendix F

1	2
NKVDPOM (5,24)	Corrected rotational speed of each constant speed line of HPC performance with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 24 – maximum number of constant speed lines of single performance)
PIKVDOM (5,10000)	Values of pressure ratio in the performance of HPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
GKVDPOM (5,10000)	Values of corrected air flow in the performance of HPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
KPDKVOM (5,10000)	Values of efficiencies in the performance of HPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle, 10000 –maximum number of values for one performance)
VNAKVDOM(5),	Values of IGV installation angles which are used for stratifying the performances of HPC with open valves (five values that correspond to five performances, which are stratified according to IGV installation angle)
KUKVDO	Sign that identifies that performance of HPC were already inputted
XRPDAT.FOR	
NAKP1M	Number of argument that determines pressure recalculation coefficient at fan discharge (primary flow)
NAKT1M	Number of argument that determines temperature recalculation coefficient at fan discharge (primary flow)
KTRPKP1	Number of points in dependence of pressure recalculation coefficient at fan discharge (primary flow)
KTRPKT1	Number of points in dependence of temperature recalculation coefficient at fan discharge (primary flow)
NAKP2M	Number of argument that determines pressure recalculation coefficient at fan discharge (secondary flow)
NAKT2M	Number of argument that determines temperature recalculation coefficient at fan discharge (secondary flow)
KTRPKP2	Number of points in dependence of pressure recalculation coefficient at fan discharge (secondary flow)
KTRPKT2	Number of points in dependence of temperature recalculation coefficient at fan discharge (secondary flow)
KP1M(10)	Array with values of pressure recalculation coefficient at fan discharge (primary flow)
KT1M(10)	Array with values of temperature recalculation coefficient at fan discharge (primary flow)
AKP1M(10)	Array of arguments that correspond to values of pressure recalculation coefficient at fan discharge (primary flow)
AKT1M(10)	Array of arguments that correspond to values of temperature recalculation coefficient at fan discharge (primary flow)
KP2M(10)	Array with values of pressure recalculation coefficient at fan discharge (secondary flow)
KT2M(10)	Array with values of temperature recalculation coefficient at fan discharge (secondary flow)

Continuation of Appendix F

1	2
AKP2M(10)	Array of arguments that correspond to values of pressure recalculation coefficient at fan discharge (secondary flow)
AKT2M(10)	Array of arguments that correspond to values of temperature recalculation coefficient at fan discharge (secondary flow)
KPRP(1),	Pressure recalculation coefficient at fan discharge (primary flow)
KTRP(1)	Temperature recalculation coefficient at fan discharge (primary flow)
KPRP(2),	Pressure recalculation coefficient at fan discharge (secondary flow)
KTRP(2)	Temperature recalculation coefficient at fan discharge (secondary flow)
XTNDAT.FOR	
NPRTNM(15)	Array with corrected rotational speeds of LPT performance
PITNDM(225)	Array with pressure ratios of LPT performance
KPDTNM(225)	Array with efficiencies of LPT performance
ATNDM(225)	Array with corrected gas flow at LPT inlet
KVETTN	Number of constant speed lines in LPT performance
KTTND	Maximum number of points in single constant speed line of LPT performance
KTTNDM(15)	Array with numbers of points in each constant speed line of LPT performance
XTSDAT.FOR	
NPRTSM(15)	Array with corrected rotational speeds of IPT performance
PITSDM(225)	Array with pressure ratios of IPT performance
KPDTSM(225)	Array with efficiencies of IPT performance
ATSDM(225)	Array with corrected gas flow at IPT inlet
KVETTS	Number of constant speed lines in IPT performance
KTTSD	Maximum number of points in single constant speed line of IPT performance
KTTSDM(15)	Array with numbers of points in each constant speed line of IPT performance
XTVDAT.FOR	
NPRTVM(11)	Array with corrected rotational speeds of HPT performance
PITVDM(155)	Array with pressure ratios of HPT performance
KPDTVM(155)	Array with efficiencies of HPT performance
ATVDM(155)	Array with corrected gas flow at HPT inlet
KVETTV	Number of constant speed lines in HPT performance
KTTVD	Maximum number of points in single constant speed line of HPT performance
KTTVDM(11)	Array with numbers of points in each constant speed line of HPT performance

APPENDIX G
Parametric representation of unit performances

Unit	Paramet rame-ters	Equation for parameterized performance	Unit perfor-mance pa-rameter	Performance parameter number
1	2	3	4	5
Intake	σ_{IN}	$\sigma_{IN} = \sigma_{IN\ 0} \times \left(1 + a \frac{G_{a\ cor} - G_{a\ cor\ 0}}{\Delta G_{a\ cor}} + bM_{fl} + c \right)$	a b c	1 2 3
Fan (when fan has common perfor-mance for both flows)	π_F^*	$\pi_F^* = \pi_{F\ 0}^* Ч \left(1 + a \frac{n_{F\ cor} - n_{F\ cor\ 0}}{\Delta n_{F\ cor}} + b\left(Z_F - Z_{F\ 0}\right) + c \right)$	a b c	4 5 6
	$G_{F\ cor}$	$G_{F\ cor} = G_{F\ cor\ 0}\left(1 + c\right)$	c	7
	η_F^*	$\eta_F^* = \eta_{F\ 0}^* Ч \left(1 + a \frac{n_{F\ cor} - n_{F\ cpr\ 0}}{\Delta n_{F\ cor}} + b\left(Z_F - Z_{F\ 0}\right) + c \right)$	a b c	8 9 10
Fan (bypass duct)	π_{F2}^*	$\pi_{F2}^* = \pi_{F2\ 0}^* Ч \left(1 + a \frac{n_{F2\ cor} - n_{F2\ cor\ 0}}{\Delta n_{F2\ cor}} + b\left(Z_{F2} - Z_{F2\ 0}\right) + c \right)$	a b c	11 12 13
	$G_{F2\ cor}$	$G_{F2\ cor} = G_{F2\ cor\ 0}\left(1 + c\right)$	c	14
	η_{F2}^*	$\eta_{F2}^* = \eta_{F2\ 0}^* Ч \left(1 + a \frac{n_{F2\ cor} - n_{F2\ cpr\ 0}}{\Delta n_{F2\ cor}} + b\left(Z_{F2} - Z_{F2\ 0}\right) + c \right)$	a b c	15 16 17
Fan (primary duct)	π_{F1}^*	$\pi_{F1}^* = \pi_{F1\ 0}^* Ч \left(1 + a \frac{n_{F1\ cor} - n_{F1\ cor\ 0}}{\Delta n_{F1\ cor}} + b\left(Z_{F1} - Z_{F1\ 0}\right) + c \right)$	a b c	18 19 20
	$G_{F1\ cor}$	$G_{F1\ cor} = G_{F1\ cor\ 0}\left(1 + c\right)$	c	21
	η_{F1}^*	$\eta_{F1}^* = \eta_{F1\ 0}^* Ч \left(1 + a \frac{n_{F1\ cor} - n_{F1\ cpr\ 0}}{\Delta n_{F1\ cor}} + b\left(Z_{F1} - Z_{F1\ 0}\right) + c \right)$	a b c	22 23 24

Continuation of Appendix G

1	2	3	4	5
Fan (boosters)	π^*_B	$\pi^*_B=\pi^*_{B\,0}\,Ч\left(1+a\dfrac{n_{B\,cor}-n_{B\,cor\,0}}{\Delta n_{B\,cor}}+b\left(Z_B-Z_{B\,0}\right)+c\right)$	a b c	25 26 27
	$G_{B\,cor}$	$G_{B\,cor}=G_{B\,cor\,0}\left(1+c\right)$	c	28
	η^*_B	$\eta^*_B=\eta^*_{B\,0}\,Ч\left(1+a\dfrac{n_{B\,cor}-n_{B\,cpr\,0}}{\Delta n_{B\,cor}}+b\left(Z_B-Z_{B\,0}\right)+c\right)$	a b c	29 30 31
Intermediate pressure compressor	π^*_{IPC}	$\pi^*_{IPC}=\pi^*_{IPC\,0}\,Ч\left(1+a\dfrac{n_{IPC\,cor}-n_{IPC\,cor\,0}}{\Delta n_{IPC\,cor}}+b\left(Z_{IPC}-Z_{IPC\,0}\right)+c\right)$	a b c	32 33 34
	$G_{IPC\,cor}$	$G_{IPC\,cor}=G_{IPC\,cor\,0}\left(1+c\right)$	c	35
	η^*_{IPC}	$\eta^*_{IPC}=\eta^*_{IPC\,0}\,Ч\left(1+a\dfrac{n_{IPC\,cor}-n_{IPC\,cpr\,0}}{\Delta n_{IPC\,cor}}+b\left(Z_{IPC}-Z_{IPC\,0}\right)+c\right)$	a b c	36 37 38
High pressure com-pressor	π^*_{HPC}	$\pi^*_{HPC}=\pi^*_{HPC\,0}\,Ч\left(1+a\dfrac{n_{HPC\,cor}-n_{HPC\,cor\,0}}{\Delta n_{HPC\,cor}}+b\left(Z_{HPC}-Z_{HPC\,0}\right)+c\right)$	a b c	39 40 41
	$G_{HPC\,cor}$	$G_{HPC\,cor}=G_{HPC\,cor\,0}\left(1+c\right)$	c	42
	η^*_{HPC}	$\eta^*_{HPC}=\eta^*_{HPC\,0}\,Ч\left(1+a\dfrac{n_{HPC\,cor}-n_{HPC\,cor\,0}}{\Delta n_{HPC\,cor}}+b\left(Z_{HPC}-Z_{HPC\,0}\right)+c\right)$	a b c	43 44 45
High pressure turbine	A_{HPT}	$A_{HPT}=A^*_{HPT}\left(1+a\dfrac{\pi^*_{HPT}-\pi^*_{HPT\,0}}{\Delta\pi^*_{HPT}}+b\dfrac{n_{HPT\,cor}-n_{HPT\,cor\,0}}{\Delta n_{HPT\,cor}}+c\right)$	a b c	46 47 48
	η^*_{HPT}	$\eta^*_{HPT}=\eta^*_{HPT\,0}\left(1+a\dfrac{\pi^*_{HPT}-\pi^*_{HPT\,0}}{\Delta\pi^*_{HPT}}+b\dfrac{n_{HPT\,cor}-n_{HPT\,cor\,0}}{\Delta n_{HPT\,cor}}+c\right)$	a b c	49 50 51

Continuation of Appendix G

1	2	3	4	5
Intermediate pressure turbine	A_{IPT}	$A_{IPT}=A^*_{IPT}\left(1+a\frac{\pi^*_{IPT}-\pi^*_{IPT\ 0}}{\Delta\pi^*_{IPT}}+b\frac{n_{IPT\ cor}-n_{IPT\ cor\ 0}}{\Delta n_{IPT\ cor}}+c\right)$	a b c	52 53 54
	η^*_{IPT}	$\eta^*_{IPT}=\eta^*_{IPT\ 0}\left(1+a\frac{\pi^*_{IPT}-\pi^*_{IPT\ 0}}{\Delta\pi^*_{IPT}}+b\frac{n_{IPT\ cor}-n_{IPT\ cor\ 0}}{\Delta n_{IPT\ cor}}+c\right)$	a b c	55 56 57
Low pressure turbine	A_{LPT}	$A_{LPT}=A^*_{LPT}\left(1+a\frac{\pi^*_{LPT}-\pi^*_{LPT\ 0}}{\Delta\pi^*_{LPT}}+b\frac{n_{LPT\ cor}-n_{LPT\ cor\ 0}}{\Delta n_{LPT\ cor}}+c\right)$	a b c	58 59 60
	η^*_{LPT}	$\eta^*_{LPT}=\eta^*_{LPT\ 0}\left(1+a\frac{\pi^*_{LPT}-\pi^*_{LPT\ 0}}{\Delta\pi^*_{LPT}}+b\frac{n_{LPT\ cor}-n_{LPT\ cor\ 0}}{\Delta n_{LPT\ cor}}+c\right)$	a b c	61 62 63
Combustion chamber	σ_{CC}	$\sigma_{CC}=\sigma_{CC\ 0}\left(1+a\frac{\Delta T^*_{CC}-\Delta T^*_{CC\ 0}}{\Delta T^*_{CC\ 0}}+b\frac{\lambda_{CC}-\lambda_{CC\ 0}}{\Delta\lambda_{CC}}+c\right)$	a b c	64 65 66
	η_{CC}	$\eta_{CC}=\eta_{CC\ 0}\left(1+a\frac{\Delta T^*_{CC}-\Delta T^*_{CC\ 0}}{\Delta T^*_{CC\ 0}}+b\frac{\lambda_{CC}-\lambda_{CC\ 0}}{\Delta\lambda_{CC}}+c\right)$	a b c	67 68 69
Transition duct between LPC and IPC	$\sigma_{trans\ 1}$	$\sigma_{trans\ 1}=\sigma_{trans\ 1\ 0}\left(1+a\frac{\lambda_{trans\ 1}-\lambda_{trans\ 1\ 0}}{\Delta\lambda_{trans\ 1}}+c\right)$	a c	82 83
Transition duct between IPC and HPC	$\sigma_{trans\ 2}$	$\sigma_{trans\ 2}=\sigma_{trans\ 2\ 0}\left(1+a\frac{\lambda_{trans\ 2}-\lambda_{trans\ 2\ 0}}{\Delta\lambda_{trans\ 2}}+c\right)$	a c	84 85
Transition duct between HPT and IPT	$\sigma_{trans\ 3}$	$\sigma_{trans\ 3}=\sigma_{trans\ 3\ 0}\left(1+a\frac{\lambda_{trans\ 3}-\lambda_{trans\ 3\ 0}}{\Delta\lambda_{trans\ 3}}+c\right)$	a c	86 87

Continuation of Appendix G

1	2	3	4	5
Transition duct between IPT and LPT	$\sigma_{trans\ 4}$	$\sigma_{trans\ 4} = \sigma_{trans\ 4\ 0}\left(1 + a\frac{\lambda_{trans\ 4} - \lambda_{trans\ 4\ 0}}{\Delta\lambda_{trans\ 4}} + c\right)$	a c	88 89
Transi-tion duct in bypass	$\sigma_{trans\ 5}$	$\sigma_{trans\ 5} = \sigma_{trans\ 5\ 0}\left(1 + a\frac{\lambda_{trans\ 5} - \lambda_{trans\ 5\ 0}}{\Delta\lambda_{trans\ 5}} + c\right)$	a c	90 91
Primary nozzle	$\sigma_{N\ 1}$	$\sigma_{N\ 1} = \sigma_{N\ 1\ 0}\left(1 + a\frac{\lambda_{N\ 1} - \lambda_{N\ 1\ 0}}{\Delta\lambda_{N\ 1}} + c\right)$	a c	92 93
	$\phi_{N\ 1}$	$\varphi_{N\ 1} = \varphi_{N\ 1\ 0}\left(1 + a\frac{\lambda_{N\ 1} - \lambda_{N\ 1\ 0}}{\Delta\lambda_{N\ 1}} + c\right)$	a c	94 95
	$\mu_{N\ 1}$	$\mu_{N\ 1} = \mu_{N\ 1\ 0}\left(1 + a\frac{\lambda_{N\ 1} - \lambda_{N\ 1\ 0}}{\Delta\lambda_{N\ 1}} + bM_{fl} + c\right)$	a b c	96 97 98
Secondary nozzle	$\sigma_{N\ 2}$	$\sigma_{N\ 2} = \sigma_{N\ 2\ 0}\left(1 + a\frac{\lambda_{N\ 2} - \lambda_{N\ 2\ 0}}{\Delta\lambda_{N\ 2}} + c\right)$	a c	99 100
	$\phi_{N\ 2}$	$\varphi_{N\ 2} = \varphi_{N\ 2\ 0}\left(1 + a\frac{\lambda_{N\ 2} - \lambda_{N\ 2\ 0}}{\Delta\lambda_{N\ 2}} + c\right)$	a c	101 102
	$\mu_{N\ 2}$	$\mu_{N\ 2} = \mu_{N\ 2\ 0}\left(1 + a\frac{\lambda_{N\ 2} - \lambda_{N\ 2\ 0}}{\Delta\lambda_{N\ 2}} + bM_{fl} + c\right)$	a b c	103 104 105
Common nozzle	σ_{N}	$\sigma_{N} = \sigma_{N\ 0}\left(1 + a\frac{\lambda_{N} - \lambda_{N\ 0}}{\Delta\lambda_{N}} + c\right)$	a c	106 107
	ϕ_{N}	$\varphi_{N} = \varphi_{N\ 0}\left(1 + a\frac{\lambda_{N} - \lambda_{N\ 0}}{\Delta\lambda_{N}} + c\right)$	a c	108 109
	μ_{N}	$\mu_{N} = \mu_{N\ 0}\left(1 + a\frac{\lambda_{N} - \lambda_{N\ 0}}{\Delta\lambda_{N}} + bM_{fl} + c\right)$	a b c	110 111 112
Mixer	σ_{MIX}	$\sigma_{MIX} = \sigma_{MIX\ 0}\left(1 + a\frac{\lambda_{N\ 1} - \lambda_{N\ 1\ 0}}{\Delta\lambda_{N\ 1}} + b\frac{\lambda_{N\ 2} - \lambda_{N\ 2\ 0}}{\Delta\lambda_{N\ 2}} + c\right)$	a b c	113 114 115

Continuation of Appendix G

1		2	3	4	5
Air heating degree		τ_F	$\tau_F=\tau_{F0}\left(1+a\frac{\pi^*_{F2}-\pi^*_{F20}}{\Delta\pi^*_{F2}}+c\right)$	a c	116 118
		τ_{IPC}	$\tau_{IPC}=\tau_{IPC0}\times\left(1+a\frac{\pi^*_{IPC}-\pi^*_{IPC0}}{\Delta\pi^*_{IPC}}+c\right)$	a c	119 121
Heating degree of bleed air from:	compressor	τ_{HPT}	$\tau_{HPT}=\tau_{HPT0}\left(1+a\frac{\pi^*_{HPT}-\pi^*_{HPT0}}{\Delta\pi^*_{HPT}}+c\right)$	a c	122 124
	combustion chamber	τ_{CC}	$\tau_{CC}=\tau_{CC0}\left(1+a\frac{\pi^*_{CC}-\pi^*_{CC0}}{\Delta\pi^*_{CC0}}+c\right)$	a c	125 126
Air bleeds		δG_1	$\delta G_1=\delta G_{10}+C$	c	127
		δG_2	$\delta G_2=\delta G_{20}+C$	c	128
		δG_3	$\delta G_3=\delta G_{30}+C$	c	129
		δG_4	$\delta G_4=\delta G_{40}+C$	c	130
		δG_5	$\delta G_5=\delta G_{50}+C$	c	131
		δG_6	$\delta G_6=\delta G_{60}+C$	c	132
		δG_7	$\delta G_7=\delta G_{70}+C$	c	133
		δG_8	$\delta G_8=\delta G_{80}+C$	c	134